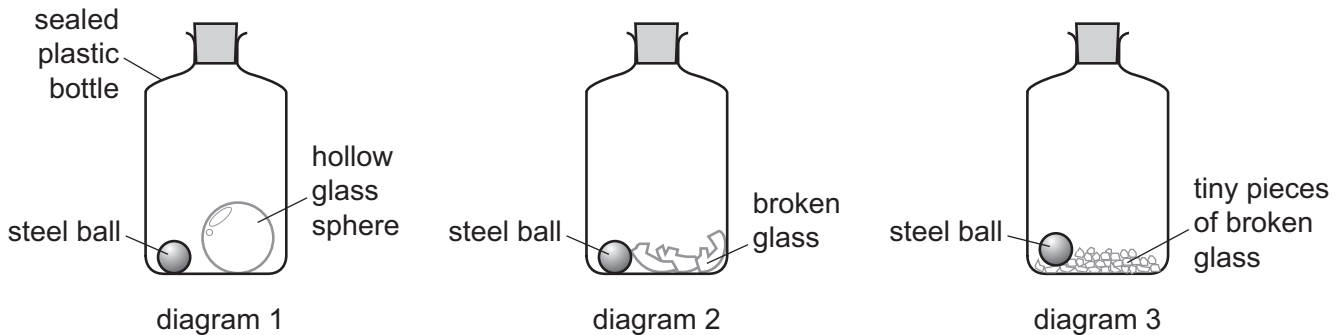


- 4 Diagram 1 shows a sealed plastic bottle containing a hollow glass sphere and a steel ball.

Diagram 2 shows the same bottle after it has been shaken.

Diagram 3 shows the same bottle after it has been shaken again until the broken glass is in tiny pieces.



The mass of the bottle and contents in diagram 1 is m_1 .

The mass of the bottle and contents in diagram 2 is m_2 .

The mass of the bottle and contents in diagram 3 is m_3 .

Which statement gives the correct relation between m_1 , m_2 and m_3 ?

- A m_1 is equal to m_2 and m_2 is equal to m_3 .
- B m_1 is greater than m_2 and m_2 is greater than m_3 .
- C m_1 is less than m_2 and m_2 is greater than m_3 .
- D m_1 is less than m_2 and m_2 is less than m_3 .

- 5 On Mars, the acceleration of free fall g is 3.7 m/s^2 .

What is the weight of a 2.0 kg mass on Mars?

- A 0.54 N B 1.9 N C 7.4 N D 20 N

- 6 The mass of an empty flask is 34 g .

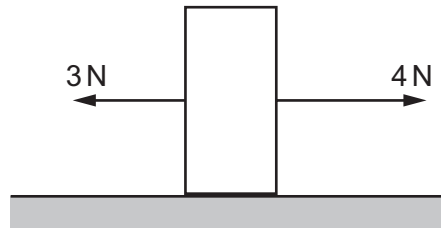
The volume of liquid added to the flask is 20 cm^3 .

The total mass of the flask and the liquid is 50 g .

What is the density of the liquid?

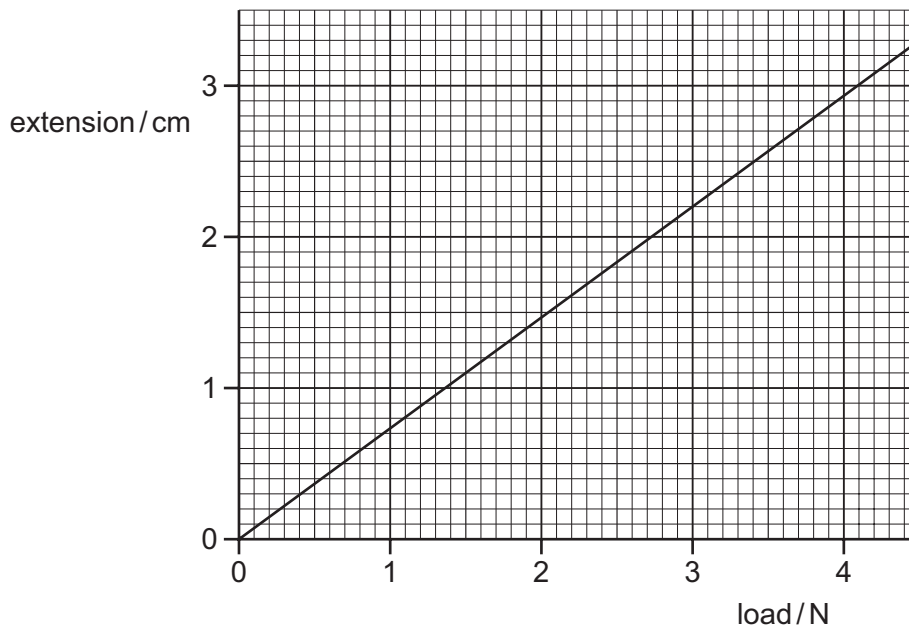
- A 0.80 g/cm^3 B 1.25 g/cm^3 C 2.50 g/cm^3 D 4.20 g/cm^3

- 7 The diagram shows a solid object on a flat surface, with two forces acting on the object.



What is the resultant force on the object?

- A 1 N to the left
 - B 1 N to the right
 - C 7 N to the left
 - D 7 N to the right
- 8 The extension–load graph for a spring is shown. The unstretched length of the spring is 17.0 cm.



When an object is suspended from the spring, the length of the spring is 19.2 cm.

What is the weight of the object?

- A 1.4 N
 - B 1.6 N
 - C 2.6 N
 - D 3.0 N
- 9 Which situation involves no work being done and no energy being transferred?
- A a car skidding to a stop on a road
 - B a crane lifting a load
 - C a heavy load hanging from a strong bar
 - D a student dragging a big box over a rough floor

- 10** A student suggests that there are several ways of transferring energy to a small, stationary block of iron on a smooth table. He makes the following suggestions.

- 1 Heat it.
- 2 Shine light on it.
- 3 Pass a current through it.

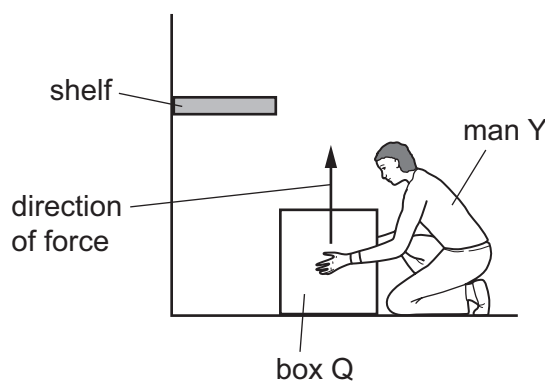
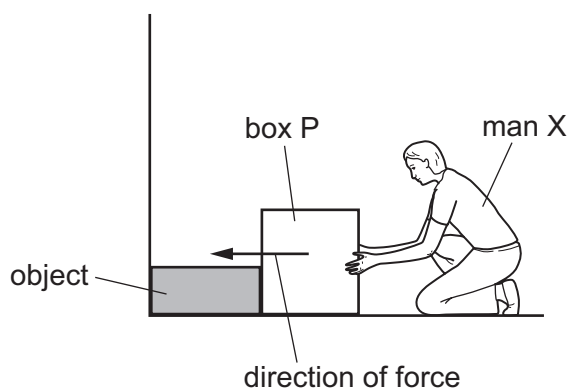
Which suggestions are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 11** Two men, X and Y, try to move identical heavy boxes, P and Q.

Man X tries to push box P along the floor. The box does not move because an object is in the way.

Man Y lifts box Q from the floor onto a shelf.



Which man does the most work on their box, and which box gains the most energy?

	man doing most work	box gaining most energy
A	X	P
B	X	Q
C	Y	P
D	Y	Q

- 12** A book has a mass of 400 g.

The surface of the book in contact with a table has dimensions $0.10 \text{ m} \times 0.20 \text{ m}$.

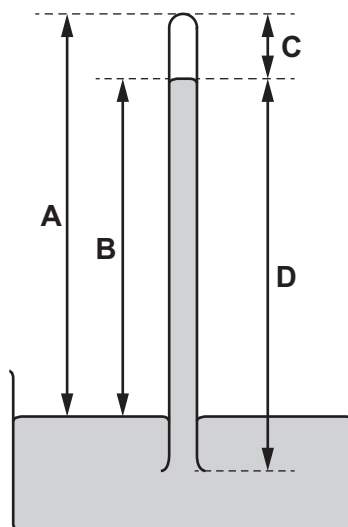
The gravitational field strength g is 10 N/kg .

What is the pressure exerted on the table due to the book?

- A** 0.08 N/m^2 **B** 8.0 N/m^2 **C** 20 N/m^2 **D** 200 N/m^2

- 13** A mercury barometer is used to measure atmospheric pressure.

Which distance gives a measure of the atmospheric pressure?



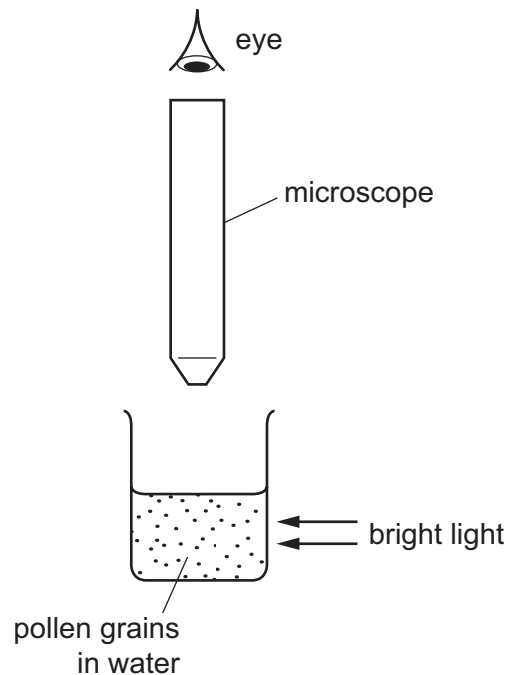
- 14** Air trapped in a sealed bottle increases in temperature.

Which row shows the changes that occur to the trapped air?

	pressure of the air	average speed of molecules in the air
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 15 Very small pollen grains are suspended in water. A bright light shines from the side.

When looked at through a microscope, small specks of light are seen to be moving in a random, jerky manner.



What are the moving specks of light?

- A pollen grains being hit by other pollen grains
 - B pollen grains being hit by water molecules
 - C water molecules being hit by other water molecules
 - D water molecules being hit by pollen grains
- 16 Why are small gaps left between the metal rails of a railway track?
- A to allow for expansion of the rails on a hot day
 - B to allow for contraction of the rails on a hot day
 - C to allow for expansion of the rails on a cold day
 - D to allow for contraction of the rails on a cold day
- 17 All thermometers require a physical property that changes with temperature.

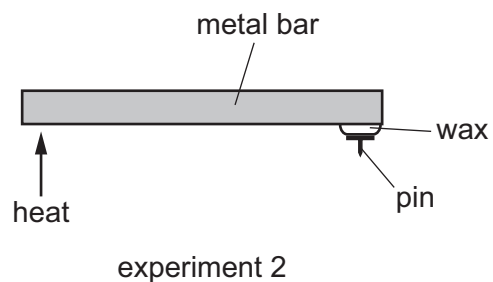
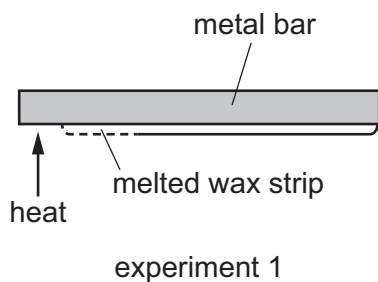
Which property would **not** be suitable for use in a thermometer?

- A pressure
- B volume
- C electrical resistance
- D mass

- 18 The thermal capacity of object Y is greater than that of object Z.

What is a consequence of this?

- A Object Y needs less thermal energy to melt it than object Z.
 - B Object Y needs less thermal energy to raise its temperature by 1°C than object Z.
 - C Object Y needs more thermal energy to melt it than object Z.
 - D Object Y needs more thermal energy to raise its temperature by 1°C than object Z.
- 19 Two students carry out different experiments to compare the abilities of different metals to conduct thermal energy.



In experiment 1, the bar is heated for one minute and the length of wax strip that melts is measured.

In experiment 2, the bar is heated and the time taken for the pin to drop off is measured.

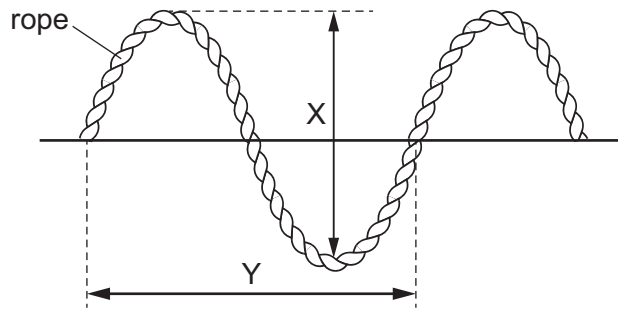
What happens to each of these measurements when a better conductor of thermal energy is tested?

	length of melted wax strip	time taken for the pin to drop
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 20 In which substances is convection a method of thermal energy transfer?

- A air and water only
- B air only
- C air, water and wood
- D water only

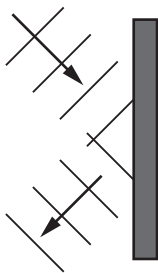
21 The diagram represents a wave on a rope.



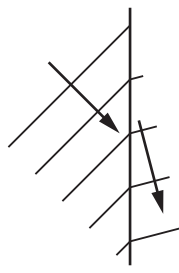
Which type of wave is shown and which labelled arrow shows the wavelength of the wave?

	type of wave	wavelength
A	longitudinal	X
B	longitudinal	Y
C	transverse	X
D	transverse	Y

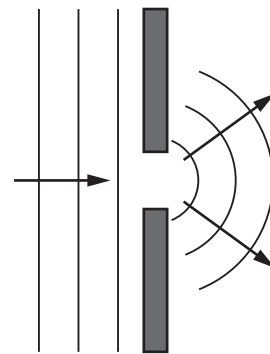
22 The diagrams show water waves in three different situations. The arrows show the direction of travel of the waves.



1



2

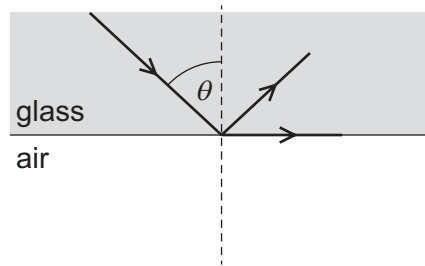


3

What does each diagram show?

	1	2	3
A	diffraction	reflection	refraction
B	diffraction	refraction	reflection
C	reflection	diffraction	refraction
D	reflection	refraction	diffraction

- 23** The diagram shows a narrow beam of light incident on a glass-air boundary. Some of the light emerges along the surface of the glass and some is reflected back into the glass.

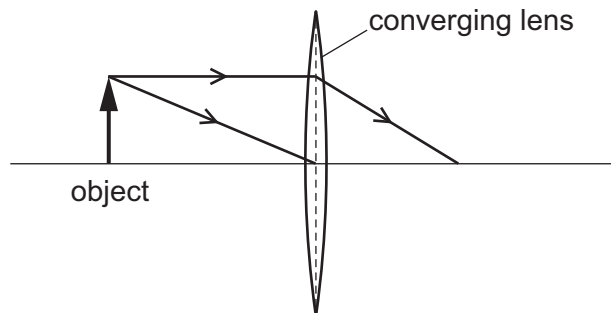


Which row is correct?

	this is an example of total internal reflection	angle θ is the critical angle
A	no	yes
B	no	no
C	yes	no
D	yes	yes

- 24** An object is placed in front of a thin converging lens.

The diagram shows the paths of two rays from the top of the object.



An image of the object is formed on a screen to the right of the lens.

How does this image compare with the object?

- A** It is larger and inverted.
- B** It is larger and the same way up.
- C** It is smaller and inverted.
- D** It is smaller and the same way up.

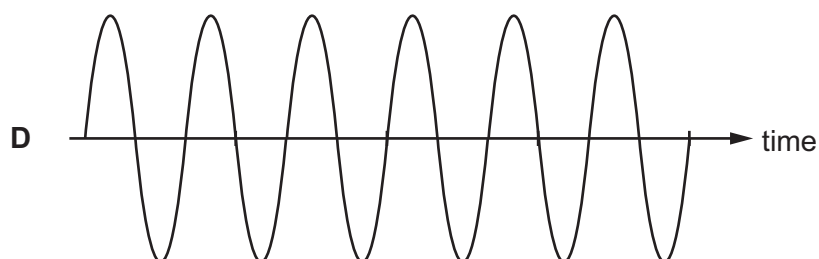
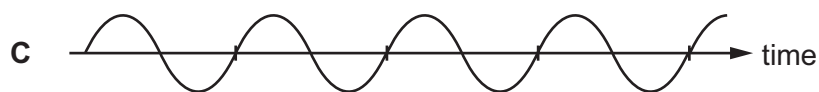
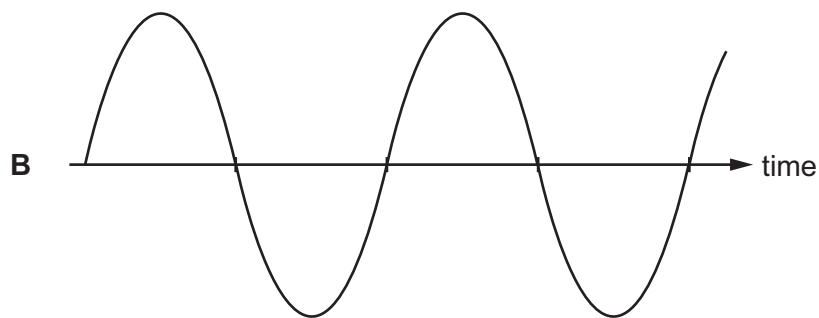
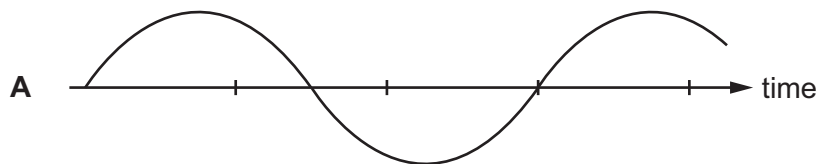
25 A student is asked to give two uses of four different types of electromagnetic radiation.

In which row are both the uses correct?

	radiation	use 1	use 2
A	radio	sterilising medical equipment	mobile phone masts
B	microwaves	mobile phones	sterilising medical equipment
C	infrared	remote controllers	intruder alarms
D	X-rays	security in airports	intruder alarms

26 The diagrams represent the waves produced by four sources of sound. The scales are the same for all the diagrams.

Which sound has the highest frequency?



- 27** Diagram 1 shows a small compass needle with its poles marked. It is not near any magnetic materials.

Diagram 2 shows a bar magnet with its poles marked. The compass needle is placed at point P.

diagram 1



diagram 2

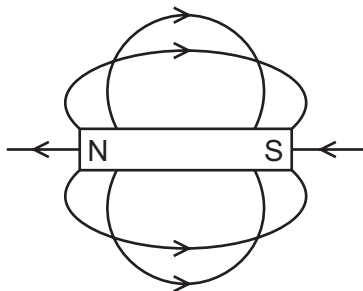


• P

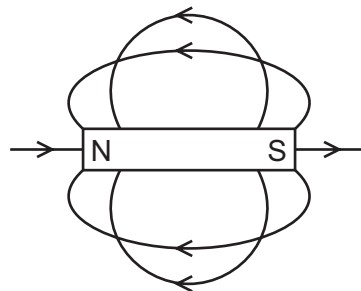
In which direction will the N pole of the compass needle point?

- A** to the left
 - B** to the right
 - C** down the page
 - D** up the page
- 28** Which diagram shows the pattern and direction of the magnetic field lines around a bar magnet?

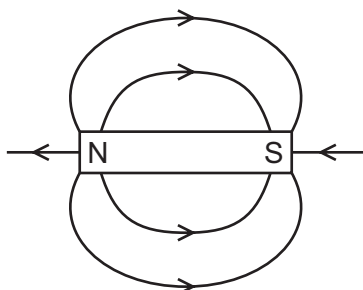
A



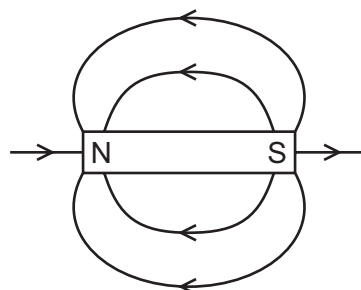
B



C



D



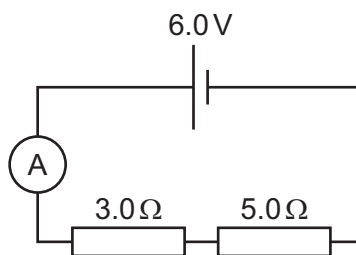
- 29 A student rubs a plastic rod with a cloth.

The rod becomes positively charged.

What has happened to the rod?

- A It has gained electrons.
- B It has gained protons.
- C It has lost electrons.
- D It has lost protons.

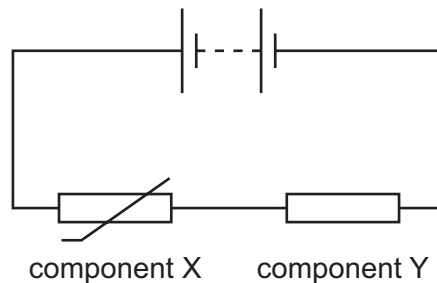
- 30 The circuit diagram shows a cell connected to an ammeter and two resistors.



What is the current in the circuit?

- A 0.75 A
- B 1.3 A
- C 12 A
- D 48 A

- 31 The diagram shows an electrical circuit.



Which row describes what happens when the temperature rises?

	resistance of component X	potential difference (p.d.) across component Y
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 32 Diagram 1 shows a resistor connected in a circuit. Diagram 2 shows an identical resistor connected in parallel with the first one.

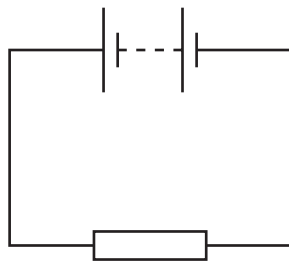


diagram 1

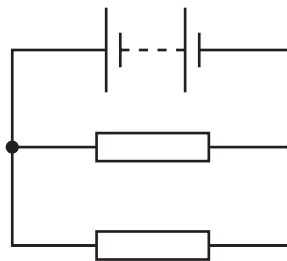


diagram 2

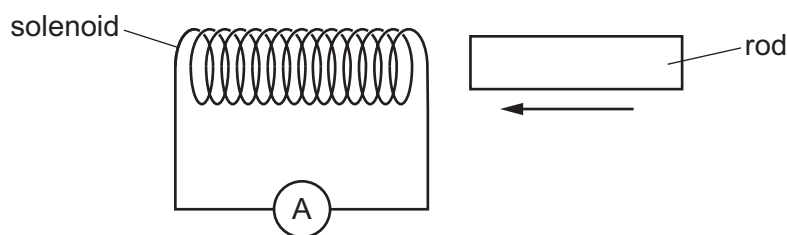
What is the combined resistance of the two resistors?

- A greater than in the circuit of diagram 1
 - B less than in the circuit of diagram 1
 - C the same as in the circuit of diagram 1
 - D zero
- 33 The current in a lamp connected on its own to the mains supply is 0.60 A.

A table decoration has three of these lamps connected in parallel.

Which rating of fuse is suitable to protect this circuit?

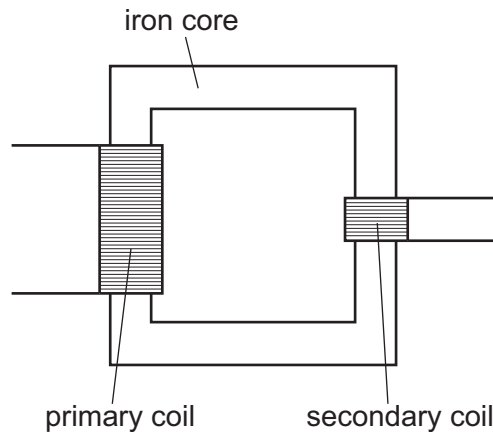
- A 0.2 A
 - B 0.6 A
 - C 1.0 A
 - D 5.0 A
- 34 A solenoid is connected to a very sensitive ammeter. A rod is inserted into one end of the solenoid. The ammeter shows that there is a small electric current in the solenoid while the rod is moving.



Which rod is being inserted?

- A a heated copper rod
- B a magnetised steel rod
- C an uncharged nylon rod
- D a radioactive uranium rod

- 35** The diagram shows a transformer. There are 460 turns on the primary coil and 24 turns on the secondary coil. The primary voltage is 230 V.



What is the secondary voltage?

- A** 2.0 V **B** 12 V **C** 48 V **D** 4400 V
- 36** A class is designing a d.c. motor. To achieve a greater turning effect, three suggestions are made.
- 1 Have a larger current in the coil of the motor.
 - 2 Have a stronger magnet in the motor.
 - 3 Put a larger number of turns on the coil.

Which suggestions will help to increase the turning effect?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only
- 37** The charge on a proton is e .

What is the charge on an electron and what is the charge on a neutron?

	electron	neutron
A	e	e
B	e	0
C	$-e$	$-e$
D	$-e$	0

38 The nuclide notation of the isotope strontium-90 is $^{90}_{38}\text{Sr}$.

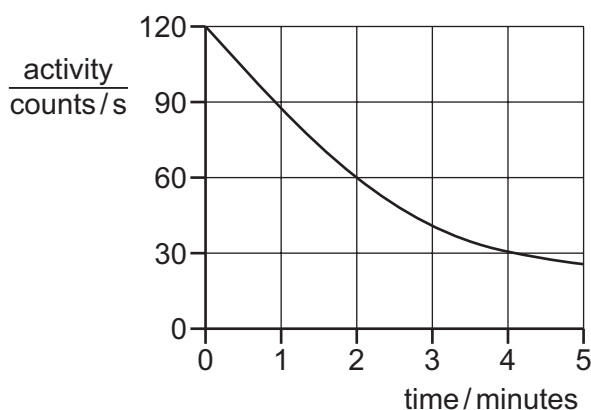
Which statement is correct?

- A A nucleus of strontium-90 has 38 neutrons.
- B A nucleus of strontium-90 has 52 neutrons.
- C A nucleus of strontium-90 has 90 electrons.
- D A nucleus of strontium-90 has 90 neutrons.

39 Which statement about α -particles and γ -rays is correct?

- A α -particles are a form of electromagnetic radiation.
- B α -particles penetrate materials more easily than γ -rays.
- C The emission of an α -particle produces a nucleus of a different element.
- D γ -rays are more ionising than α -particles.

40 The graph shows the activity of a radioactive source over a period of time.



What is the half-life of the source?

- A 1.0 minute
- B 2.0 minutes
- C 2.5 minutes
- D 4.0 minutes

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Question	Answer	Marks
1	C	1
2	A	1
3	B	1
4	A	1
5	B	1
6	D	1
7	D	1
8	D	1
9	B	1
10	A	1
11	D	1
12	D	1
13	A	1
14	A	1
15	B	1
16	C	1
17	A	1
18	D	1
19	C	1
20	A	1
21	B	1
22	C	1
23	B	1
24	A	1
25	C	1
26	D	1
27	A	1
28	D	1

Question	Answer	Marks
29	C	1
30	C	1
31	C	1
32	B	1
33	D	1
34	B	1
35	C	1
36	C	1
37	D	1
38	A	1
39	C	1
40	B	1



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Question	Answer	Marks
1	A	1
2	C	1
3	A	1
4	A	1
5	B	1
6	D	1
7	B	1
8	D	1
9	A	1
10	A	1
11	D	1
12	D	1
13	D	1
14	C	1
15	B	1
16	A	1
17	D	1
18	D	1
19	C	1
20	B	1
21	D	1
22	B	1
23	A	1
24	B	1
25	A	1
26	D	1
27	D	1
28	A	1

Question	Answer	Marks
29	C	1
30	C	1
31	C	1
32	C	1
33	B	1
34	B	1
35	C	1
36	A	1
37	D	1
38	D	1
39	A	1
40	B	1



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Question	Answer	Marks
1	C	1
2	A	1
3	A	1
4	A	1
5	C	1
6	A	1
7	B	1
8	D	1
9	C	1
10	D	1
11	D	1
12	D	1
13	B	1
14	D	1
15	B	1
16	A	1
17	D	1
18	D	1
19	C	1
20	A	1
21	D	1
22	D	1
23	A	1
24	A	1
25	C	1
26	D	1
27	B	1
28	C	1

Question	Answer	Marks
29	C	1
30	A	1
31	B	1
32	B	1
33	D	1
34	B	1
35	B	1
36	A	1
37	D	1
38	B	1
39	C	1
40	B	1



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0625/13

Paper 1 Multiple Choice (Core)

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **20** pages. Blank pages are indicated.



- 1 Diagram 1 shows a solid, rectangular-sided block.

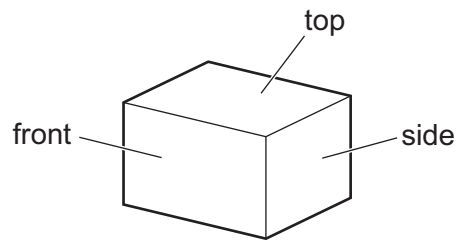


diagram 1

Diagram 2 shows the same block from the front and from the side.

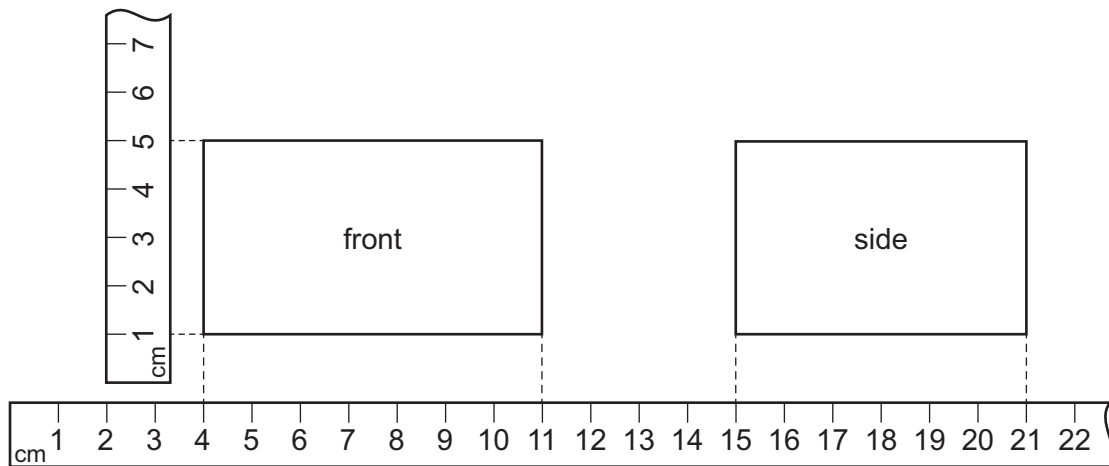


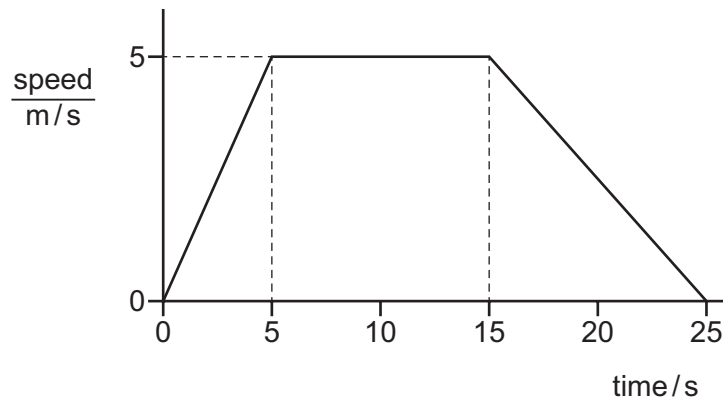
diagram 2

Metre rules have been shown close to the edges of the block.

What is the volume of the block?

- A** 120 cm^3 **B** 168 cm^3 **C** 264 cm^3 **D** 1155 cm^3

- 2 The speed–time graph shows the motion of an object.



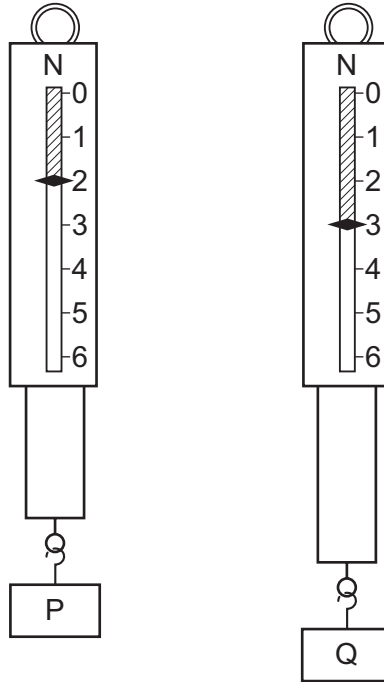
How far does the object travel at constant speed?

- A** 25 m **B** 50 m **C** 75 m **D** 125 m

3 Which statement about acceleration is correct?

- A It is related to the changing speed of an object.
- B It is the distance an object travels in one second.
- C It is the force acting on an object divided by the distance it travels in one second.
- D It is the force acting on an object when it is near to the Earth.

4 Two metal blocks P and Q have identical dimensions. They hang on identical spring balances.



Which statement about P and Q is correct?

- A They have different volumes and different weights.
- B They have different volumes and equal weights.
- C They have equal volumes and equal weights.
- D They have equal volumes and different weights.

5 Which row gives the correct weight for the mass shown?

The value of g is 10 N/kg .

	mass / kg	weight / N
A	2	20
B	10	1
C	10	10
D	20	2

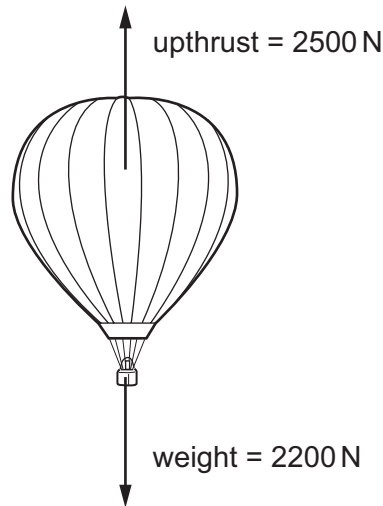
- 6 A rectangular gymnasium is 50 m long, 25 m wide and 8.0 m high.

The density of air is 1.2 kg/m^3 .

What is the best estimate of the mass of air in the gymnasium?

- A** 0.00012 kg **B** 100 kg **C** 8300 kg **D** 12 000 kg

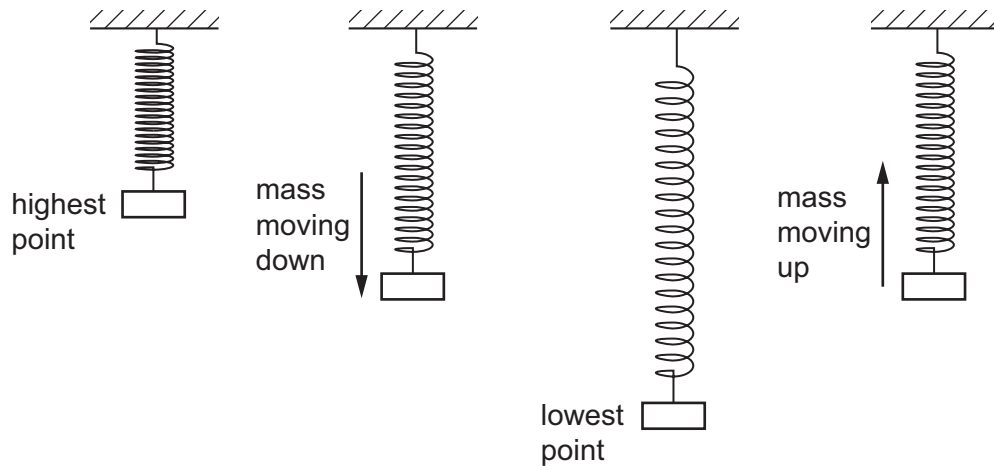
- 7 The diagram shows two of the three vertical forces acting on a hot-air balloon. The hot-air balloon is moving upwards at constant speed.



What is the air resistance acting on the hot-air balloon?

- A** 300 N downwards
B 300 N upwards
C 4700 N downwards
D 4700 N upwards
- 8 Which statement about the moment of a force is **not** correct?
- A** If an object is balanced about a pivot the resultant moment on the object must be zero.
B The moment of a force is a measure of its turning effect.
C The moment of a force about a point is equal to: force $\times$ perpendicular distance from the point.
D The moment of a force about a point increases when the perpendicular distance of the force from the point decreases.

- 9 A mass bounces up and down on a steel spring. The diagram shows the mass and the spring at different points during the motion.



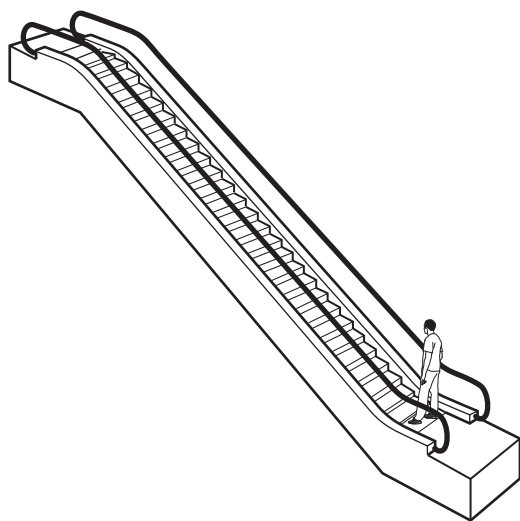
At which point does the mass have the least gravitational potential energy and at which point is the most elastic energy stored in the spring?

	least amount of gravitational potential energy	most elastic energy stored in spring
A	mass moving down	mass moving up
B	mass moving down	lowest point
C	lowest point	mass moving up
D	lowest point	lowest point

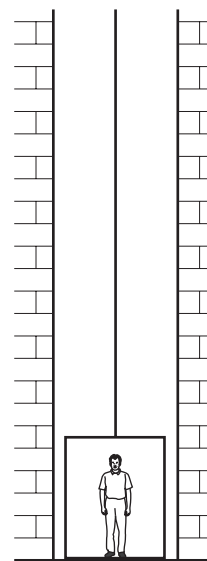
- 10 Which energy resource stores kinetic energy?

- A** coal
- B** nuclear fission
- C** solar
- D** wind

- 11 A man can either take an escalator or a lift to travel up between two floors in a hotel.



escalator



lift

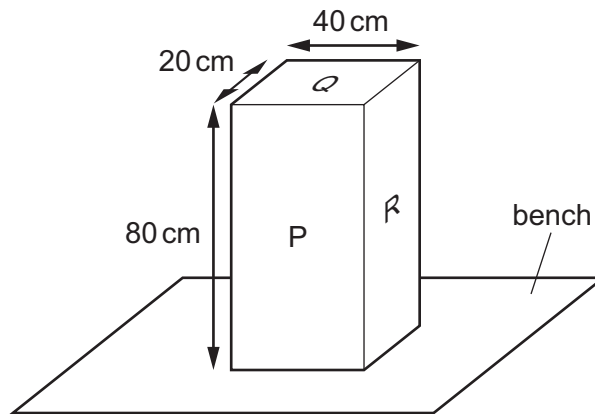
The escalator takes 20 seconds to carry the man between the two floors. The useful work done against gravity is W . The useful power developed is P .

The lift takes 30 seconds to carry the same man between the same two floors.

How much useful work against gravity is done by the lift, and how much useful power is developed by the lift?

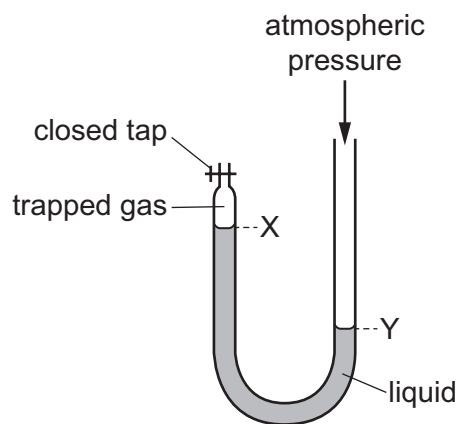
	useful work done against gravity by lift	useful power developed by lift
A	more than W	less than P
B	more than W	P
C	W	less than P
D	W	P

- 12 The diagram shows a solid block resting on a bench. The dimensions of the block are shown.



On which labelled surface should the block rest to produce the smallest pressure on the bench?

- A P
 - B Q
 - C R
 - D P, Q and R produce the same pressure
- 13 The diagram shows a U-shaped glass tube, closed at one end by a tap. The glass tube contains a liquid as shown.

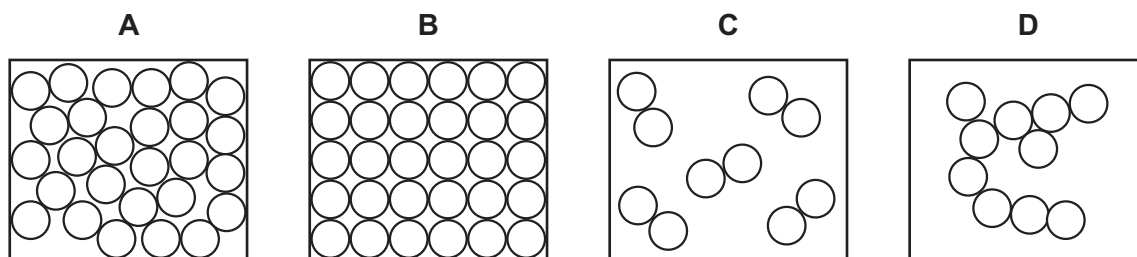


Some of the trapped gas is removed.

What will happen to the levels X and Y?

	level X	level Y
A	higher	lower
B	higher	higher
C	lower	higher
D	lower	lower

14 What is the correct particle diagram for a solid?



15 A sealed box contains a fixed mass of gas.

Which action results in each molecule of the gas colliding with the walls of the container less frequently and with a smaller force?

- A Decrease the temperature of the gas.
- B Decrease the volume of the container.
- C Increase the temperature of the gas.
- D Increase the volume of the container.

16 The diagram shows a liquid-in-glass thermometer.



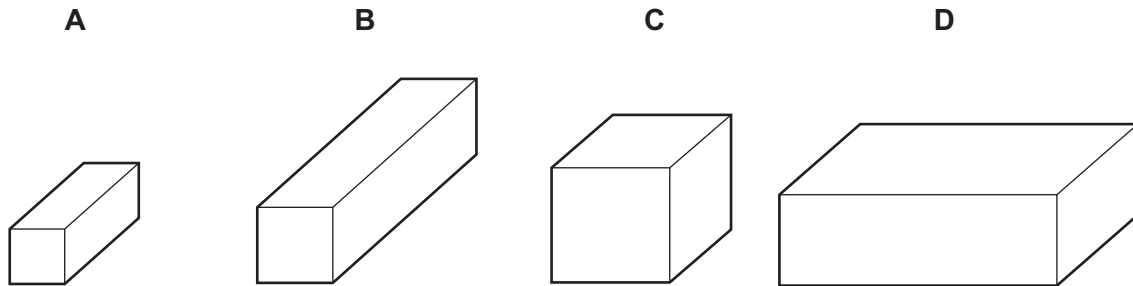
Which row gives the correct labels for the thermometer?

	X	Y
A	water	narrow tube of uniform diameter
B	alcohol	narrow tube of uniform diameter
C	water	this end immersed in substance to be measured
D	alcohol	this end immersed in substance to be measured

- 17 The diagrams show four blocks of steel. The blocks are all drawn to the same scale.

The same quantity of thermal energy is given to each block.

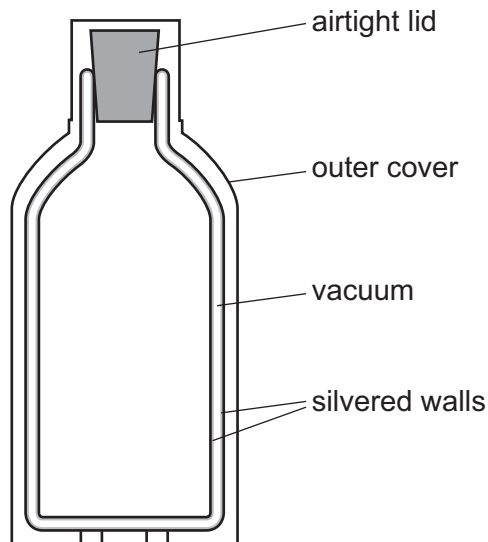
Which block shows the greatest rise in temperature?



- 18 A gas is cooled so that its molecules move more slowly, come closer together and do not move freely.

Which process is being described?

- A boiling
 - B condensing
 - C freezing
 - D melting
- 19 The diagram shows a vacuum flask used to keep a liquid warm.



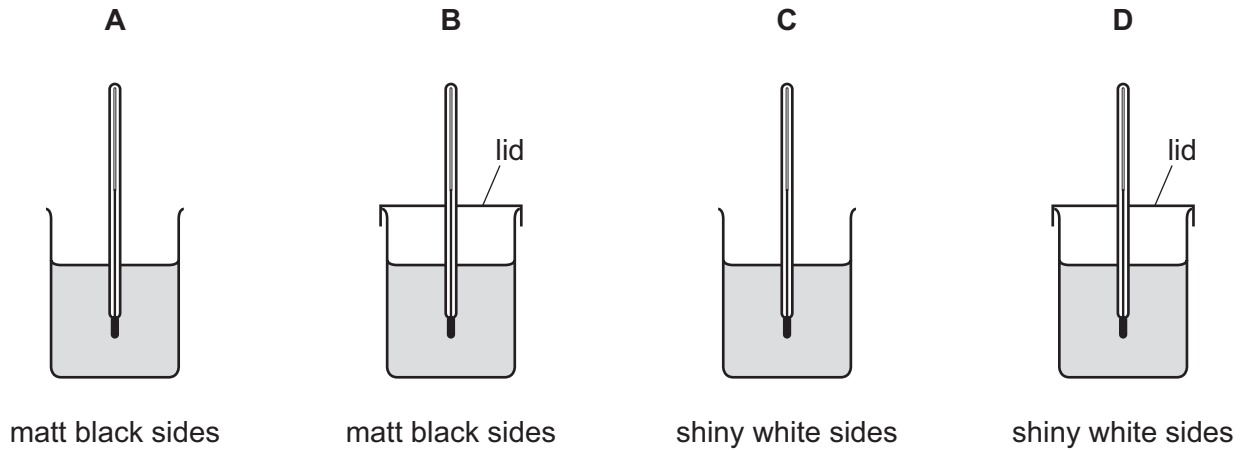
Which methods of heat loss are reduced by the **vacuum** between the silvered walls?

- A conduction only
- B conduction and convection only
- C convection and radiation only
- D conduction, convection and radiation

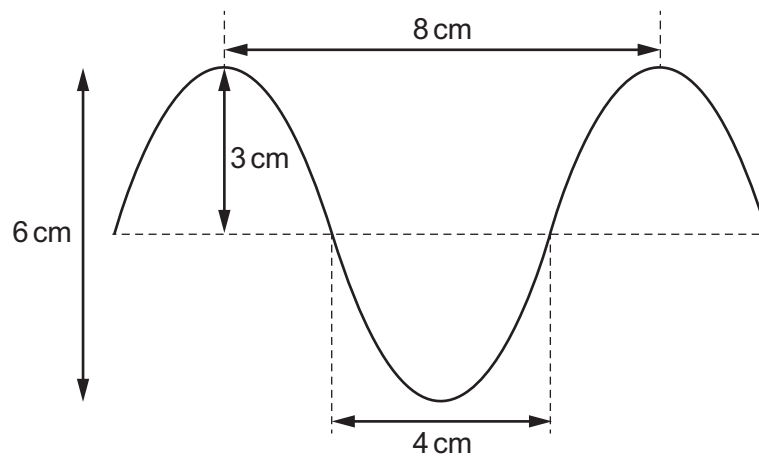
- 20** Equal volumes of water at 100°C are put in four containers. Two containers have matt black sides and two containers have shiny white sides. One of each type of container has a lid.

The containers are left for two minutes.

Which container has the highest temperature?



- 21** The diagram shows a wave.



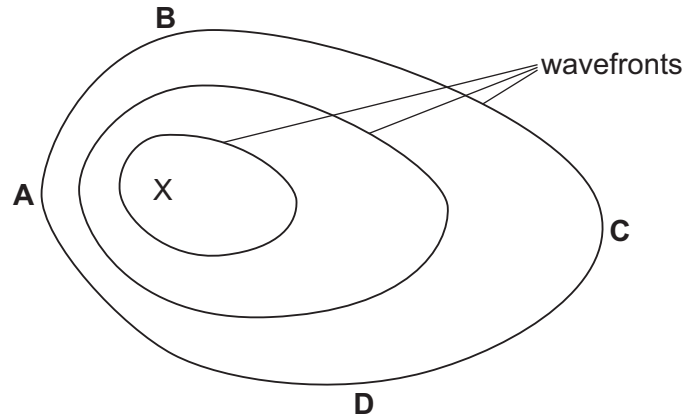
What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

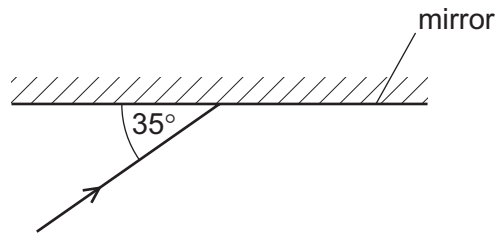
- 22** Waves travel more quickly on the surface of water when the water is deep.

A stone is dropped at point X into a pool of varying depth. The diagram shows the first three wavefronts on the surface of the pool.

The region between X and which labelled point is likely to be the deepest?



- 23** The diagram shows a ray of light incident on a plane mirror.



The angle between the ray and the mirror is 35° .

The ray is reflected by the mirror.

What is the angle of reflection?

- A** 35° **B** 55° **C** 70° **D** 110°
- 24** A beam of white light is split into a spectrum of seven colours.

Which name is given to this process?

- A** diffraction
B dispersion
C reflection
D refraction

- 25** An intruder alarm sensor detects that a person is warmer than his surroundings.

Which type of electromagnetic wave does the sensor detect?

- A** infrared
- B** radio
- C** ultraviolet
- D** visible light

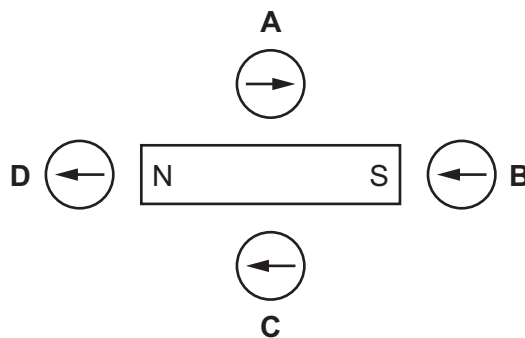
- 26** A tuning fork produces a sound when it vibrates.

What is the effect on the sound produced when the tuning fork vibrates more times every second and with a larger amplitude?

- A** higher pitch and less loud
- B** higher pitch and louder
- C** lower pitch and less loud
- D** lower pitch and louder

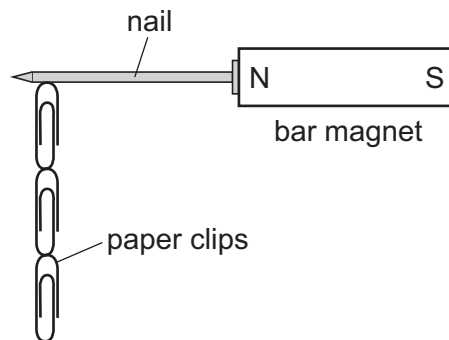
- 27** The diagram shows a magnet with some plotting compasses. The compasses show the direction of the magnetic field of the magnet.

Which plotting compass has the needle pointing in the wrong direction?



28 Four nails **A**, **B**, **C** and **D** are tested to find which makes the strongest permanent magnet.

One of the nails is placed against a bar magnet and the number of paper clips which the nail can support is recorded.

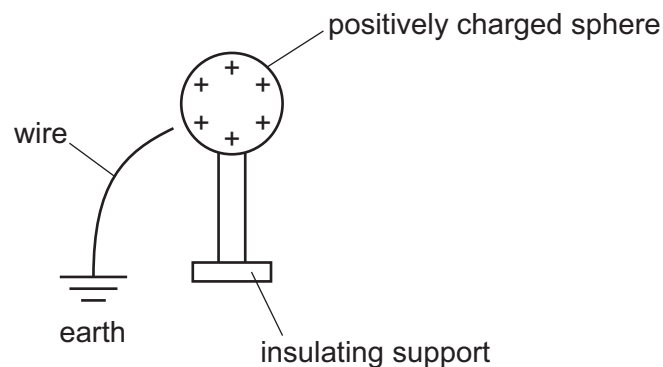


The bar magnet is then removed and the number of paper clips remaining attached to the nail is recorded. Each nail is tested individually.

Which nail becomes the strongest permanent magnet?

	number of paper clips attached to the nail	
	bar magnet present	bar magnet removed
A	2	0
B	2	1
C	4	3
D	5	2

29 The diagram shows a positively charged conducting sphere and a wire connected to earth.



What happens when the wire is touched onto the sphere?

- A** Electrons flow from earth to the sphere.
- B** Electrons flow from the sphere to earth.
- C** Positive charges flow from earth to the sphere.
- D** Positive charges flow from the sphere to earth.

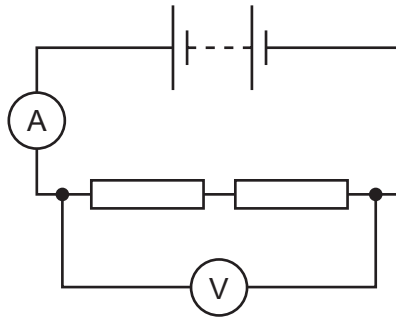
30 Three electrical quantities are listed.

- potential difference
- electromotive force
- current

Which quantities are measured in volts?

- A** potential difference only
B potential difference and current only
C potential difference and electromotive force only
D potential difference, electromotive force and current

31 A student uses the circuit shown to determine the resistance of two identical resistors.

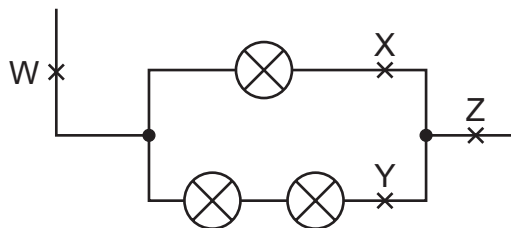


The voltmeter reading is 2.2 V and the ammeter reading is 0.25 A.

What is the resistance of each resistor?

- A** $0.275\ \Omega$ **B** $0.55\ \Omega$ **C** $4.4\ \Omega$ **D** $8.8\ \Omega$

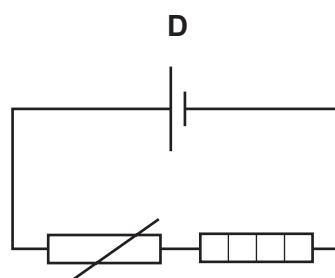
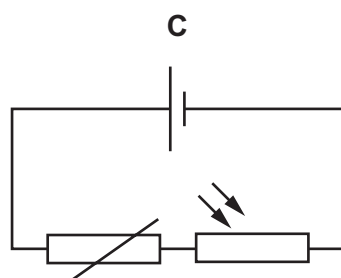
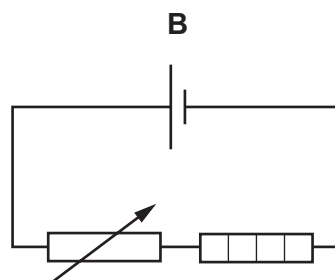
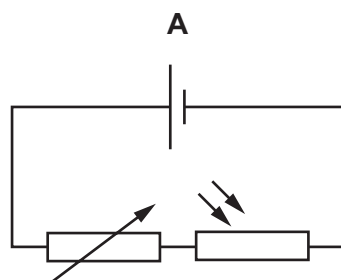
32 The diagram shows part of a circuit containing three identical lamps.



At which two points do the currents have the same value?

- A** W and X **B** W and Z **C** X and Y **D** Y and Z

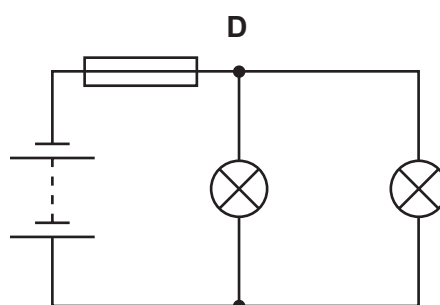
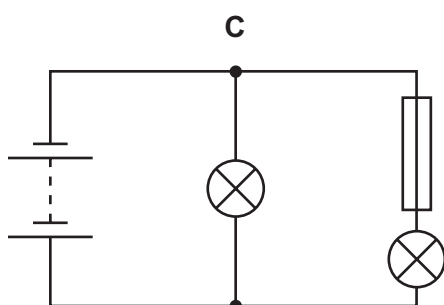
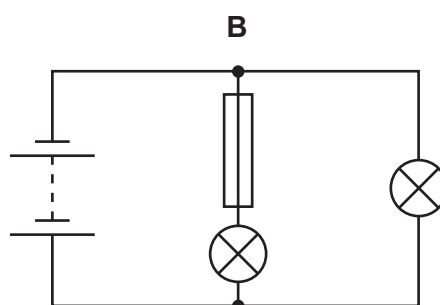
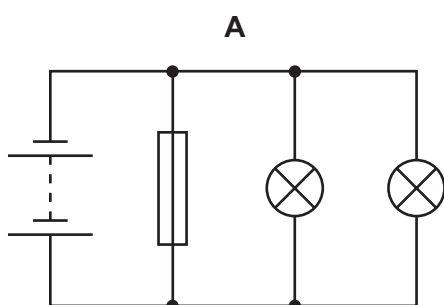
33 Which circuit contains a thermistor and a heater?



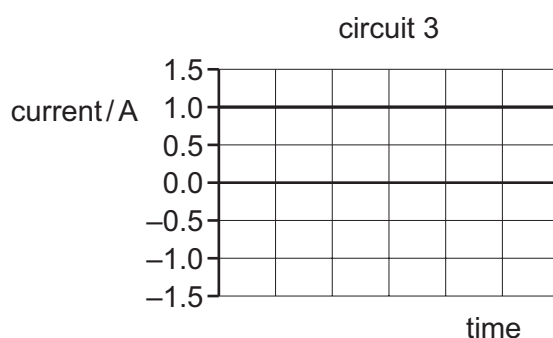
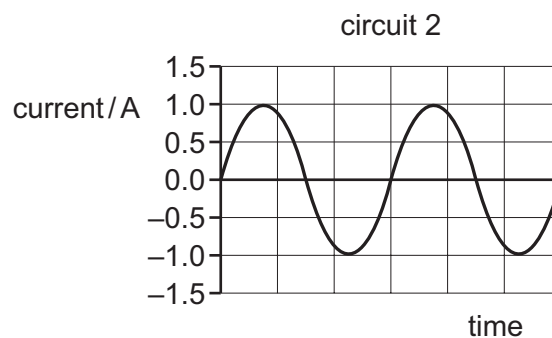
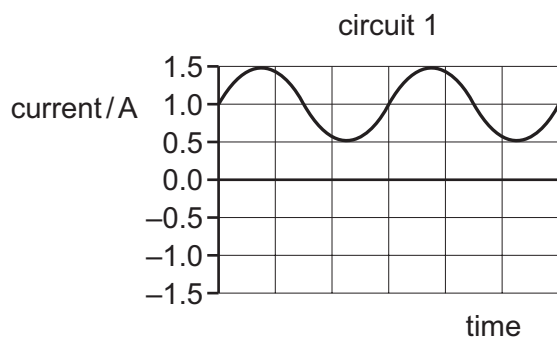
34 A student constructs four circuits, each containing a fuse.

The fuse blows in one circuit and both lamps in the circuit go out.

In which circuit does the fuse blow and both lamps go out?



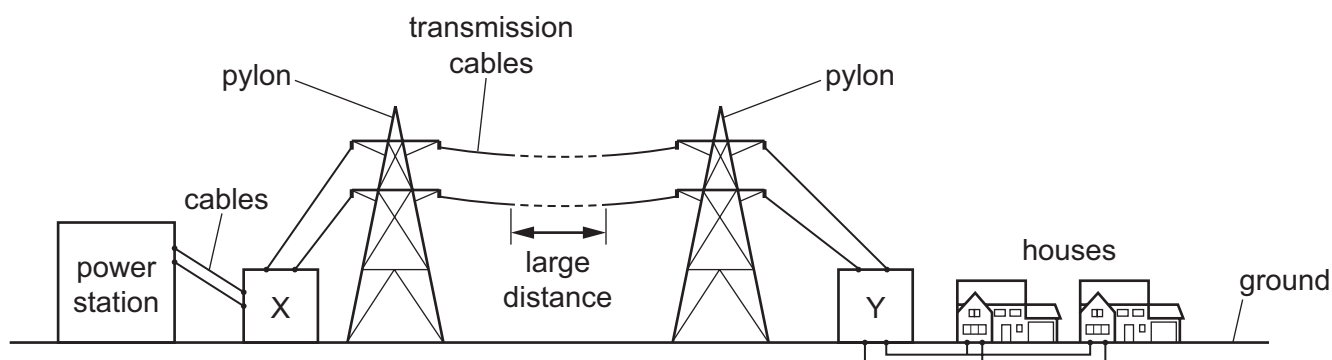
35 The graphs show how the currents in three circuits vary with time.



In which circuits is there a direct current?

- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

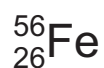
36 The diagram represents the transmission of electricity from a power station to homes that are many kilometres away. Two transformers are labelled X and Y.



What type of transformers are X and Y?

	X	Y
A	step-down transformer	step-down transformer
B	step-down transformer	step-up transformer
C	step-up transformer	step-down transformer
D	step-up transformer	step-up transformer

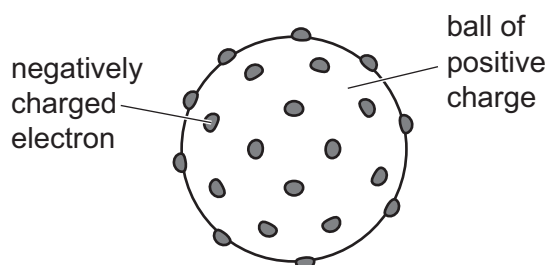
37 A nuclide of the element iron has the symbol shown.



What does a neutral atom of this nuclide contain?

	protons	neutrons	electrons
A	26	30	26
B	26	56	30
C	30	26	56
D	56	26	30

38 The diagram shows an early model of the structure of an atom.

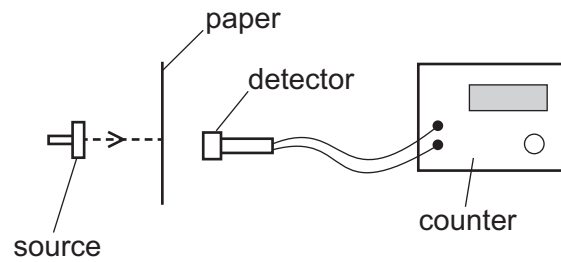


This early model is different from the atomic model accepted today.

Which statement about the early model is **not** included in the model accepted today?

- A** The atom is mainly filled with a ball of positive charge.
- B** The electrons are negatively charged.
- C** There are positive and negative charges.
- D** There are small particles called electrons.

- 39 A thin sheet of paper is placed between a radioactive source and a radiation detector. The count rate falls to a very low reading.



From this result, which type of radiation is the source emitting?

- A α -particles
 - B β -particles
 - C γ -rays
 - D X-rays
- 40 Why should all radioactive materials be handled carefully?
- A They all make anything they touch radioactive.
 - B They all catch fire very easily.
 - C They all emit ionising radiation.
 - D They all have long half-lives.

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Cambridge IGCSE™

PHYSICS

0625/12

Paper 1 Multiple Choice (Core)

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **20** pages. Blank pages are indicated.



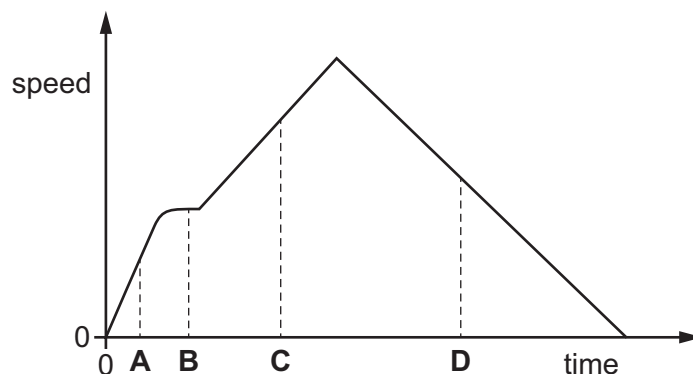
- 1 Five athletes P, Q, R, S and T compete in a race. The table shows the finishing times for the athletes.

athlete	P	Q	R	S	T
finishing time / s	22.50	24.40	25.20	26.50	23.20

Which statement is correct?

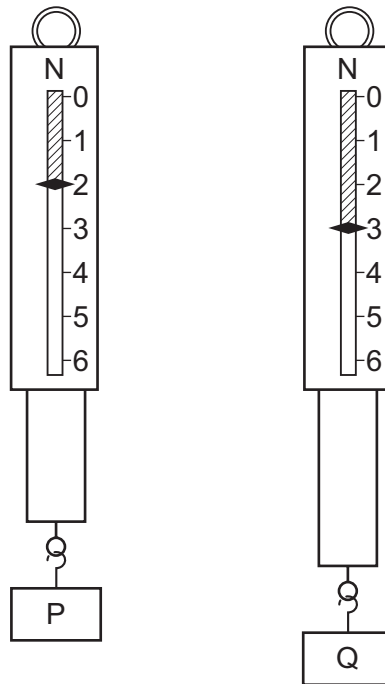
- A** Athlete P won the race and was 0.70 s ahead of the athlete in second place.
B Athlete P won the race and was 1.90 s ahead of the athlete in second place.
C Athlete S won the race and was 1.30 s ahead of the athlete in second place.
D Athlete S won the race and was 2.10 s ahead of the athlete in second place.
- 2 The graph shows how the speed of an object varies with time.

At which labelled time is the acceleration greatest?



- 3 Which statement about acceleration is correct?
- A** It is related to the changing speed of an object.
B It is the distance an object travels in one second.
C It is the force acting on an object divided by the distance it travels in one second.
D It is the force acting on an object when it is near to the Earth.

- 4 Two metal blocks P and Q have identical dimensions. They hang on identical spring balances.



Which statement about P and Q is correct?

- A They have different volumes and different weights.
 - B They have different volumes and equal weights.
 - C They have equal volumes and equal weights.
 - D They have equal volumes and different weights.
- 5 A space probe is taken from the Earth to Mars.

The force of gravity on the surface of Mars is less than the force of gravity on the surface of the Earth.

How do the weight and the mass of a space probe on the surface of Mars compare to their values when the probe is on the surface of the Earth?

	weight on Mars	mass on Mars
A	decreased	decreased
B	decreased	unchanged
C	unchanged	decreased
D	unchanged	unchanged

- 6 Water has a density of 1000 kg/m^3 .

A rectangular swimming pool has an average depth of 1.6 m.

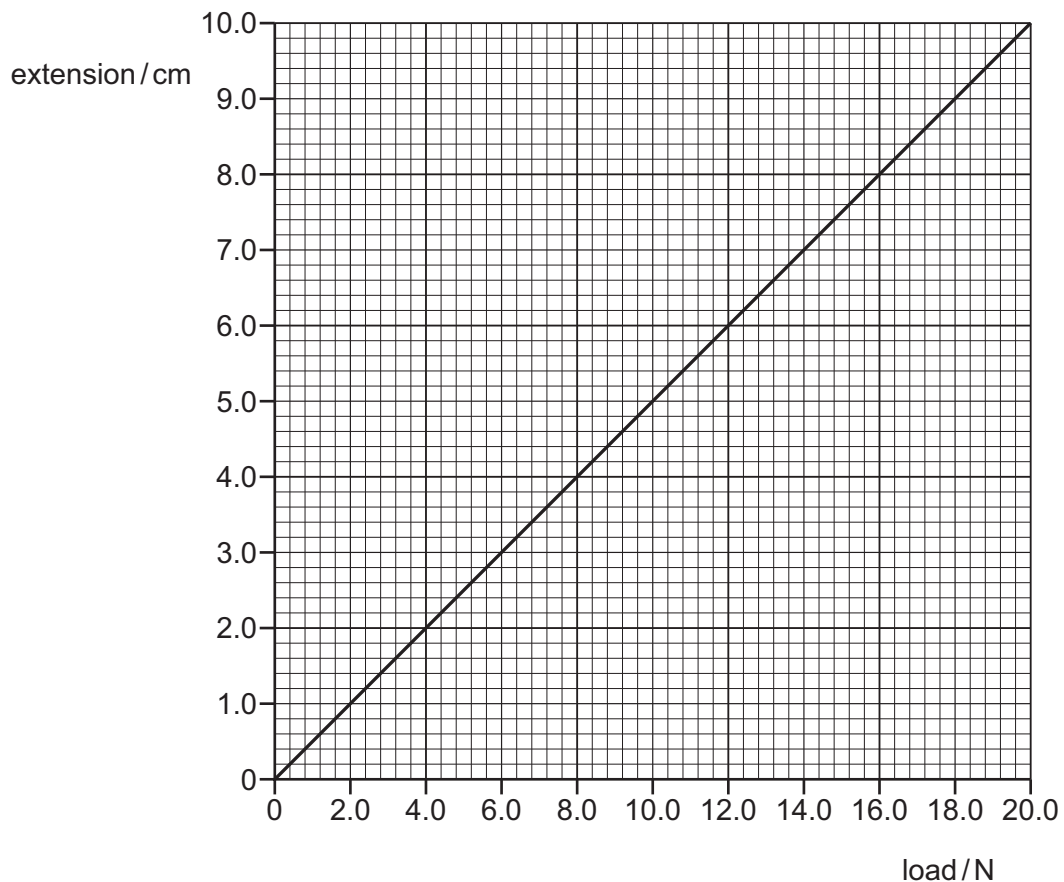
The length of the pool is 25 m.

The width of the pool is 10 m.

What is the mass of the water in the swimming pool?

- A** 2.5 kg **B** 400 kg **C** 400 000 kg **D** 800 000 kg

- 7 The diagram shows an extension–load graph for a spring.



An empty can of weight 3.0 N is suspended from the spring.

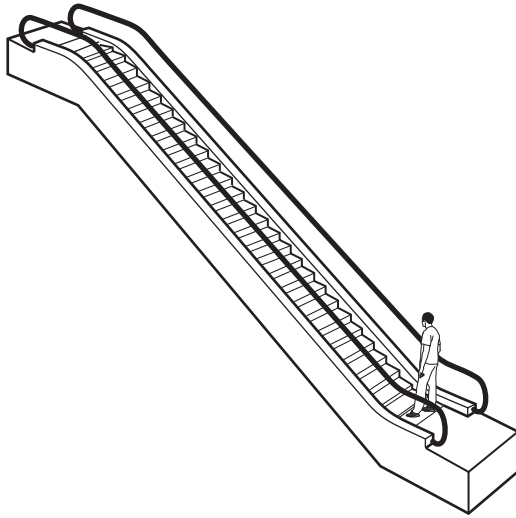
Liquid is poured into the can until the extension is 8.0 cm.

What is the weight of the liquid?

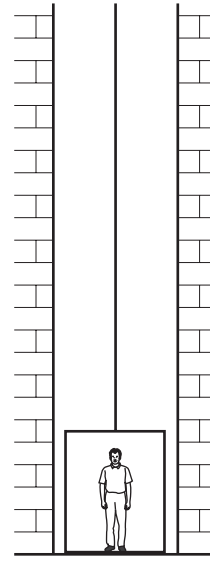
- A** 4.0 N **B** 10.0 N **C** 13.0 N **D** 16.0 N

- 8 What is meant by the moment of a force on an object?
- A the magnitude of the force on the object
 - B the direction of the force on the object
 - C the time for which the force acts on the object
 - D the turning effect of the force on the object
- 9 Which energy resource is **not** renewable?
- A geothermal
 - B nuclear fission
 - C solar
 - D wind
- 10 Two motors X and Y lift loads of the same weight through the same vertical distance.
- Motor X is more efficient than motor Y.
- Which statement about the motors is correct?
- A The useful energy output of motor X is larger than that of motor Y.
 - B The useful energy output of motor X is smaller than that of motor Y.
 - C The energy input of motor X is larger than that of motor Y.
 - D The energy input of motor X is smaller than that of motor Y.

- 11 A man can either take an escalator or a lift to travel up between two floors in a hotel.



escalator



lift

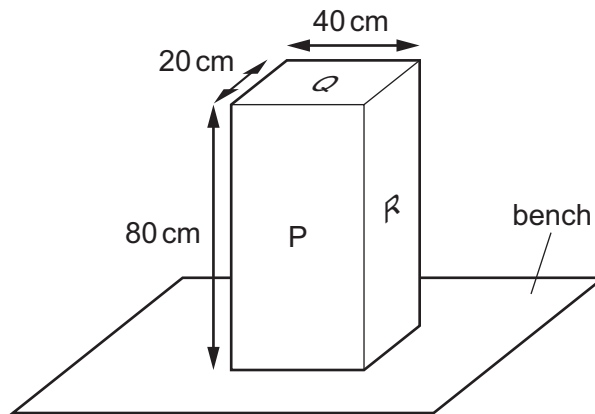
The escalator takes 20 seconds to carry the man between the two floors. The useful work done against gravity is W . The useful power developed is P .

The lift takes 30 seconds to carry the same man between the same two floors.

How much useful work against gravity is done by the lift, and how much useful power is developed by the lift?

	useful work done against gravity by lift	useful power developed by lift
A	more than W	less than P
B	more than W	P
C	W	less than P
D	W	P

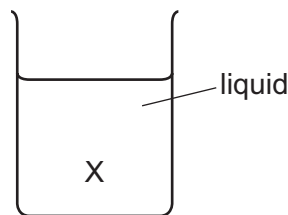
- 12 The diagram shows a solid block resting on a bench. The dimensions of the block are shown.



On which labelled surface should the block rest to produce the smallest pressure on the bench?

- A P
- B Q
- C R
- D P, Q and R produce the same pressure

- 13 A beaker contains a liquid.



On what does the liquid pressure at position X depend?

- A both the density of the liquid and the depth of X below the surface
 - B both the surface area of the liquid and the depth of X below the surface
 - C both the surface area of the liquid and the volume of the liquid
 - D the depth of X below the surface only
- 14 Evaporation occurs from the surface of a pool of water.

Which statement describes this change of state?

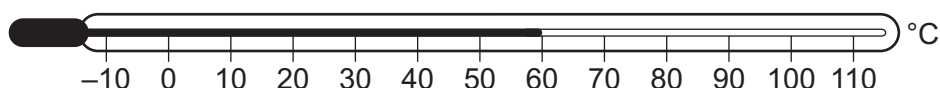
- A Electrons move from the liquid and become a gas.
- B Molecules that move from the liquid have the same energy as those that stay in the liquid.
- C The more energetic molecules escape the liquid.
- D The more energetic molecules remain in the liquid.

- 15** A gas, in a sealed container, is compressed slowly so that its temperature does not change.

What happens to the molecules of the gas as a result of this compression?

- A** The average speed of the molecules increases.
- B** The average force in a collision between a molecule and the container increases.
- C** There are more frequent collisions between molecules and the container.
- D** The volume of each molecule decreases.

- 16** The diagram shows a liquid-in-glass thermometer.



What is the temperature difference between the two fixed points on the Celsius scale?

- A** 10 °C
- B** 100 °C
- C** 110 °C
- D** 120 °C

- 17** A metal block is left overnight in a cool, shady room. In the morning, the metal block is moved into warm surroundings.

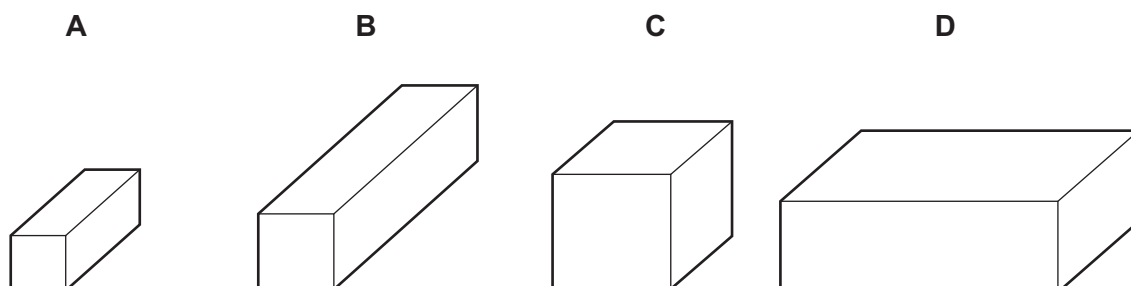
Which statement about the metal block is correct in the morning?

- A** The internal energy of the metal block increases.
- B** The temperature of the metal block decreases.
- C** Convection transfers energy throughout the metal block.
- D** The metal contracts slightly.

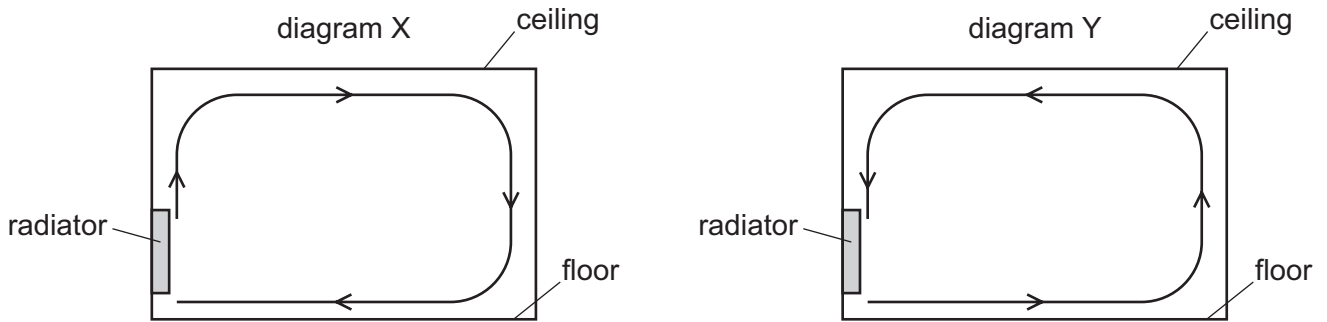
- 18** The diagrams show four blocks of steel. The blocks are all drawn to the same scale.

The same quantity of thermal energy is given to each block.

Which block shows the greatest rise in temperature?



- 19** A room is heated by a radiator. The diagrams X and Y show two possible circulations of hot air, which heat the room.



Which diagram and reason explain the heating of the room by convection?

	diagram	reason
A	X	air density decreases when air is heated
B	X	air density increases when air is heated
C	Y	air density decreases when air is heated
D	Y	air density increases when air is heated

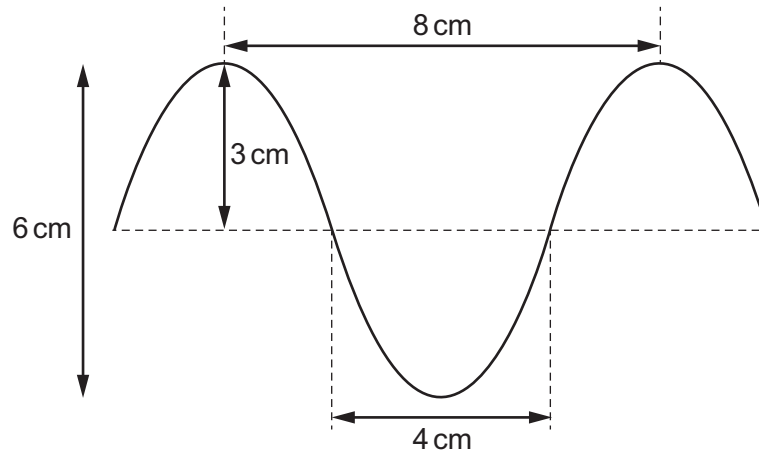
- 20** Two similar liquid-in-glass thermometers P and Q are placed in direct sunlight.

The bulb of thermometer P is painted white. The bulb of thermometer Q is painted black.

How and why would the thermometer readings differ?

- A** P would read higher than Q because black is a good absorber of radiation.
- B** P would read higher than Q because black is a poor absorber of radiation.
- C** P would read lower than Q because black is a good absorber of radiation.
- D** P would read lower than Q because black is a poor absorber of radiation.

21 The diagram shows a wave.



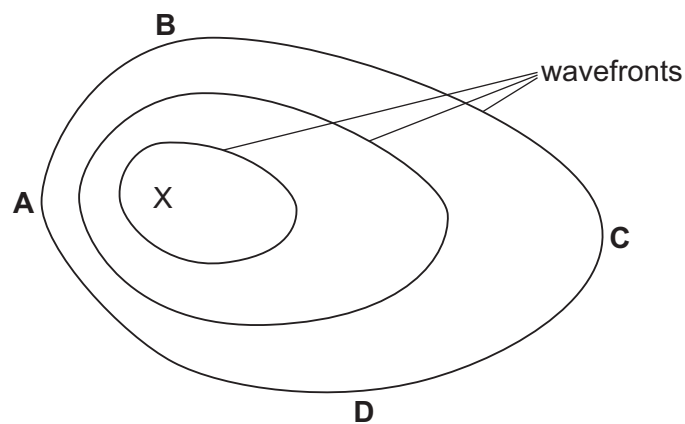
What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

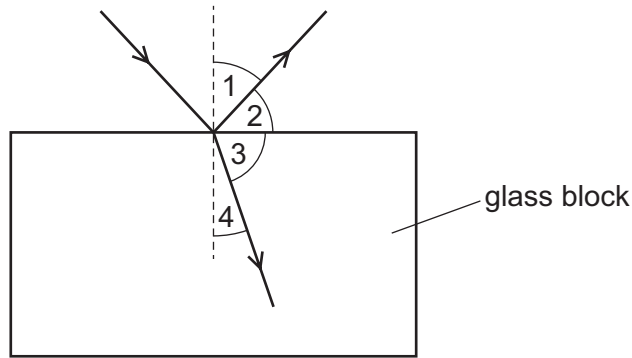
22 Waves travel more quickly on the surface of water when the water is deep.

A stone is dropped at point X into a pool of varying depth. The diagram shows the first three wavefronts on the surface of the pool.

The region between X and which labelled point is likely to be the deepest?



- 23 The diagram shows a ray of light incident on the surface of a glass block.



The ray of light is partially reflected back into the air and partially refracted into the glass block.

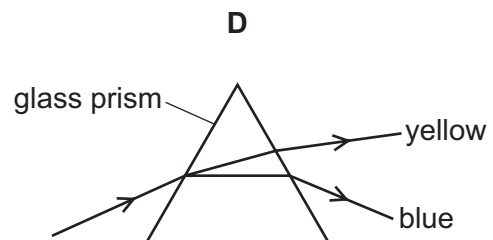
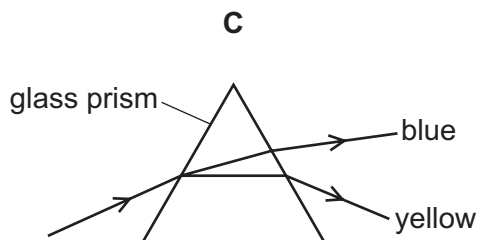
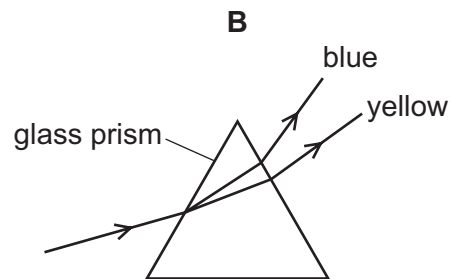
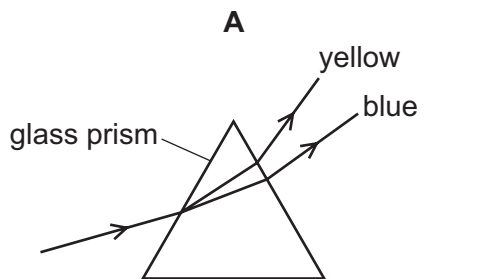
Which row correctly identifies the angle of reflection and the angle of refraction?

	angle of reflection	angle of refraction
A	1	3
B	1	4
C	2	3
D	2	4

- 24 A beam of light consists of yellow and blue light.

The beam of light is incident on a glass prism.

Which diagram is correct?



- 25** An intruder alarm sensor detects that a person is warmer than his surroundings.

Which type of electromagnetic wave does the sensor detect?

- A** infrared
- B** radio
- C** ultraviolet
- D** visible light

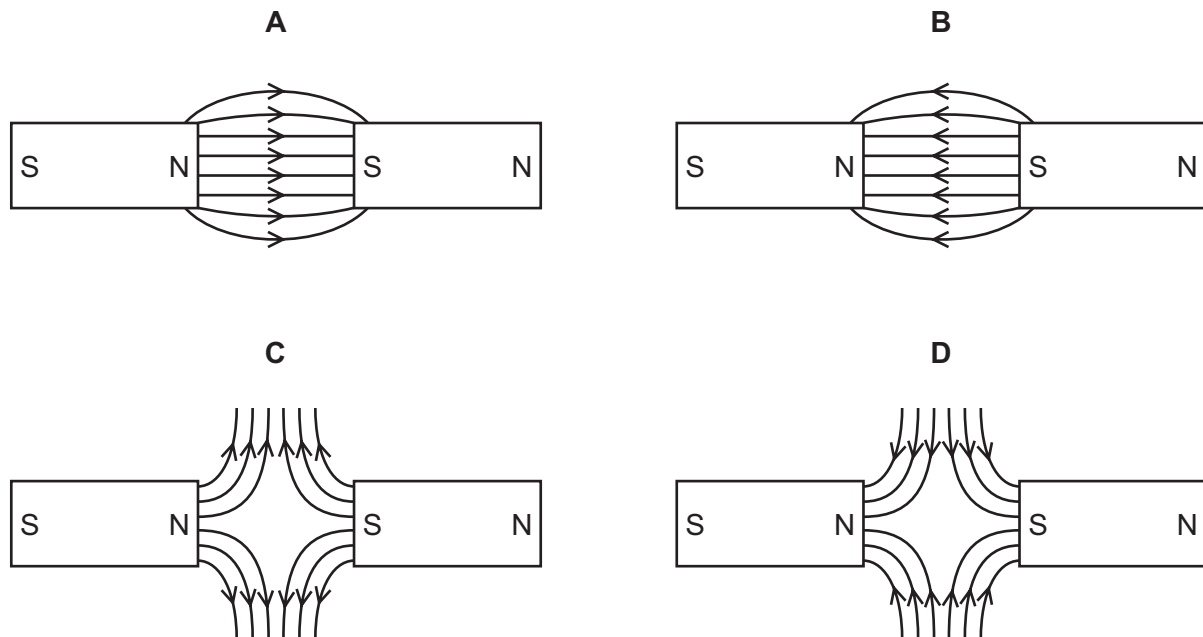
- 26** A dolphin sends out a sound wave. An echo returns 0.010 s later from a fish which is 7.5 m from the dolphin.

What is the speed of the sound wave in water?

- A** 0.075 m/s **B** 0.15 m/s **C** 750 m/s **D** 1500 m/s

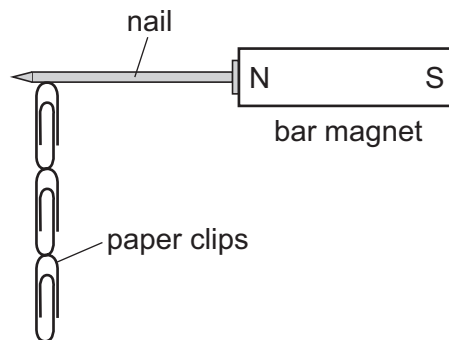
- 27** The diagrams show two bar magnets which are attracting each other.

Which diagram shows the magnetic field pattern between the poles?



28 Four nails **A**, **B**, **C** and **D** are tested to find which makes the strongest permanent magnet.

One of the nails is placed against a bar magnet and the number of paper clips which the nail can support is recorded.



The bar magnet is then removed and the number of paper clips remaining attached to the nail is recorded. Each nail is tested individually.

Which nail becomes the strongest permanent magnet?

	number of paper clips attached to the nail	
	bar magnet present	bar magnet removed
A	2	0
B	2	1
C	4	3
D	5	2

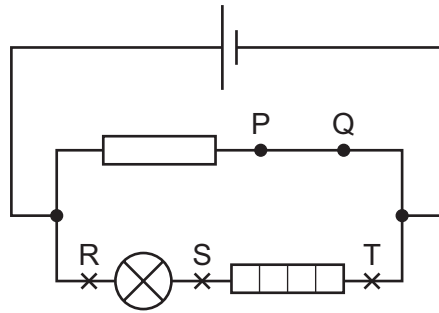
29 A cloth is used to rub an uncharged plastic rod.

Both the rod and the cloth become charged.

Why does the plastic rod become negatively charged and the cloth become positively charged?

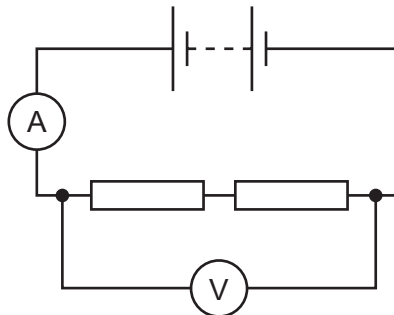
- A** The rod gains electrons and the cloth gains positive charges.
- B** The rod gains electrons and the cloth loses electrons.
- C** The rod loses electrons and the cloth gains electrons.
- D** The rod loses electrons and the cloth loses positive charges.

- 30 The diagram shows a circuit. The wire between P and Q can be removed and replaced by a circuit component.



Where should a voltmeter be connected to measure the potential difference (p.d.) across the lamp?

- A between P and Q in place of the wire
 - B in parallel with R and S
 - C in parallel with R and T
 - D in parallel with S and T
- 31 A student uses the circuit shown to determine the resistance of two identical resistors.

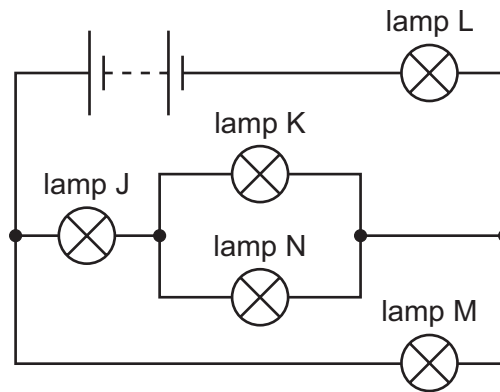


The voltmeter reading is 2.2 V and the ammeter reading is 0.25 A.

What is the resistance of each resistor?

- A $0.275\ \Omega$
- B $0.55\ \Omega$
- C $4.4\ \Omega$
- D $8.8\ \Omega$

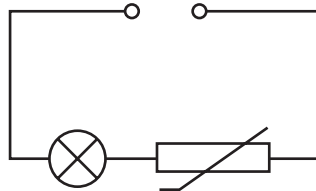
- 32 The circuit shown contains five lamps J, K, L, M and N. All the lamps are glowing.



One lamp is removed and two other lamps go out.

Which lamp is removed?

- A lamp J
 - B lamp K
 - C lamp L
 - D lamp M
- 33 The diagram shows a control circuit. The lamp is lit.



The temperature of the surroundings increases.

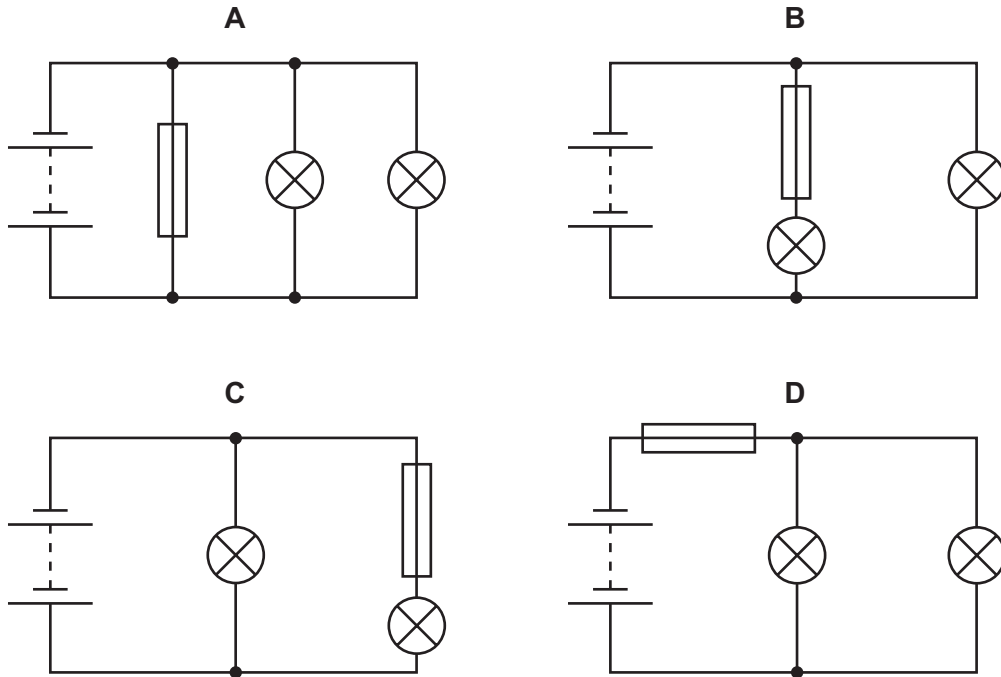
What will happen to the brightness of the lamp?

- A It will be brighter.
- B It will be less bright.
- C It will not change.
- D It will become brighter and then less bright.

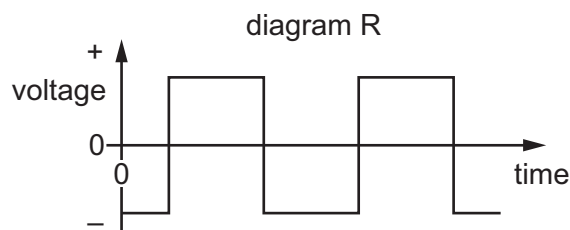
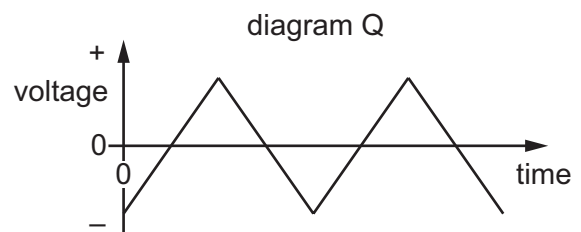
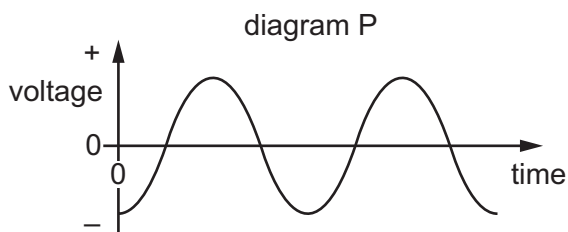
34 A student constructs four circuits, each containing a fuse.

The fuse blows in one circuit and both lamps in the circuit go out.

In which circuit does the fuse blow and both lamps go out?



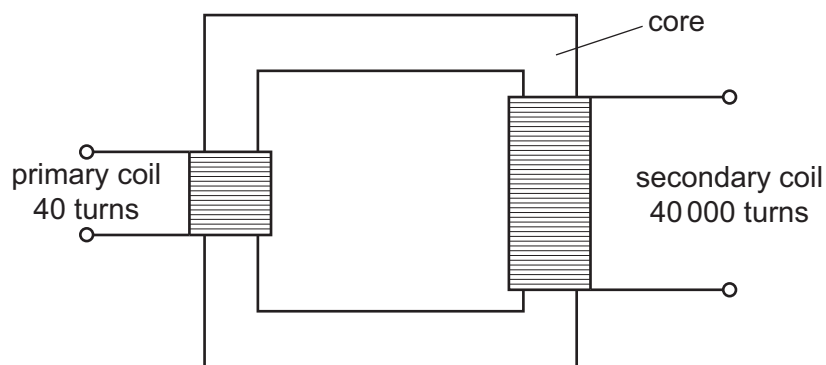
35 The diagrams P, Q and R show three voltage–time graphs.



Which graphs show an alternating voltage?

A P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

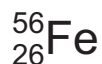
36 The diagram shows a transformer.



Which statement about this transformer is correct?

- A** It can operate from a 12 V battery.
- B** It has a core which is made of steel.
- C** It steps the input voltage up by a factor of 1000.
- D** It steps the input voltage down by a factor of 1000.

37 A nuclide of the element iron has the symbol shown.



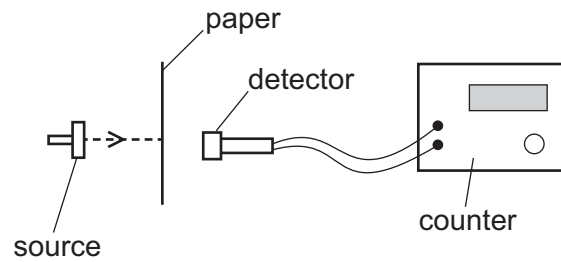
What does a neutral atom of this nuclide contain?

	protons	neutrons	electrons
A	26	30	26
B	26	56	30
C	30	26	56
D	56	26	30

38 Which statement about the nuclei of all atoms is correct?

- A** They all contain electrons.
- B** They are all always stable.
- C** They all contain protons and electrons.
- D** They all have a positive charge.

- 39** A thin sheet of paper is placed between a radioactive source and a radiation detector. The count rate falls to a very low reading.



From this result, which type of radiation is the source emitting?

- A** α -particles
 - B** β -particles
 - C** γ -rays
 - D** X-rays
- 40** A radioactive isotope has a half-life of 120 minutes.
- It emits radiation at a rate of 100 particles per second.
- How long does it take for the rate of emission to fall to 25 particles per second?
- A** 30 minutes **B** 45 minutes **C** 90 minutes **D** 240 minutes

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Cambridge IGCSE™

PHYSICS

0625/11

Paper 1 Multiple Choice (Core)

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **20** pages. Blank pages are indicated.

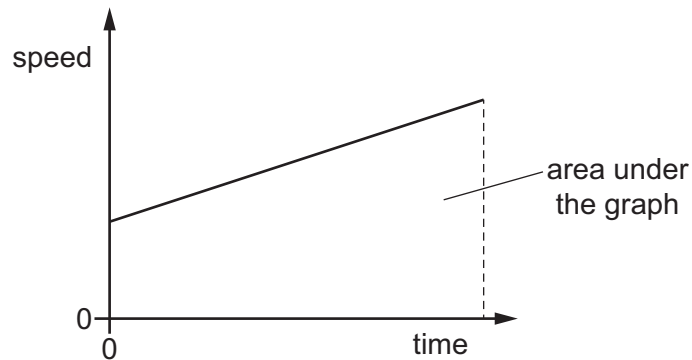


- 1 A pendulum makes 50 complete swings in 2 min 40 s.

What is the time period for 1 complete swing?

- A 1.6 s B 3.2 s C 4.8 s D 6.4 s

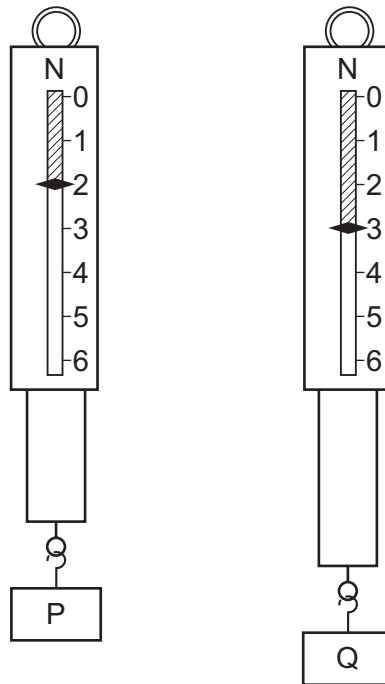
- 2 The motion of an object is represented by the speed–time graph shown.



Which quantity is equal to the area under the graph?

- A acceleration
B average speed
C distance travelled
D kinetic energy
- 3 Which statement about acceleration is correct?
- A It is related to the changing speed of an object.
B It is the distance an object travels in one second.
C It is the force acting on an object divided by the distance it travels in one second.
D It is the force acting on an object when it is near to the Earth.

- 4 Two metal blocks P and Q have identical dimensions. They hang on identical spring balances.



Which statement about P and Q is correct?

- A They have different volumes and different weights.
 - B They have different volumes and equal weights.
 - C They have equal volumes and equal weights.
 - D They have equal volumes and different weights.
- 5 A mass of 6.0 kg rests on the surface of a planet.

On this planet, $g = 20 \text{ N/kg}$.

What is the weight of the object?

- A 0.30 N B 0.60 N C 60 N D 120 N

- 6 The mass of a measuring cylinder is 190 g.

400 cm³ of liquid is put into the measuring cylinder.

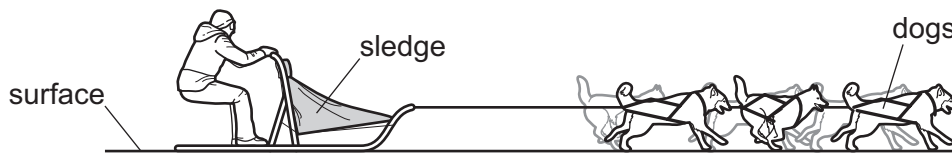
The total mass of the measuring cylinder and the liquid is 560 g.

Four solid objects are lowered in turn into the liquid. The densities of the objects are shown.

- 1 0.40 g/cm³
- 2 0.90 g/cm³
- 3 1.2 g/cm³
- 4 2.7 g/cm³

Which objects will float in the liquid?

- A** 1 only **B** 1 and 2 only **C** 1, 2 and 3 **D** 3 and 4 only
- 7 A sledge is pulled in a straight line by dogs, as shown.



The dogs produce a total horizontal driving force of 600 N.

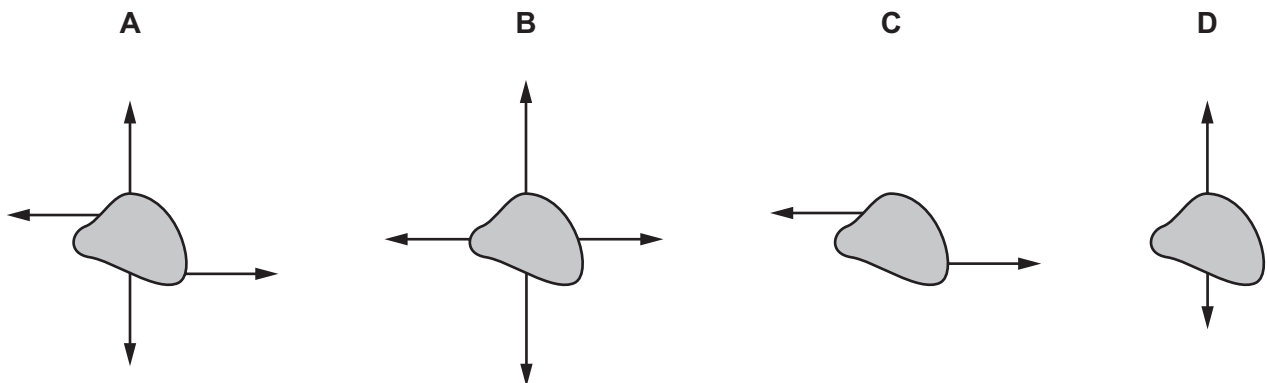
The frictional force between the sledge and the surface is 150 N and the air resistance on the sledge is 450 N.

What is the resultant force acting on the sledge?

- A** 0 N **B** 300 N **C** 900 N **D** 1200 N
- 8 Forces are applied to four identical objects.

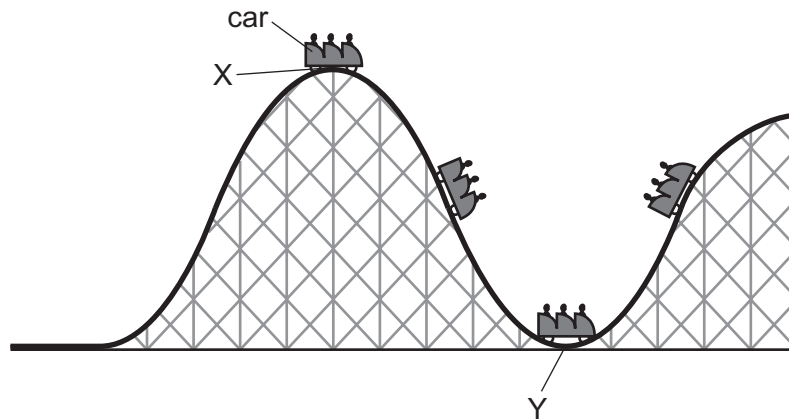
The lengths of the arrows indicate the magnitude of each force.

Which object is in equilibrium?



- 9 The diagram shows part of a rollercoaster ride with the car at different positions.

The car runs freely down from position X to position Y and up the hill on the other side.



What happens to the kinetic energy and to the gravitational potential energy of the car as it moves from position X to position Y?

	kinetic energy	gravitational potential energy
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

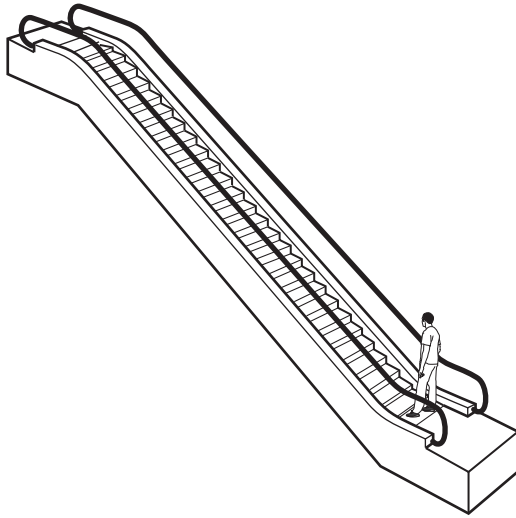
- 10 Student P uses a force of 35 N to push a box 3.0 m across the floor.

Student Q uses a force of 22 N to push another box 1.8 m across the floor.

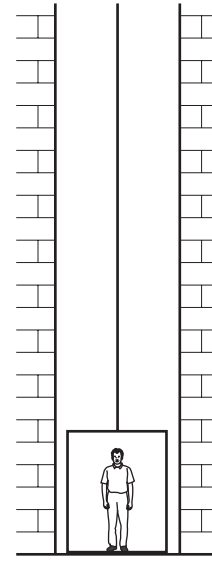
Which statement gives a full explanation why student P uses more energy than student Q?

- A** Student P pushes his box a greater distance than student Q.
- B** Student P pushes his box a greater distance and uses a bigger force than student Q.
- C** Student P uses a bigger force than student Q.
- D** Student P pushes a heavier box than student Q.

- 11 A man can either take an escalator or a lift to travel up between two floors in a hotel.



escalator



lift

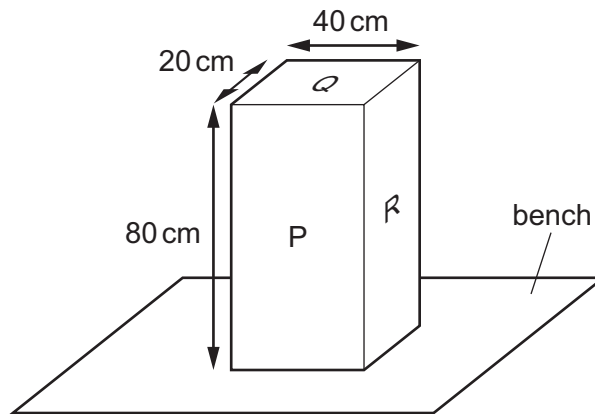
The escalator takes 20 seconds to carry the man between the two floors. The useful work done against gravity is W . The useful power developed is P .

The lift takes 30 seconds to carry the same man between the same two floors.

How much useful work against gravity is done by the lift, and how much useful power is developed by the lift?

	useful work done against gravity by lift	useful power developed by lift
A	more than W	less than P
B	more than W	P
C	W	less than P
D	W	P

12 The diagram shows a solid block resting on a bench. The dimensions of the block are shown.

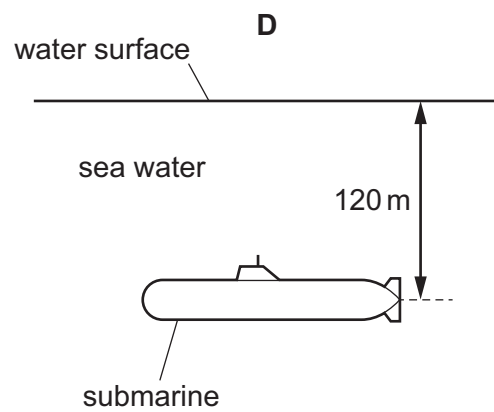
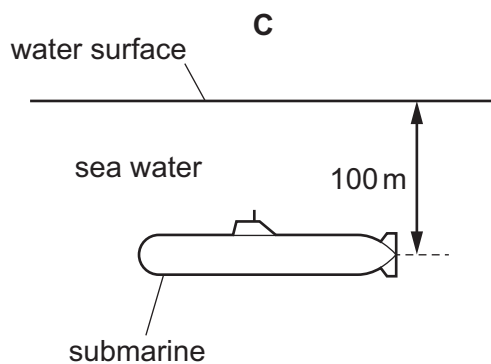
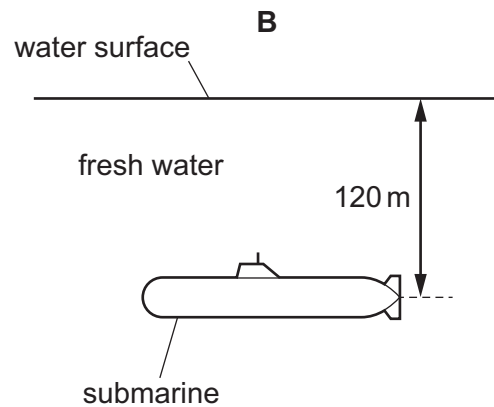
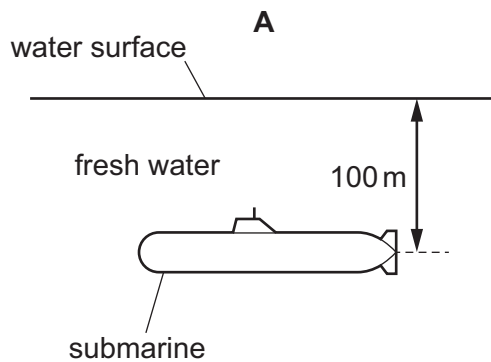


On which labelled surface should the block rest to produce the smallest pressure on the bench?

- A P
- B Q
- C R
- D P, Q and R produce the same pressure

- 13** Four submarines are submerged. The density of fresh water is 1000 kg/m^3 and the density of sea water is 1020 kg/m^3 .

Which submarine experiences the greatest pressure due to the water?



- 14** A gas is heated in a sealed container.

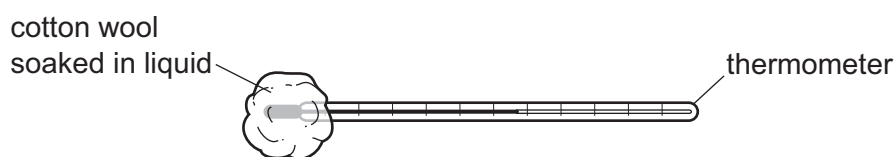
The volume of the container does not change.

What happens to the molecules of the gas?

- A** The average distance between molecules increases.
- B** The average kinetic energy of the molecules increases.
- C** The mass of each molecule increases.
- D** The volume of each molecule increases.

- 15** A student is investigating evaporation. She soaks a piece of cotton wool in a liquid and attaches this to the bulb of a thermometer.

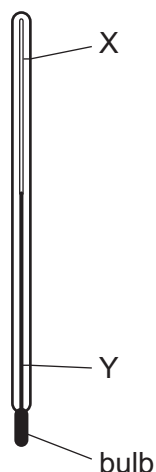
As the liquid evaporates the temperature reading on the thermometer changes.



Which statement is correct?

- A** The more energetic molecules leave the liquid and the temperature reading decreases.
 - B** The more energetic molecules leave the liquid and the temperature reading increases.
 - C** The less energetic molecules leave the liquid and the temperature reading decreases.
 - D** The less energetic molecules leave the liquid and the temperature reading increases.
- 16** The thermometer in the diagram has no scale.

Before this thermometer can be used to measure temperature, two standard temperatures known as fixed points are needed. These are labelled X and Y.



Which row describes these fixed points on the Celsius scale?

	X	Y
A	temperature of pure boiling water	normal body temperature
B	temperature of pure boiling water	temperature of pure melting ice
C	normal body temperature	temperature of pure boiling water
D	temperature of pure melting ice	temperature of pure boiling water

- 17 When a hot gas is left to cool, its internal energy decreases.

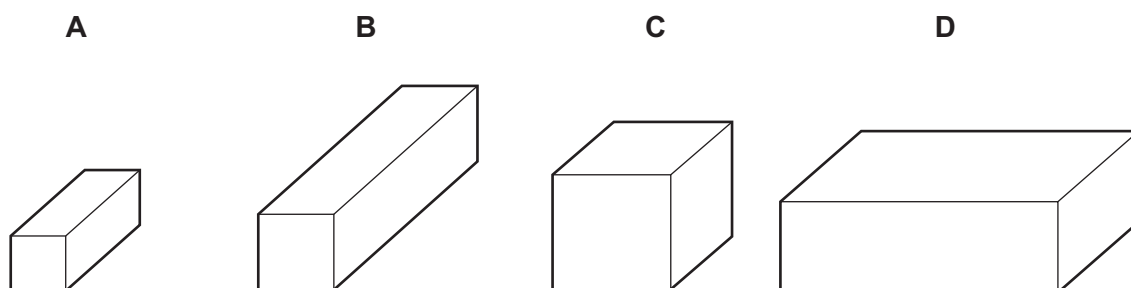
What causes this?

- A a decrease in the kinetic energy of the gas particles
- B a decrease in the gravitational potential energy of the gas particles
- C an increase in the average speed of the gas particles
- D an increase in the average distance of separation of the gas particles

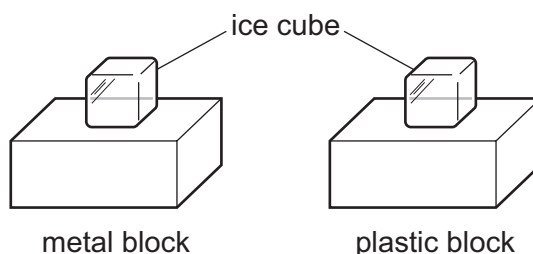
- 18 The diagrams show four blocks of steel. The blocks are all drawn to the same scale.

The same quantity of thermal energy is given to each block.

Which block shows the greatest rise in temperature?



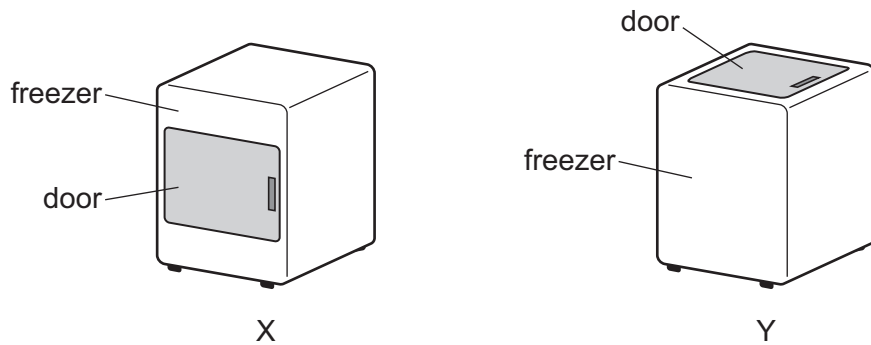
- 19 One ice cube is placed on a metal block. An identical ice cube is placed on a plastic block. The blocks are left next to each other on a table in a laboratory.



Which ice cube melts first and why?

- A The ice cube on the plastic block melts first because plastic is a good insulator of thermal energy.
- B The ice cube on the plastic block melts first because plastic is a good conductor of thermal energy.
- C The ice cube on the metal block melts first because metal is a good conductor of thermal energy.
- D The ice cube on the metal block melts first because metal is a good insulator of thermal energy.

- 20 Two freezers X and Y are identical except that one has a door opening at the front and the other has a door opening at the top.

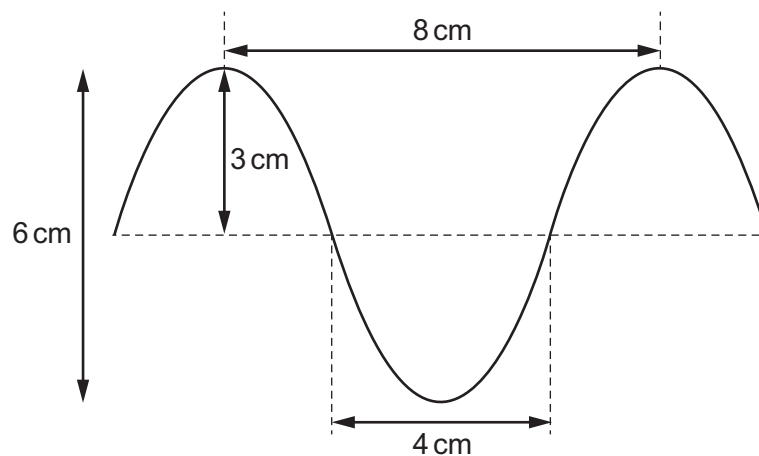


Both doors are the same size and are opened for the same amount of time.

Which freezer gains the least amount of thermal energy in this time and why?

	freezer gaining the least thermal energy	reason
A	X	cold air falls
B	X	warm air falls
C	Y	cold air falls
D	Y	warm air falls

- 21 The diagram shows a wave.



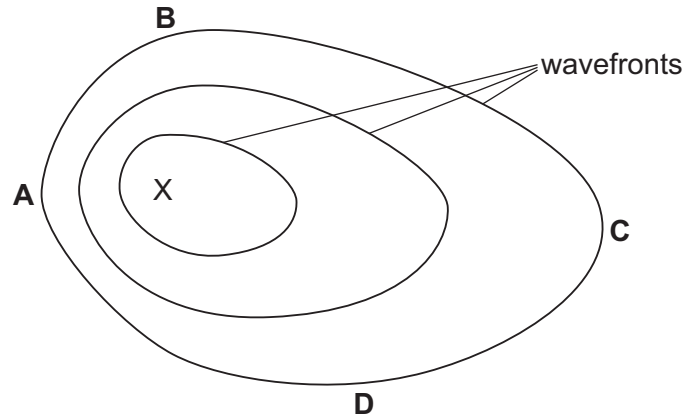
What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

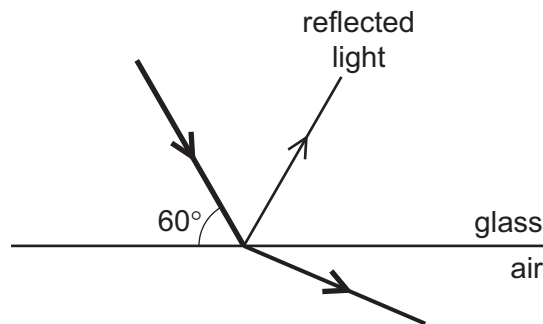
- 22** Waves travel more quickly on the surface of water when the water is deep.

A stone is dropped at point X into a pool of varying depth. The diagram shows the first three wavefronts on the surface of the pool.

The region between X and which labelled point is likely to be the deepest?



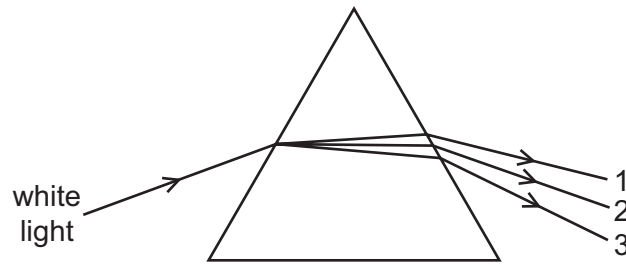
- 23** The diagram shows a beam of light travelling through glass and meeting a glass-air interface.



Which row correctly describes what is happening at the glass-air interface?

	angle of incidence at the interface	observation
A	30°	some internal reflection
B	30°	total internal reflection
C	60°	some internal reflection
D	60°	total internal reflection

- 24** A narrow beam of white light passes through a prism and is dispersed into a spectrum.



Which row is correct?

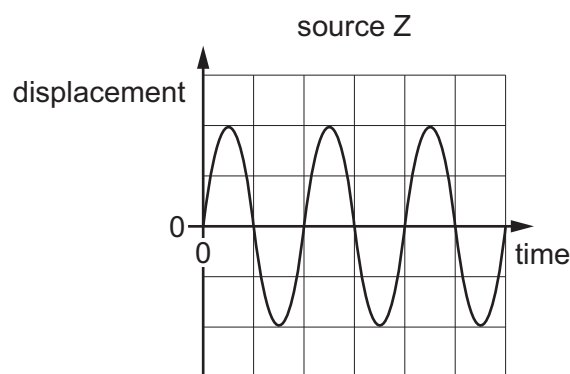
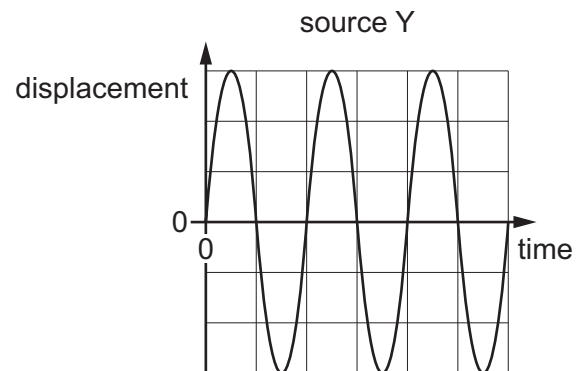
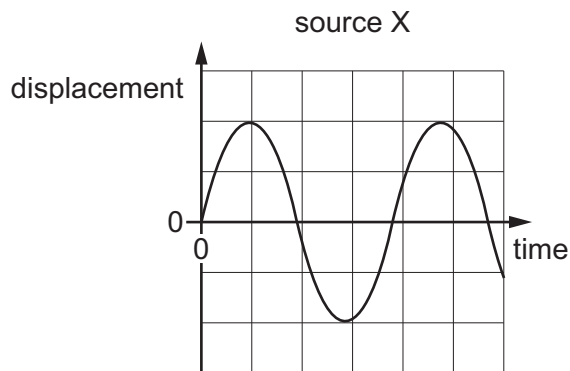
	colour 1	colour 2	colour 3
A	blue	yellow	red
B	red	blue	yellow
C	red	yellow	blue
D	yellow	blue	red

- 25** An intruder alarm sensor detects that a person is warmer than his surroundings.

Which type of electromagnetic wave does the sensor detect?

- A** infrared
- B** radio
- C** ultraviolet
- D** visible light

- 26** The graphs show the displacement of particles in sound waves from three sources X, Y and Z. The scales on the graphs are all identical.



Which sources are producing sound waves with the same pitch?

- A** X and Y only **B** Y and Z only **C** X and Z only **D** X, Y and Z

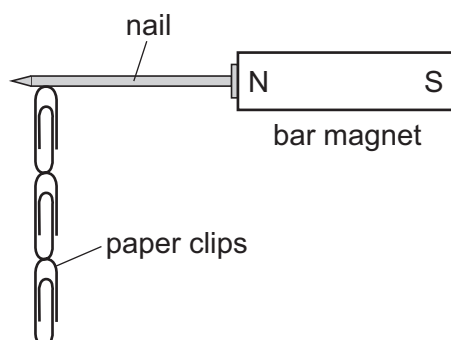
- 27** A permanent magnet is to be made.

Which row gives the correct material to use and the correct reason for choosing this material?

	material	reason
A	soft iron	easy to demagnetise
B	soft iron	hard to demagnetise
C	steel	easy to demagnetise
D	steel	hard to demagnetise

28 Four nails **A**, **B**, **C** and **D** are tested to find which makes the strongest permanent magnet.

One of the nails is placed against a bar magnet and the number of paper clips which the nail can support is recorded.



The bar magnet is then removed and the number of paper clips remaining attached to the nail is recorded. Each nail is tested individually.

Which nail becomes the strongest permanent magnet?

	number of paper clips attached to the nail	
	bar magnet present	bar magnet removed
A	2	0
B	2	1
C	4	3
D	5	2

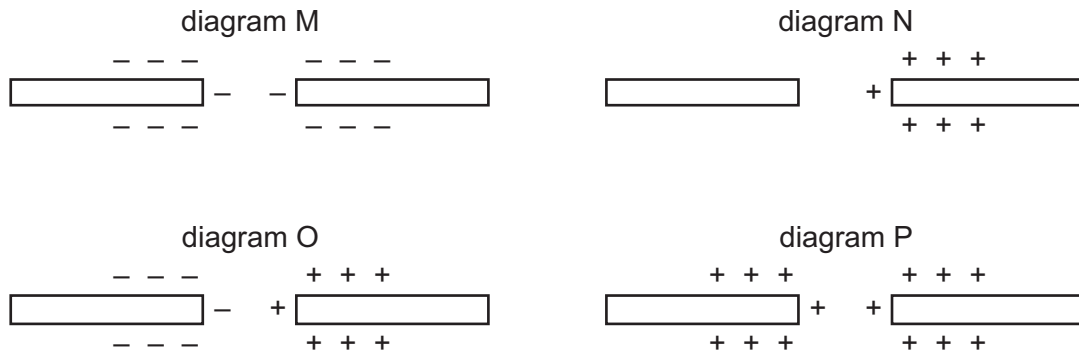
29 Three statements about electric charge are given.

- 1 An ammeter directly measures how much electric charge is in an object.
- 2 A moving electric charge can be detected by an ammeter.
- 3 A flow of electric charge is an electric current.

Which statements are correct?

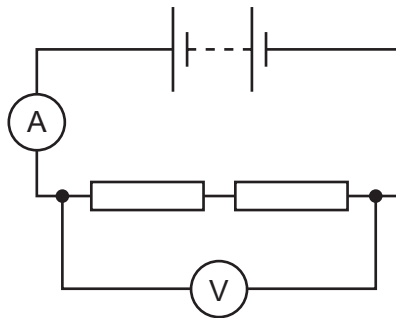
- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 30 Each of the four diagrams M, N, O and P shows a separate pair of insulating rods. Each rod is charged as shown.



In which **two** arrangements do the pairs of rods experience a force of repulsion?

- A** M and N **B** O and P **C** M and P **D** N and O
- 31 A student uses the circuit shown to determine the resistance of two identical resistors.

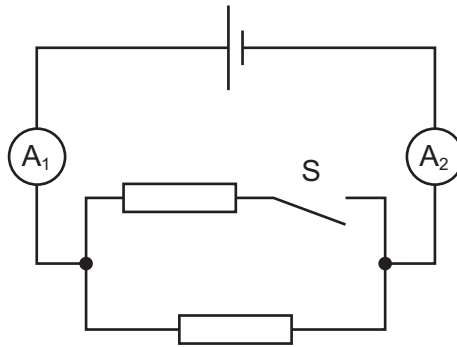


The voltmeter reading is 2.2 V and the ammeter reading is 0.25 A.

What is the resistance of each resistor?

- A** $0.275\ \Omega$ **B** $0.55\ \Omega$ **C** $4.4\ \Omega$ **D** $8.8\ \Omega$

32 In the circuit shown, A_1 and A_2 are ammeters.



Switch S is closed.

Which row is correct?

	the resistance of the whole circuit	reading of A_1	reading of A_2
A	decreases	stays the same	increases
B	decreases	increases	increases
C	increases	stays the same	stays the same
D	increases	decreases	decreases

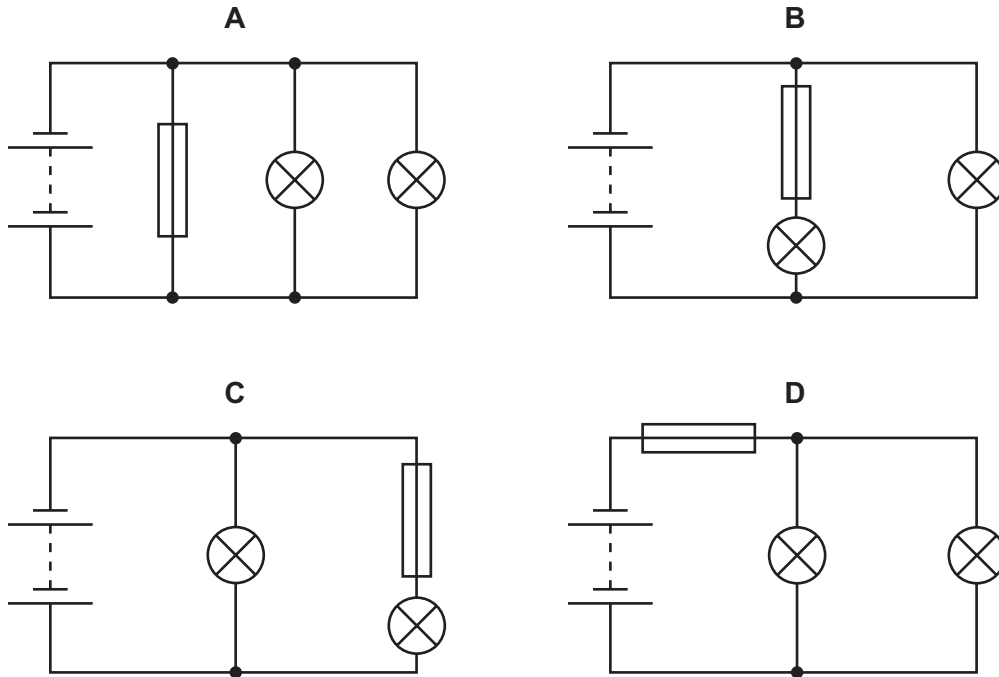
33 What happens to the resistance of an LDR when the brightness of light falling on it increases?

- A** Its resistance decreases.
- B** Its resistance increases.
- C** Its resistance increases then decreases.
- D** Its resistance stays the same.

- 34** A student constructs four circuits, each containing a fuse.

The fuse blows in one circuit and both lamps in the circuit go out.

In which circuit does the fuse blow and both lamps go out?



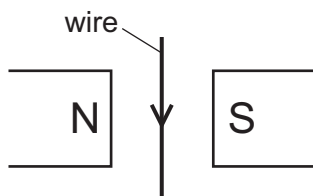
- 35** A transformer has 200 turns on its primary coil and is connected to a 240 V a.c. supply.

The output voltage of the transformer is 60 V a.c..

How many turns are on the secondary coil of the transformer?

- A** 20 **B** 50 **C** 72 **D** 800

- 36** A conducting wire is placed between the poles of a magnet. When an electric current in the wire is in the direction shown, then the force on the wire acts out of the page.



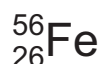
Three statements of different conditions and how the wire is affected are given.

- 1 The current is towards the top of the page and the direction of the magnetic field is unchanged then the force produced acts into the page.
- 2 The current is towards the bottom of the page and the magnetic field is reversed then the force produced acts into the page.
- 3 The current in the wire is alternating and the wire vibrates into and out of the page.

Which statements are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 37** A nuclide of the element iron has the symbol shown.



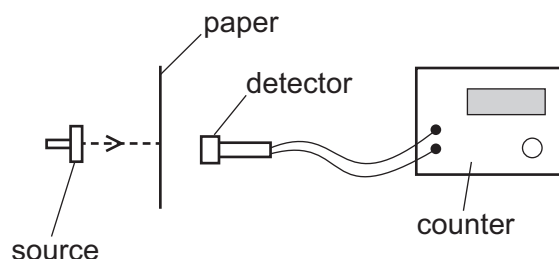
What does a neutral atom of this nuclide contain?

	protons	neutrons	electrons
A	26	30	26
B	26	56	30
C	30	26	56
D	56	26	30

- 38** What is the nucleon number of a nuclide?

- A** the number of neutrons in the nucleus
B the number of protons in the nucleus
C the number of protons minus the number of neutrons in the nucleus
D the number of protons plus the number of neutrons in the nucleus

- 39 A thin sheet of paper is placed between a radioactive source and a radiation detector. The count rate falls to a very low reading.



From this result, which type of radiation is the source emitting?

- A α -particles
 - B β -particles
 - C γ -rays
 - D X-rays
- 40 In 1986 the Chernobyl nuclear power station in Ukraine suffered a meltdown.

This caused background radiation in many countries, thousands of kilometres from Chernobyl, to increase.

What was transported in the atmosphere to these countries to cause this rise in background radiation?

- A α -particles
- B β -particles
- C γ -rays
- D radioactive isotopes

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PHYSICS

0625/13

Paper 1 Multiple Choice (Core)

May/June 2020

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This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	B	1
3	A	1
4	D	1
5	A	1
6	D	1
7	A	1
8	D	1
9	D	1
10	D	1
11	C	1
12	A	1
13	A	1
14	B	1
15	A	1
16	B	1
17	A	1
18	B	1
19	B	1
20	D	1
21	B	1
22	C	1
23	B	1
24	B	1
25	A	1
26	B	1
27	C	1
28	C	1

Question	Answer	Marks
29	A	1
30	C	1
31	C	1
32	B	1
33	D	1
34	D	1
35	B	1
36	C	1
37	A	1
38	A	1
39	A	1
40	C	1



Cambridge IGCSE™

PHYSICS

0625/12

Paper 1 Multiple Choice (Core)

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

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This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	A	1
3	A	1
4	D	1
5	B	1
6	C	1
7	C	1
8	D	1
9	B	1
10	D	1
11	C	1
12	A	1
13	A	1
14	C	1
15	C	1
16	B	1
17	A	1
18	A	1
19	A	1
20	C	1
21	B	1
22	C	1
23	B	1
24	D	1
25	A	1
26	D	1
27	A	1
28	C	1

Question	Answer	Marks
29	B	1
30	B	1
31	C	1
32	A	1
33	A	1
34	D	1
35	D	1
36	C	1
37	A	1
38	D	1
39	A	1
40	D	1



Cambridge IGCSE™

PHYSICS**0625/11**

Paper 1 Multiple Choice (Core)

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

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Question	Answer	Marks
1	B	1
2	C	1
3	A	1
4	D	1
5	D	1
6	B	1
7	A	1
8	B	1
9	C	1
10	B	1
11	C	1
12	A	1
13	D	1
14	B	1
15	A	1
16	B	1
17	A	1
18	A	1
19	C	1
20	C	1
21	B	1
22	C	1
23	A	1
24	C	1
25	A	1
26	B	1
27	D	1
28	C	1

Question	Answer	Marks
29	C	1
30	C	1
31	C	1
32	B	1
33	A	1
34	D	1
35	B	1
36	D	1
37	A	1
38	D	1
39	A	1
40	D	1



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

PHYSICS

Paper 1 Multiple Choice (Core)

0625/13

May/June 2019

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

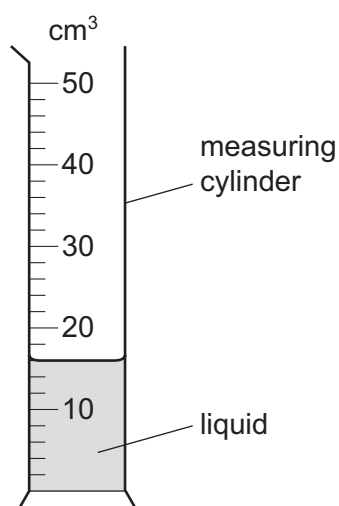
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

This syllabus is regulated for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **20** printed pages.



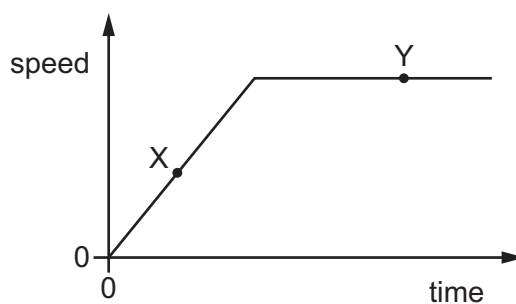
- 1 The diagram shows a measuring cylinder containing liquid.



What is the reading for the volume of liquid in the cylinder?

- A** 13.0 cm³ **B** 13.5 cm³ **C** 16.0 cm³ **D** 17.0 cm³

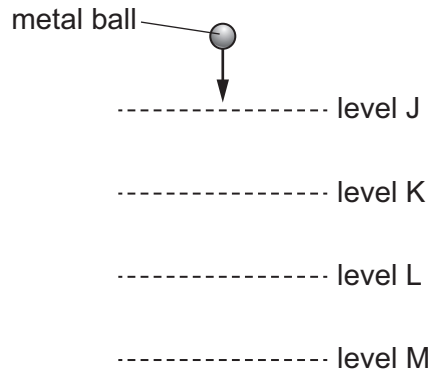
- 2 The diagram shows the speed-time graph for a car.



Which row describes the motion of the car at point X and at point Y?

	point X	point Y
A	at rest	moving with constant speed
B	moving with constant speed	at rest
C	moving with changing speed	at rest
D	moving with changing speed	moving with constant speed

- 3 A heavy metal ball falls vertically downwards through air past four equally spaced levels J, K, L and M.



The times taken to fall from one level to the next are measured.

Where is the speed of the ball greatest and which time is shortest?

	speed is greatest between	time is shortest between
A	J and K	J and K
B	J and K	L and M
C	L and M	J and K
D	L and M	L and M

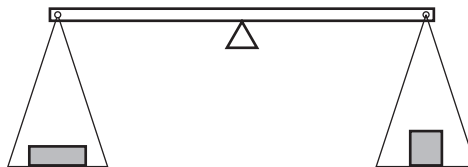
- 4 An object which has a mass of 600 kg is on the planet Mars.

The acceleration due to gravity on Mars is 4.0 m/s^2 .

What is the weight of the object on Mars?

- A** 60 N **B** 150 N **C** 2400 N **D** 6000 N

- 5 Two objects are placed on a balance, one on each side:

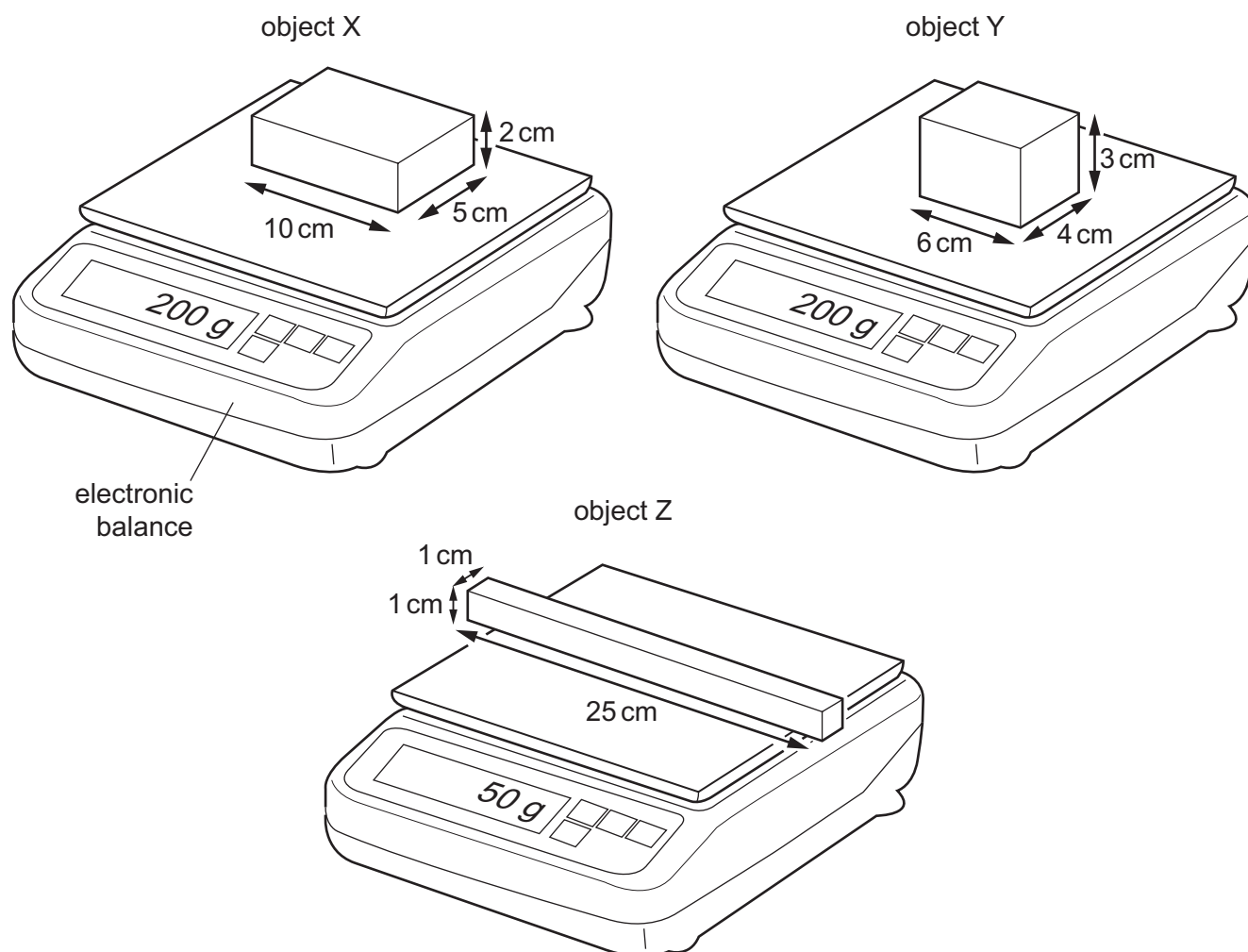


Which properties of the objects can be compared using the balance?

- A** weight, mass and volume
B weight and mass only
C volume and density
D density only

6 X, Y and Z are three regularly shaped solid objects.

Their dimensions and masses are shown in the diagrams.

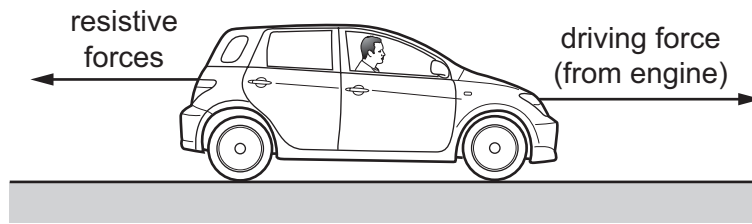


Which objects have the same density?

- A** X, Y and Z **B** X and Y only **C** X and Z only **D** Y and Z only

- 7 A car is driven from rest on a long straight road. The car engine exerts a constant driving force.

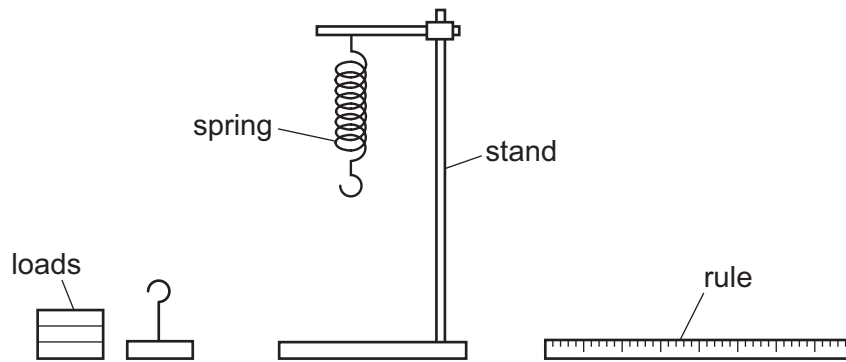
The diagram shows the horizontal forces acting on the car. The resistive forces are proportional to the speed of the car.



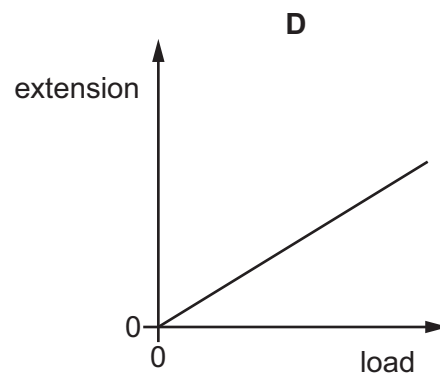
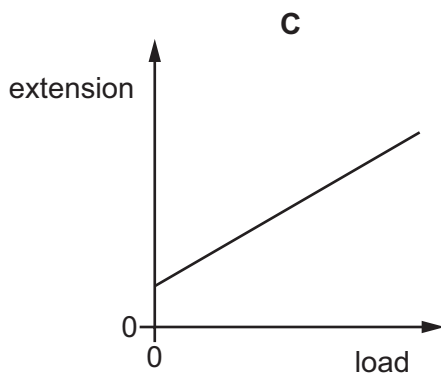
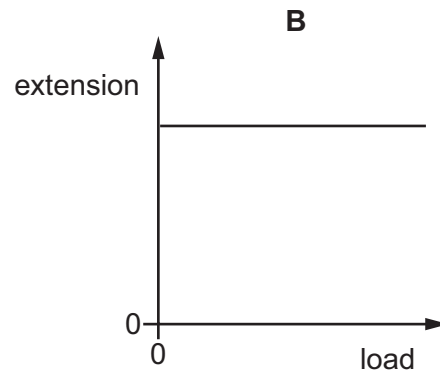
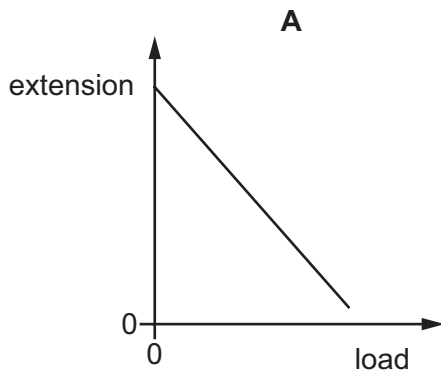
Why does the car eventually reach a maximum speed?

- A The resistive forces decrease to make the acceleration of the car negative.
- B The resistive forces decrease to make the acceleration of the car zero.
- C The resistive forces increase to make the acceleration of the car negative.
- D The resistive forces increase to make the acceleration of the car zero.

- 8 A spring is suspended from a stand. Loads are added and the extensions are measured.

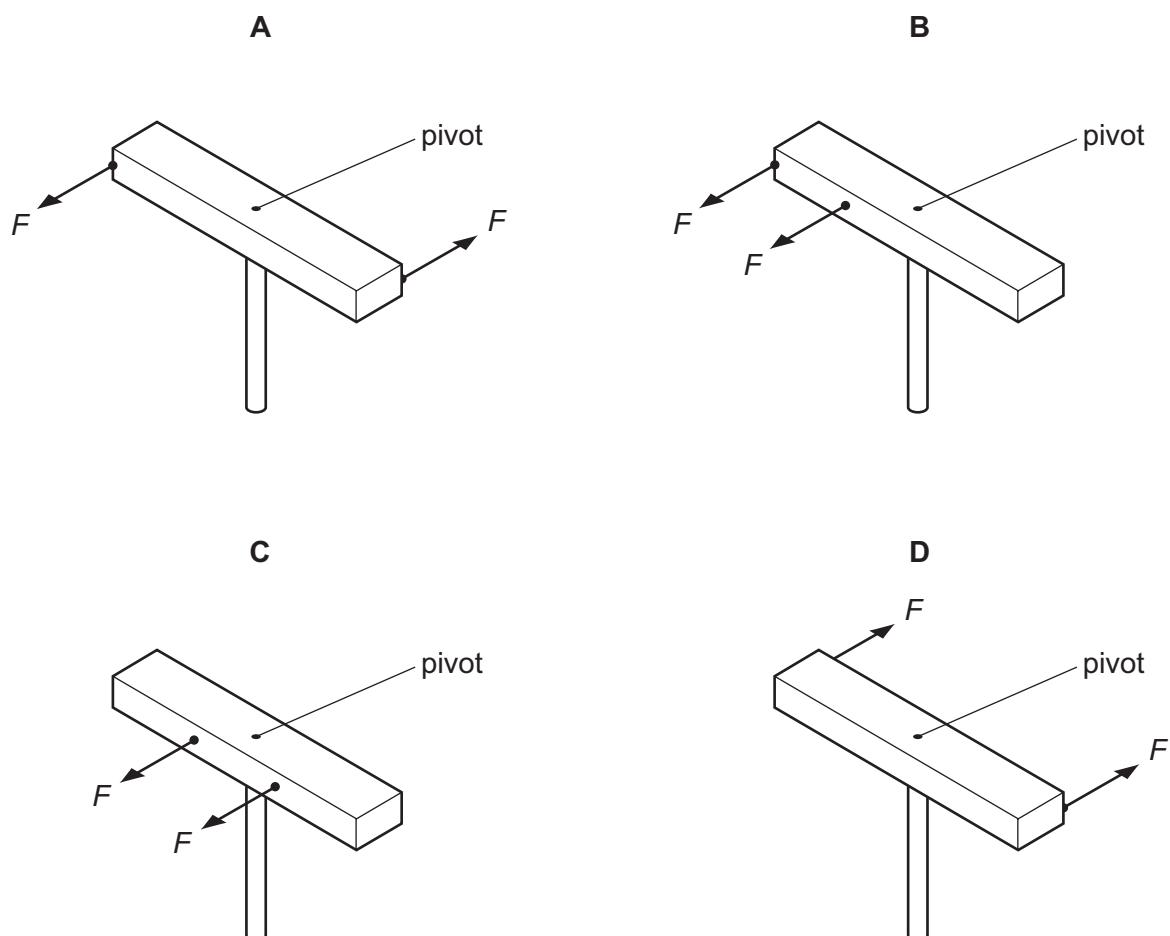


Which graph shows the result of plotting extension against load?



- 9 A wooden bar is pivoted at its centre so that it can rotate freely. Two equal forces F are applied to the bar.

In which diagram is the turning effect greatest?

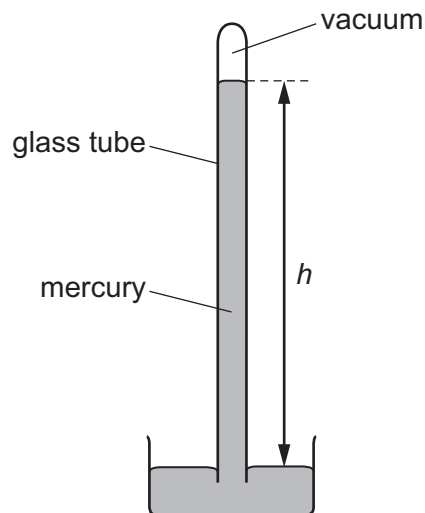


- 10 A machine is very efficient.

What does this mean?

- A It produces a large amount of power.
 - B It uses very little energy.
 - C It wastes very little energy.
 - D It works very quickly.
- 11 A body is lifted against gravity.
- Which equation shows the work done on the body?
- A work done = change in distance moved by the body
 - B work done = change in energy of the body
 - C work done = change in force on the body
 - D work done = change in power of the body

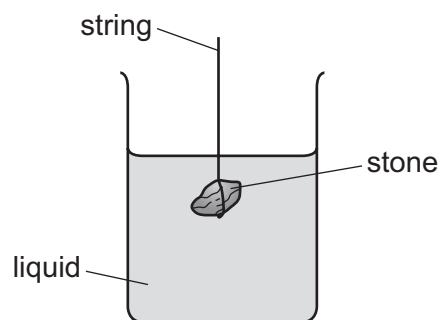
12 The diagram shows a simple mercury barometer.



A change is made and the vertical height h of the column increases.

Which change causes this increase?

- A A liquid with a lower density than mercury has been used.
 - B Atmospheric pressure decreases.
 - C A narrower glass tube is used.
 - D The glass tube is tilted.
- 13 The diagram shows a stone suspended on a string under the surface of a liquid. The stone experiences a pressure caused by the liquid.



What would increase the pressure on the stone?

- A decreasing the surface area of the stone
- B increasing the mass of the stone
- C lowering the stone deeper into the liquid
- D using a liquid with a lower density

14 Which statement is correct?

- A** A solid can flow.
- B** A solid can be compressed easily.
- C** A solid has a fixed shape.
- D** A solid takes the shape of its container.

15 On a hot summer day, the level of the water in a pond falls.

Which statement explains this?

- A** The least energetic water molecules escape from the surface and do not return.
- B** The least energetic water molecules escape from the surface and then return.
- C** The most energetic water molecules escape from the surface and do not return.
- D** The most energetic water molecules escape from the surface and then return.

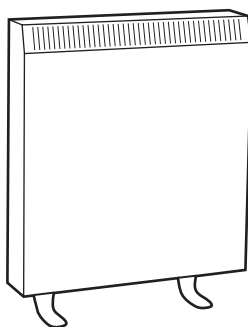
16 Here are three statements about a liquid-in-glass thermometer with a Celsius scale.

- 1 The lower fixed point is the temperature at which pure water freezes.
- 2 The upper fixed point is the temperature at which pure water boils.
- 3 A scale is made by dividing the distance between the fixed points into equal divisions.

Which statements are correct?

- A** 1 and 2 only **B** 2 and 3 only **C** 1 and 3 only **D** 1, 2 and 3

- 17 A night storage heater contains a large block of material that is heated electrically during the night. During the day the block cools down, releasing thermal energy into the room.



Which thermal capacity and which night-time temperature increase will cause the most energy to be stored by the block?

	thermal capacity of block	night-time temperature increase
A	large	large
B	large	small
C	small	large
D	small	small

- 18 A student investigates the melting point of a pure substance. She heats the substance slowly and takes readings of its temperature as the substance starts to melt and when it finishes melting.

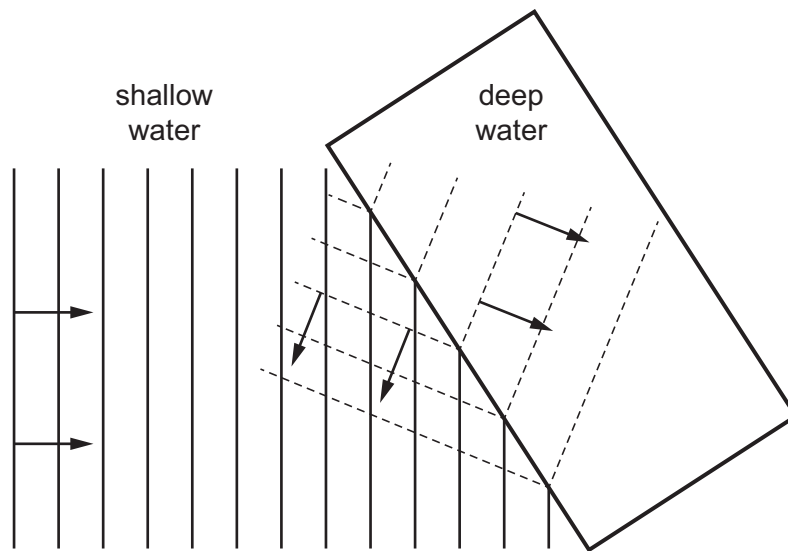
Which statement is correct?

- A** The temperature decreases slightly as the substance melts.
 - B** The temperature fluctuates as the substance melts.
 - C** The temperature increases as the substance melts.
 - D** The temperature stays the same as the substance melts.
- 19 A person holds an empty glass beaker and pours hot water into it.
- Why does it take a few seconds before his hand starts to feel hot?
- A** Glass is a poor conductor of heat.
 - B** Water is a poor conductor of heat.
 - C** Glass is a better conductor of heat than water.
 - D** Water is a better conductor of heat than glass.

- 20** The metal surface of a kettle is hot.

What happens to the cool air outside the kettle when it comes into contact with the hot kettle?

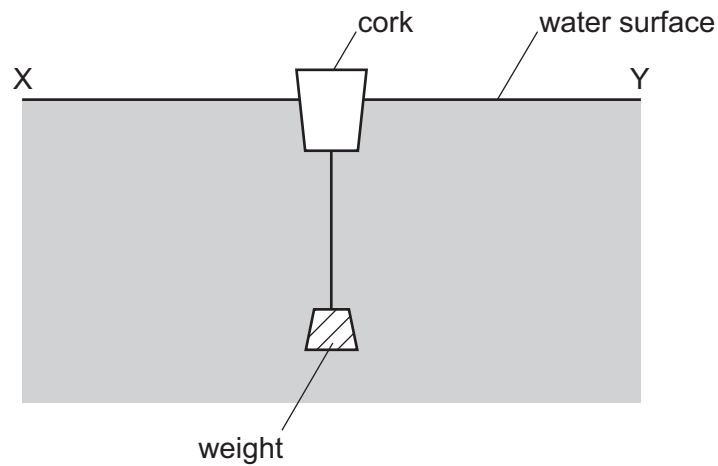
- A** The density of the air decreases and the air falls.
 - B** The density of the air decreases and the air rises.
 - C** The density of the air increases and the air falls.
 - D** The density of the air increases and the air rises.
- 21** A student draws a diagram to show two different properties of a water wave. The arrows show the wave directions.



Which two wave properties does the diagram show?

- A** refraction and diffraction
- B** reflection and dispersion
- C** reflection and diffraction
- D** reflection and refraction

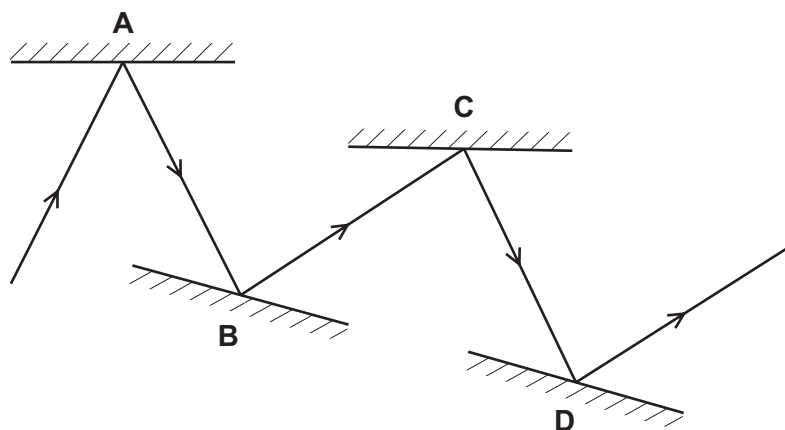
- 22 The diagram shows a cork with a weight attached so that the cork floats upright in water.



Transverse waves travel across the water from X to Y.

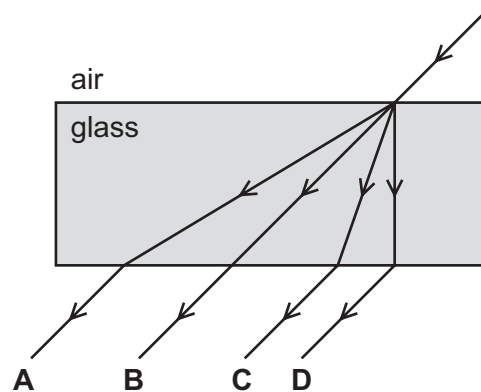
In which direction do the waves make the cork move?

- A → ← right and left
 - B ↑↓ up and down
 - C → only to the right
 - D ← only to the left
- 23 A student draws a ray diagram to show how a ray of light is reflected by a number of mirrors.
- Which reflection has **not** been drawn correctly?

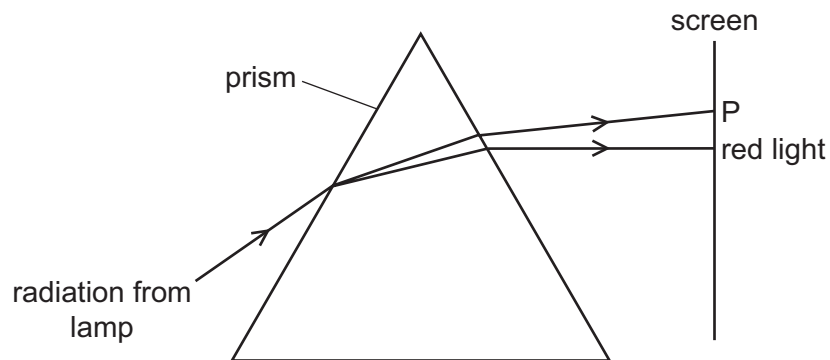


- 24 Light passes through a glass block.

What is the path of the light?

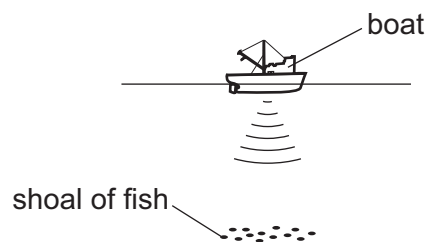


- 25 The diagram shows radiation from a lamp passing through a prism.



Which type of radiation is found at P?

- A γ -rays
 - B infrared
 - C ultraviolet
 - D X-rays
- 26 A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from a shoal of fish. The sound reaches the boat again after 1.2 s. The speed of sound in the water is 1500 m/s.



How far below the bottom of the boat is the shoal of fish?

- A 450 m
- B 900 m
- C 1800 m
- D 3600 m

27 Which statement about ultrasound is correct?

- A It has a higher frequency than audible sound, and it is a longitudinal wave.
- B It has a higher frequency than audible sound, and it is a transverse wave.
- C It has a lower frequency than audible sound, and it is a longitudinal wave.
- D It has a lower frequency than audible sound, and it is a transverse wave.

28 A metal bar is placed inside a current-carrying coil, as shown in diagram 1.

There is a small current in the coil. The bar holds a few nails, as shown in diagram 2.

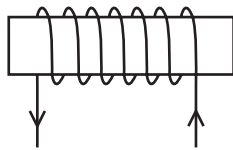


diagram 1

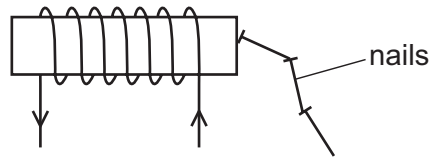


diagram 2

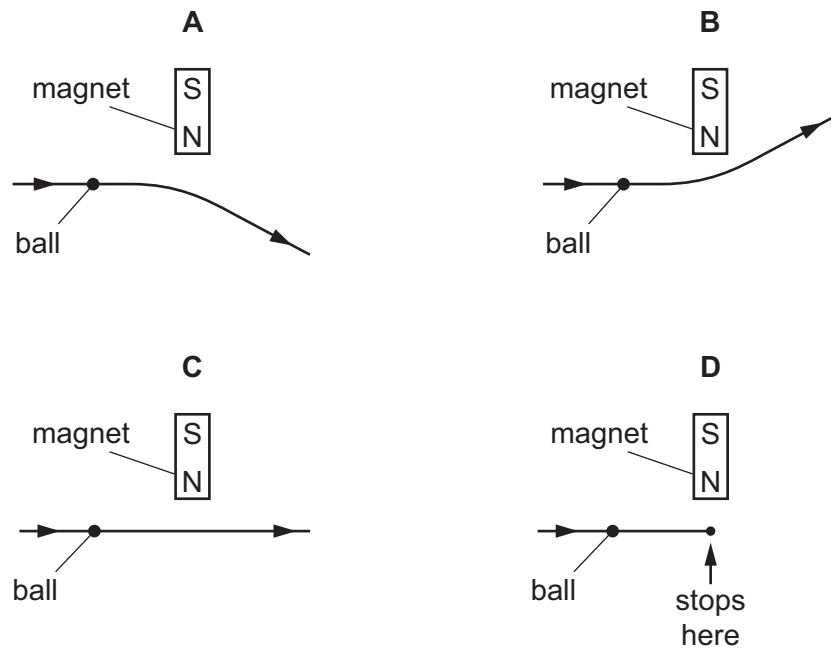
When there is no current in the coil, the nails drop off.

Which row is correct?

	metal from which the bar is made	effect of a larger current in the coil
A	soft iron	it makes no difference
B	soft iron	the bar holds more nails
C	steel	it makes no difference
D	steel	the bar holds more nails

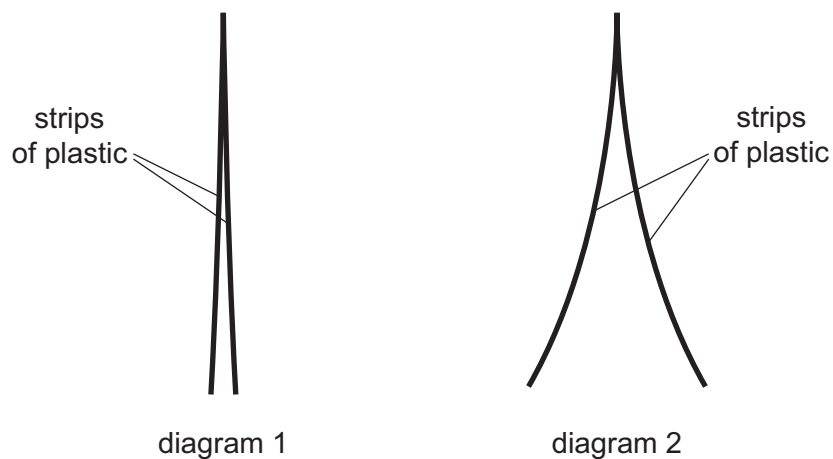
- 29** An iron ball on a horizontal wooden table rolls near the north pole of a bar magnet which is lying on the table.

Which diagram shows the most likely path of the ball, as seen from above the table?



- 30** Diagram 1 shows two thin, uncharged strips of plastic.

Diagram 2 shows the same strips after they have been rubbed with a dry cloth.



Which row describes the charge on the strips after rubbing, and the force between the strips after rubbing?

	charge on strips	force between strips
A	opposite	attraction
B	opposite	repulsion
C	the same	attraction
D	the same	repulsion

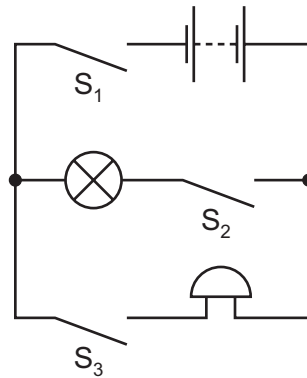
- 31** A battery stores chemical potential energy. The battery is connected to a resistor.

As the battery runs down, what happens to its chemical energy?

	The energy is transferred by	and ends up as
A	electrical working	internal energy in the resistor
B	electrical working	potential energy in the resistor
C	mechanical working	internal energy in the resistor
D	mechanical working	potential energy in the resistor

- 32** The diagram shows a circuit including a lamp, an electric bell and three switches S_1 , S_2 and S_3 .

The lamp and bell are not faulty.



The bell is ringing but the lamp is not lit.

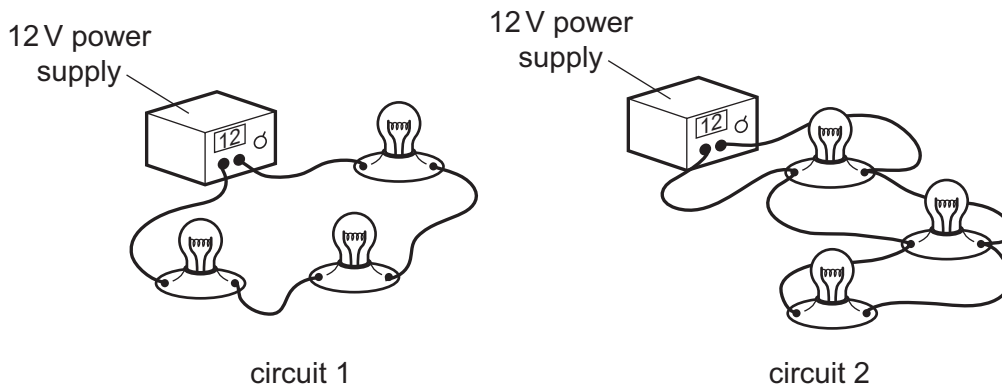
Which switches are closed?

- A** S_1 only
- B** S_1 and S_2 only
- C** S_1 and S_3 only
- D** S_1 , S_2 and S_3

- 33** A student is designing a lighting circuit for a dolls' house. He sets up two different circuits.

Each circuit contains a 12 V power supply and three identical lamps.

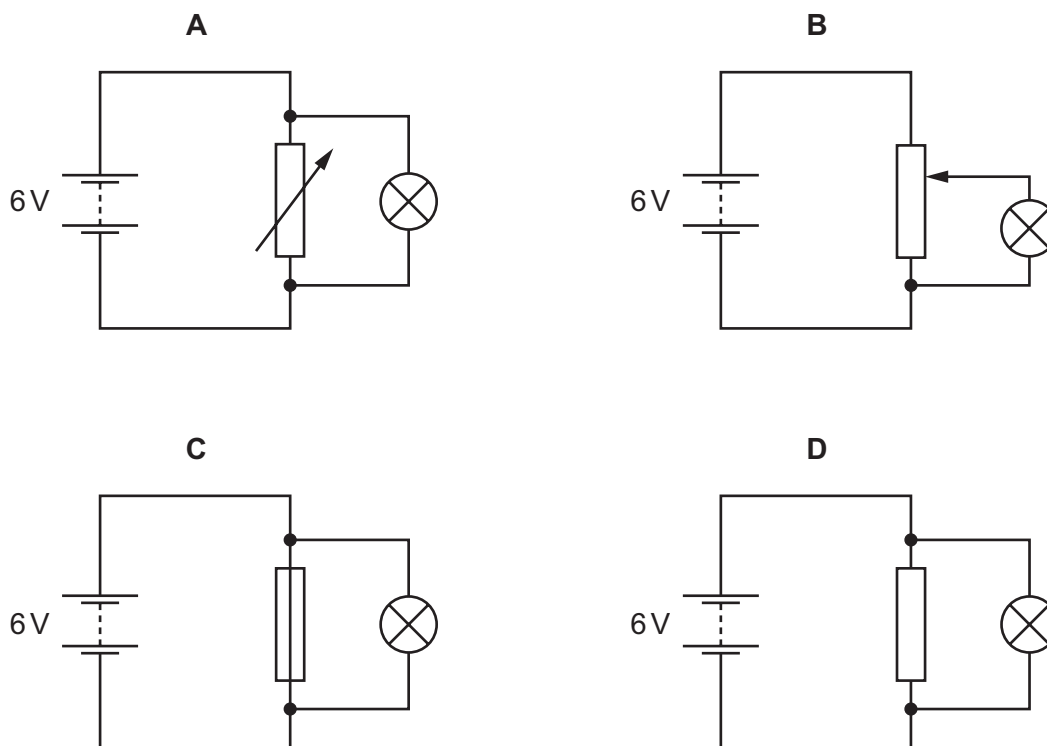
Each lamp is designed to operate at normal brightness when connected individually to a 12 V supply.



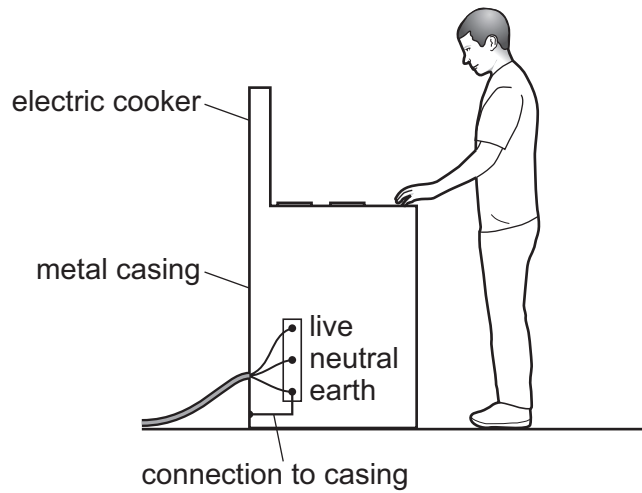
Which statement is correct?

- A** In circuit 1, each of the lamps is at normal brightness.
 - B** In circuit 1, if one lamp fails, the other lamps remain lit.
 - C** In circuit 2, if one lamp fails, the other lamps remain lit.
 - D** In circuit 2, the current from the power supply is less than in circuit 1.
- 34** A lamp is to be connected in a circuit so that the potential difference (p.d.) across it can be varied from 0 to 6 V.

Which circuit would be most suitable?



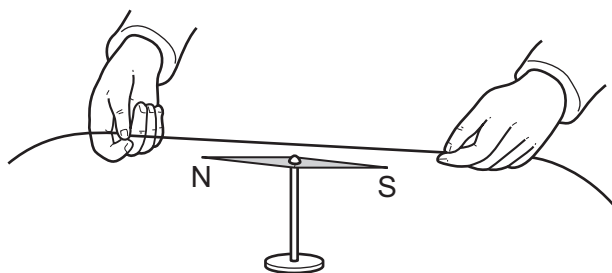
35 A simple wiring diagram for an electric cooker is shown.



Why is there a wire connecting the metal case of the cooker to earth?

- A** It improves the efficiency of the cooker.
- B** It prevents the metal case from becoming too hot when the cooker is left on.
- C** It reduces the risk of an electric shock if the live wire touches the metal case.
- D** The electric cooker will not switch on without it.

36 In an experiment, a wire is held above a compass needle as shown.



An electric current is switched on in the wire and the compass needle is deflected.

Which row explains why this happens and then describes what happens when the current is reversed?

	why this happens	what happens when the current is reversed
A	there is a magnetic field inside the wire	the compass needle deflects in the opposite direction
B	there is a magnetic field inside the wire	there is no effect on the compass needle
C	there is a magnetic field around the wire	the compass needle deflects in the opposite direction
D	there is a magnetic field around the wire	there is no effect on the compass needle

37 Which device relies upon the magnetic effect of an electric current?

- A** fuse
- B** relay
- C** thermistor
- D** variable resistor

38 Which statement about the structure of an atom is correct?

- A** It contains positively charged particles only.
- B** It contains negatively charged particles only.
- C** It contains no charged particles.
- D** It contains positively charged particles and negatively charged particles.

39 Which statement about γ -radiation is correct?

- A It consists of very small charged particles.
- B It is a form of electromagnetic radiation.
- C It is less penetrating than β -radiation.
- D It is more highly ionising than α -radiation.

40 A radioactive element undergoes α -decay.

Which statement is **not** correct?

- A A different element is formed.
- B Radiation is emitted from the nucleus.
- C The decay is a random process.
- D The number of particles in the nucleus stays the same.

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PHYSICS

Paper 1 Multiple Choice (Core)

0625/12

May/June 2019

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

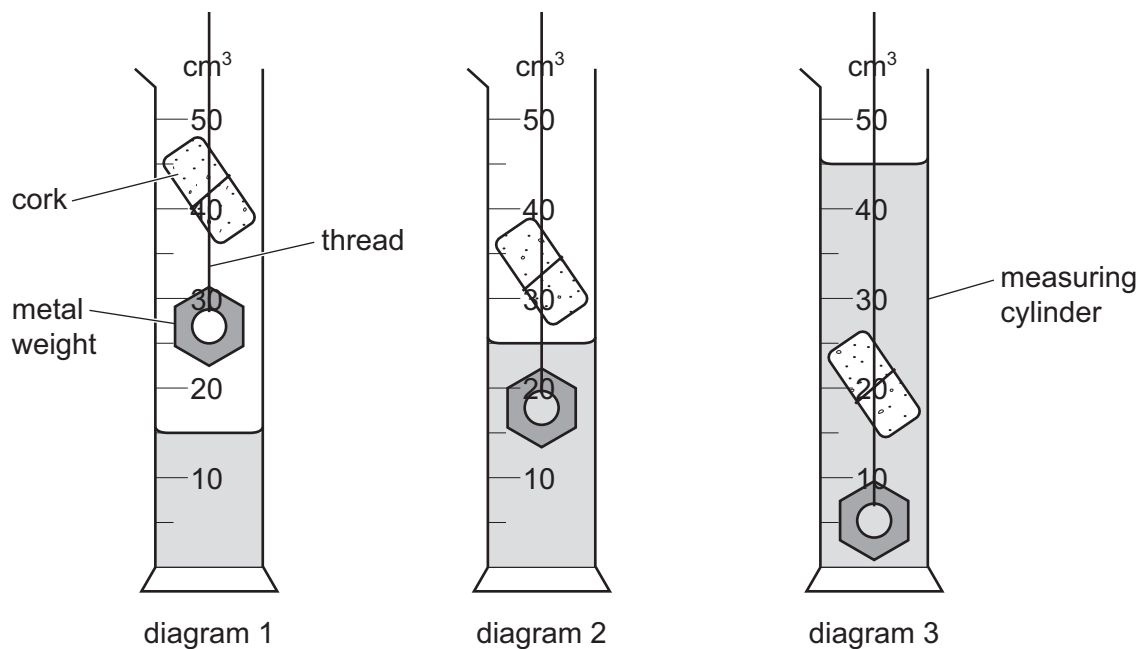
This syllabus is regulated for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **20** printed pages.

- 1 Diagram 1 shows a measuring cylinder containing water. A metal weight with a cork attached by a thread is held above the water.

Diagram 2 shows the apparatus after the weight has been lowered into the water.

Diagram 3 shows the apparatus after the weight and the cork have been submerged.

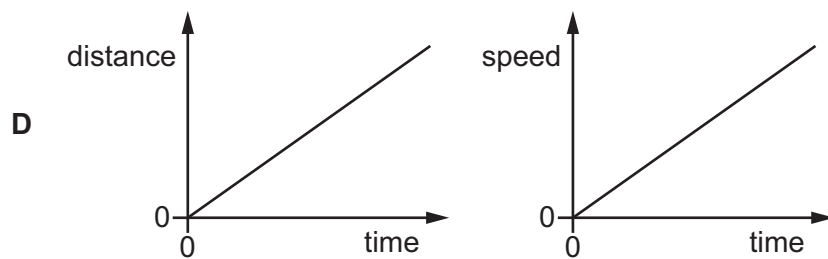
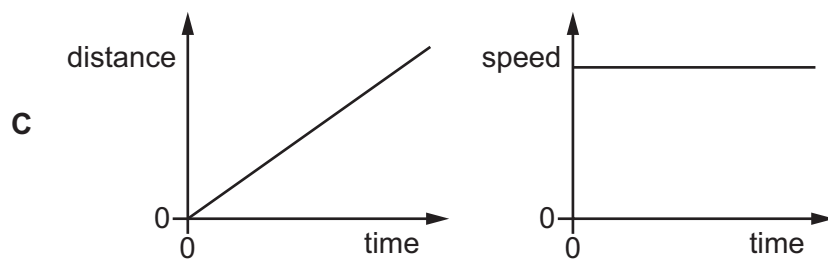
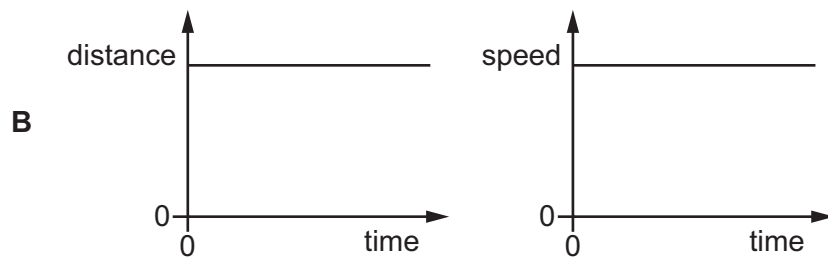
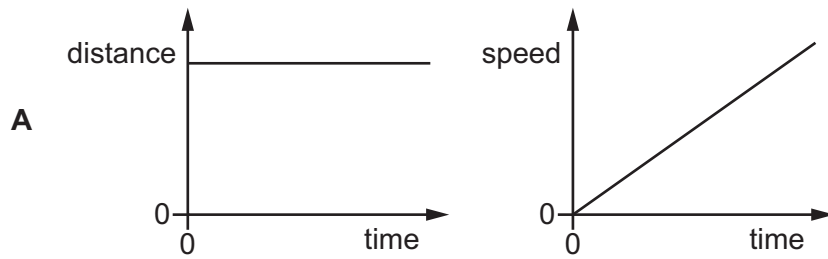


What is the volume of the cork?

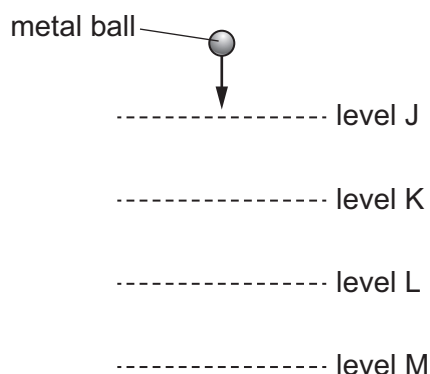
- A** 20 cm^3 **B** 30 cm^3 **C** 45 cm^3 **D** 70 cm^3

- 2 A car travels at constant speed.

Which pair of graphs show both how the distance travelled by the car and the car's speed vary with time?



- 3 A heavy metal ball falls vertically downwards through air past four equally spaced levels J, K, L and M.



The times taken to fall from one level to the next are measured.

Where is the speed of the ball greatest and which time is shortest?

	speed is greatest between	time is shortest between
A	J and K	J and K
B	J and K	L and M
C	L and M	J and K
D	L and M	L and M

- 4 Which row contains two correct statements about the mass and the weight of an object?

	mass of an object	weight of an object
A	is measured using a measuring cylinder	is measured using a balance
B	is the gravitational force exerted on the object	is the amount of matter in the object
C	is measured in newtons	is measured in kilograms
D	is the same everywhere	can vary from place to place

- 5 The table gives approximate values of the acceleration due to gravity and the atmospheric pressure on three planets.

	Earth	Venus	Mars
<u>acceleration due to gravity</u> m/s^2	10	9	4
atmospheric pressure / kPa	100	9000	1

A body has a mass of 10 kg on Earth.

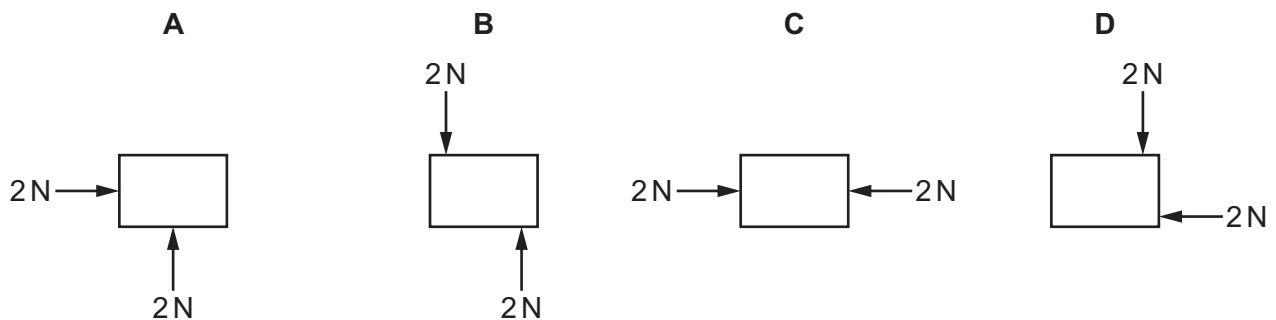
Which statement about the weight of the body is correct?

- A** The weight is greatest on Earth.
B The weight is greatest on Mars.
C The weight is greatest on Venus.
D The weight is the same on each planet.
- 6 A metal has a density of 8.0 g/cm^3 . A solid cube of mass 1.0 kg is made from this metal.

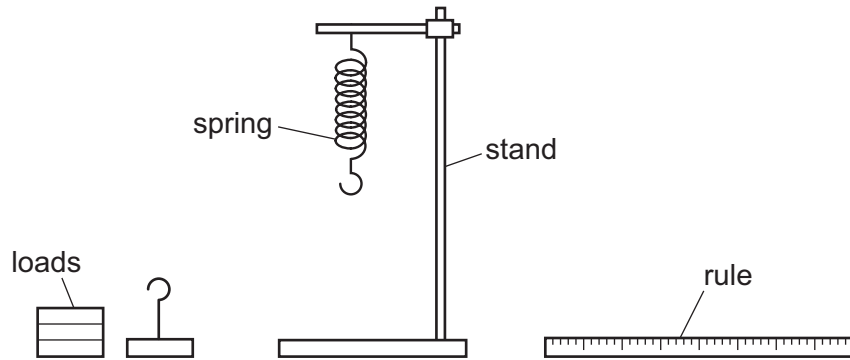
How long is each side of the cube?

- A** 0.50 cm **B** 2.0 cm **C** 5.0 cm **D** 42 cm

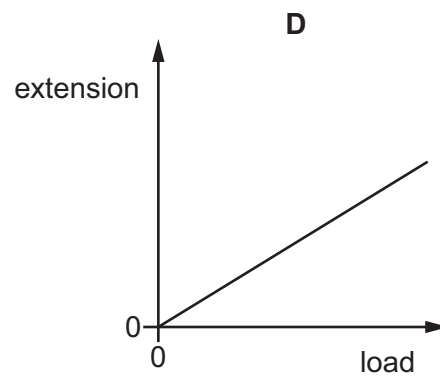
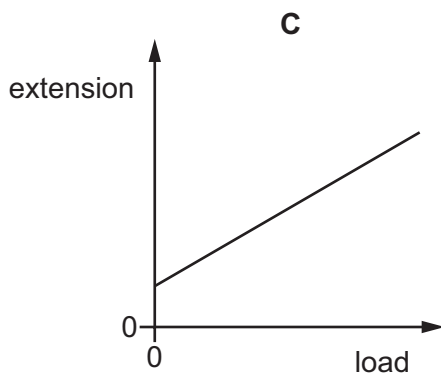
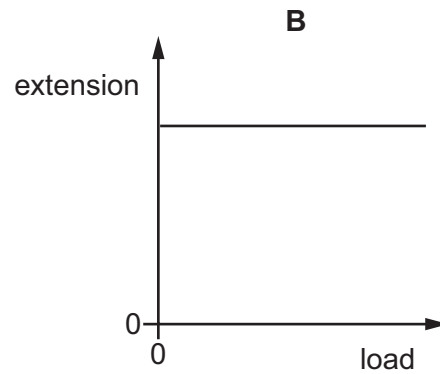
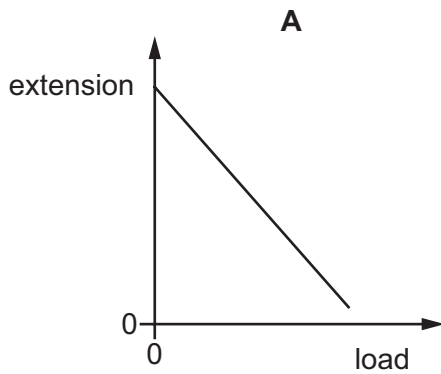
- 7 Which object is in equilibrium?



- 8 A spring is suspended from a stand. Loads are added and the extensions are measured.

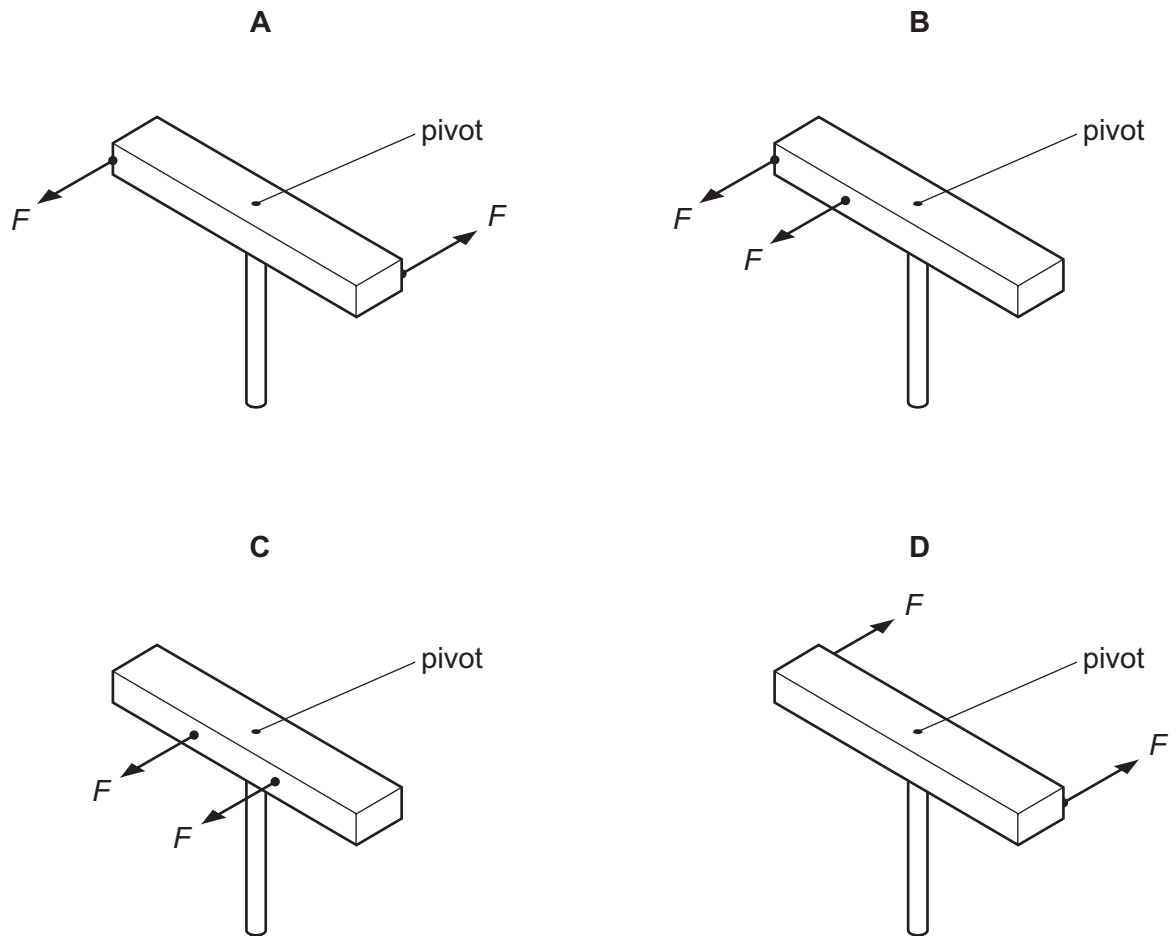


Which graph shows the result of plotting extension against load?



- 9 A wooden bar is pivoted at its centre so that it can rotate freely. Two equal forces F are applied to the bar.

In which diagram is the turning effect greatest?



- 10 A machine is very efficient.

What does this mean?

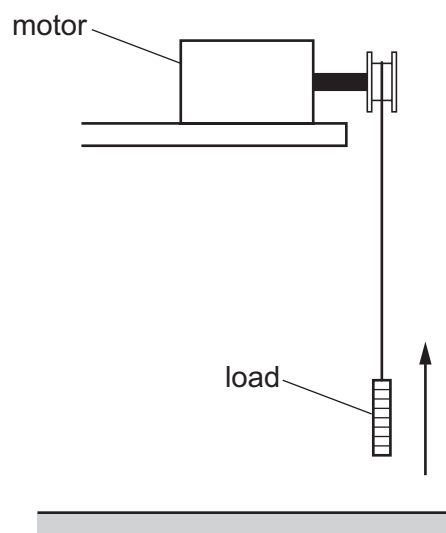
- A It produces a large amount of power.
- B It uses very little energy.
- C It wastes very little energy.
- D It works very quickly.

11 An object falls under gravity.

What happens to the gravitational potential energy and to the kinetic energy of the object?

	gravitational potential energy	kinetic energy
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

12 A student is testing four different electric motors. He measures the time it takes for a motor to lift either a heavy load or a light load through a height of 1 metre.



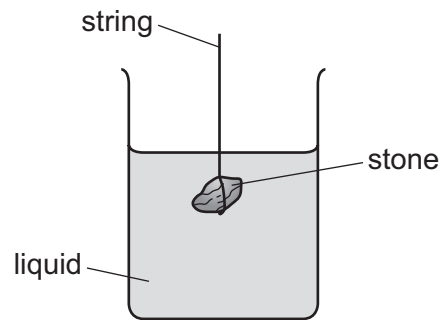
He makes a similar measurement for the other three motors.

The table shows his results.

Which motor produces the most power?

	load	time taken / s
A	heavy	12
B	heavy	16
C	light	12
D	light	16

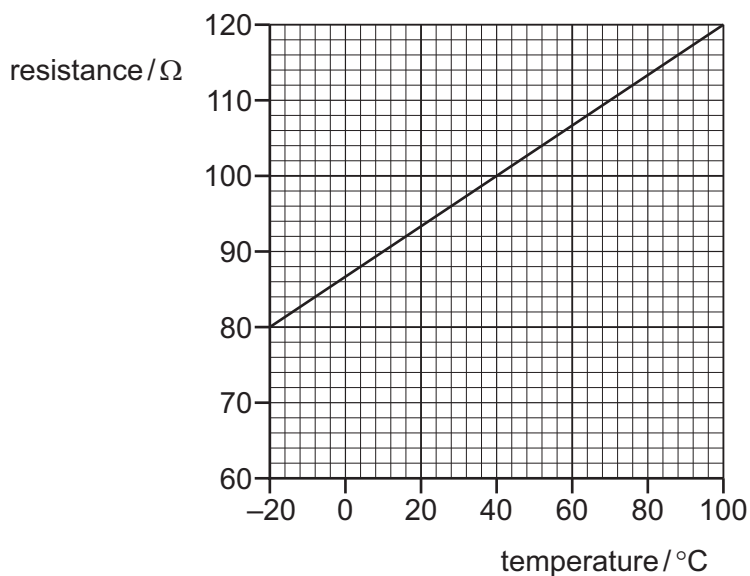
- 13** The diagram shows a stone suspended on a string under the surface of a liquid. The stone experiences a pressure caused by the liquid.



What would increase the pressure on the stone?

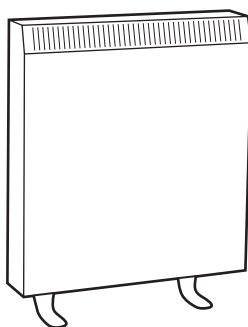
- A** decreasing the surface area of the stone
 - B** increasing the mass of the stone
 - C** lowering the stone deeper into the liquid
 - D** using a liquid with a lower density
- 14** A sample of mercury is heated.
- In which states of matter will its volume increase as its temperature rises?
- A** gas only
 - B** liquid and gas only
 - C** solid and liquid only
 - D** solid, liquid and gas
- 15** When water evaporates, what escapes from the surface of the water?
- A** individual atoms
 - B** individual molecules
 - C** individual protons
 - D** tiny drops of water

- 16 The resistance of a resistor varies linearly with temperature as shown on the graph.



Which statement is correct?

- A** When immersed in boiling water, the resistance is $120\ \Omega$.
- B** The resistance at the lower fixed point is $80\ \Omega$.
- C** When the resistance is $100\ \Omega$ the temperature is $120\ ^\circ\text{C}$.
- D** The resistor can only be used at temperatures between $0\ ^\circ\text{C}$ and $100\ ^\circ\text{C}$.
- 17 A night storage heater contains a large block of material that is heated electrically during the night. During the day the block cools down, releasing thermal energy into the room.



Which thermal capacity and which night-time temperature increase will cause the most energy to be stored by the block?

	thermal capacity of block	night-time temperature increase
A	large	large
B	large	small
C	small	large
D	small	small

18 Which row describes the process of condensation?

	change of state	separation of molecules
A	gas to liquid	decreases
B	gas to liquid	increases
C	liquid to gas	decreases
D	liquid to gas	increases

19 The metal surface of a kettle is hot.

What happens to the cool air outside the kettle when it comes into contact with the hot kettle?

- A** The density of the air decreases and the air falls.
- B** The density of the air decreases and the air rises.
- C** The density of the air increases and the air falls.
- D** The density of the air increases and the air rises.

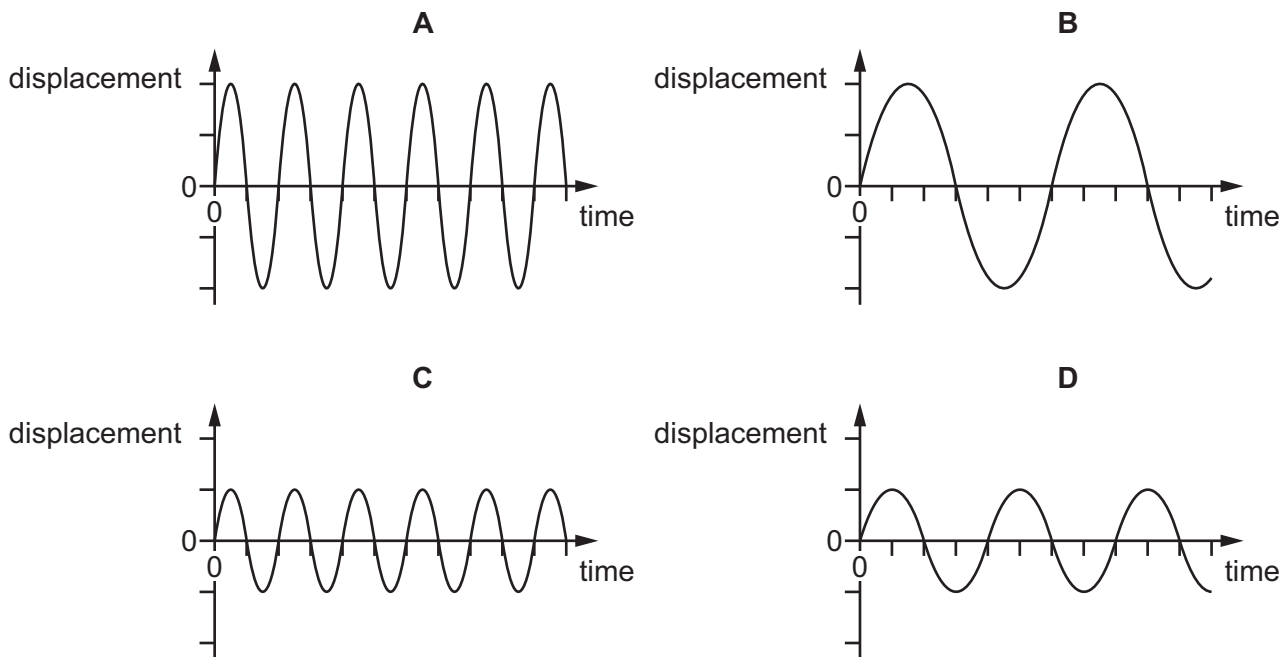
20 Vacuum flasks usually have silvered walls that help to keep the contents of the flask hot.

Why are the walls silvered?

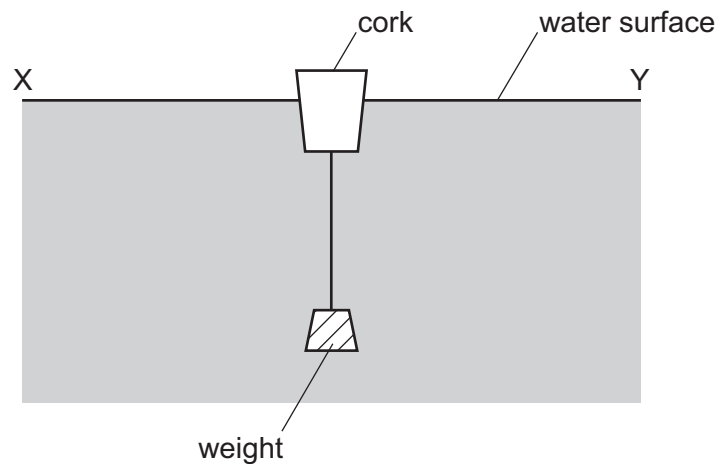
- A** to absorb thermal energy from the air around the flask
- B** to increase the rate of convection inside the flask
- C** to reduce energy loss to the surroundings by conduction
- D** to reflect thermal radiation back into the flask

- 21** The diagrams show graphs of displacement against time for four waves. All the graphs are drawn to the same scale.

Which wave has the largest amplitude and the highest frequency?



- 22** The diagram shows a cork with a weight attached so that the cork floats upright in water.



Transverse waves travel across the water from X to Y.

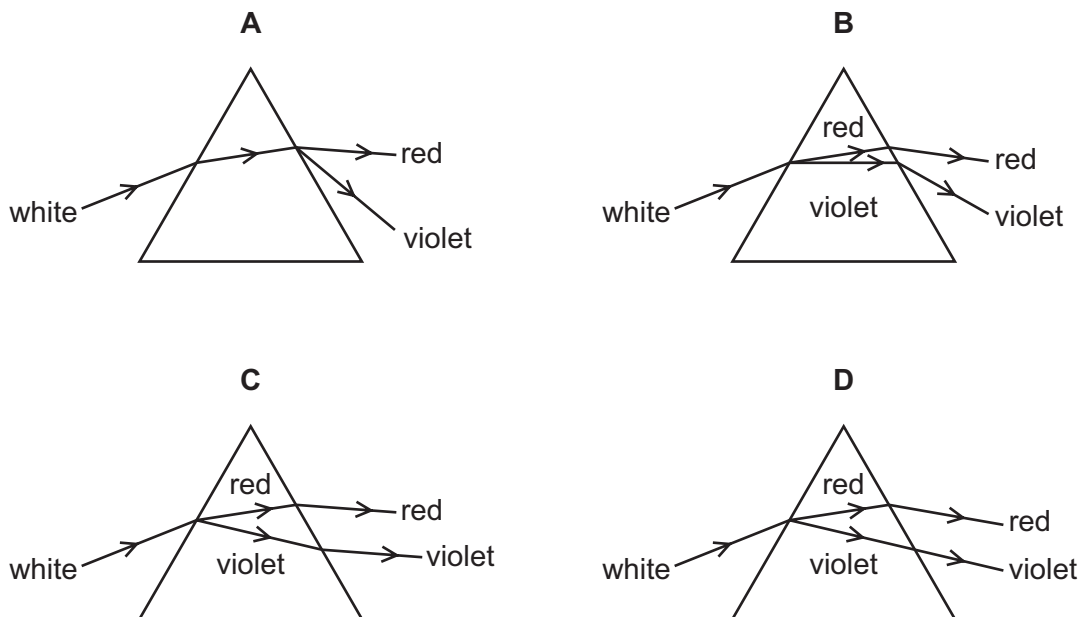
In which direction do the waves make the cork move?

- A** → ← right and left
- B** ↑ ↓ up and down
- C** → only to the right
- D** ← only to the left

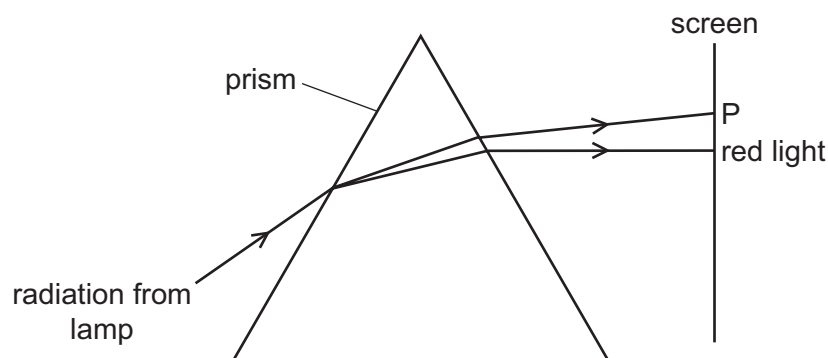
23 Which conditions are necessary for light to be totally internally reflected?

	the incident light is in	angle of incidence
A	the less dense medium	less than the critical angle
B	the less dense medium	greater than the critical angle
C	the more dense medium	less than the critical angle
D	the more dense medium	greater than the critical angle

24 Which diagram shows the dispersion of white light by a glass prism?



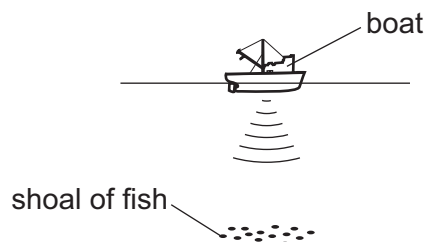
25 The diagram shows radiation from a lamp passing through a prism.



Which type of radiation is found at P?

- A** γ -rays
- B** infrared
- C** ultraviolet
- D** X-rays

- 26** A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from a shoal of fish. The sound reaches the boat again after 1.2 s. The speed of sound in the water is 1500 m/s.

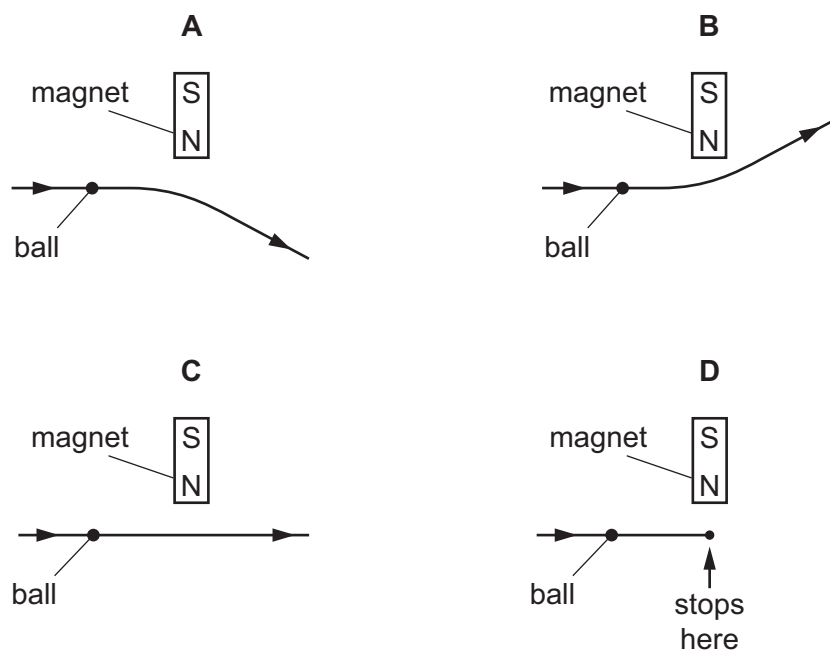


How far below the bottom of the boat is the shoal of fish?

- A** 450 m **B** 900 m **C** 1800 m **D** 3600 m
- 27** An observer stands at the finish line of a 100 m race. He wants to time the winner's run. He starts his stop-watch as soon as he sees the smoke from the starting gun instead of when he hears the bang.
- What is the reason for doing this?
- A** Light travels much faster than sound.
B There is a risk he might respond to an echo from a wall.
C Humans react slower to sound than to light.
D Humans react more quickly to sound than to light.
- 28** A soft iron bar is a long way from any magnetic field.
- How can the material of the bar be described?
- A** It is magnetic and strongly magnetised.
B It is magnetic and unmagnetised.
C It is non-magnetic and strongly magnetised.
D It is non-magnetic and unmagnetised.

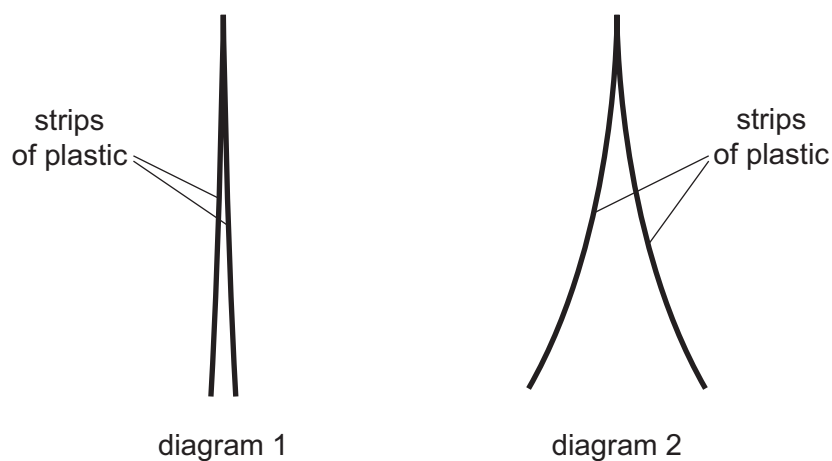
- 29** An iron ball on a horizontal wooden table rolls near the north pole of a bar magnet which is lying on the table.

Which diagram shows the most likely path of the ball, as seen from above the table?



- 30** Diagram 1 shows two thin, uncharged strips of plastic.

Diagram 2 shows the same strips after they have been rubbed with a dry cloth.

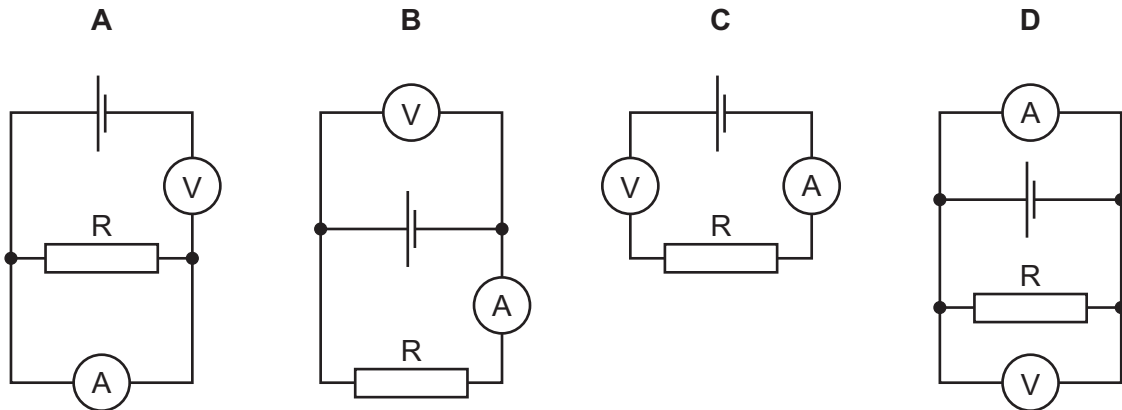


Which row describes the charge on the strips after rubbing, and the force between the strips after rubbing?

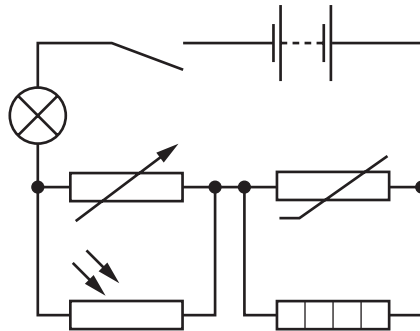
	charge on strips	force between strips
A	opposite	attraction
B	opposite	repulsion
C	the same	attraction
D	the same	repulsion

31 An ammeter and a voltmeter are used to determine the resistance of a resistor.

Which circuit diagram shows the ammeter and the voltmeter correctly connected?



32 The diagram shows a circuit.

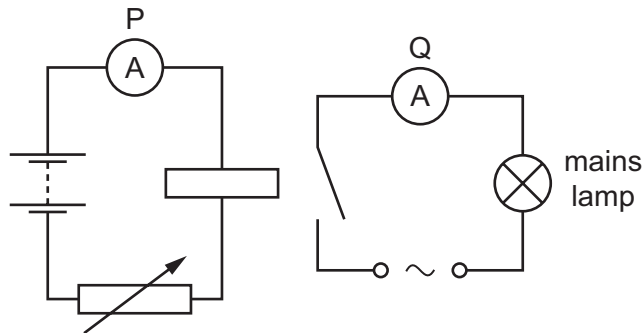


What is connected in parallel with the thermistor?

- A** heater
- B** lamp
- C** light-dependent resistor
- D** variable resistor

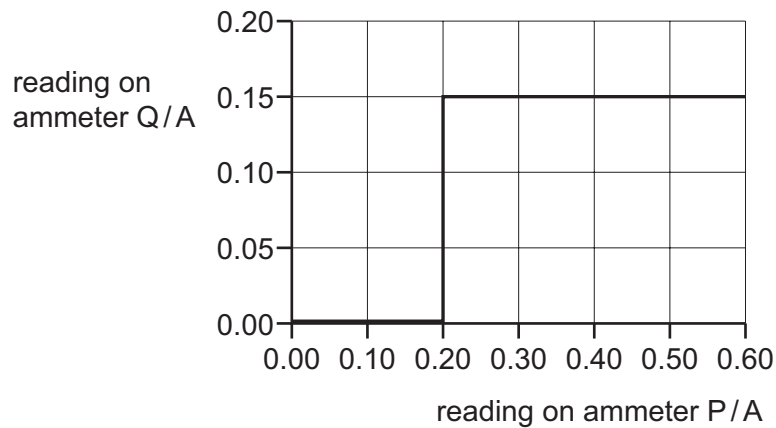
33 The circuit shows a relay being used to operate a mains lamp.

Two ammeters are labelled P and Q.



The variable resistor is used to vary the current in the relay coil. The mains lamp switches on when there is a large enough current in the relay coil.

The graph shows how the reading on ammeter Q changes as the reading on ammeter P increases.

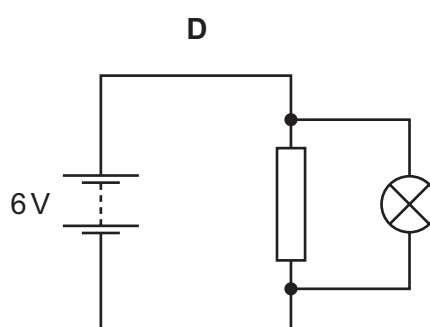
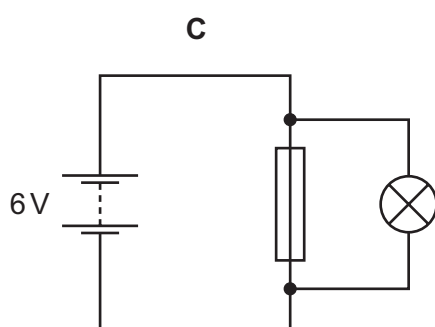
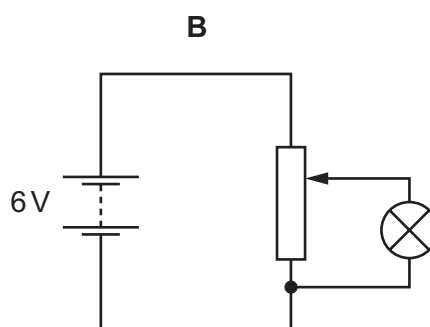
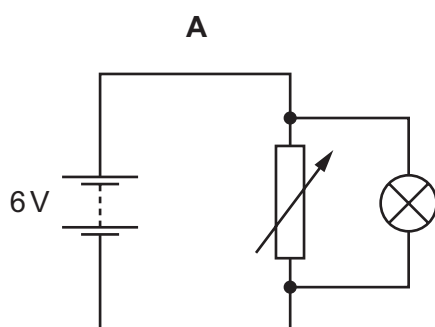


What is the minimum current needed in the relay coil to switch on the mains lamp?

- A** 0.15 A **B** 0.20 A **C** 0.35 A **D** 0.60 A

- 34** A lamp is to be connected in a circuit so that the potential difference (p.d.) across it can be varied from 0 to 6 V.

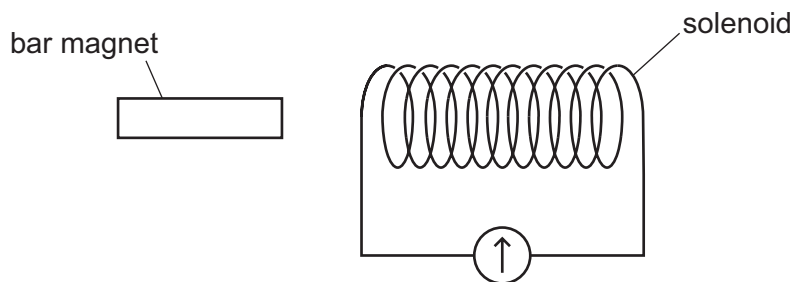
Which circuit would be most suitable?



- 35** Which components are designed to improve the safe working of a mains electrical supply?

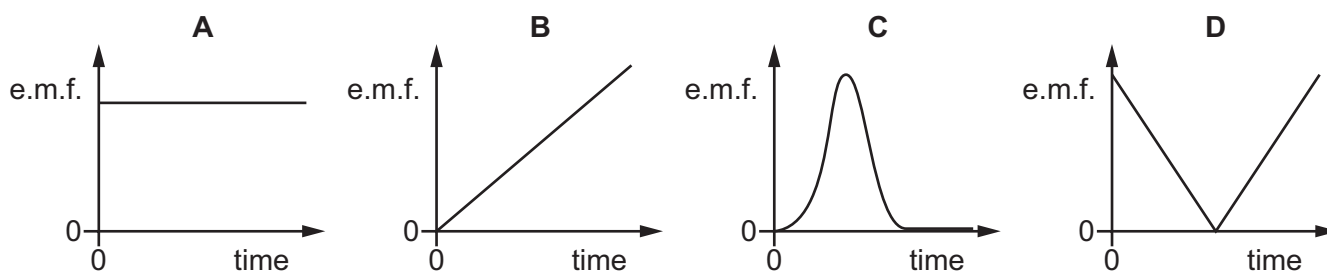
	circuit breaker	earth wire	fuse
A	✓	✓	✗
B	✓	✗	✓
C	✗	✓	✓
D	✓	✓	✓

36 A bar magnet is held near a solenoid. The coil is connected to a galvanometer.



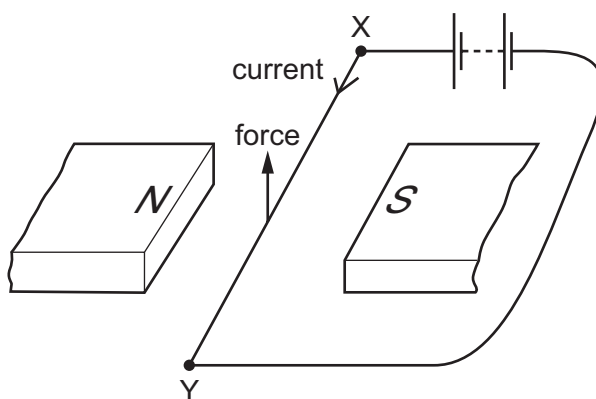
The magnet is moved into the coil of wire and then held stationary inside the coil.

Which graph shows how the induced electromotive force (e.m.f.) varies with time?



37 A wire XY lies between the poles of a magnet.

The diagram shows the upward force on the wire XY caused when there is an electric current in the direction XY as shown.



Three tests are made using this apparatus.

- 1 The current direction is reversed.
- 2 The N and S poles are swapped around.
- 3 The current is switched off.

Which will result in **no change** in the size of the force on the wire?

- A** 1 and 2 only **B** 1 only **C** 2 only **D** 3 only

38 A simple model of the atom consists of small particles orbiting a central nucleus.

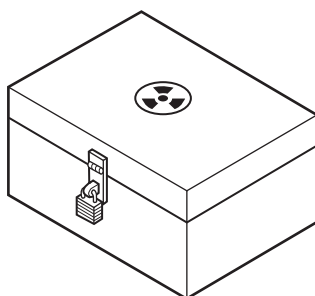
Which row is correct?

	charge on nucleus	charge on orbiting particles
A	negative	negative
B	negative	positive
C	positive	negative
D	positive	positive

39 Which statement explains the meaning of the half-life of a radioactive isotope?

- A** half the time taken for one nucleus of the isotope to decay
- B** half the time taken for the isotope to decay completely
- C** the time taken for half of the nuclei of the isotope to decay
- D** the time taken for one nucleus of the isotope to split in half

40 The diagram shows a lead-lined box used for storing radioactive sources.



Why is the inside of the box lined with lead?

- A** It helps the sources to stay radioactive for longer.
- B** It makes the box heavier.
- C** It makes the radioactive sources more stable.
- D** It reduces the amount of radiation that can escape from the box.

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Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

PHYSICS

Paper 1 Multiple Choice (Core)

0625/11

May/June 2019

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

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This document consists of **19** printed pages and **1** blank page.

- 1 A digital stop-clock measures time in minutes and seconds.

The stop-clock reads 00:50 when it is started (i.e. 00 minutes 50 seconds).

It reads 02:10 when it is stopped.

What is the shortest possible time that has elapsed between starting and stopping the stop-clock?

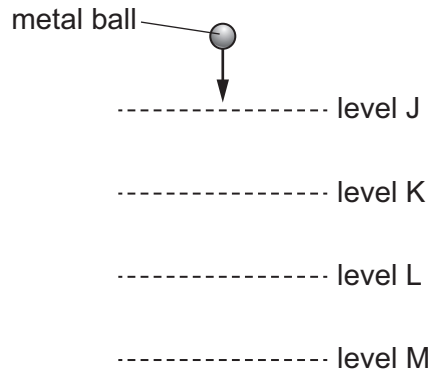
- A 1 minute 20 seconds
- B 2 minutes 00 seconds
- C 2 minutes 10 seconds
- D 3 minutes 00 seconds

- 2 A long-distance runner wishes to calculate her average speed for a race.

Which calculation should she use?

- A $\text{average speed} = \frac{\text{total distance}}{\text{total time}}$
- B $\text{average speed} = \text{total distance} \times \text{total time}$
- C $\text{average speed} = \frac{\text{total time}}{\text{total distance}}$
- D $\text{average speed} = \text{total distance} + \text{total time}$

- 3 A heavy metal ball falls vertically downwards through air past four equally spaced levels J, K, L and M.



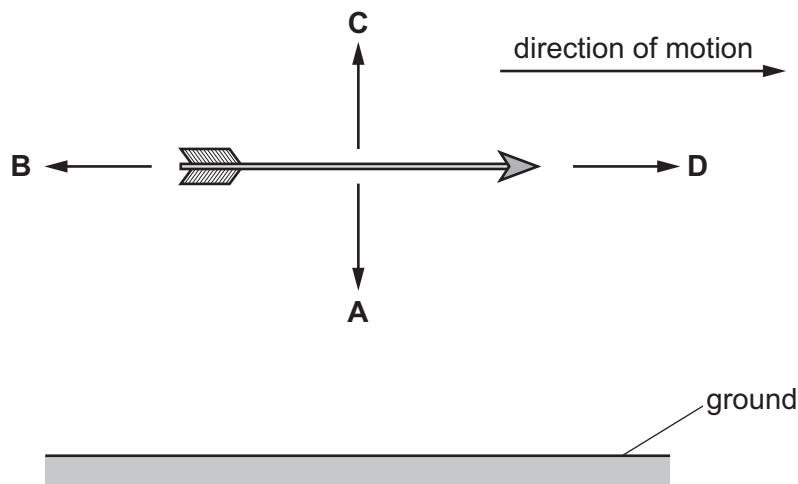
The times taken to fall from one level to the next are measured.

Where is the speed of the ball greatest and which time is shortest?

	speed is greatest between	time is shortest between
A	J and K	J and K
B	J and K	L and M
C	L and M	J and K
D	L and M	L and M

- 4 An arrow travels horizontally in a straight line at constant speed.

In which direction does the weight act?



- 5 Two rectangular blocks consist of different materials.

Four different methods are suggested to compare the two masses.

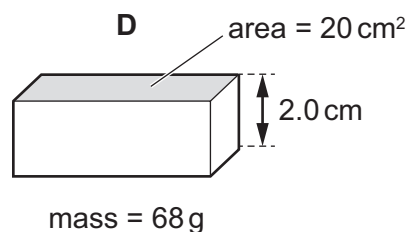
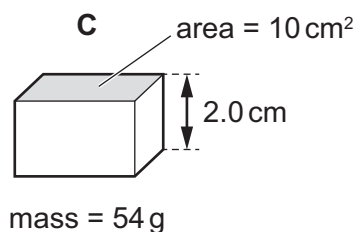
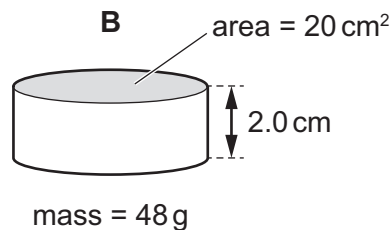
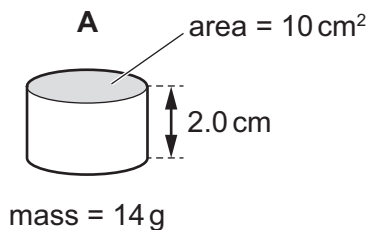
- 1 Compare the accelerations with which they fall freely.
- 2 Compare the values of their lengths $\times$ breadths $\times$ heights.
- 3 Hang each in turn from the same spring. Compare the extensions.
- 4 Place one in the right-hand pan of a beam balance and the other in the left-hand pan.

Which methods give a comparison of the two masses?

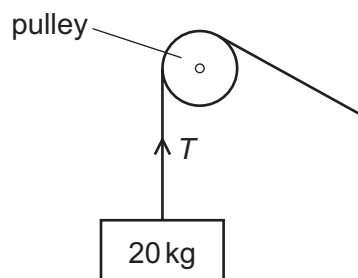
- A** 1, 2 and 3 **B** 1 and 2 only **C** 3 and 4 only **D** 4 only

- 6 The diagrams show four solid blocks with their dimensions and masses.

Which block has the greatest density?



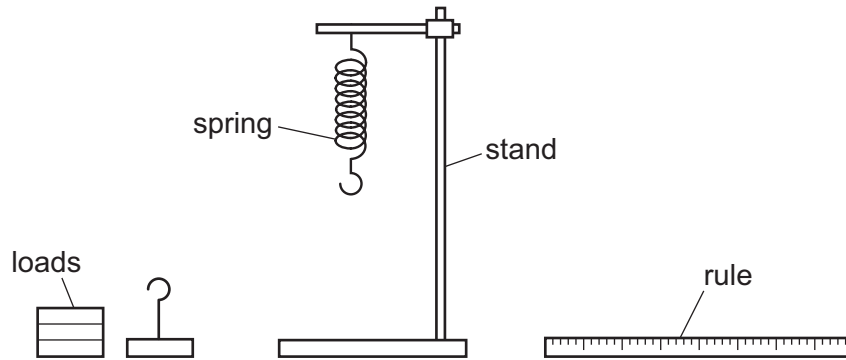
- 7 A mass of 20 kg is held stationary by a rope passing over a frictionless pulley.



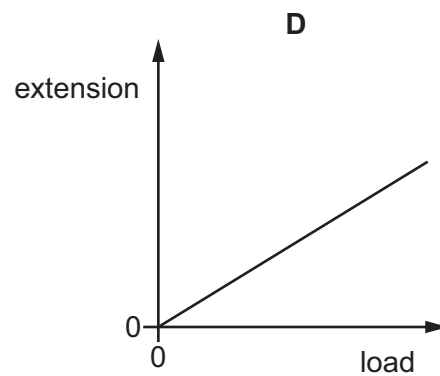
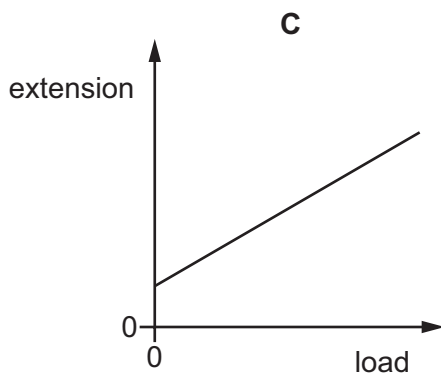
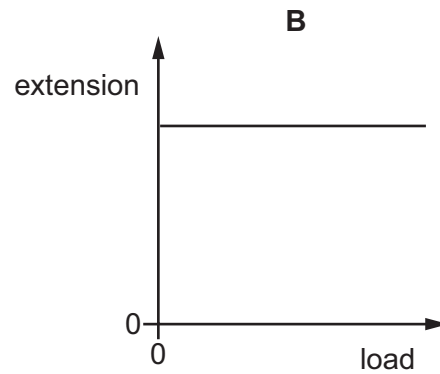
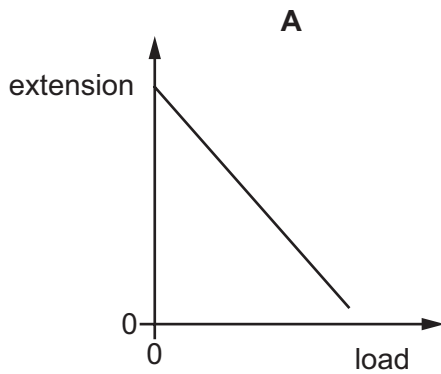
What is the tension T in the rope?

- A** 10 kg **B** 20 kg **C** 100 N **D** 200 N

- 8 A spring is suspended from a stand. Loads are added and the extensions are measured.

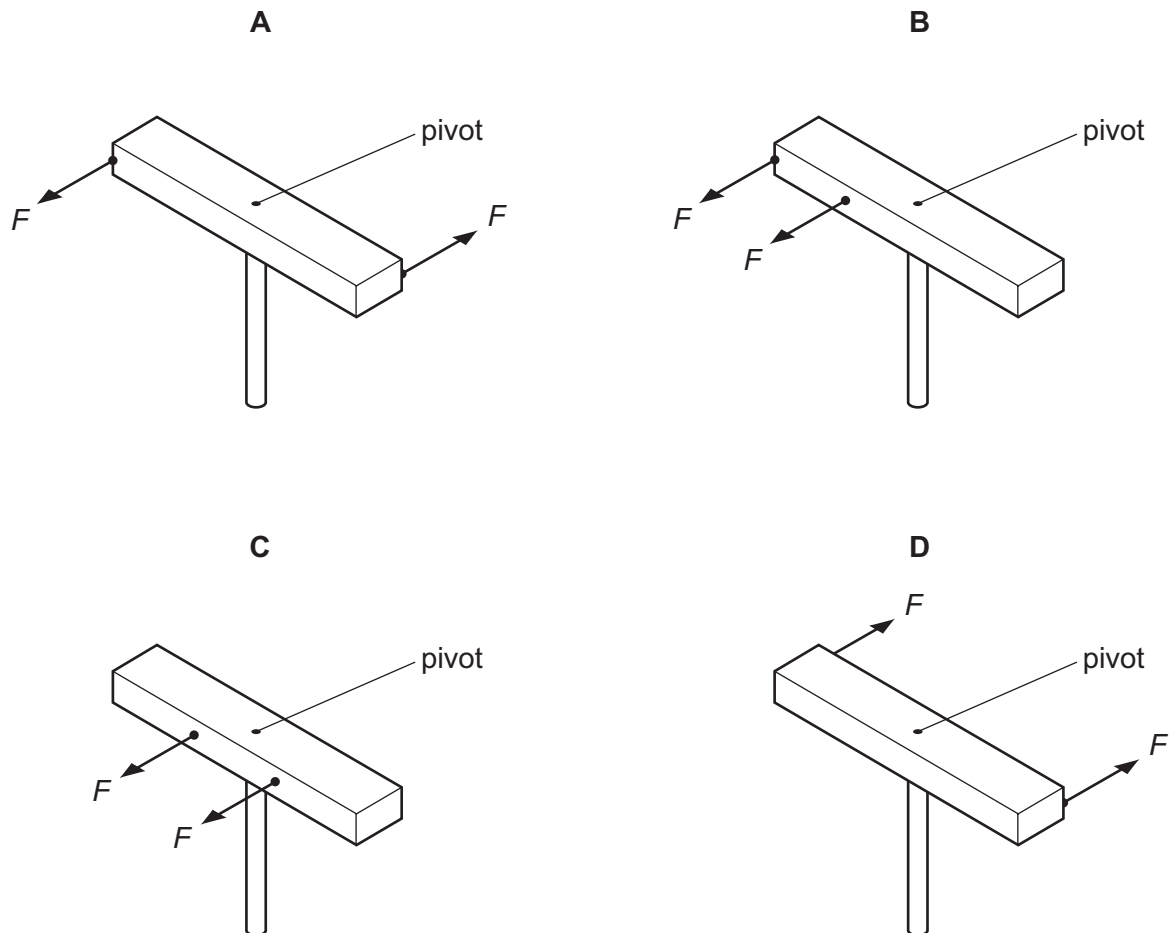


Which graph shows the result of plotting extension against load?



- 9 A wooden bar is pivoted at its centre so that it can rotate freely. Two equal forces F are applied to the bar.

In which diagram is the turning effect greatest?



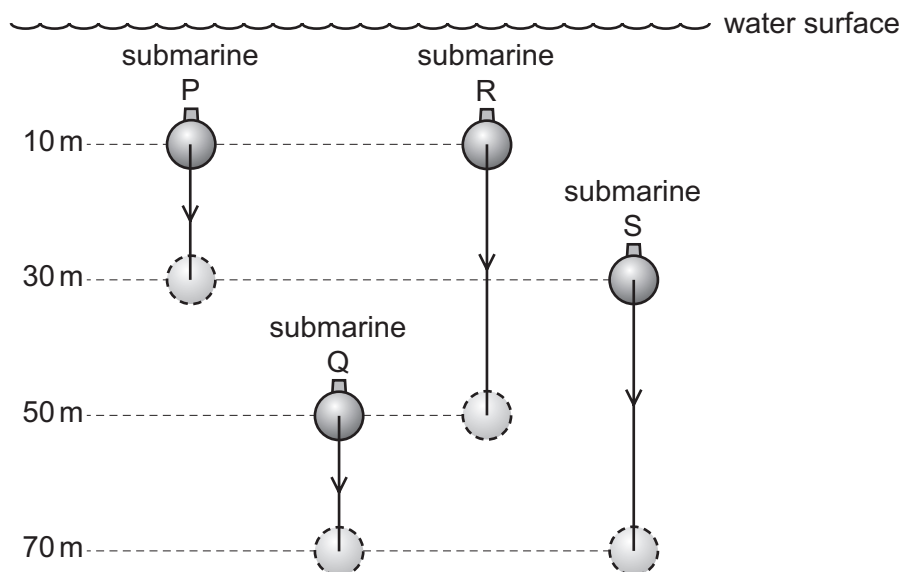
- 10 A machine is very efficient.

What does this mean?

- A It produces a large amount of power.
 - B It uses very little energy.
 - C It wastes very little energy.
 - D It works very quickly.
- 11 An object is lifted vertically by a motor.
- In which example is the power produced the greatest?
- A lifting it a shorter distance in a longer time
 - B lifting it the same distance in a longer time
 - C lifting it a shorter distance in the same time
 - D lifting it the same distance in a shorter time

- 12 Four identical submarines P, Q, R and S are lowered from one depth to another in water of a constant density.

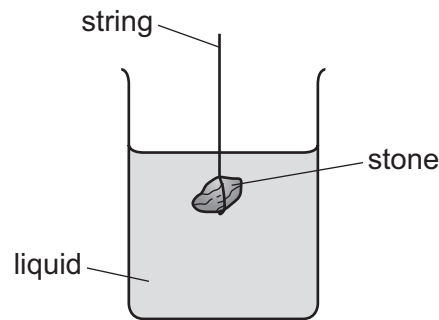
The initial and final depths of each submarine are shown in the diagram.



Which row is correct for the difference in pressure from the initial to final depth of each submarine?

	least change in pressure	greatest change in pressure
A	submarines P and Q	submarines R and S
B	submarines P and Q	submarine R only
C	submarine P only	submarines R and S
D	submarine P only	submarine R only

- 13** The diagram shows a stone suspended on a string under the surface of a liquid. The stone experiences a pressure caused by the liquid.



What would increase the pressure on the stone?

- A** decreasing the surface area of the stone
 - B** increasing the mass of the stone
 - C** lowering the stone deeper into the liquid
 - D** using a liquid with a lower density
- 14** Why can a gas be compressed easily into a smaller volume?
- A** The molecules are far apart.
 - B** The molecules do not attract each other.
 - C** The molecules move randomly.
 - D** The volume of each molecule can be reduced.
- 15** When a liquid evaporates, some molecules escape. The temperature of the remaining liquid changes.

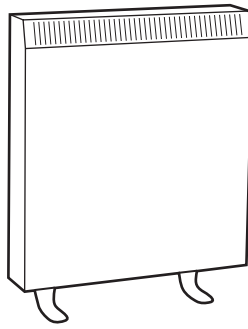
What is the effect on the temperature and from where do the molecules escape?

	temperature of liquid	molecules escape from
A	decreases	everywhere within the liquid
B	decreases	the surface only
C	increases	everywhere within the liquid
D	increases	the surface only

- 16** What happens when a metal block is heated?

- A** Its breadth, height and length all increase.
- B** Its width increases only.
- C** Its height increases only.
- D** Its length increases only.

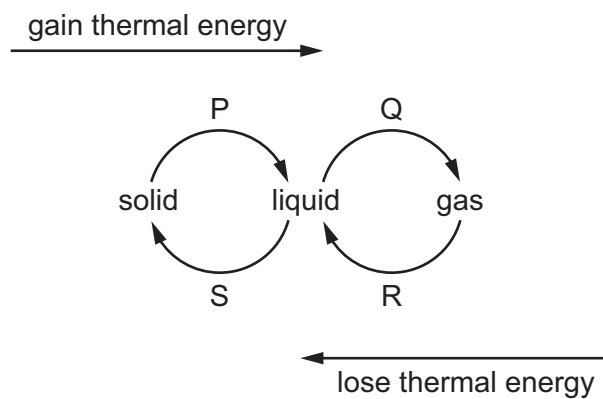
- 17 A night storage heater contains a large block of material that is heated electrically during the night. During the day the block cools down, releasing thermal energy into the room.



Which thermal capacity and which night-time temperature increase will cause the most energy to be stored by the block?

	thermal capacity of block	night-time temperature increase
A	large	large
B	large	small
C	small	large
D	small	small

- 18 The diagram shows the changes of state P, Q, R and S that occur in solids, liquids and gases when they gain or lose thermal energy.



What is the name of change R?

- A** condensation
- B** solidification
- C** boiling
- D** melting

19 In which does thermal conduction **not** occur?

- A** a gas
- B** a liquid
- C** a solid
- D** a vacuum

20 The metal surface of a kettle is hot.

What happens to the cool air outside the kettle when it comes into contact with the hot kettle?

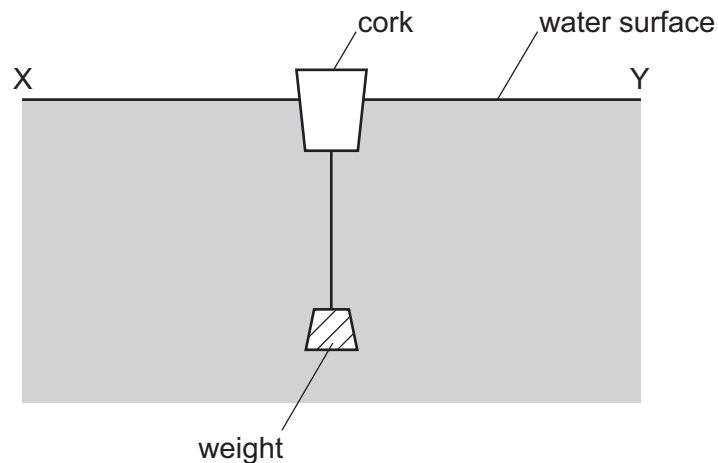
- A** The density of the air decreases and the air falls.
- B** The density of the air decreases and the air rises.
- C** The density of the air increases and the air falls.
- D** The density of the air increases and the air rises.

21 Ultrasound is used in a hospital to scan a patient. Ultrasound refracts at the boundary between muscle and bone because it travels at a greater speed in bone.

Which change takes place when the ultrasound travels from muscle into bone?

- A** The frequency of the wave decreases.
- B** The frequency of the wave increases.
- C** The wavelength of the wave decreases.
- D** The wavelength of the wave increases.

- 22** The diagram shows a cork with a weight attached so that the cork floats upright in water.



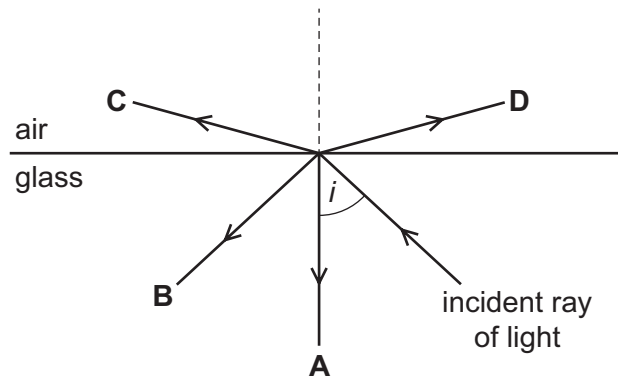
Transverse waves travel across the water from X to Y.

In which direction do the waves make the cork move?

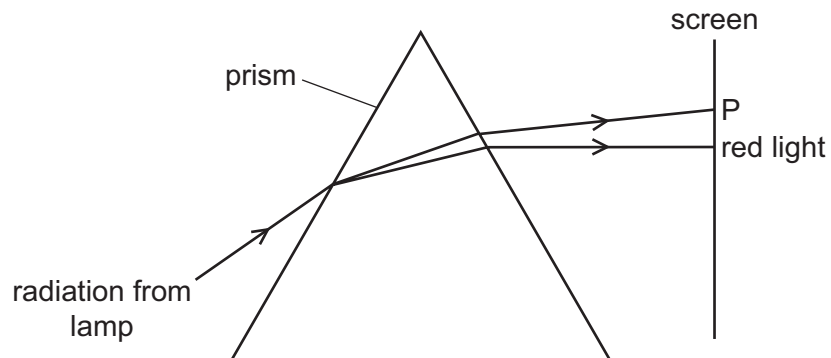
- A** → ← right and left
 - B** ↑↓ up and down
 - C** → only to the right
 - D** ← only to the left
- 23** An object is placed 30 cm in front of a plane mirror.
- Which statement describes the image of the object?
- A** The image is the same size and 30 cm from the object.
 - B** The image is the same size and 60 cm from the object.
 - C** The image is smaller and 30 cm from the object.
 - D** The image is smaller and 60 cm from the object.

- 24** The diagram shows light incident at a glass-air boundary. The angle of incidence i of the ray is greater than the critical angle.

Which line shows the path of the light after it meets the boundary?



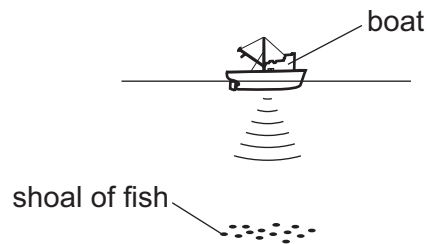
- 25** The diagram shows radiation from a lamp passing through a prism.



Which type of radiation is found at P?

- A** γ -rays
- B** infrared
- C** ultraviolet
- D** X-rays

- 26** A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from a shoal of fish. The sound reaches the boat again after 1.2 s. The speed of sound in the water is 1500 m/s.

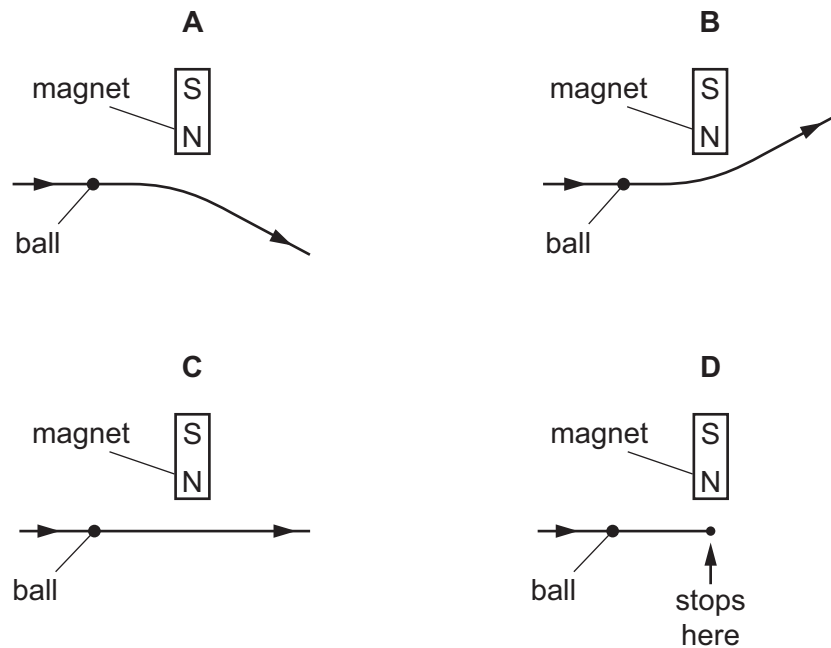


How far below the bottom of the boat is the shoal of fish?

- A** 450 m **B** 900 m **C** 1800 m **D** 3600 m
- 27** Which range is approximately correct for the audio frequencies that can be detected by a healthy human ear?
- A** 2 Hz to 2000 Hz
B 2 Hz to 20 000 Hz
C 20 Hz to 2000 Hz
D 20 Hz to 20 000 Hz
- 28** Why is soft iron used for the core of an electromagnet?
- A** Soft iron easily becomes a permanent magnet.
B Soft iron is a good electrical conductor.
C Soft iron is a poor thermal conductor.
D Soft iron loses its magnetism when the current in the coil is switched off.

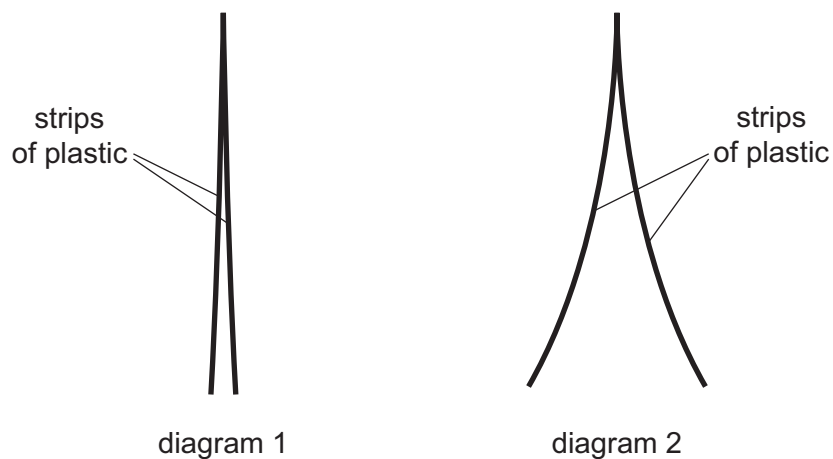
- 29** An iron ball on a horizontal wooden table rolls near the north pole of a bar magnet which is lying on the table.

Which diagram shows the most likely path of the ball, as seen from above the table?



- 30** Diagram 1 shows two thin, uncharged strips of plastic.

Diagram 2 shows the same strips after they have been rubbed with a dry cloth.

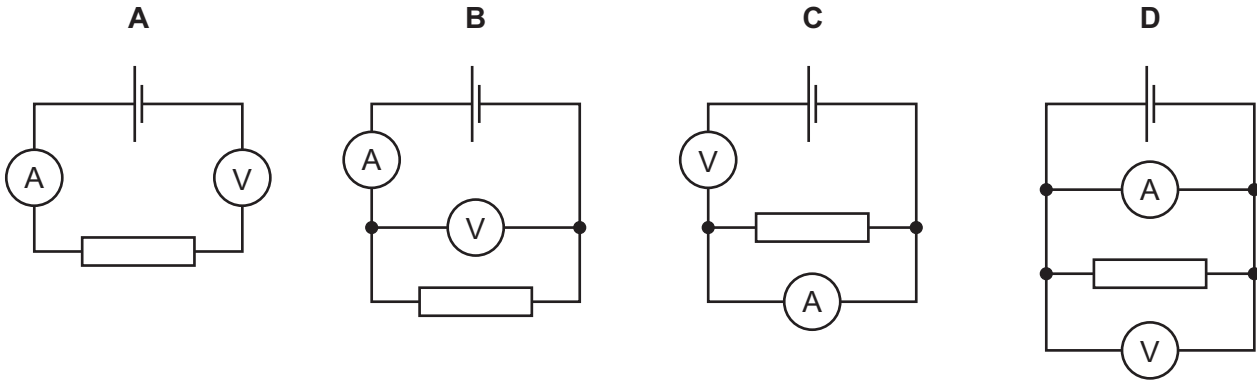


Which row describes the charge on the strips after rubbing, and the force between the strips after rubbing?

	charge on strips	force between strips
A	opposite	attraction
B	opposite	repulsion
C	the same	attraction
D	the same	repulsion

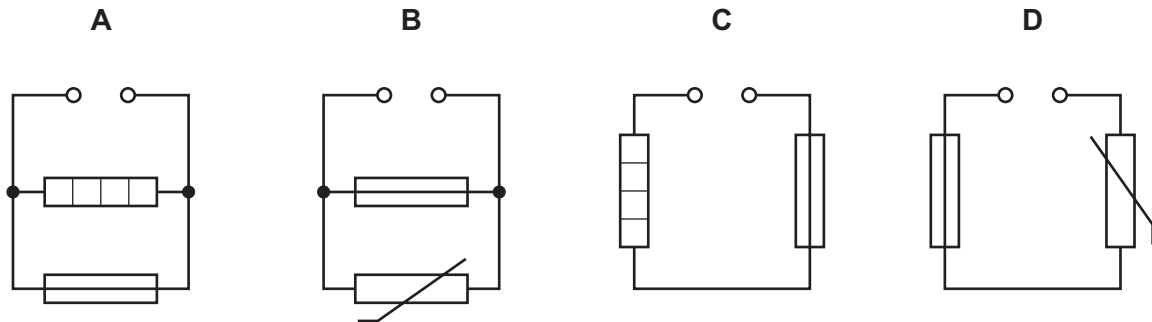
31 A voltmeter and an ammeter are used to measure the resistance of a resistor.

Which diagram shows the voltmeter and the ammeter correctly connected?



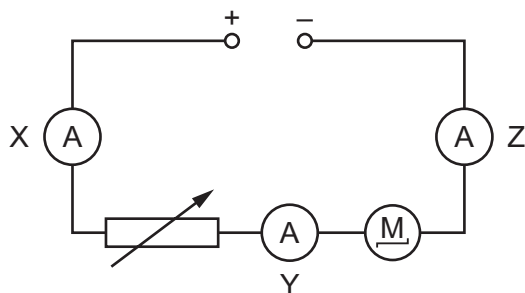
32 A student sets up four circuits.

In which circuit is there a heater in series with a fuse?



- 33** The diagram shows a circuit containing a d.c. power supply, a motor and a variable resistor.

Three ammeters X, Y and Z show the current in different parts of the circuit.

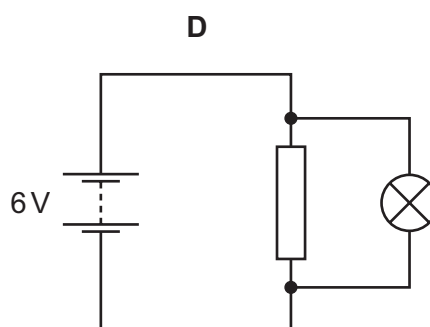
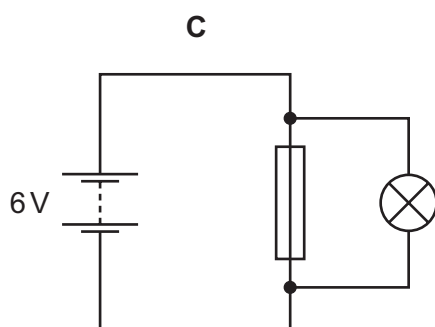
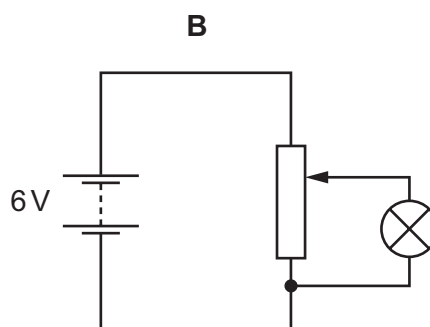
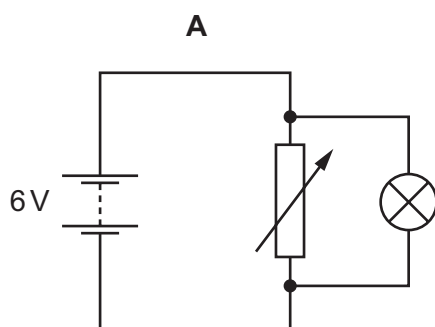


The reading on X is 4.0 A.

Which statement is correct?

- A** The readings on Y and Z are both less than 4.0 A.
 - B** The readings on Y and Z are both equal to 4.0 A.
 - C** The readings on Y and Z are both greater than 4.0 A.
 - D** The reading on Z is zero.
- 34** A lamp is to be connected in a circuit so that the potential difference (p.d.) across it can be varied from 0 to 6 V.

Which circuit would be most suitable?



- 35** An electric heater is plugged into the mains supply using a fused plug.

The current in the heater is 10 A.

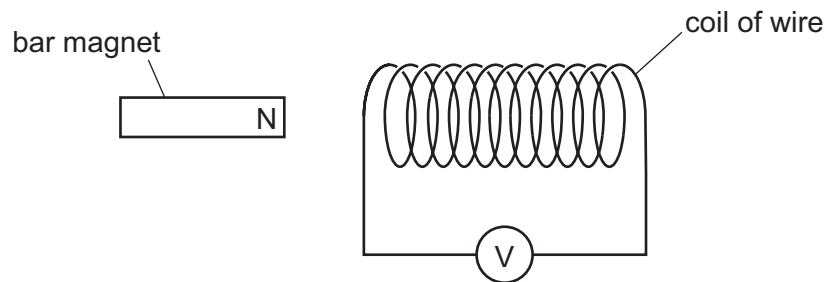
The cable attached to the heater is rated at 15 A.

The fuses available are rated at 1 A, 3 A, 5 A and 13 A.

Which fuse should be used?

- A** 1 A **B** 3 A **C** 5 A **D** 13 A

- 36** A bar magnet is held near a coil of wire. The coil is connected to a sensitive voltmeter.



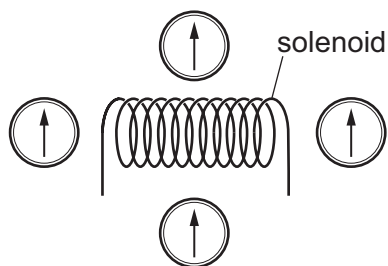
The N-pole of the magnet is moved quickly towards the coil. The voltmeter shows a reading of +10 mV.

The N-pole of the magnet is then moved slowly away from the same end of the coil. The reading on the voltmeter is observed.

Which voltmeter reading is possible?

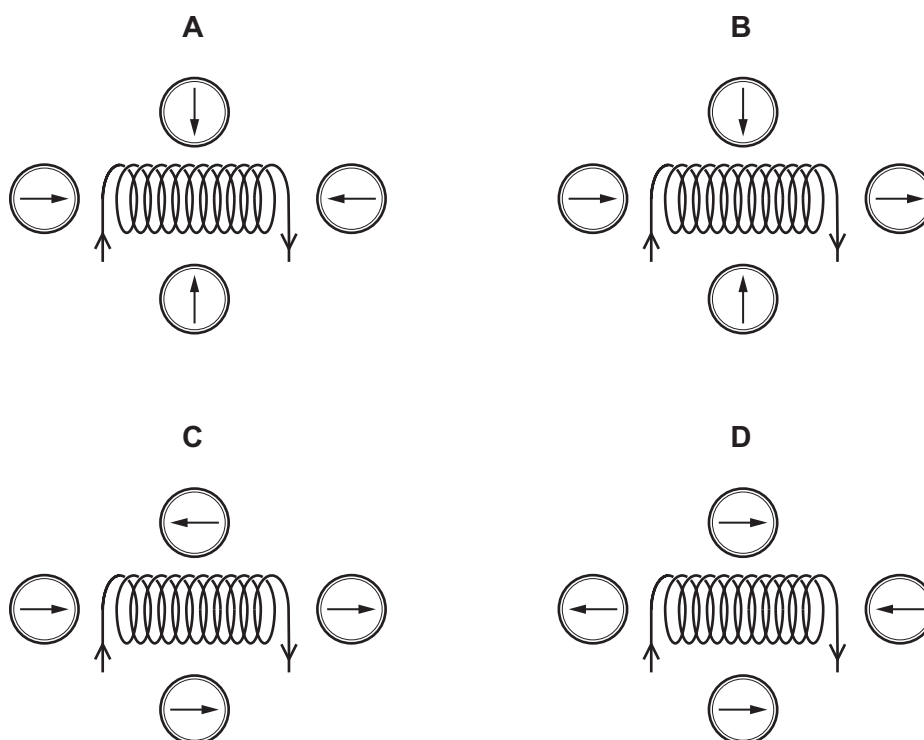
- A** -15 mV **B** -5 mV **C** 0 mV **D** +5 mV

37 Four small compasses are placed around a solenoid.



A current is now switched on in the solenoid.

Which diagram shows possible new directions of the compass needles?



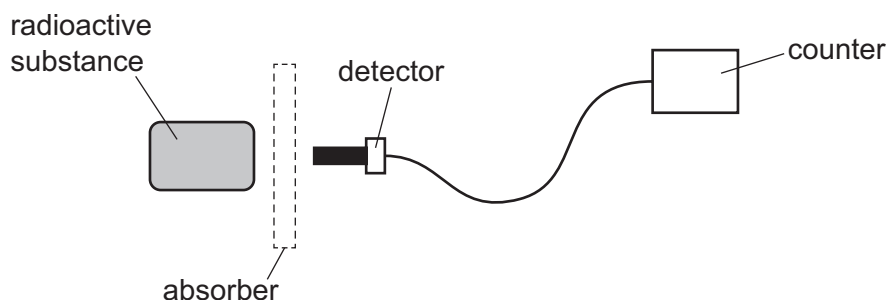
38 Four nuclides are represented below.



Which pair of nuclides are isotopes of the same element?

- A** E and G **B** E and L **C** G and L **D** G and M

- 39 A student measures the level of radiation emitted from a radioactive substance. He places a detector very close to the substance. He puts different absorbers between the radioactive substance and the detector.



The student's results are shown. These results are corrected for background radiation.

absorber	<u>counter reading</u> counts per minute
none	95
thin paper	52
few mm of aluminium	52
several cm of lead	12

Which types of radiation are being emitted by the substance?

- A α -particles and β -particles only
 - B α -particles and γ -rays only
 - C β -particles and γ -rays only
 - D α -particles, β -particles and γ -rays
- 40 The nuclei of the atoms in a substance are changing randomly and emitting radiation.

What is happening to the substance?

- A It is undergoing electromagnetic induction.
- B It is undergoing magnetisation.
- C It is undergoing solidification.
- D It is undergoing radioactive decay.

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PHYSICS

0625/13

Paper 1 Multiple Choice (Core)

May/June 2019

MARK SCHEME

Maximum Mark: 40

Published

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	D	1
3	D	1
4	C	1
5	B	1
6	C	1
7	D	1
8	D	1
9	A	1
10	C	1
11	B	1
12	A	1
13	C	1
14	C	1
15	C	1
16	D	1
17	A	1
18	D	1
19	A	1
20	B	1
21	D	1
22	B	1
23	C	1
24	C	1
25	B	1
26	B	1
27	A	1
28	B	1

Question	Answer	Marks
29	B	1
30	D	1
31	A	1
32	C	1
33	C	1
34	B	1
35	C	1
36	C	1
37	B	1
38	D	1
39	B	1
40	D	1



PHYSICS

0625/12

Paper 1 Multiple Choice (Core)

May/June 2019

MARK SCHEME

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Question	Answer	Marks
1	A	1
2	C	1
3	D	1
4	D	1
5	A	1
6	C	1
7	C	1
8	D	1
9	A	1
10	C	1
11	B	1
12	A	1
13	C	1
14	D	1
15	B	1
16	A	1
17	A	1
18	A	1
19	B	1
20	D	1
21	A	1
22	B	1
23	D	1
24	B	1
25	B	1
26	B	1
27	A	1
28	B	1

Question	Answer	Marks
29	B	1
30	D	1
31	B	1
32	A	1
33	B	1
34	B	1
35	D	1
36	C	1
37	A	1
38	C	1
39	C	1
40	D	1



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

PHYSICS

0625/11

Paper 1 Multiple Choice (Core)

May/June 2019

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Cambridge Assessment
International Education

Question	Answer	Marks
1	A	1
2	A	1
3	D	1
4	A	1
5	C	1
6	C	1
7	D	1
8	D	1
9	A	1
10	C	1
11	D	1
12	A	1
13	C	1
14	A	1
15	B	1
16	A	1
17	A	1
18	A	1
19	D	1
20	B	1
21	D	1
22	B	1
23	B	1
24	B	1
25	B	1
26	B	1
27	D	1
28	D	1

Question	Answer	Marks
29	B	1
30	D	1
31	B	1
32	C	1
33	B	1
34	B	1
35	D	1
36	B	1
37	D	1
38	A	1
39	B	1
40	D	1



Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS

Paper 1 Multiple Choice (Core)

0625/13

May/June 2018

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

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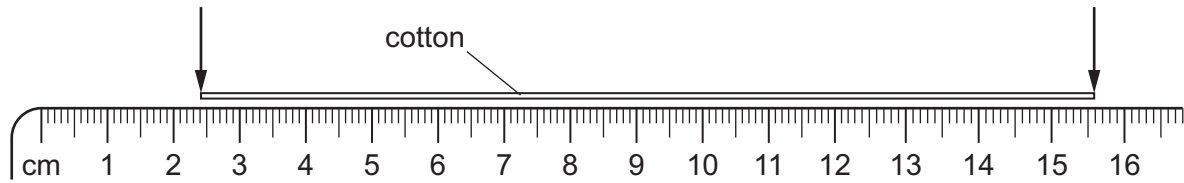
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

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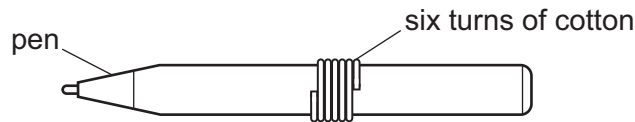
This document consists of **16** printed pages.



- 1 A length of cotton is measured between two points on a ruler.



When the length of cotton is wound closely around a pen, it goes round six times.



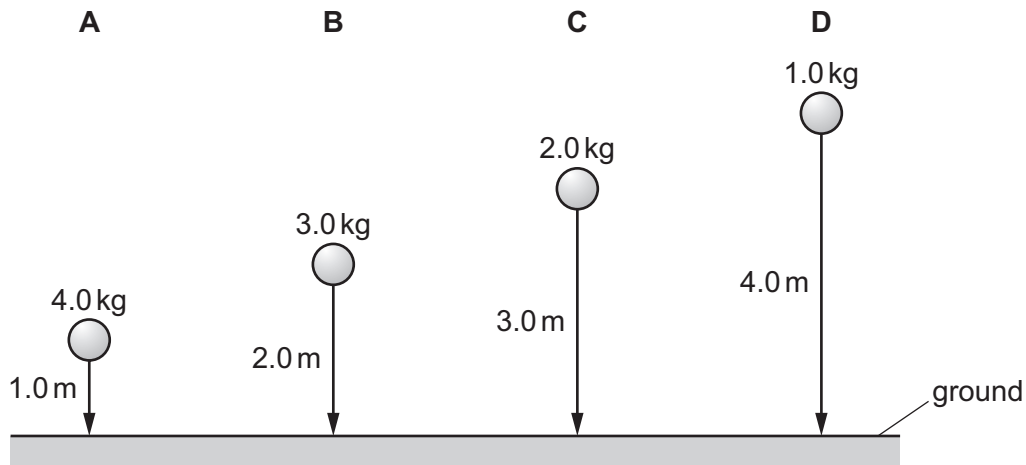
What is the distance once round the pen?

- A** 2.2 cm **B** 2.6 cm **C** 13.2 cm **D** 15.6 cm

- 2 Four balls with different masses are dropped from the heights shown.

Air resistance may be ignored.

Which ball has the greatest average speed?



- 3 A car takes 15 minutes to travel along a road that is 20 km long.

What is the average speed of the car?

- A** 0.75 km/h **B** 5.0 km/h **C** 80 km/h **D** 300 km/h

- 4 A person steps onto a bathroom scales.

The bathroom scales records both mass and weight.

Which row shows the readings on the scales?

	mass	weight
A	60 N	600 kg
B	60 kg	600 N
C	600 kg	60 N
D	600 N	60 kg

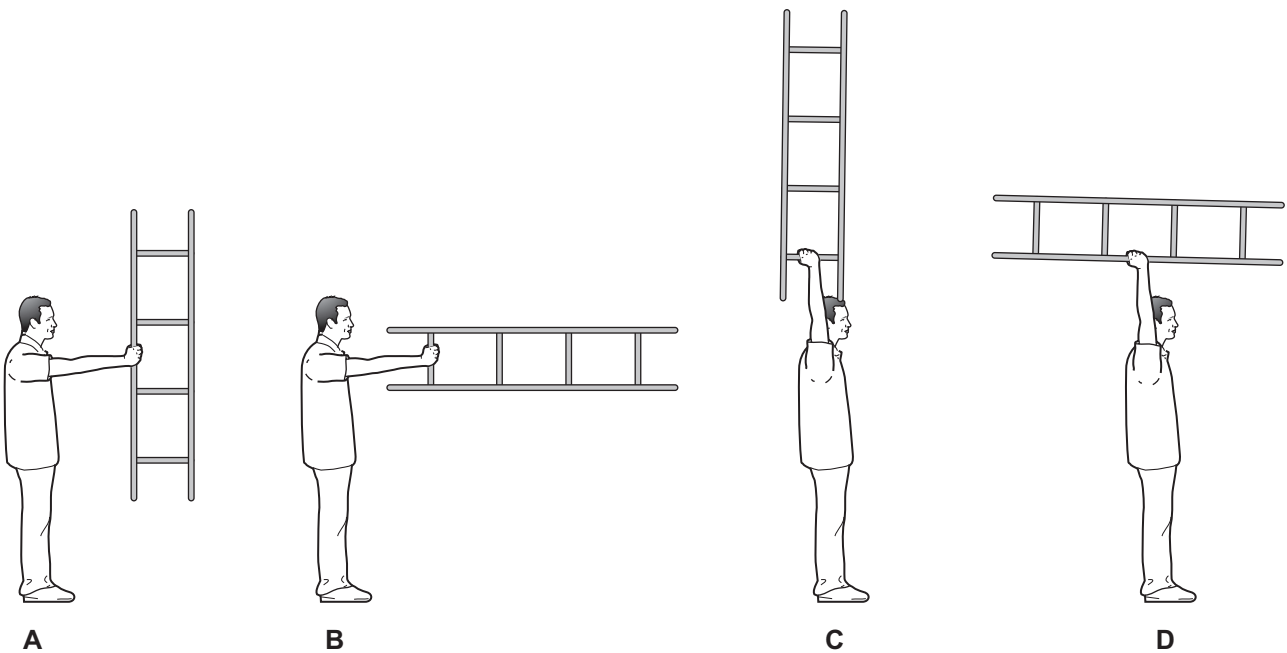
- 5 Which properties of an object can be changed by a force?

- A** direction of motion, mass and speed
- B** direction of motion, shape and speed
- C** direction of motion and speed only
- D** mass, shape and speed

- 6 A man holds a ladder in four different positions, pivoting around his shoulder.

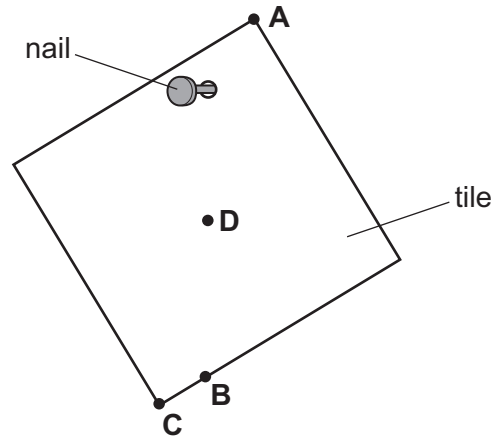
The weight of the ladder causes a moment about the man's shoulder.

In which position is the moment greatest?



- 7 A hole is drilled in a square tile. The diagram shows the tile hanging freely on a nail.

Where is the centre of mass of the tile?



- 8 Which method of drying clothes has the least impact on the environment?

- A Evaporate the water in them in an electrically heated tumble dryer.
- B Hang them on a washing line in direct sunlight.
- C Remove the water from them in an electric spin dryer.
- D Suspend them close to a coal fire.

- 9 A student does work by pulling a box across a horizontal floor.

She now pulls a second box along the same floor.

Which row indicates that the student is now doing twice as much work?

	force used to pull box	distance the box is pulled
A	is doubled	is doubled
B	is doubled	is halved
C	stays the same	is doubled
D	stays the same	is halved

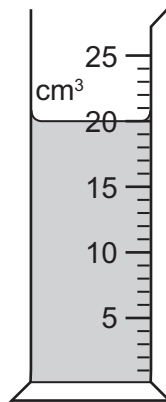
- 10 A simple barometer includes a column of mercury.

Which property of this column of mercury is used to give a measurement of atmospheric pressure?

- A its cross-sectional area
- B its height
- C its temperature
- D its thermal capacity

- 11 The diagram shows some liquid in a measuring cylinder.

The mass of the liquid is 16 g.



What is the density of the liquid?

- A 0.80 g/cm^3 B 1.25 g/cm^3 C 36 g/cm^3 D 320 g/cm^3

- 12 Brownian motion of particles is observed.

Which statements describe the movement of the particles?

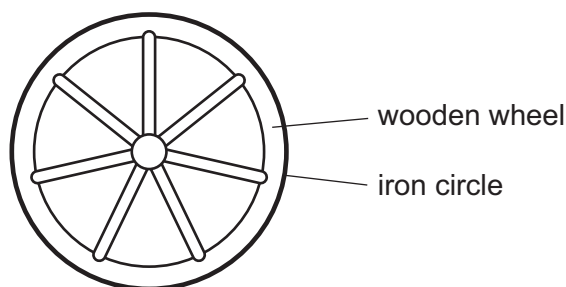
- 1 The particles all travel along a curved path.
- 2 The particles move randomly.
- 3 The particles all travel in the same direction.

- A 1 and 3 B 1 only C 2 and 3 D 2 only

- 13** A pure liquid is left in an open beaker and some of the liquid molecules escape by evaporation.

Which statement about this process is correct?

- A** None of the escaping molecules return to the liquid.
 - B** The escaping molecules are generally the more energetic ones.
 - C** The rate of escape of the molecules can be increased by increasing the depth of the liquid in the beaker.
 - D** The temperature of the remaining liquid is unaffected by the escape of the molecules.
- 14** A wooden wheel can be strengthened by putting a tight circle of iron around it.



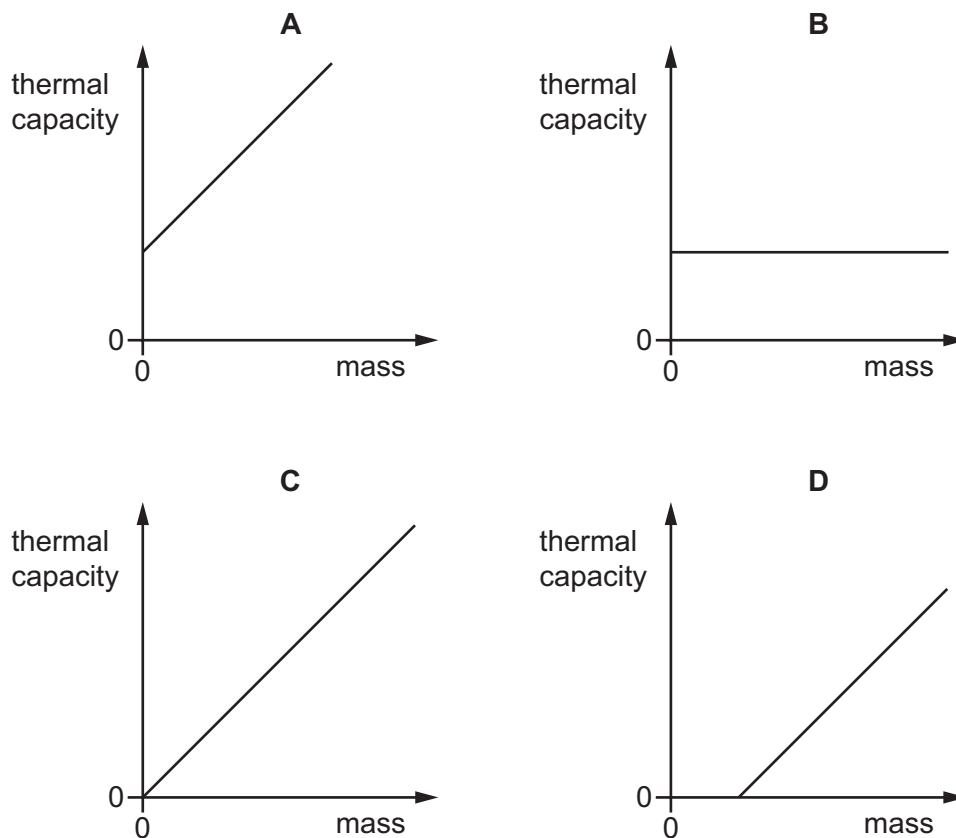
Which action would make it easier to fit the circle over the wood?

- A** cooling the iron circle
 - B** heating the iron circle
 - C** heating the wooden wheel and cooling the iron circle
 - D** heating the wooden wheel but not heating or cooling the iron circle
- 15** A student wishes to calibrate a mercury-in-glass thermometer with a °C scale.

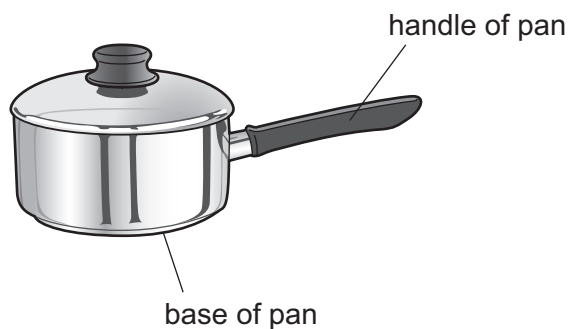
Which values should she use for the lower fixed point and for the upper fixed point?

	lower fixed point	upper fixed point
A	melting point of ice	boiling point of mercury
B	melting point of ice	boiling point of water
C	melting point of mercury	boiling point of mercury
D	melting point of mercury	boiling point of water

16 Which diagram shows how the thermal capacity of a silver object depends on its mass?



17 The diagram shows a pan used for cooking food.



Which row is correct for the materials used to make the base and the handle of the pan?

	base of pan	handle of pan
A	good thermal conductor	good thermal conductor
B	good thermal conductor	poor thermal conductor
C	poor thermal conductor	good thermal conductor
D	poor thermal conductor	poor thermal conductor

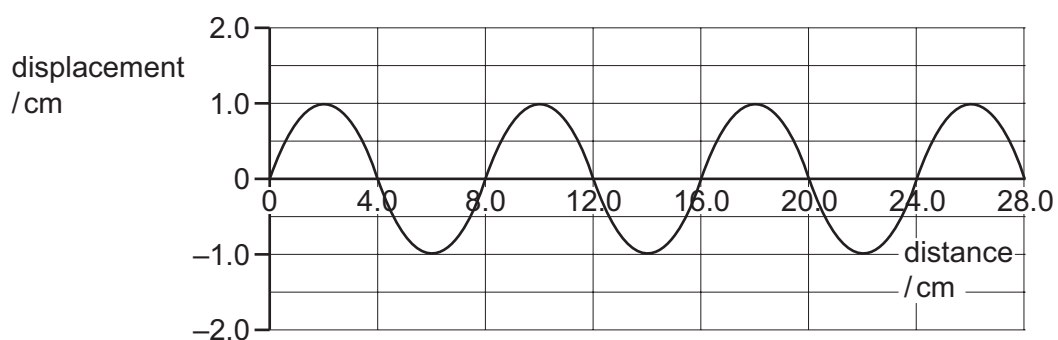
18 A saucepan of water is heated from its base on an electric hob.

After five minutes the water near the top of the saucepan is becoming hot.

What is the main process of heat transfer within the water?

- A** condensation
- B** conduction
- C** convection
- D** radiation

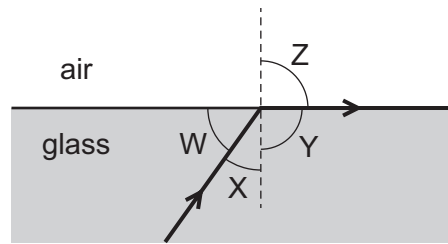
19 The diagram shows a wave.



Which row is correct?

	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

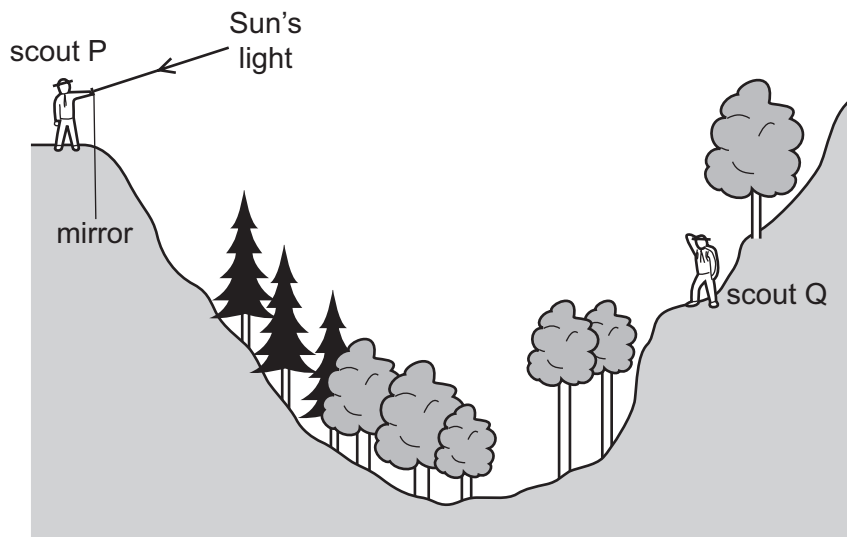
- 20 The diagram shows a ray of light passing from glass to air, at the critical angle.



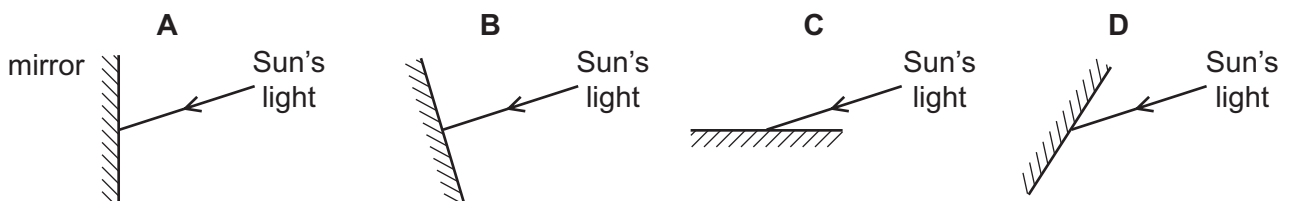
Which angles are the angle of incidence and the angle of refraction?

	angle of incidence	angle of refraction
A	W	Y
B	W	Z
C	X	Y
D	X	Z

- 21 Scout P signals to scout Q on the other side of a valley by using a mirror to reflect the Sun's light.



Which mirror position allows the Sun's light to be reflected to scout Q?

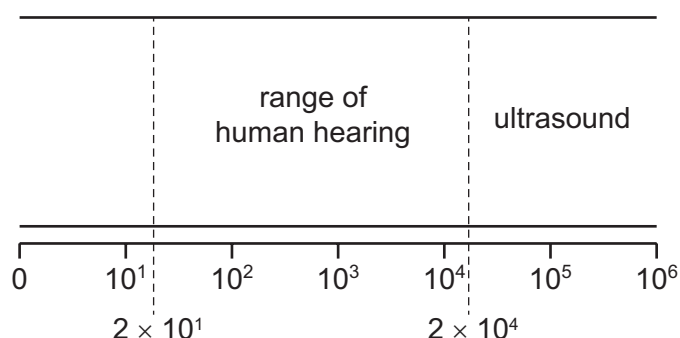


- 22** It takes 0.24 seconds for a microwave signal to travel from a transmitting station on Earth to a satellite and then back to the station on Earth. Microwaves travel at 3.0×10^8 m/s.

How far is the transmitting station from the satellite?

- A** 36 000 km
- B** 72 000 km
- C** 36 000 000 km
- D** 72 000 000 km

- 23** The diagram shows the ranges of human hearing and of ultrasound waves.



To which characteristic of sound waves do the numbers on the diagram refer?

- A** amplitude in cm
 - B** frequency in Hz
 - C** speed in metres/second
 - D** wavelength in metres
- 24** A permanent magnet is placed close to a bar of soft iron.

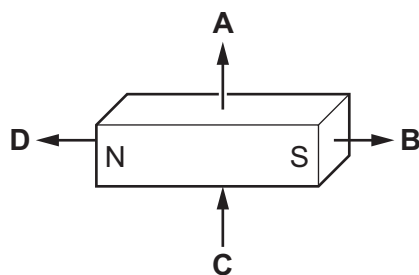


What are the polarities of end P and of end Q?

	end P	end Q
A	N	N
B	N	S
C	S	N
D	S	S

- 25** The diagram shows a permanent bar magnet placed on a wooden table. A student draws arrows to show the direction of the magnetic field at four points.

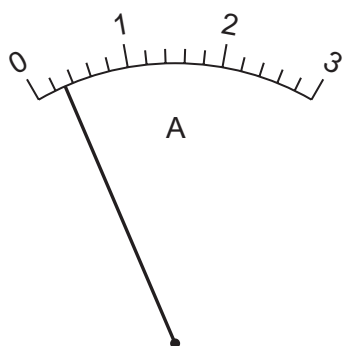
Which arrow shows the direction of the magnetic field at that position?



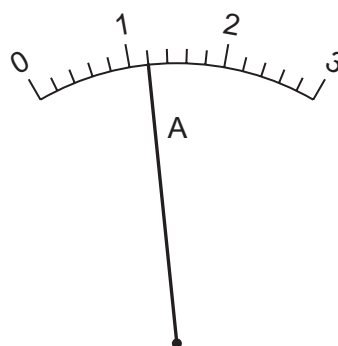
- 26** Which statement about electric charges is correct?

- A** Like charges attract and unlike charges attract.
- B** Like charges attract and unlike charges repel.
- C** Like charges repel and unlike charges attract.
- D** Like charges repel and unlike charges repel.

- 27** The diagrams show two readings on the same ammeter. Reading 1 is taken before the ammeter is connected in a circuit. Reading 2 shows the reading when the ammeter is connected in a circuit.



reading 1
not connected in a circuit



reading 2
connected in a circuit

What is the current in the circuit?

- A** 0.80 A
- B** 0.90 A
- C** 0.95 A
- D** 1.20 A

- 28 A student measures the potential difference across a device and the current in the device.

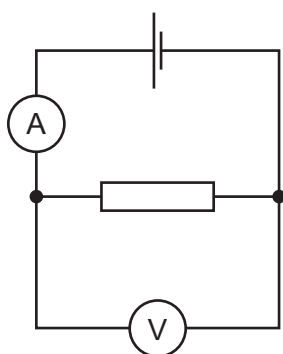
Which calculation gives the resistance of the device?

- A current + potential difference
 - B current $\div$ potential difference
 - C potential difference $\div$ current
 - D potential difference $\times$ current
- 29 A battery in an electric circuit is connected to a lamp.
- Chemical energy in the battery is transferred to which other types of energy?
- A internal energy only
 - B kinetic energy and sound energy
 - C light energy and thermal energy
 - D light energy only
- 30 Which component is represented by the symbol shown?

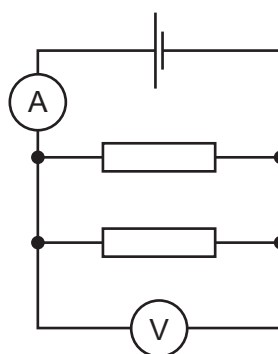


- A fixed resistor
- B fuse
- C thermistor
- D variable resistor

- 31 A student sets up a circuit which she calls circuit 1. She records the value of the current I_1 , and calculates the resistance R_1 of the circuit.



circuit 1



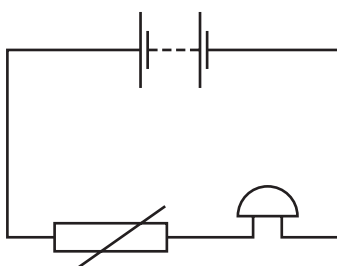
circuit 2

She then connects an identical resistor in parallel with the original resistor. She calls this circuit 2. She records current I_2 and calculates the total resistance R_2 of this circuit.

Which row correctly compares the two currents and the two resistances in the circuits?

	I_2	R_2
A	greater than I_1	greater than R_1
B	greater than I_1	less than R_1
C	less than I_1	greater than R_1
D	less than I_1	less than R_1

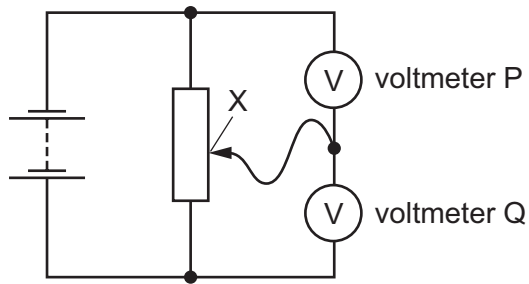
- 32 A student sets up this circuit.



What is the purpose of the circuit?

- A** to allow a lamp to be made dimmer or brighter as required
- B** to amplify the sound of a voice
- C** to light a lamp in the dark
- D** to sound a bell when the temperature rises

- 33 The diagram shows two voltmeters P and Q connected to a potential divider.

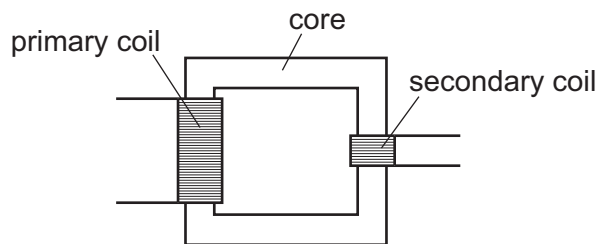


The sliding connection at point X is moved towards the top of the diagram.

What happens to the reading on P and to the reading on Q?

	reading on P	reading on Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 34 The diagram shows a transformer.



Which materials are the most suitable for the core and for the coils?

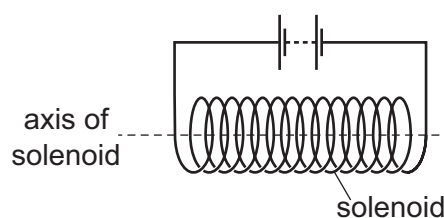
	core material	coil material
A	copper	copper
B	copper	iron
C	iron	copper
D	iron	iron

35 A current-carrying conductor placed in a magnetic field experiences a force.

Which changes result in reversing the direction of the force?

- A** Decrease the current, keeping the field direction constant.
- B** Increase the current and reverse the field direction.
- C** Increase the current, keeping the field direction constant.
- D** Reverse the current and reverse the field direction.

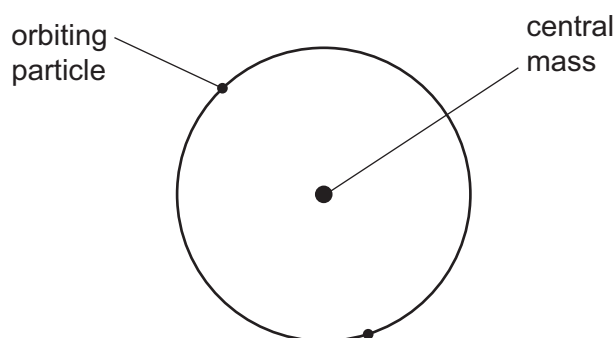
36 A solenoid is connected to a battery.



Which statement about the magnetic field at the centre of the solenoid is correct?

- A** The magnetic field along the axis is zero.
- B** The direction of the magnetic field is at an angle of 45° to the axis.
- C** The direction of the magnetic field is parallel to the axis.
- D** The direction of the magnetic field is perpendicular to the axis.

37 In the atomic model, an atom consists of a central mass, orbited by much smaller particles.



What is the name of the central mass and of the orbiting particles?

	central mass	orbiting particles
A	neutron	α -particles
B	neutron	electrons
C	nucleus	α -particles
D	nucleus	electrons

38 One isotope of lead is represented by the nuclide symbol $^{214}_{82}\text{Pb}$.

How many neutrons and how many protons does one nucleus of this isotope contain?

	number of neutrons	number of protons
A	82	132
B	82	214
C	132	82
D	214	82

39 Three sources of background radiation are listed.

- 1 cosmic rays
- 2 medical X-rays
- 3 radioactive emissions from radon gas from the ground

Which of these sources are naturally occurring?

A 1 and 3 only **B** 1 only **C** 2 and 3 only **D** 2 only

40 The rates of emission from four radioactive sources are measured at 20 minute intervals.

Each row in the table shows the results for one of the radioactive sources.

Which source has the longest half-life?

	rate of emission / emissions per minute		
	time 0	time 20 min	time 40 min
A	120	60	30
B	120	110	101
C	240	60	15
D	240	170	122

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Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS

Paper 1 Multiple Choice (Core)

0625/12

May/June 2018

45 minutes

Additional Materials: Multiple Choice Answer Sheet
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Soft pencil (type B or HB recommended)



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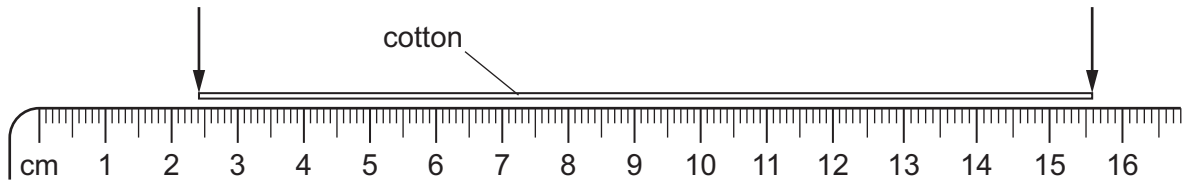
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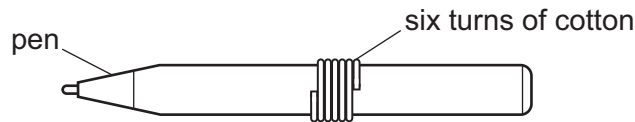
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- 1 A length of cotton is measured between two points on a ruler.



When the length of cotton is wound closely around a pen, it goes round six times.



What is the distance once round the pen?

- A** 2.2 cm **B** 2.6 cm **C** 13.2 cm **D** 15.6 cm
- 2 A ball is dropped in an evacuated tube. A series of photographs is taken at equal time intervals from the time of release. Another ball of the same size but twice the mass is also dropped in the same evacuated tube and photographed.

Which diagram shows the motion of the heavier ball?

	heavier ball (mass $\times 2$)			
first ball	A	B	C	D
				
				
				
				
				
				
				
				
				

- 3 A car takes 15 minutes to travel along a road that is 20 km long.

What is the average speed of the car?

- A** 0.75 km/h **B** 5.0 km/h **C** 80 km/h **D** 300 km/h

- 4 Which statement about the mass and the weight of an object is correct?

- A** They are both affected by changes in the acceleration of free fall.
B They are both forces.
C They have different units.
D Weight is calculated by dividing mass by the acceleration of free fall.

- 5 Two objects P and Q are placed in a beaker containing a liquid.

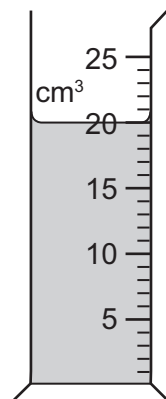
Object P floats in the liquid and object Q sinks.

Which row for the densities of object P, object Q and the liquid is possible?

	density of object P g/cm^3	density of object Q g/cm^3	density of liquid g/cm^3
A	1.2	0.6	0.8
B	1.2	1.4	1.0
C	11.3	8.9	13.6
D	11.3	19.3	13.6

- 6 The diagram shows some liquid in a measuring cylinder.

The mass of the liquid is 16 g.



What is the density of the liquid?

- A** 0.80 g/cm^3 **B** 1.25 g/cm^3 **C** 36 g/cm^3 **D** 320 g/cm^3

- 7 A spacecraft is travelling in space with no resultant force and no resultant moment acting on it.

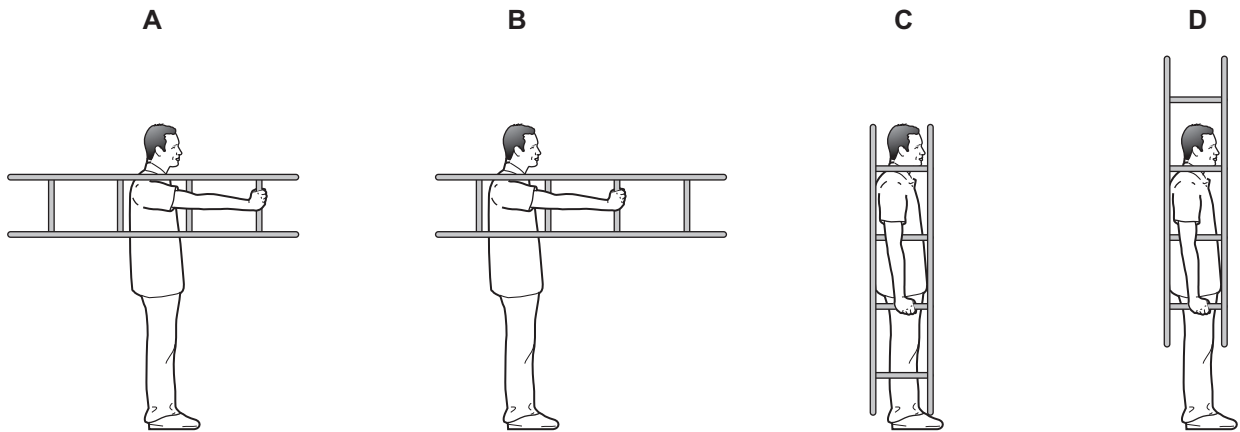
Which statement about the spacecraft is correct?

- A Its direction is changing.
- B It is in equilibrium.
- C Its speed is decreasing.
- D Its speed is increasing.

- 8 A man holds a short ladder in four different positions.

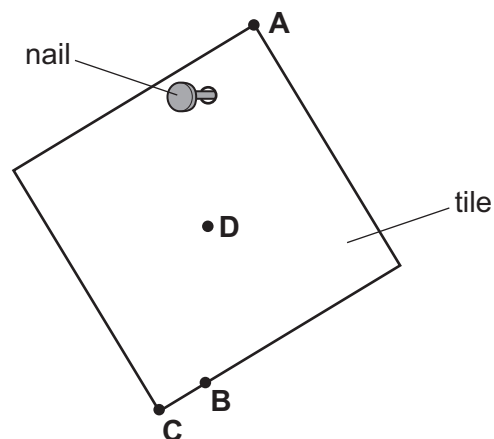
The weight of the ladder causes a moment about the man's shoulder.

In which position is the moment greatest?



- 9 A hole is drilled in a square tile. The diagram shows the tile hanging freely on a nail.

Where is the centre of mass of the tile?



10 Which source of energy is renewable?

- A coal
- B natural gas
- C oil
- D wind

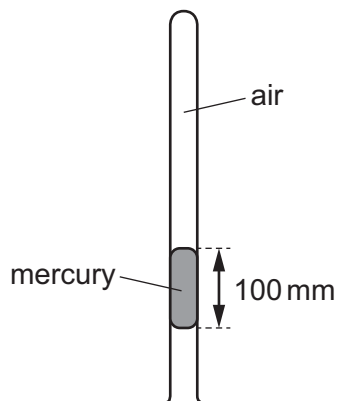
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12 Air is trapped in a closed tube by a thread of mercury. The mercury thread is 100 mm long. The tube is held as shown.



Atmospheric pressure is 770 mm Hg.

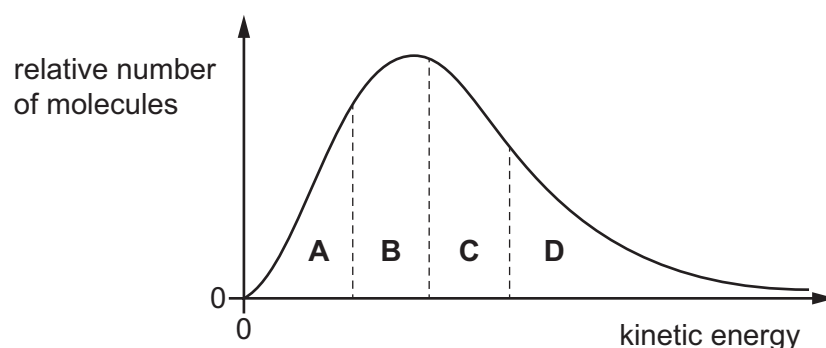
What is the pressure of the trapped air?

- A 100 mm Hg
- B 670 mm Hg
- C 770 mm Hg
- D 870 mm Hg

- 13** Brownian motion is the random motion of particles due to molecular bombardment.

In which states of matter is Brownian motion observed?

- A** gases, liquids and solids
 - B** gases and liquids only
 - C** gases and solids only
 - D** liquids and solids only
- 14** The diagram shows the relative number of molecules in a liquid that have a given kinetic energy. The graph is divided into sections so that each section contains the same number of molecules. From which section does the greatest number of molecules escape from the liquid per unit time?

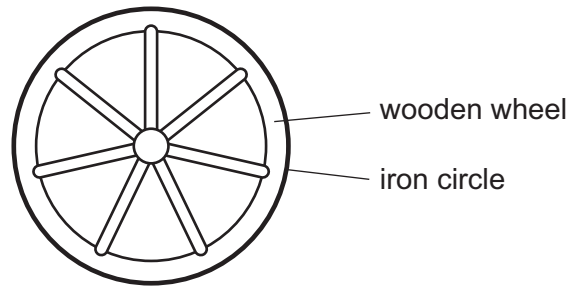


- 15** An electric kettle heats some water. The same kettle then heats a different liquid. The temperature of the liquid rises more rapidly than the temperature of the water.

What is a possible explanation of this difference?

- A** The liquid condenses on the cooler parts of the kettle less than the water does.
- B** The liquid expands more than the water as it heats up.
- C** The liquid has a lower boiling point than the water.
- D** The liquid has a smaller thermal capacity than the water.

- 16** A wooden wheel can be strengthened by putting a tight circle of iron around it.



Which action would make it easier to fit the circle over the wood?

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Which values should she use for the lower fixed point and for the upper fixed point?

	lower fixed point	upper fixed point
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B	melting point of ice	boiling point of water
C	melting point of mercury	boiling point of mercury
D	melting point of mercury	boiling point of water

- 18** Four thermometers, with their bulbs painted different colours, are placed at equal distances from a radiant heater.

Which thermometer shows the slowest temperature rise when the heater is first switched on?

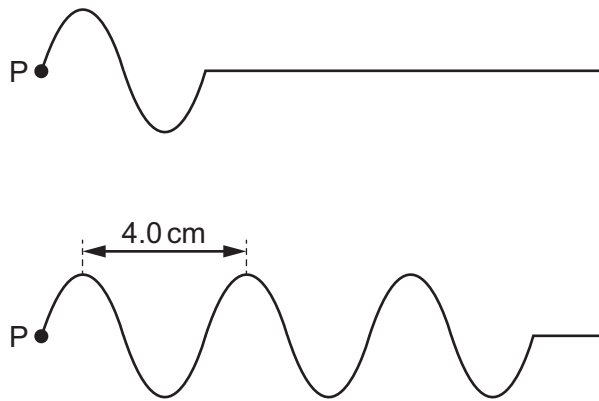
- A** matt black
- B** matt white
- C** shiny black
- D** shiny white

- 19 A tank contains water. Ripples are produced on the surface of the water.

What causes the ripples to refract?

- A The cold water in the tank is replaced by warm water.
- B The ripples change speed as they move from deep to shallow water.
- C The ripples hit the wall of the tank.
- D The ripples pass through a narrow gap.

- 20 The diagrams show a wave on the surface of the water in a tank at times 1.0 s apart. The wave is produced at P and travels to the right.

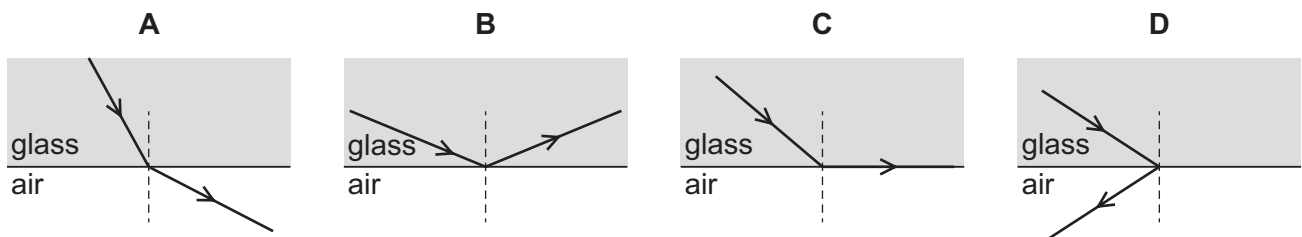


Which row gives the frequency and the speed of this water wave?

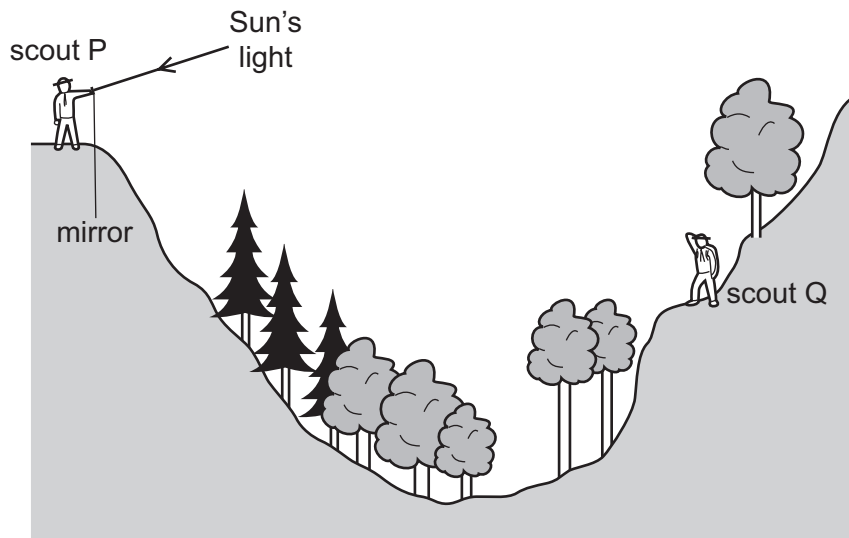
	frequency / Hz	<u>speed</u> cm/s
A	2.0	4.0
B	2.0	8.0
C	4.0	4.0
D	4.0	8.0

- 21 Light passes from glass into air.

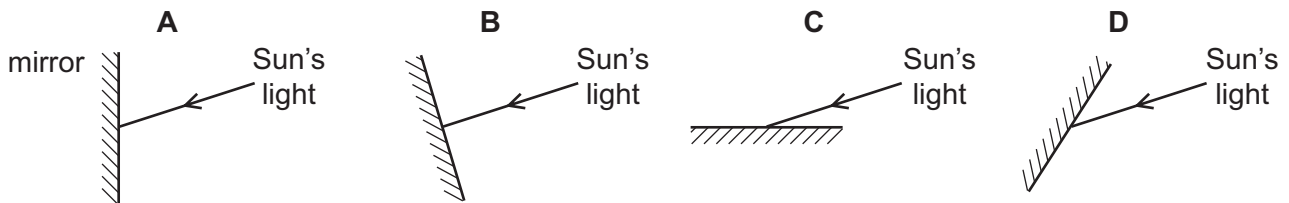
Which diagram shows a ray of light incident at the critical angle on the air-glass boundary?



- 22 Scout P signals to scout Q on the other side of a valley by using a mirror to reflect the Sun's light.



Which mirror position allows the Sun's light to be reflected to scout Q?



- 23 The table describes white light that passes through a prism and forms a spectrum.

Which row is correct?

	colour refracted the most	colour next to the red
A	red	orange
B	red	yellow
C	violet	orange
D	violet	yellow

- 24 Where do all types of electromagnetic waves travel at the same speed?

- A** air
- B** a vacuum
- C** glass
- D** water

- 25** A siren is emitting a sound. As time passes, the sound becomes louder and higher pitched.

What is happening to the amplitude and to the frequency of the emitted sound wave?

	amplitude	frequency
A	decreasing	decreasing
B	decreasing	increasing
C	increasing	decreasing
D	increasing	increasing

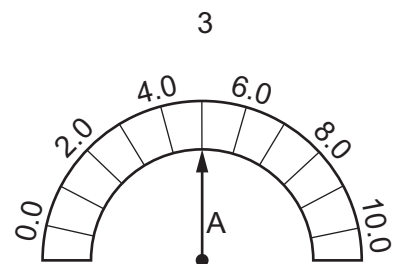
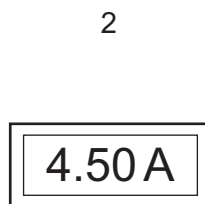
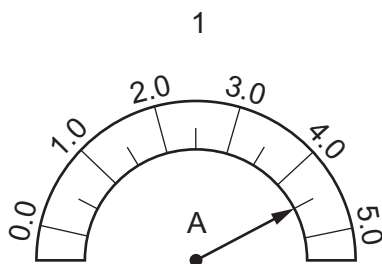
- 26** A permanent magnet is placed close to a bar of soft iron.



What are the polarities of end P and of end Q?

	end P	end Q
A	N	N
B	N	S
C	S	N
D	S	S

- 27** The diagrams show three ammeters.



Which ammeters show the same value of current?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 28 A student measures the potential difference across a device and the current in the device.

Which calculation gives the resistance of the device?

- A current + potential difference
 - B current $\div$ potential difference
 - C potential difference $\div$ current
 - D potential difference $\times$ current
- 29 A lamp is connected across one cell, then across two cells. The potential difference (p.d.) across the lamp and the current in it are measured in each case.

The results are shown.

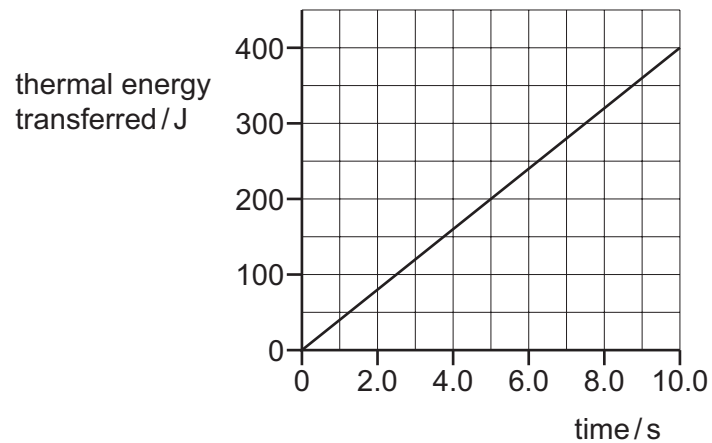
number of cells	p.d./V	current/A
1	2.8	0.25
2	5.4	0.40

What is the change in the resistance of the lamp when the number of cells is increased from one to two?

- A It decreases by 0.015Ω .
- B It increases by 1.5Ω .
- C It increases by 2.3Ω .
- D It increases by 17Ω .

- 30** An electrical heater transfers thermal energy to the surroundings.

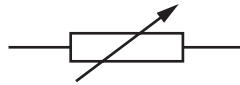
The graph shows how the amount of thermal energy transferred varies with time.



The heater continues to transfer energy at the same rate.

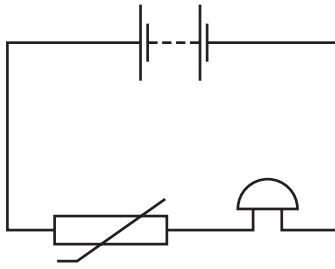
How much thermal energy is transferred by the heater in 5.0 minutes?

- A** 200 J **B** 400 J **C** 2000 J **D** 12 000 J
- 31** Which electrical component does the symbol represent?



- A** a fuse
B a relay coil
C a thermistor
D a variable resistor

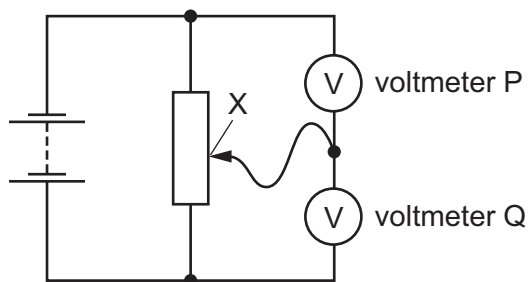
32 A student sets up this circuit.



What is the purpose of the circuit?

- A** to allow a lamp to be made dimmer or brighter as required
- B** to amplify the sound of a voice
- C** to light a lamp in the dark
- D** to sound a bell when the temperature rises

33 The diagram shows two voltmeters P and Q connected to a potential divider.



The sliding connection at point X is moved towards the top of the diagram.

What happens to the reading on P and to the reading on Q?

	reading on P	reading on Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

34 A simple electric generator induces an electromotive force (e.m.f.).

Which modification would increase the induced e.m.f.?

- A** Increase the number of turns in the coil of the generator.
- B** Increase the distance between the magnetic poles.
- C** Reduce the strength of the magnetic field around the coil.
- D** Reverse the direction of the magnetic field.

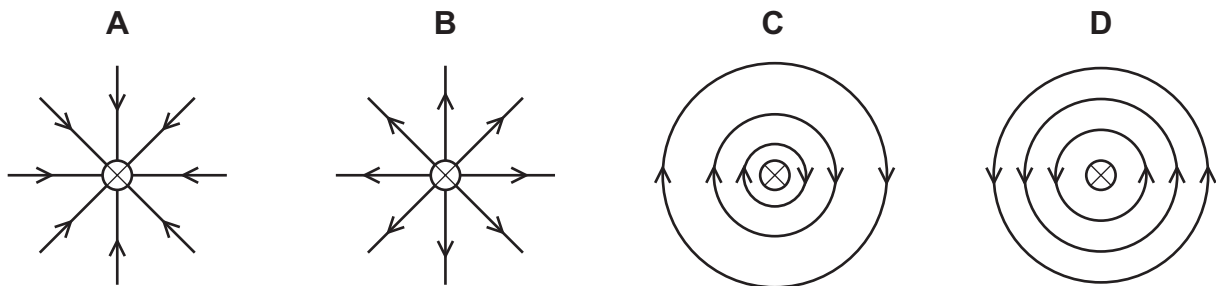
- 35** A transformer has N_p turns in the primary coil and N_s turns in the secondary coil.

Which row gives the values of N_p and N_s for a transformer that steps up a voltage of 1200 V to 36 000 V?

	N_p	N_s
A	2 000	60 000
B	2 000	600 000
C	60 000	2 000
D	600 000	2 000

- 36** A straight wire is perpendicular to the paper. It carries a current into the paper.

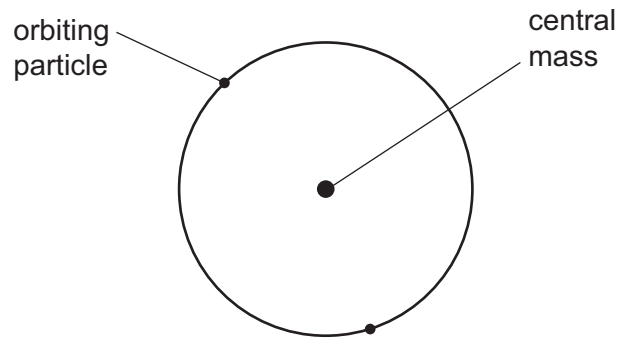
What is the magnetic field pattern and its direction near the wire?



key

⊗ wire with current into the page

37 In the atomic model, an atom consists of a central mass, orbited by much smaller particles.



What is the name of the central mass and of the orbiting particles?

	central mass	orbiting particles
A	neutron	α -particles
B	neutron	electrons
C	nucleus	α -particles
D	nucleus	electrons

38 The table shows the composition of three different nuclei.

nucleus	number of protons	number of neutrons
X	3	3
Y	3	4
Z	4	3

Which nuclei are isotopes of the same element?

A X, Y and Z **B** X and Y only **C** X and Z only **D** Y and Z only

39 The table compares the penetrating abilities and ionising effects of α -radiation and of γ -radiation.

Which row is correct?

	least penetrating	most ionising
A	α	α
B	α	γ
C	γ	α
D	γ	γ

40 Radioactive materials must be handled in a safe way.

What is **not** a safety procedure?

- A** Monitor exposure time to radioactive materials.
- B** Store radioactive materials in cardboard boxes.
- C** Use tongs to pick up the radioactive source.
- D** Wear protective clothing.

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Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS

Paper 1 Multiple Choice (Core)

0625/11

May/June 2018

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

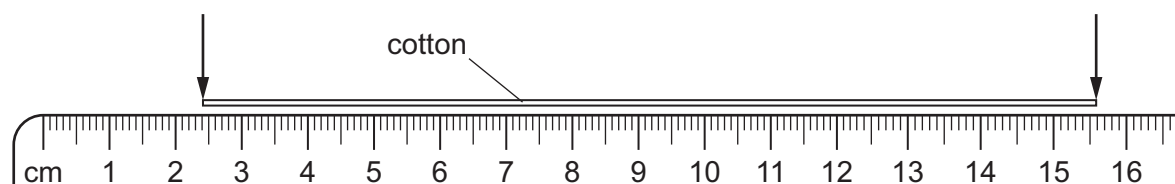
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

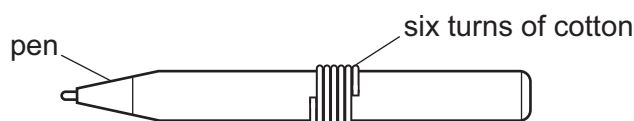
This document consists of **15** printed pages and **1** blank page.



- 1 A length of cotton is measured between two points on a ruler.



When the length of cotton is wound closely around a pen, it goes round six times.

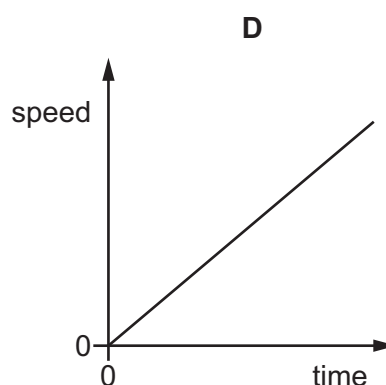
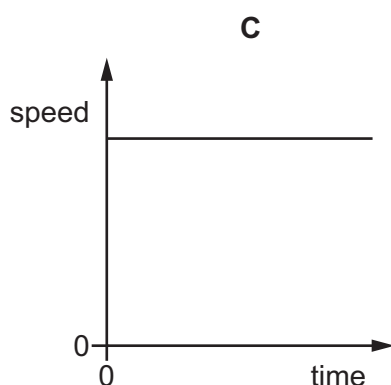
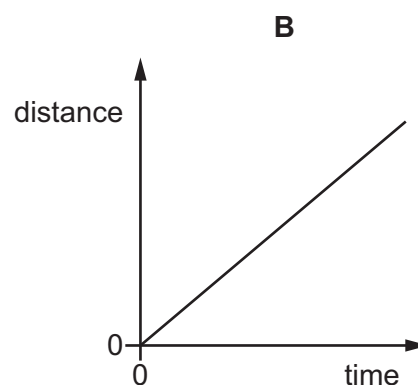
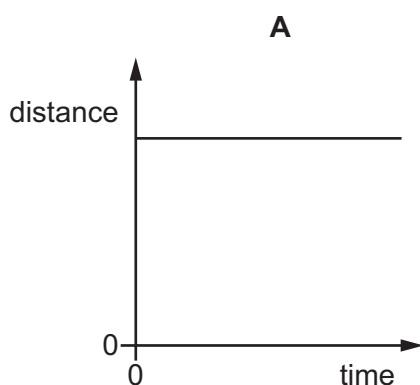


What is the distance once round the pen?

- A** 2.2 cm **B** 2.6 cm **C** 13.2 cm **D** 15.6 cm

- 2 A car is moving along a straight, level road, with a constant acceleration.

Which graph shows the motion of the car?



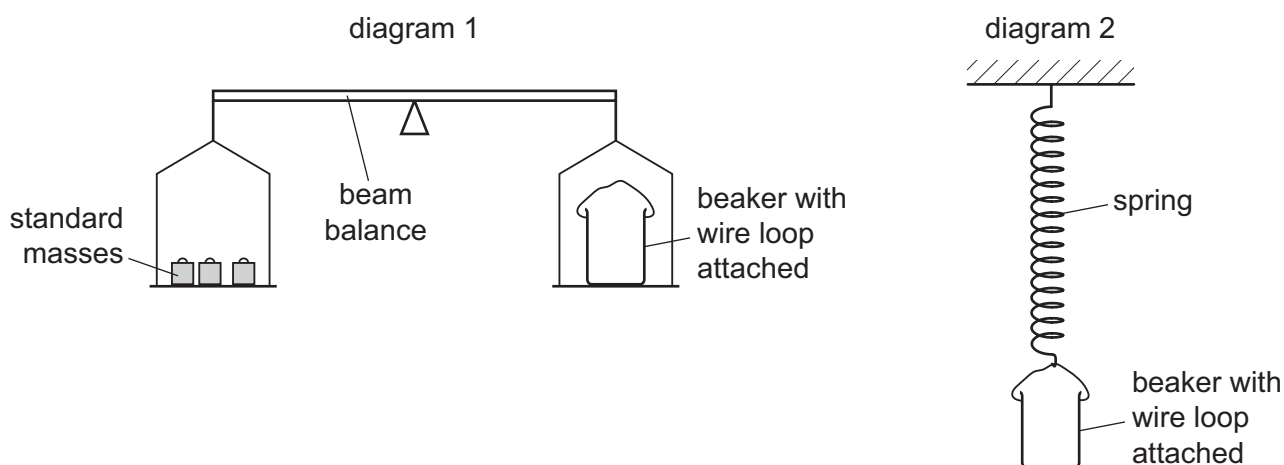
- 3 A car takes 15 minutes to travel along a road that is 20 km long.

What is the average speed of the car?

- A** 0.75 km/h **B** 5.0 km/h **C** 80 km/h **D** 300 km/h

- 4 Diagram 1 shows a beam balance. A beaker with a wire loop balances the standard masses.

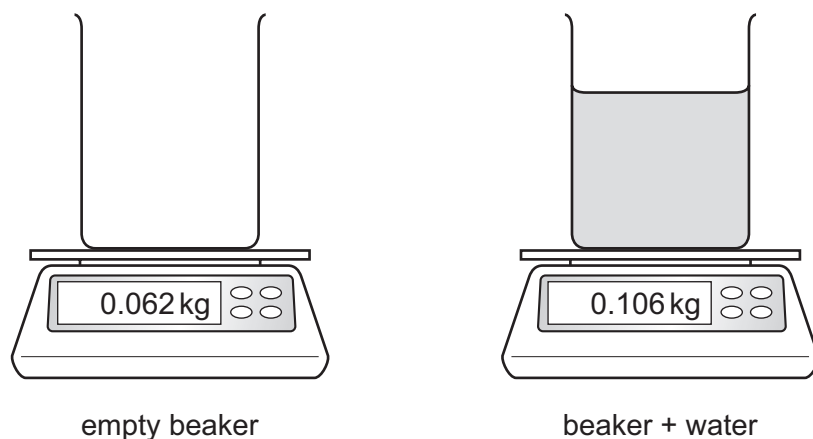
The beaker is then removed and hung from a spring. The spring extends by 5.0 cm, as in diagram 2.



The experiment is repeated with the same apparatus on the Moon, where the acceleration of free fall is less than on Earth.

Which statement describes what happens on the Moon?

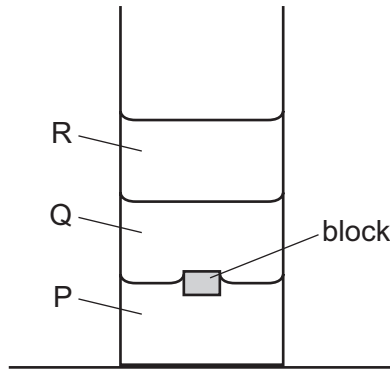
- A The beam balance is balanced and the spring extends by 5.0 cm.
 - B The beam balance is balanced and the spring extends by less than 5.0 cm.
 - C The right-hand balance pan is higher and the spring extends by 5.0 cm.
 - D The right-hand balance pan is higher and the spring extends by less than 5.0 cm.
- 5 An empty beaker is placed on a top-pan balance. Some water is now poured into the beaker.



What is the weight of the water?

- A 0.044 kg
- B 0.168 kg
- C 0.0044 N
- D 0.44 N

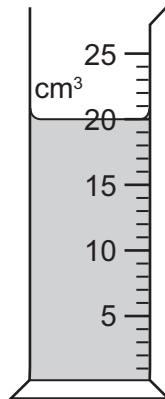
- 6 Three liquids P, Q and R have different densities and do not mix. The liquids are placed in a measuring cylinder and allowed to settle. A small block is then dropped into the measuring cylinder and comes to rest, as shown.



Which statement about the density of the block is correct?

- A It is equal to the density of Q.
 - B It is greater than the density of P.
 - C It is greater than the density of R.
 - D It is less than the density of Q.
- 7 The diagram shows some liquid in a measuring cylinder.

The mass of the liquid is 16 g.



What is the density of the liquid?

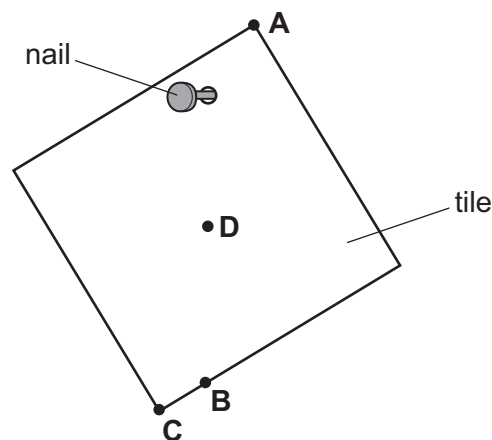
- A 0.80 g/cm³
- B 1.25 g/cm³
- C 36 g/cm³
- D 320 g/cm³

- 8 A car is moving in a straight line on a level road. Its engine provides a forward force on the car. A second force of equal size acts on the car due to resistive forces.

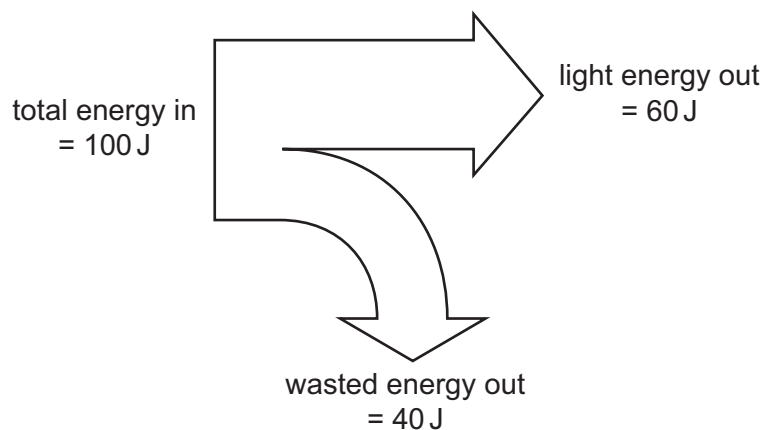
Which statement describes what happens?

- A The car changes direction.
 - B The car moves at a constant speed.
 - C The car slows down.
 - D The car speeds up.
- 9 A hole is drilled in a square tile. The diagram shows the tile hanging freely on a nail.

Where is the centre of mass of the tile?



- 10 The diagram shows the energy transferred in a lamp in one second.



Which type of wasted energy is produced by the lamp?

- A chemical potential energy
- B electrical energy
- C gravitational potential energy
- D thermal energy

11 Which energy resource is **not** renewable?

- A fossil fuel
- B sunlight
- C tides
- D wind

12 A student does work by pulling a box across a horizontal floor.

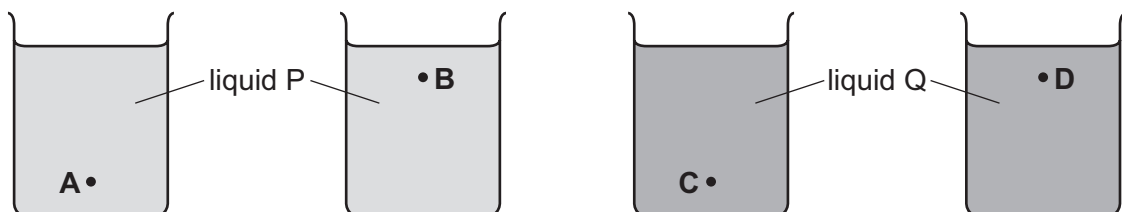
She now pulls a second box along the same floor.

Which row indicates that the student is now doing twice as much work?

	force used to pull box	distance the box is pulled
A	is doubled	is doubled
B	is doubled	is halved
C	stays the same	is doubled
D	stays the same	is halved

13 Four identical beakers are filled with equal volumes of liquids P or Q, as shown. Liquid P is more dense than liquid Q.

At which point is the pressure the least?



14 A woman has a weight of 600 N. She stands on a horizontal floor. The area of her feet in contact with the floor is 0.050 m^2 .

What is the pressure she exerts on the floor?

- A $1.2 \times 10^3 \text{ N/m}^2$
- B $2.4 \times 10^3 \text{ N/m}^2$
- C $1.2 \times 10^4 \text{ N/m}^2$
- D $2.4 \times 10^4 \text{ N/m}^2$

- 15** On a warm day, a carton of fresh milk is covered with a wet cloth.

Why does this help to reduce the temperature of the milk?

- A** Some water evaporates from the cloth so the remaining water becomes cooler.
- B** The water has a very high thermal capacity.
- C** The water insulates the milk from the warm air around it.
- D** Water is always colder than the air around it.

- 16** Air is trapped in a cylinder by a piston.

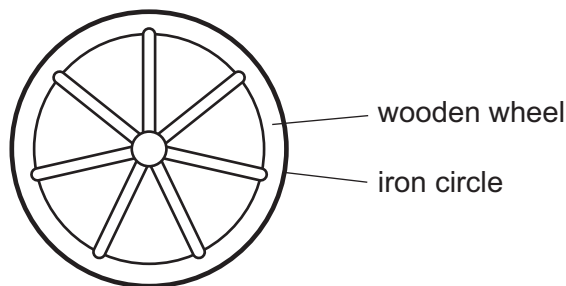
The piston is pushed inwards and the volume of the air is reduced.

The temperature of the trapped air remains constant.

Which row describes how the average speed of the air molecules and the average distance between them changes?

	average speed of molecules	average distance between molecules
A	increases	decreases
B	increases	unchanged
C	unchanged	decreases
D	unchanged	increases

- 17** A wooden wheel can be strengthened by putting a tight circle of iron around it.



Which action would make it easier to fit the circle over the wood?

- A** cooling the iron circle
- B** heating the iron circle
- C** heating the wooden wheel and cooling the iron circle
- D** heating the wooden wheel but not heating or cooling the iron circle

- 18** A student wishes to calibrate a mercury-in-glass thermometer with a °C scale.

Which values should she use for the lower fixed point and for the upper fixed point?

	lower fixed point	upper fixed point
A	melting point of ice	boiling point of mercury
B	melting point of ice	boiling point of water
C	melting point of mercury	boiling point of mercury
D	melting point of mercury	boiling point of water

- 19** Which row gives the correct name for each change of state shown?

	change of state		
	gas to liquid	liquid to solid	solid to liquid
A	condensation	melting	solidification
B	condensation	solidification	melting
C	evaporation	melting	solidification
D	evaporation	solidification	melting

- 20** On a cold day, a metal front-door knob X and a similar plastic knob Y are at the same temperature.

Why does X feel cooler to the touch than Y?

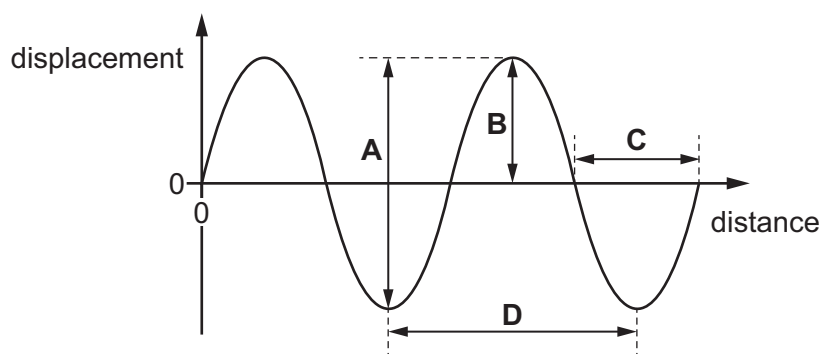
- A** X convects thermal energy better than Y.
- B** X is a better thermal conductor than Y.
- C** X is a better insulator than Y.
- D** X is a better radiator of thermal energy than Y.

- 21** A liquid is heated and it expands.

How does this lead to the formation of a convection current?

- A** The density of the heated liquid decreases.
- B** The density of the heated liquid increases.
- C** The mass of the heated liquid molecules decreases.
- D** The mass of the heated liquid molecules increases.

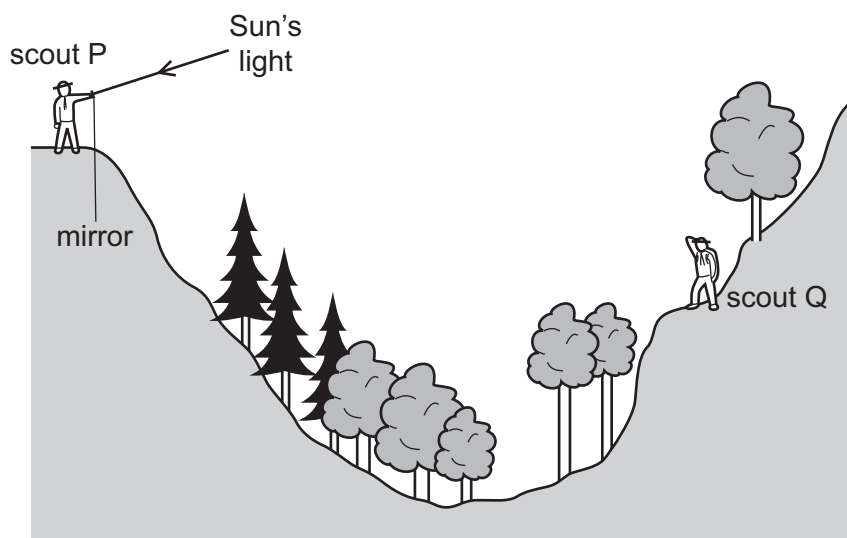
22 Which arrow on the graph shows the amplitude of the wave?



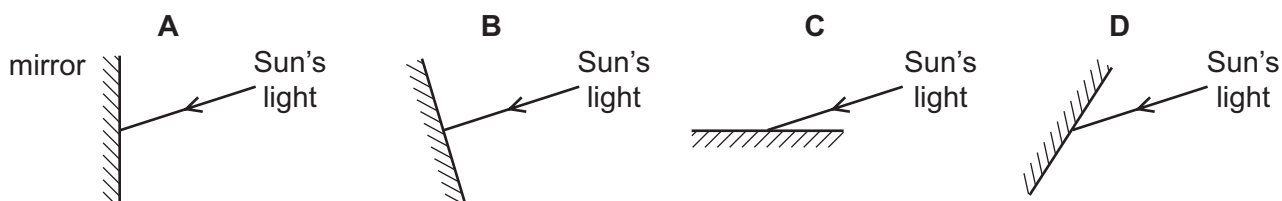
23 In which situation is the wavelength of the wave changed?

- A light from the Sun passing from air into water
- B radio waves travelling from an Earth satellite to the Moon
- C sound reflecting from a wall
- D water waves passing through a narrow gap

24 Scout P signals to scout Q on the other side of a valley by using a mirror to reflect the Sun's light.



Which mirror position allows the Sun's light to be reflected to scout Q?



- 25** The diagram shows the electromagnetic spectrum. The numbers indicate the approximate wavelength at the boundaries between the various regions of the spectrum.

For a device to be able to make use of electromagnetic radiation, it needs an aerial of approximately the same size as the radiation it is designed to work with.

P	Q	R	S	T	U	V
	1 m	10^{-3} m	7×10^{-7} m	4×10^{-7} m	10^{-8} m	10^{-11} m

Which statement is correct?

- A** A mobile phone uses radiation from region P.
B A television satellite dish uses radiation from region Q.
C The receptor cells in an eye use radiation from region R.
D The remote controller for a television uses radiation from region U.
- 26** A dolphin has a range of audible frequencies of 150 Hz–150 kHz.

Which range of frequencies can be heard both by humans with good hearing and by dolphins?

- A** 20 Hz–150 Hz
B 20 Hz–150 kHz
C 20 kHz–150 kHz
D 150 Hz–20 kHz
- 27** A permanent magnet is placed close to a bar of soft iron.



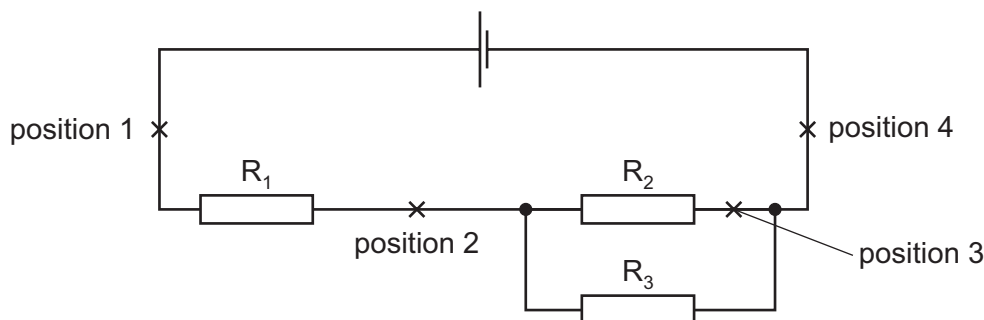
What are the polarities of end P and of end Q?

	end P	end Q
A	N	N
B	N	S
C	S	N
D	S	S

28 Which metal is used for the core of an electromagnet?

- A copper
- B iron
- C magnesium
- D steel

29 The diagram shows a cell connected to three resistors R_1 , R_2 and R_3 .



A student connects an ammeter first in position 1, then in position 2, 3 and 4 in turn.

In which positions does the ammeter show the current in R_1 ?

- A 1, 2 and 4 B 1 and 2 only C 3 only D 4 only

30 A plastic rod is rubbed with a cloth. The rod becomes positively charged.

What happens to the plastic rod and what is the charge on the cloth?

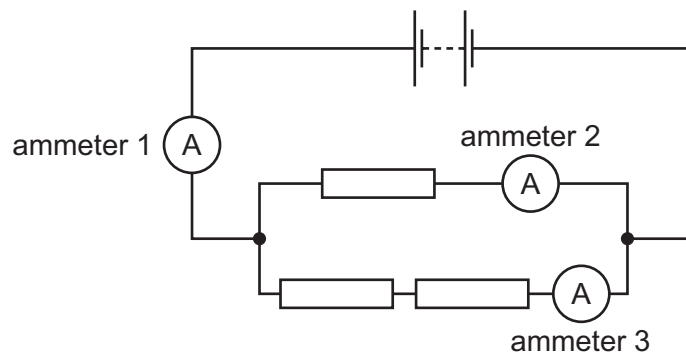
	plastic rod	charge on cloth
A	gains electrons	negative
B	gains electrons	positive
C	loses electrons	negative
D	loses electrons	positive

31 A student measures the potential difference across a device and the current in the device.

Which calculation gives the resistance of the device?

- A current + potential difference
- B current $\div$ potential difference
- C potential difference $\div$ current
- D potential difference $\times$ current

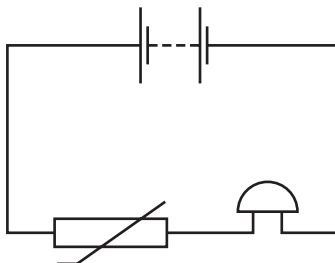
- 32** The diagram shows three identical resistors, three ammeters and a battery, connected in a circuit.



What is the order of the magnitudes of the readings on the ammeters from smallest to largest?

	smallest	intermediate	largest
A	ammeter 1	ammeter 2	ammeter 3
B	ammeter 1	ammeter 3	ammeter 2
C	ammeter 2	ammeter 3	ammeter 1
D	ammeter 3	ammeter 2	ammeter 1

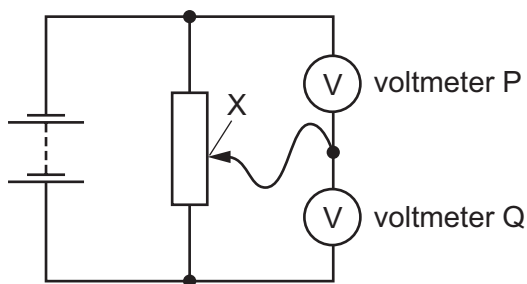
- 33** A student sets up this circuit.



What is the purpose of the circuit?

- A** to allow a lamp to be made dimmer or brighter as required
- B** to amplify the sound of a voice
- C** to light a lamp in the dark
- D** to sound a bell when the temperature rises

- 34 The diagram shows two voltmeters P and Q connected to a potential divider.



The sliding connection at point X is moved towards the top of the diagram.

What happens to the reading on P and to the reading on Q?

	reading on P	reading on Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 35 Which statement about electromagnetic induction is correct?

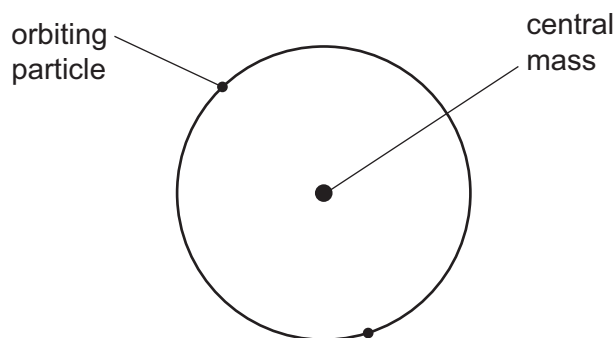
- A** A strong magnet that is held stationary near a stationary conductor causes a greater effect than a weak magnet.
- B** The effect occurs when a magnet and a conductor are both moved with the same speed and in the same direction.
- C** The effect occurs when a magnet is moved away from a nearby conductor.
- D** The effect only occurs when a magnet is moved towards a conductor.

- 36 An electrical device changes the voltage of an electrical supply from 240 V a.c. to 20 V a.c.

What is this device?

- A** a generator
- B** a relay
- C** a transformer
- D** a voltmeter

- 37 In the atomic model, an atom consists of a central mass, orbited by much smaller particles.



What is the name of the central mass and of the orbiting particles?

	central mass	orbiting particles
A	neutron	α -particles
B	neutron	electrons
C	nucleus	α -particles
D	nucleus	electrons

- 38 A neutral atom of argon-40 (${}^{40}_{18}\text{A}$) and a neutral atom of potassium-39 (${}^{39}_{19}\text{K}$) are compared.

Which atom has more electrons, and which atom has more protons?

	more electrons	more protons
A	argon	argon
B	argon	potassium
C	potassium	argon
D	potassium	potassium

- 39 Which statement about α -particles and β -particles is correct?

- A** α -particles are less ionising than β -particles.
- B** α -particles are more penetrating than β -particles.
- C** α -particles have greater mass than β -particles.
- D** α -particles have the same charge as β -particles.

- 40** An explosion in a nuclear reactor spread the isotope caesium-137 across a large area.

Ninety years after the explosion, the quantity of caesium-137 present will be 12.5% of its original level.

What is the half-life of caesium-137?

- A** 11.25 years **B** 22.5 years **C** 30.0 years **D** 45.0 years

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PHYSICS

0625/13

Paper 1 Multiple Choice (Core)

May/June 2018

MARK SCHEME

Maximum Mark: 40

Published

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This document consists of **3** printed pages.



Question	Answer	Marks
1	A	1
2	D	1
3	C	1
4	B	1
5	B	1
6	B	1
7	D	1
8	B	1
9	C	1
10	B	1
11	A	1
12	D	1
13	B	1
14	B	1
15	B	1
16	C	1
17	B	1
18	C	1
19	B	1
20	D	1
21	A	1
22	A	1
23	B	1
24	C	1
25	D	1
26	C	1
27	B	1
28	C	1

Question	Answer	Marks
29	C	1
30	B	1
31	B	1
32	D	1
33	B	1
34	C	1
35	B	1
36	C	1
37	D	1
38	C	1
39	A	1
40	B	1



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PHYSICS

0625/12

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May/June 2018

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Question	Answer	Marks
1	A	1
2	B	1
3	C	1
4	C	1
5	D	1
6	A	1
7	B	1
8	B	1
9	D	1
10	D	1
11	C	1
12	B	1
13	B	1
14	D	1
15	D	1
16	B	1
17	B	1
18	D	1
19	B	1
20	B	1
21	C	1
22	A	1
23	C	1
24	B	1
25	D	1
26	C	1
27	B	1
28	C	1

Question	Answer	Marks
29	C	1
30	D	1
31	D	1
32	D	1
33	B	1
34	A	1
35	A	1
36	C	1
37	D	1
38	B	1
39	A	1
40	B	1



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PHYSICS

0625/11

Paper 1 Multiple Choice (Core)

May/June 2018

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Question	Answer	Marks
1	A	1
2	D	1
3	C	1
4	B	1
5	D	1
6	C	1
7	A	1
8	B	1
9	D	1
10	D	1
11	A	1
12	C	1
13	D	1
14	C	1
15	A	1
16	C	1
17	B	1
18	B	1
19	B	1
20	B	1
21	A	1
22	B	1
23	A	1
24	A	1
25	B	1
26	D	1
27	C	1
28	B	1

Question	Answer	Marks
29	A	1
30	C	1
31	C	1
32	D	1
33	D	1
34	B	1
35	C	1
36	C	1
37	D	1
38	D	1
39	C	1
40	C	1



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PHYSICS

Paper 1 Multiple Choice (Core)

0625/13

May/June 2017

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

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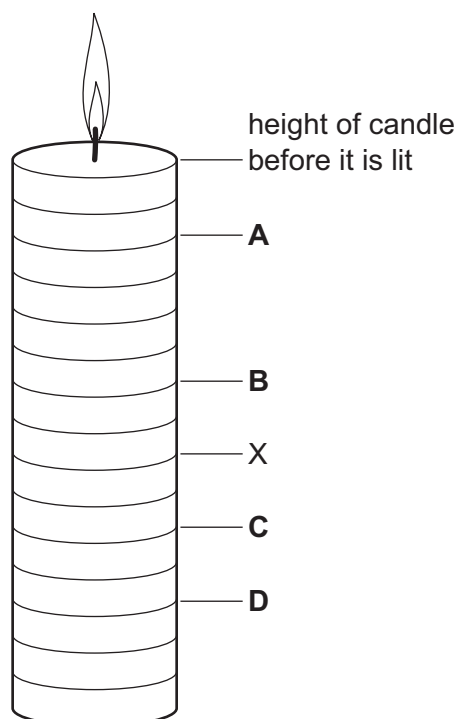
This document consists of **17** printed pages and **3** blank pages.



- 1 A candle burns evenly. It is used as a timer.

The candle is lit and burns down to point X in 2 hours.

To which labelled point does the candle burn down after a further 30 minutes?



- 2 A pendulum is swinging. Five students each measure the time it takes to swing through ten complete swings.

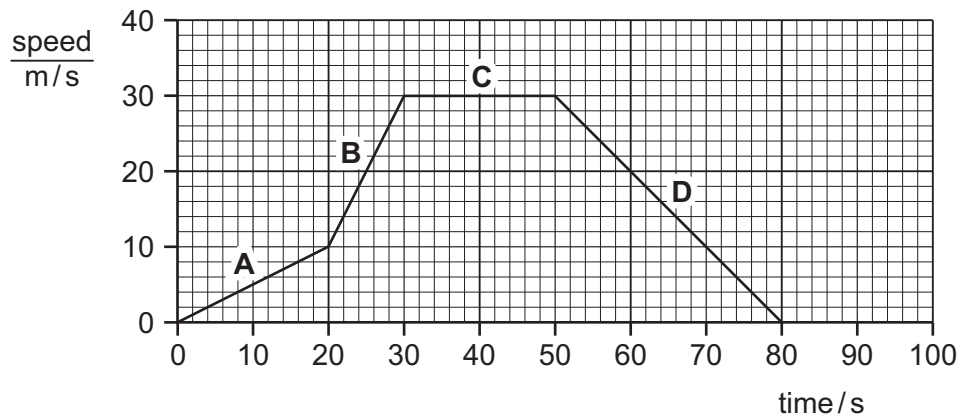
Three students measure the time as 17.2 s. Another student measures it as 16.9 s, and the fifth student measures it as 17.0 s.

What is the average period of the pendulum?

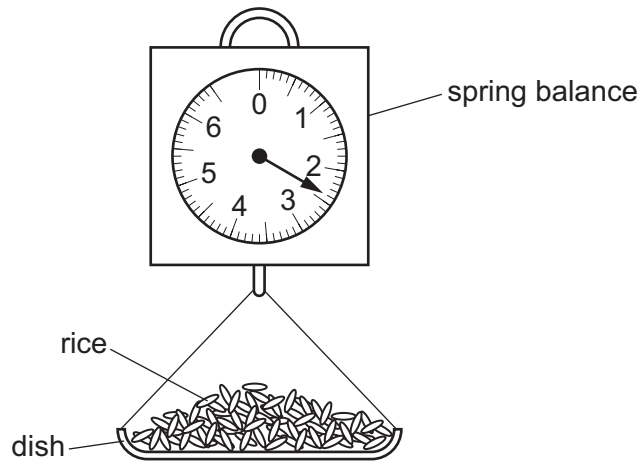
- A** 1.69 s **B** 1.70 s **C** 1.71 s **D** 1.72 s

- 3 The speed-time graph represents a motorcycle journey.

In which part of the graph is the acceleration equal to zero?



- 4 A shopkeeper pours rice into a dish that hangs from a spring balance. He records the reading.



A customer buys some pasta. The shopkeeper notices that the reading on the spring balance, with just pasta in the dish, is the same as it was with just rice in the dish.

Which quantity **must** be the same for the rice and for the pasta?

- A density
- B temperature
- C volume
- D weight

- 5 Four identical steel blocks weigh 120 N in total.

The gravitational field strength g is 10 N/kg.

What is the mass of one steel block?

- A** 3.0 kg **B** 12 kg **C** 30 kg **D** 48 kg

- 6 A steel ball bearing has a mass of 24 g and a density of 8.0 g/cm^3 . It is lowered into a measuring cylinder containing 12 cm^3 of water.

What is the new water level in the cylinder?

- A** 3.0 cm^3 **B** 4.0 cm^3 **C** 15 cm^3 **D** 16 cm^3

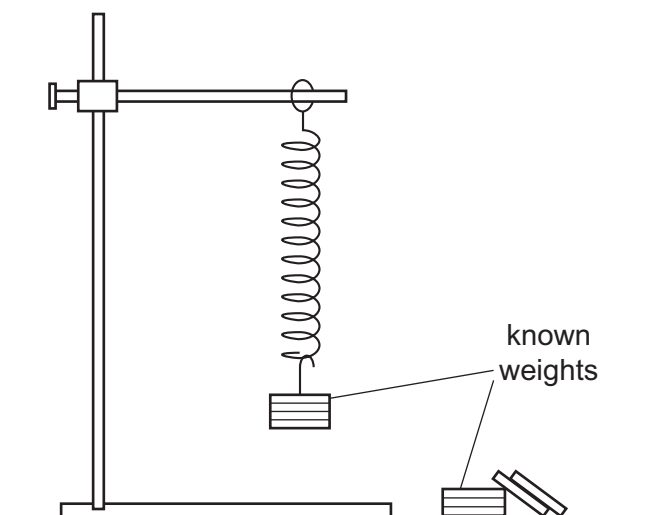
- 7 The diagram shows an object being acted upon by two forces.



What is the size of the resultant force on the object?

- A** 2.0 N **B** 3.0 N **C** 9.0 N **D** 18 N

- 8 A student is asked to investigate the extension of a spring using the apparatus shown in the diagram.



Which other piece of equipment is needed?

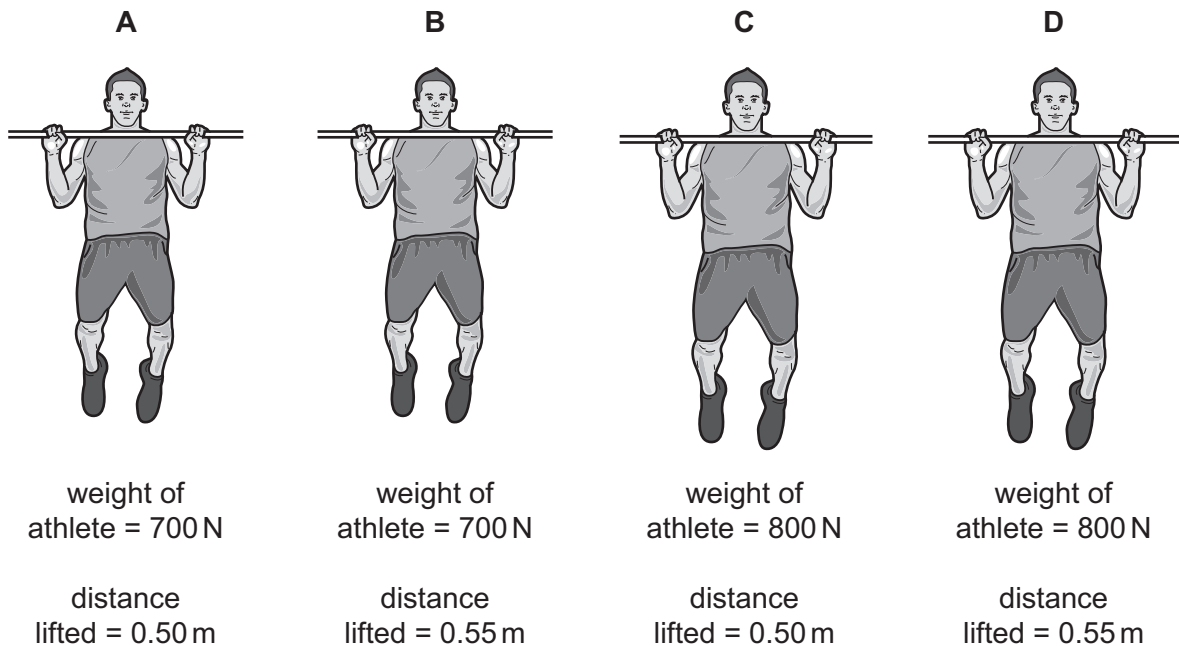
- A** measuring cylinder
B metre rule
C stopwatch
D protractor

9 In which power station are atoms of one element changed to atoms of other lighter elements?

- A a coal-fired power station
- B a hydroelectric power station
- C a nuclear power station
- D a solar power station

10 The diagrams show four different athletes training by doing pull-ups.

Which athlete does the most work?



11 A student uses her thumb to push in a drawing pin (thumb tack) into a notice board.

The pin goes into the board but does not penetrate her thumb.

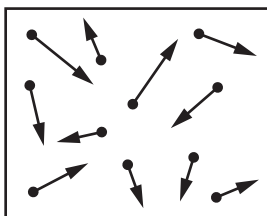
Which statement explains this?

- A The force exerted by the pin on her thumb is greater than the force exerted on the notice board.
- B The force exerted by the pin on the notice board is greater than the force exerted on her thumb.
- C The pressure of the pin on her thumb is greater than the pressure on the notice board.
- D The pressure of the pin on the notice board is greater than the pressure on her thumb.

- 12** A solid cube has sides 0.50 m long and a mass of 120 kg. It stands on the ground on one face.
What pressure does the cube exert on the ground?

A 480 kg/m³ **B** 960 kg/m³ **C** 4800 N/m² **D** 9600 N/m²

- 13** The diagram represents molecules of a gas inside a closed container of constant volume.



What happens to the molecules of the gas when the container is heated?

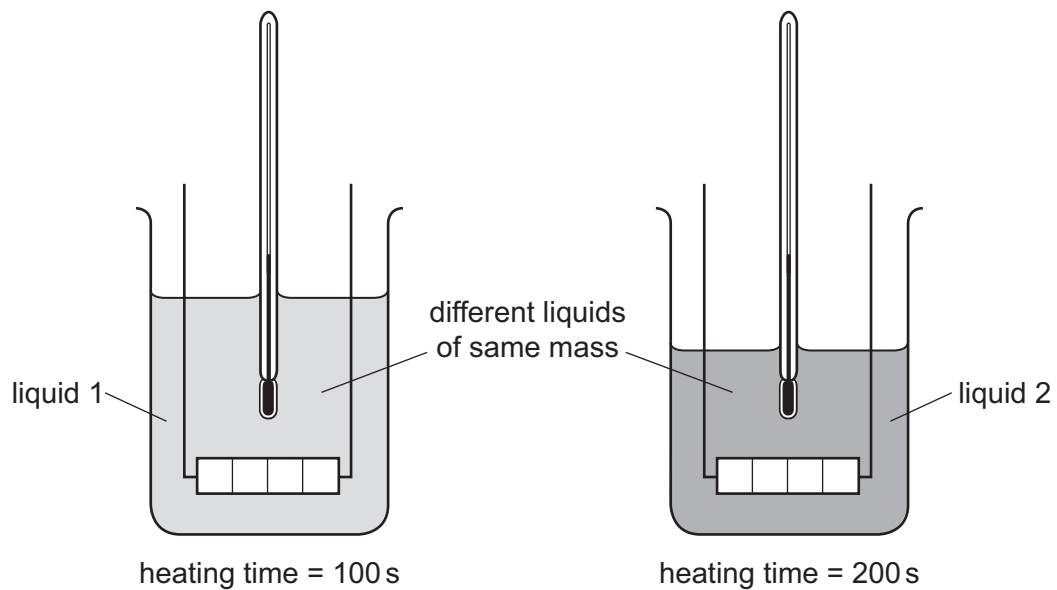
- A** They expand.
B They get closer together.
C They hit the container walls with less force.
D They move faster.
- 14** Brownian motion is observed when using a microscope to look at smoke particles in air.
What causes the smoke particles to move at random?
- A** Smoke particles are hit by air molecules.
B Smoke particles are moved by convection currents in the air.
C Smoke particles have different weights and fall at different speeds.
D Smoke particles hit the walls of the container.
- 15** The temperature of a bridge rises from 5 °C on a cold night to 25 °C at midday.
What happens to the bridge?

- A** It becomes heavier.
B It becomes more dense.
C Its length increases.
D Its mass increases.

16 Equal masses of two different liquids are put into identical beakers.

Liquid 1 is heated for 100 s and liquid 2 is heated for 200 s by heaters of the **same power**.

Each liquid has the same rise in temperature.

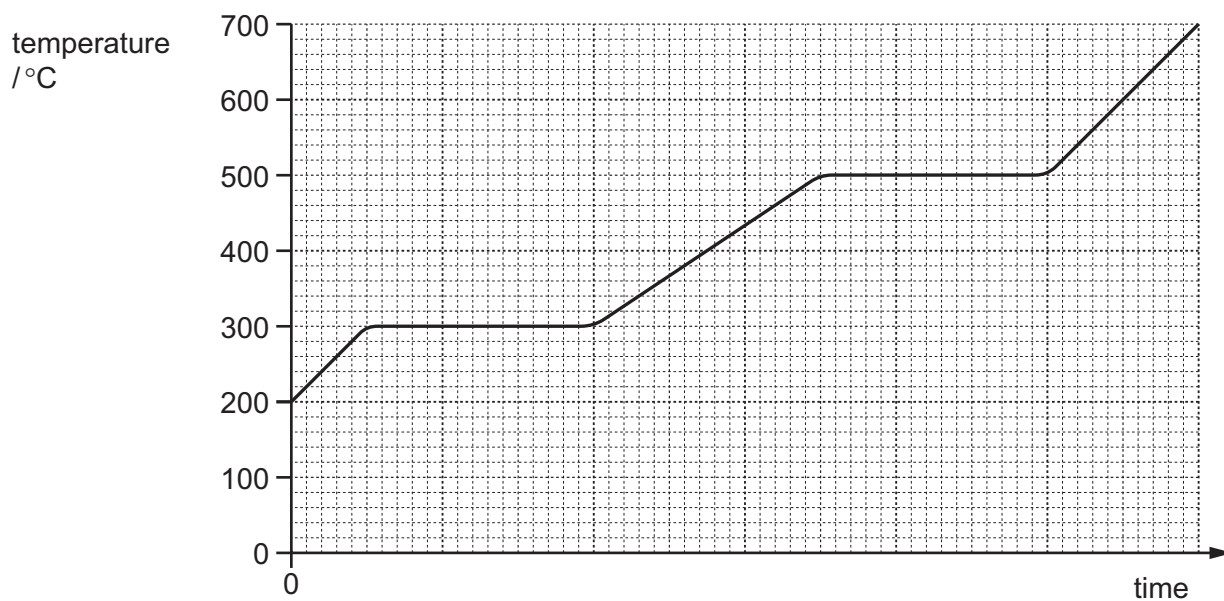


Which statement is correct?

- A** Each beaker of liquid has the same thermal capacity.
- B** Each beaker of liquid receives the same energy.
- C** Liquid 1 receives more energy than liquid 2.
- D** The thermal capacity of liquid 1 is less than the thermal capacity of liquid 2.

- 17 A solid substance is heated at a constant rate. The solid changes into a liquid and then into a gas.

The graph shows how the temperature of the substance changes.

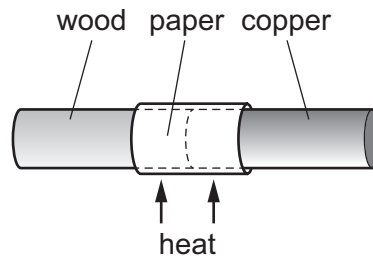


What is the melting point and what is the boiling point of the substance?

	melting point/°C	boiling point/°C
A	200	300
B	200	500
C	300	500
D	300	700

- 18** A copper bar and a wooden bar are joined. A piece of paper is wrapped tightly around the join.

The bar is heated strongly at the centre for a short time, and the paper goes brown on one side only.



Which side goes brown, and what does this show about wood and copper?

	brown side	wood	copper
A	copper	conductor	insulator
B	copper	insulator	conductor
C	wood	conductor	insulator
D	wood	insulator	conductor

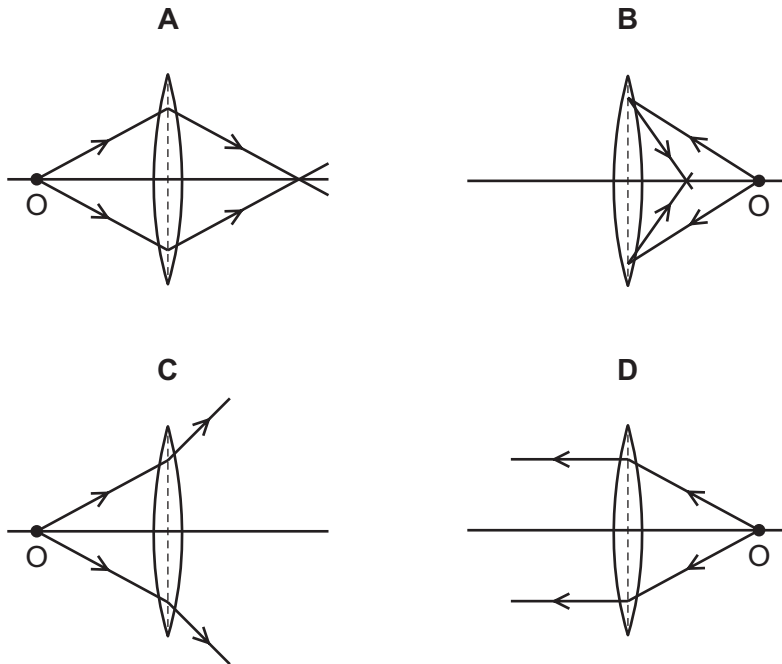
- 19** Which types of thermal energy transfer require a medium?

- A** conduction and convection
- B** conduction only
- C** convection and radiation
- D** radiation only

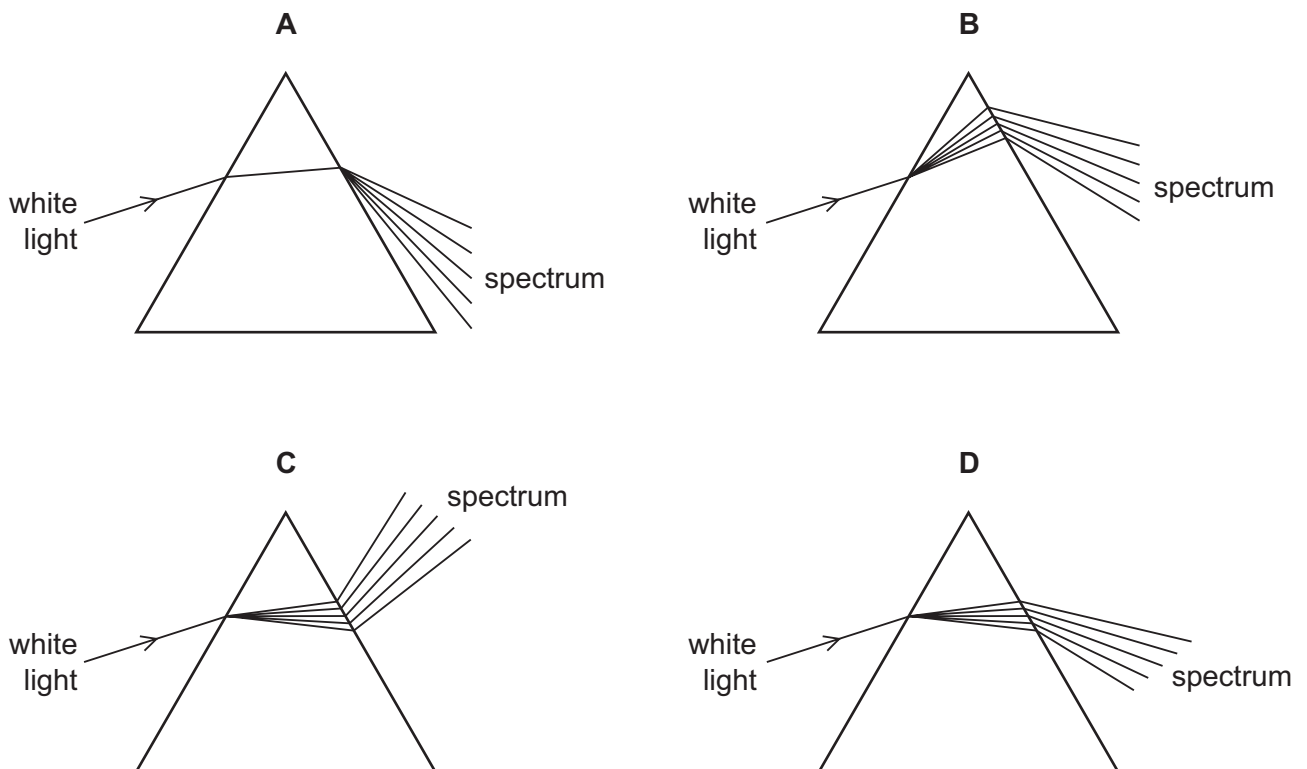
- 20** What causes the change in direction when light travels from air into glass?

- A** The amplitude of the light changes.
- B** The colour of the light changes.
- C** The frequency of the light changes.
- D** The speed of the light changes.

21 Which diagram shows how a converging lens forms a real image of an object O?



22 Which diagram shows what happens when a ray of white light passes through a prism?



23 The table shows different types of wave in the electromagnetic spectrum.

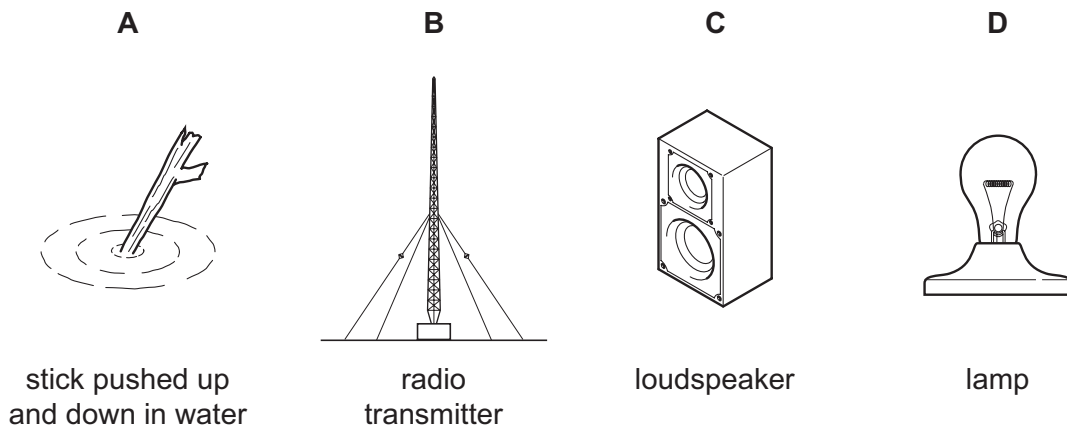
radio waves	micro-waves	infra-red waves	visible light	ultraviolet waves	X-rays	gamma rays
-------------	-------------	-----------------	---------------	-------------------	--------	------------

Where do all the waves travel at the same speed?

- A** in a vacuum
- B** in diamond
- C** in glass
- D** in water

24 The diagrams show four sources of waves.

Which source produces longitudinal waves?



25 A fire alarm is not loud enough and the pitch is too low. An engineer adjusts the alarm so that it produces a louder note of a higher pitch.

What effect does this have on the amplitude and on the frequency of the sound?

	amplitude	frequency
A	larger	greater
B	larger	smaller
C	smaller	greater
D	smaller	smaller

26 From which materials are the coil and the core of an electromagnet made?

	coil	core
A	copper	soft iron
B	copper	steel
C	soft iron	copper
D	steel	soft iron

27 In which way are a bar magnet and an electromagnet similar?

- A** A bar magnet and an electromagnet are always magnetised when stored.
- B** A bar magnet and an electromagnet can both be used to separate magnetic and non-magnetic materials.
- C** A bar magnet can be made of steel and an electromagnet uses a steel core.
- D** The magnetic field strength of a bar magnet and of an electromagnet can both be varied.

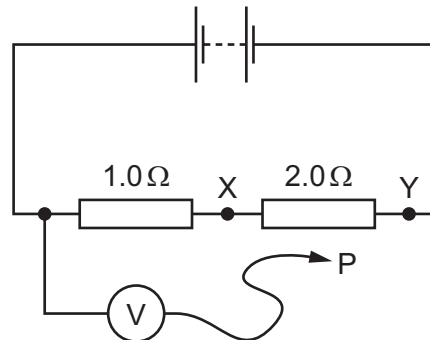
28 Which material is a conductor of electricity?

- A** brass
- B** glass
- C** plastic
- D** wood

- 29** The diagram shows a circuit containing two resistors of resistance $1.0\ \Omega$ and $2.0\ \Omega$.

A voltmeter is connected across the $1.0\ \Omega$ resistor by connecting P to X.

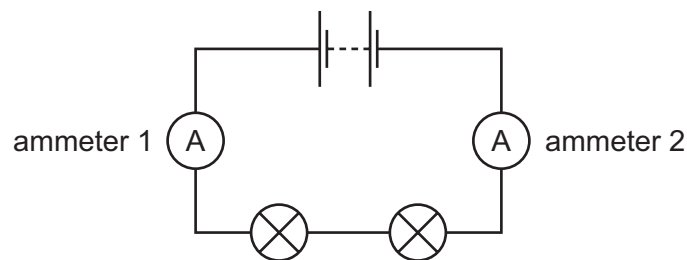
The reading on the voltmeter is 6.0 V .



P is moved to point Y in the circuit.

What is the new reading on the voltmeter?

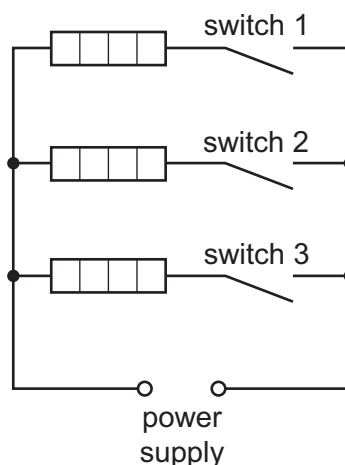
- A** 3.0 V **B** 6.0 V **C** 12 V **D** 18 V
- 30** The circuit diagram shows a simple circuit with a battery of electromotive force (e.m.f.) 6.0 V and two bulbs each of resistance $2.0\ \Omega$.



Which row gives the readings on the ammeters?

	reading on ammeter 1 / A	reading on ammeter 2 / A
A	1.5	0
B	1.5	1.5
C	3.0	0
D	3.0	1.5

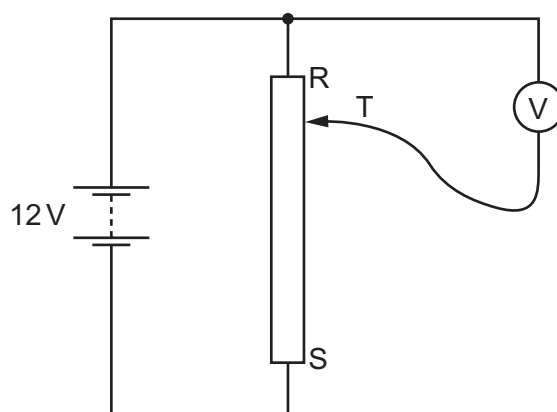
- 31 The diagram shows three identical heating elements connected to a power supply.



Which arrangement of switches causes **most** power to be used?

	switch 1	switch 2	switch 3
A	closed	closed	closed
B	closed	open	closed
C	closed	open	open
D	open	closed	closed

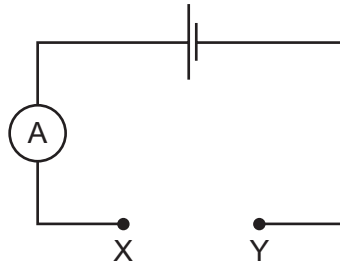
- 32 A student connects a variable potential divider (potentiometer) circuit.



What happens to the reading on the voltmeter as the sliding terminal T is moved from R to S?

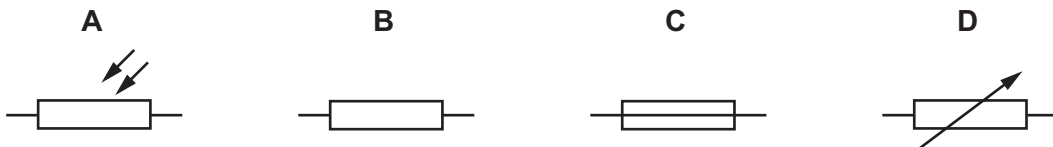
- A** It decreases from 12V to 0V.
- B** It increases from 0V to 12V.
- C** It remains at 0V.
- D** It remains at 12V.

- 33 The diagram shows a circuit used to make a light detector.



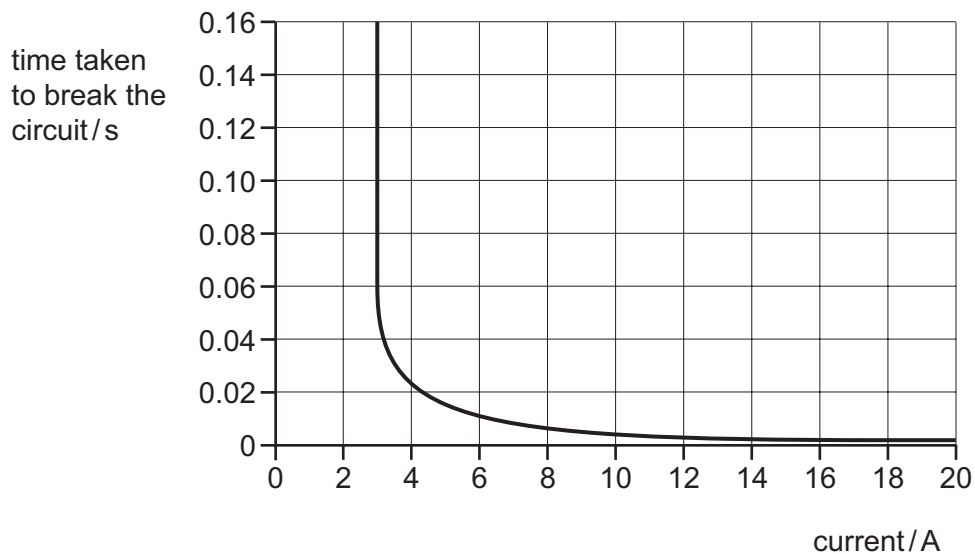
One component is connected between X and Y.

Which component causes the ammeter reading to increase when the light gets brighter?



- 34 A circuit-breaker is designed to protect a circuit which usually carries a current of 2 A.

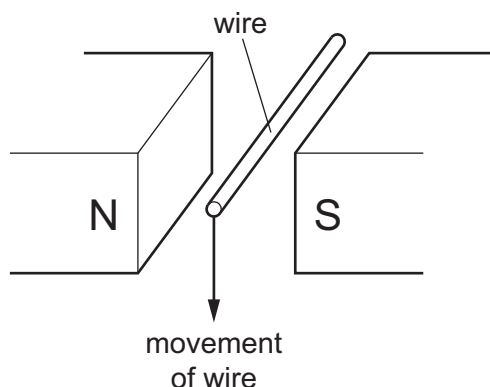
The time taken to break the circuit depends on the current, as shown in the graph.



What happens when the current in the circuit is 2 A and what happens when the current is 18 A?

	when the current is 2 A	when the current is 18 A
A	the circuit breaks in less than 0.01 s	the circuit breaks in less than 0.01 s
B	the circuit breaks in less than 0.01 s	the circuit does not break
C	the circuit does not break	the circuit breaks in less than 0.01 s
D	the circuit does not break	the circuit does not break

- 35 A wire moves in the direction shown between the poles of a magnet.



Which change increases the magnitude of the induced electromotive force (e.m.f.) across the ends of the wire?

- A increasing the gap between the poles of the magnet
- B moving the wire faster
- C using a shorter wire
- D using a thinner wire

- 36 What is the purpose of a relay?

- A to change a large voltage into a small voltage
- B to change a small voltage into a large voltage
- C to use a large current to switch on a small current
- D to use a small current to switch on a large current

- 37 Which description of a neutral atom of copper is correct?

- A a nucleus surrounded by electrons
- B a nucleus surrounded by molecules
- C electrons surrounded by a nucleus
- D electrons surrounded by molecules

- 38 A nuclide of iodine is represented by $^{131}_{53}\text{I}$.

How many neutrons are in a nucleus of this nuclide?

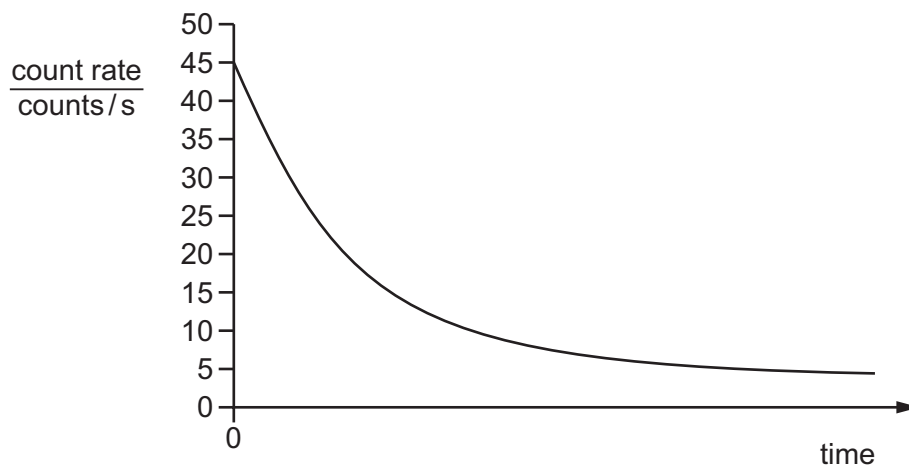
- A 53 B 78 C 131 D 184

39 A sample of radioactive isotope is decaying.

The nuclei of which atoms will decay first?

- A** It is impossible to know because radioactive decay is random.
- B** It is impossible to know unless the age of the material is known.
- C** The atoms near the centre will decay first because they are surrounded by more atoms.
- D** The atoms near the surface will decay first because the radiation can escape more easily.

40 The diagram shows a decay curve for a radioactive substance.



According to the curve shown, what is the background radiation count?

- A** 40 counts/s
- B** 20 counts/s
- C** 5 counts/s
- D** 0 counts/s

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Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS

Paper 1 Multiple Choice (Core)

0625/12

May/June 2017

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

READ THESE INSTRUCTIONS FIRST

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There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

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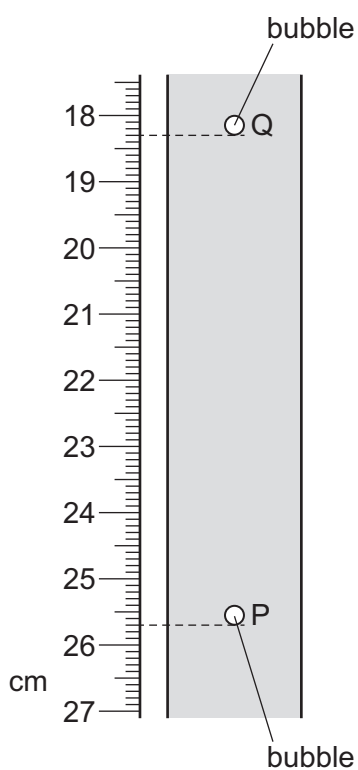
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

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- 1 Which device is used to measure the time it takes for a 10 cm^3 block of ice to melt in a laboratory at room temperature?
- A measuring cylinder
B ruler
C stopwatch
D thermometer
- 2 A student determines the average speed of a bubble rising through a liquid at constant speed.
- When the student starts the stopwatch the bubble is at position P.
- After 2.0 s the bubble is at position Q.

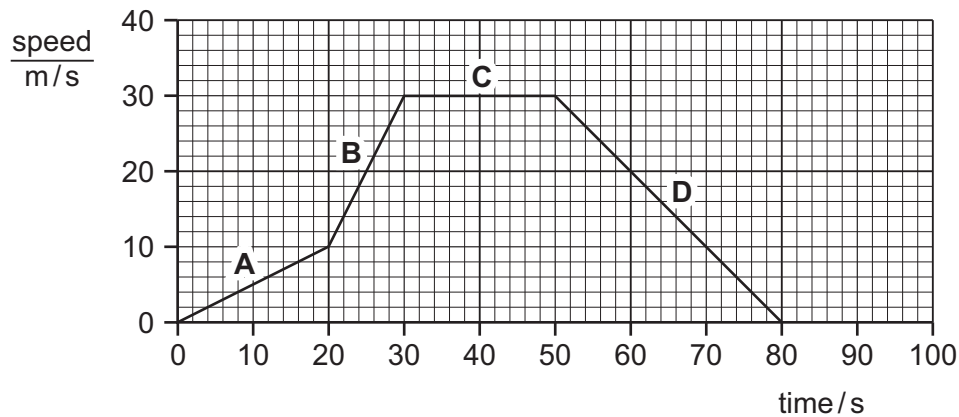


What is the speed of the bubble between P and Q?

- A 3.2 cm/s B 3.7 cm/s C 6.4 cm/s D 7.4 cm/s

- 3 The speed-time graph represents a motorcycle journey.

In which part of the graph is the acceleration equal to zero?



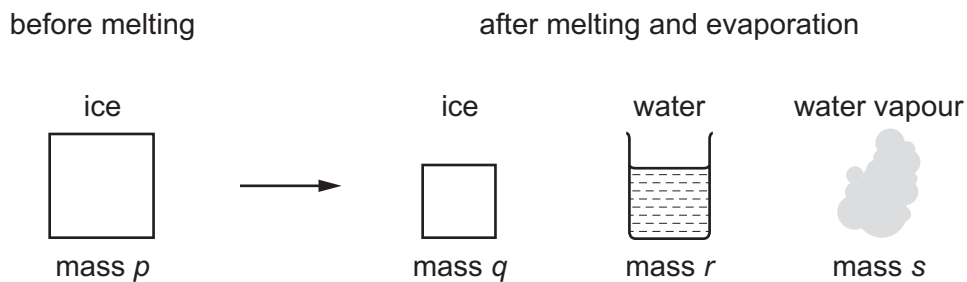
- 4 Which expression is used to find gravitational field strength g ?

- A mass $\times$ density
- B mass $\div$ weight
- C weight $\times$ mass
- D weight $\div$ mass

- 5 A block of ice is removed from a freezer. Some of the ice melts to produce water. Some of the water that is produced evaporates.

The original mass of the ice is p . The mass of the ice that has not yet melted is q . The mass of the water is r . The mass of the water vapour is s .

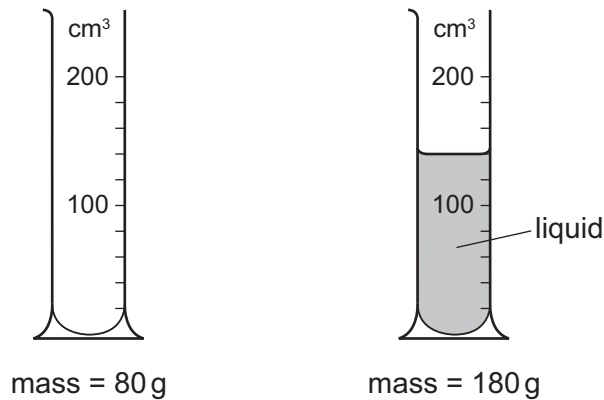
The diagram shows these changes.



Which equation gives the relationship between p , q , r and s ?

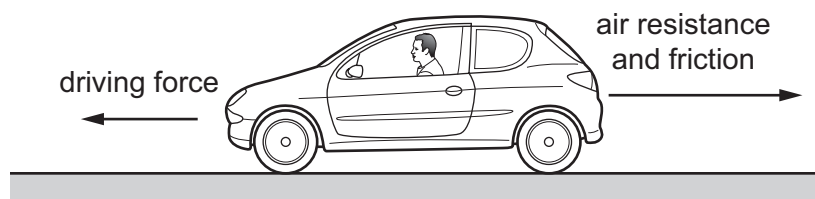
- A $p = q + r$
- B $p = q + r + s$
- C $p = q + r - s$
- D $p = q + s$

- 6 The masses of a measuring cylinder before and after pouring some liquid into it are shown in the diagram.



What is the density of the liquid?

- A $\frac{100}{120} \text{ g/cm}^3$ B $\frac{100}{140} \text{ g/cm}^3$ C $\frac{180}{120} \text{ g/cm}^3$ D $\frac{180}{140} \text{ g/cm}^3$
- 7 A car travels forwards along a straight horizontal road. Only the horizontal forces acting on it are shown.



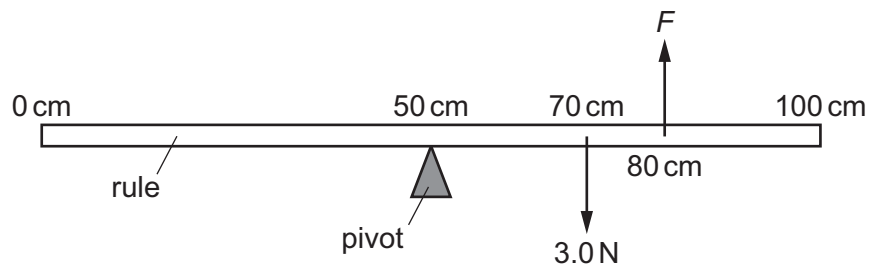
The length of each arrow represents the size of each force.

How do these forces affect the motion of the car?

- A The car moves at constant speed.
 B The car moves backwards.
 C The car slows down.
 D The car's forward speed increases.

- 8 The centre of a uniform metre rule rests on a pivot. A load of weight 3.0 N is placed at the 70 cm mark.

A force F acts upwards at the 80 cm mark. The rule is in equilibrium.



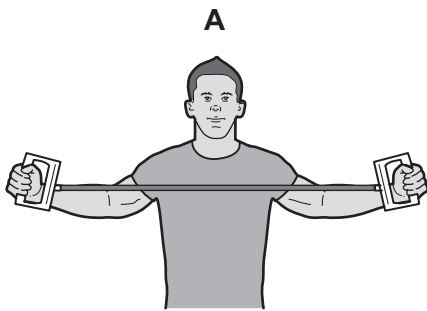
What is the magnitude of F ?

- A** 2.0 N **B** 2.6 N **C** 3.0 N **D** 4.5 N
- 9 An energy resource is used to generate electrical energy.
- Which energy resource uses a transfer of gravitational potential energy to generate this electrical energy?
- A** geothermal
B hydroelectric
C solar
D wind

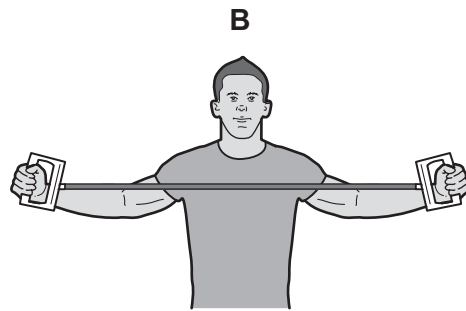
10 The diagrams show athletes training by stretching springs.

Each spring has the same stiffness.

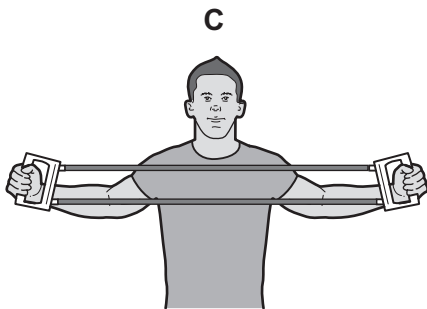
Which athlete does the most work?



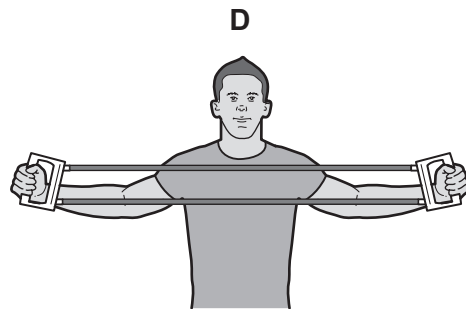
one spring stretched
by 0.60 m



one spring stretched
by 0.80 m



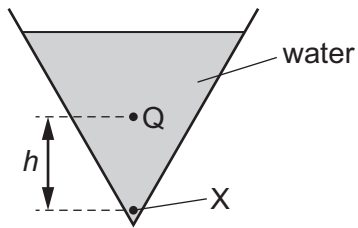
two springs stretched
by 0.60 m



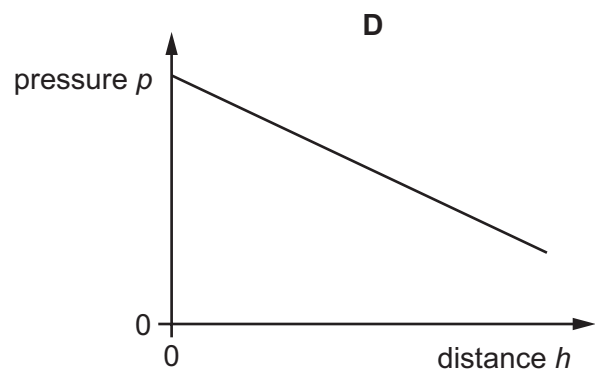
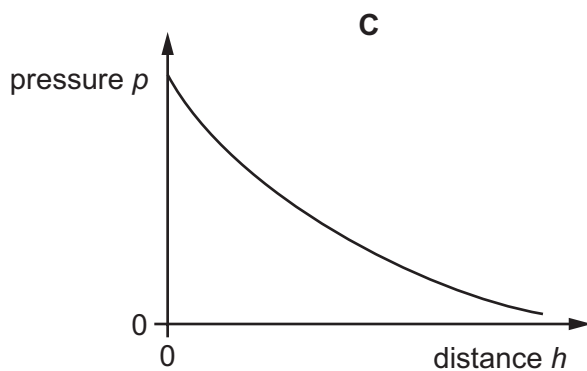
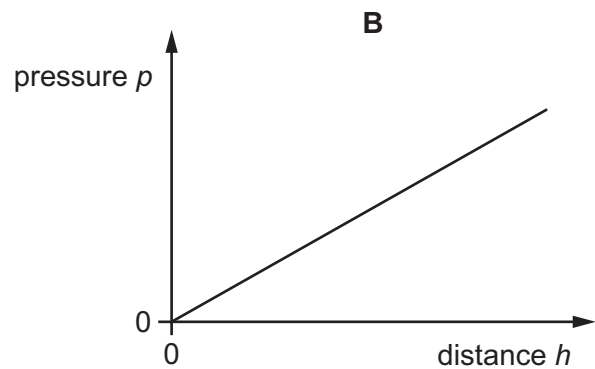
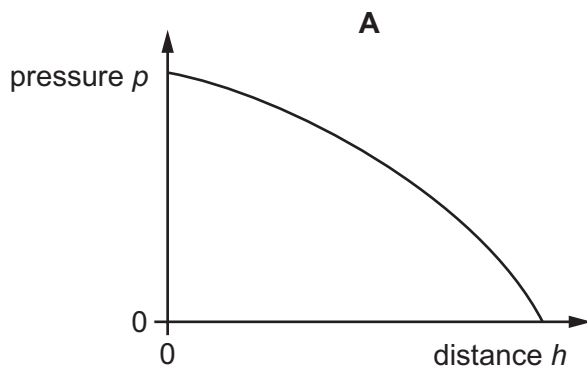
two springs stretched
by 0.80 m

11 The diagram shows a conical vessel full of water.

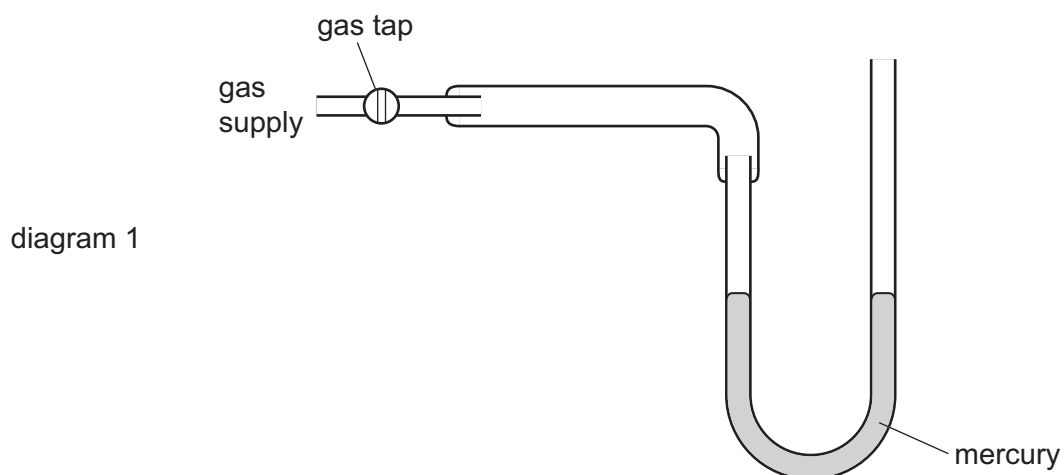
The pressure at point X due to the water is p . A point Q is a distance h above point X.



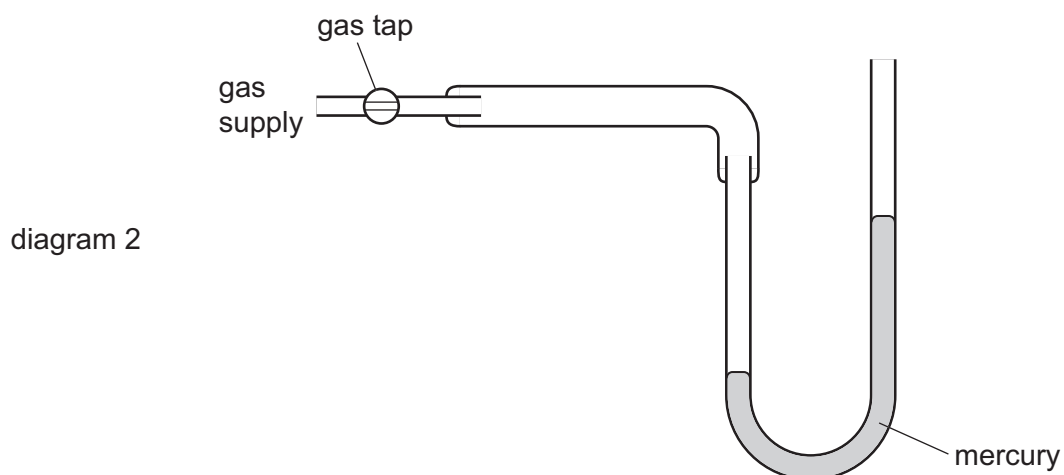
Which graph shows how the pressure due to the water at Q varies with distance h ?



- 12** In diagram 1 a manometer containing mercury (Hg) is connected to a gas tap. The tap is turned off and the mercury is at the same level on both sides.



When the gas supply is turned on, the mercury in the tube connected to the supply falls by 40 mm and the mercury in the tube open to the atmosphere rises by 40 mm, as shown in diagram 2.



What is the pressure of the gas in the gas supply?

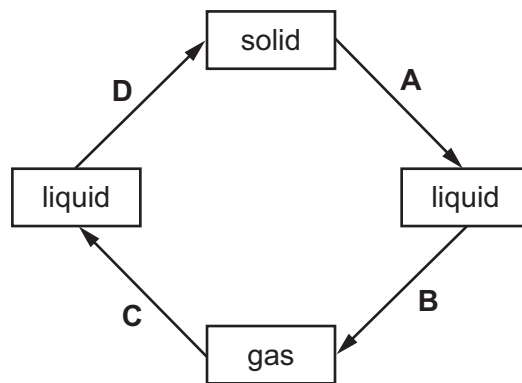
- A** 40 mm Hg above atmospheric pressure
- B** 40 mm Hg below atmospheric pressure
- C** 80 mm Hg above atmospheric pressure
- D** 80 mm Hg below atmospheric pressure

- 13** Brownian motion is observed when using a microscope to look at smoke particles in air.

What causes the smoke particles to move at random?

- A** Smoke particles are hit by air molecules.
- B** Smoke particles are moved by convection currents in the air.
- C** Smoke particles have different weights and fall at different speeds.
- D** Smoke particles hit the walls of the container.

- 14** Which change is evaporation?



- 15** A glass bottle has a metal cap. The cap fits very tightly and is difficult to remove.

The cap and the neck of the bottle are dipped in a bowl of hot water. The cap can be removed more easily.

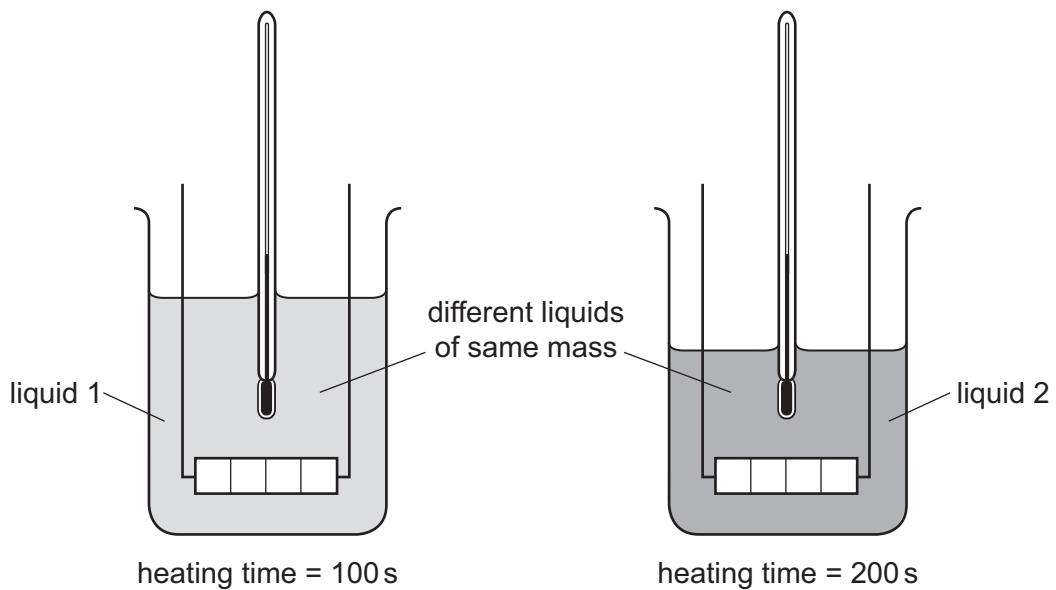
What happens to allow the cap to be removed more easily from the bottle?

- A** The cap contracts.
- B** The cap expands.
- C** The glass bottle contracts.
- D** The glass bottle expands.

16 Equal masses of two different liquids are put into identical beakers.

Liquid 1 is heated for 100 s and liquid 2 is heated for 200 s by heaters of the **same power**.

Each liquid has the same rise in temperature.

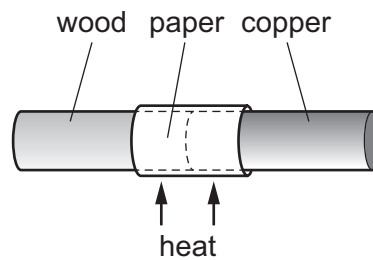


Which statement is correct?

- A** Each beaker of liquid has the same thermal capacity.
- B** Each beaker of liquid receives the same energy.
- C** Liquid 1 receives more energy than liquid 2.
- D** The thermal capacity of liquid 1 is less than the thermal capacity of liquid 2.

- 17** A copper bar and a wooden bar are joined. A piece of paper is wrapped tightly around the join.

The bar is heated strongly at the centre for a short time, and the paper goes brown on one side only.

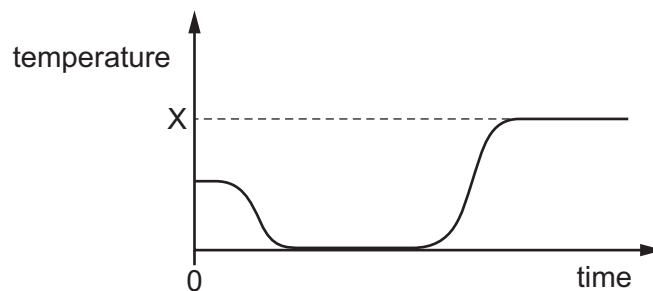


Which side goes brown, and what does this show about wood and copper?

	brown side	wood	copper
A	copper	conductor	insulator
B	copper	insulator	conductor
C	wood	conductor	insulator
D	wood	insulator	conductor

- 18** A glass of water is taken out of a refrigerator. Several ice cubes are put into the glass of water. The glass is then left in the room for several hours.

The graph shows how the temperature of the water in the glass varies from the time it is taken out of the refrigerator.



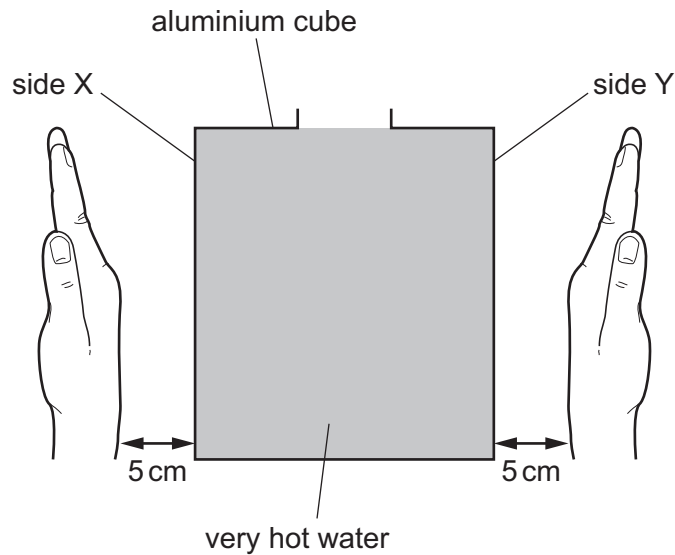
What does the temperature marked X represent?

- A** boiling point of water
- B** melting point of ice
- C** room temperature
- D** temperature inside refrigerator

- 19** A hollow aluminium cube is filled with very hot water.

Side X of the cube is opposite side Y of the cube. One of these two sides is black and one is white.

A student holds the back of one hand 5 cm from side X, and then immediately holds the back of the other hand 5 cm from side Y.



The hand held near side Y feels warmer than the hand held near side X.

Which row identifies the black side and correctly compares the rate of emission of thermal radiation from each side?

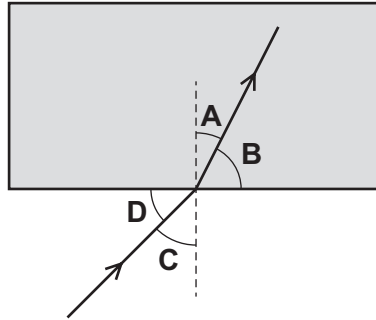
	black side	rate of emission of thermal radiation
A	X	greater for X
B	X	the same for X and Y
C	Y	greater for Y
D	Y	the same for X and Y

- 20** What causes the change in direction when light travels from air into glass?

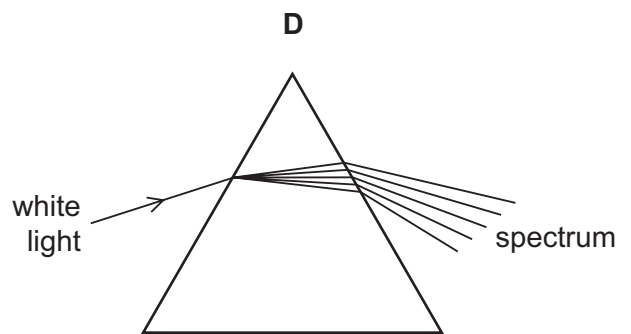
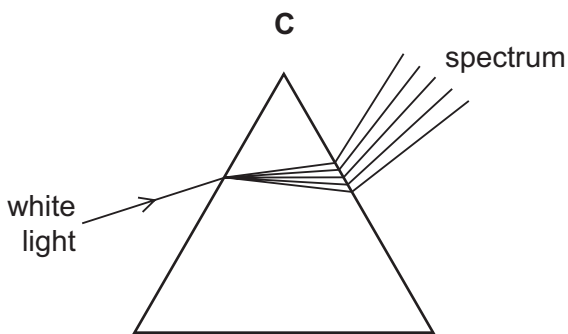
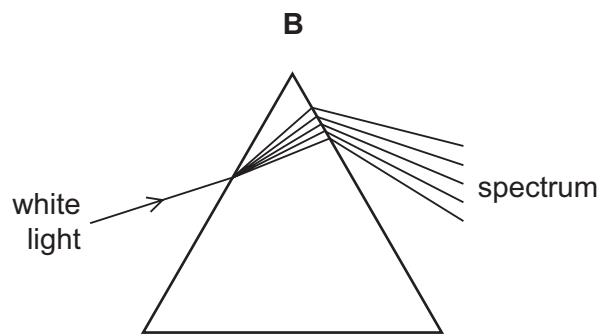
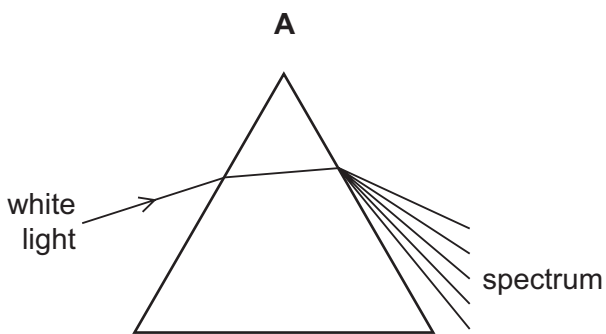
- A** The amplitude of the light changes.
- B** The colour of the light changes.
- C** The frequency of the light changes.
- D** The speed of the light changes.

21 Light passes into a glass block.

Which is the angle of refraction?

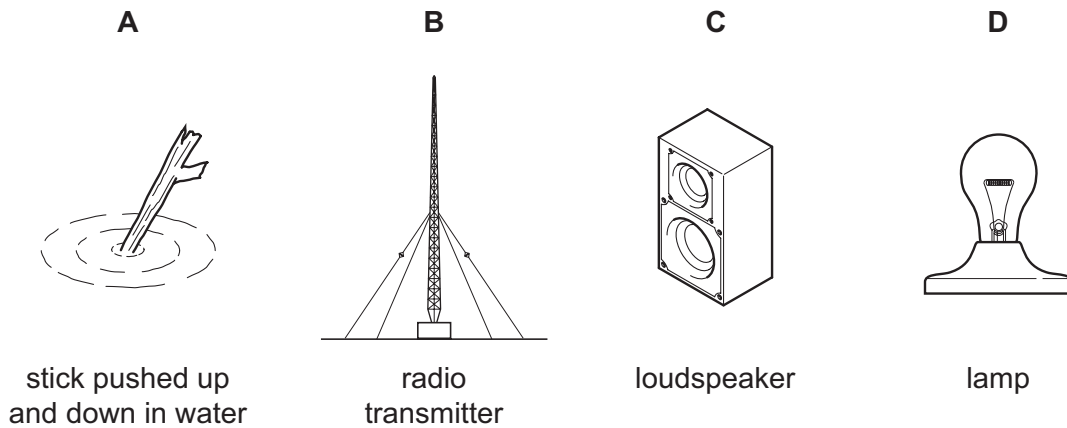


22 Which diagram shows what happens when a ray of white light passes through a prism?



23 The diagrams show four sources of waves.

Which source produces longitudinal waves?



24 The table shows different types of wave in the electromagnetic spectrum.

radio waves	micro-waves	infra-red waves	visible light	ultraviolet waves	X-rays	gamma rays
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Where do all the waves travel at the same speed?

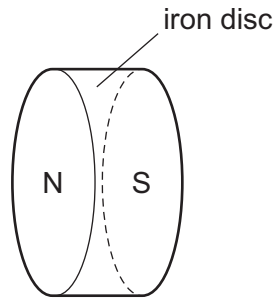
- A** in a vacuum
- B** in diamond
- C** in glass
- D** in water

25 A fire alarm is not loud enough and the pitch is too low. An engineer adjusts the alarm so that it produces a louder note of a higher pitch.

What effect does this have on the amplitude and on the frequency of the sound?

	amplitude	frequency
A	larger	greater
B	larger	smaller
C	smaller	greater
D	smaller	smaller

26 What is wrong with this labelled diagram of a permanent magnet?



- A The cross-section should be rectangular.
 - B The length should be greater than the diameter.
 - C The magnet should be made of steel.
 - D The N-pole and the S-pole should be reversed.
- 27 Which statement about magnetism is correct?
- A An unmagnetised iron bar becomes magnetised when it is placed near a magnet.
 - B An unmagnetised steel bar can be magnetised by passing a current through it.
 - C Steel is used as the core of an electromagnet.
 - D When an iron bar has been magnetised, it is difficult to demagnetise it.
- 28 A student tests the electrical conduction of four materials.

aluminium

iron

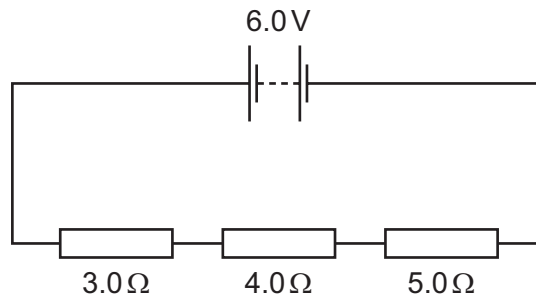
plastic

silver

Which materials conduct electricity?

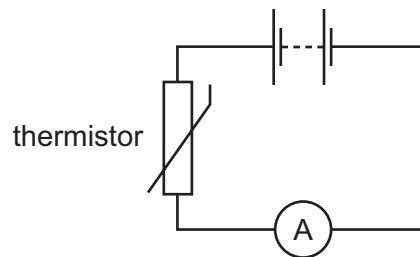
- A aluminium, iron and silver only
- B aluminium and silver only
- C iron, silver and plastic only
- D plastic only

- 29 The circuit diagram shows three resistors connected in series across a 6.0 V supply.



What is the potential difference (p.d.) across the $4.0\ \Omega$ resistor?

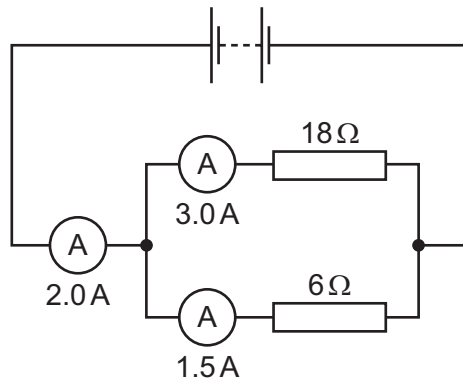
- A** 0.67 V **B** 1.5 V **C** 2.0 V **D** 6.0 V
- 30 A thermistor is connected in series with a sensitive ammeter and a battery.



Which change will give a larger ammeter reading?

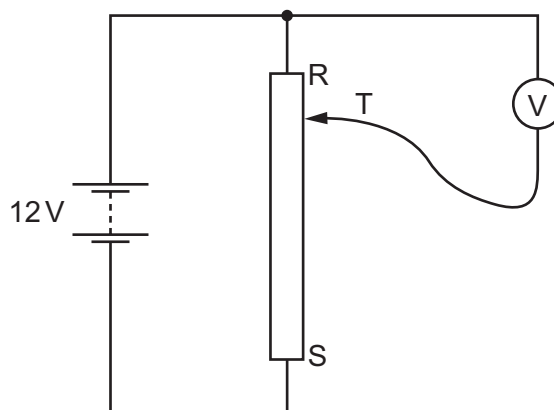
- A** adding another thermistor in series
B cooling the thermistor
C heating the thermistor
D reducing the number of cells in the battery

- 31 Three ammeters measure the currents in different parts of the circuit shown. The diagram indicates the reading on the ammeters.



How do we know that at least one of the ammeters must be faulty?

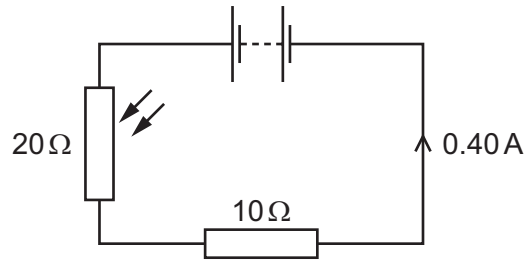
- A All three ammeters must read the same value.
 - B All the current takes the easier path through the $6\ \Omega$ resistor.
 - C The current from the battery must be equal to the sum of the currents in the two resistors.
 - D The current in the two parallel resistors must be the same.
- 32 A student connects a variable potential divider (potentiometer) circuit.



What happens to the reading on the voltmeter as the sliding terminal T is moved from R to S?

- A It decreases from 12 V to 0 V.
- B It increases from 0 V to 12 V.
- C It remains at 0 V.
- D It remains at 12 V.

33 The diagram shows an electric circuit.

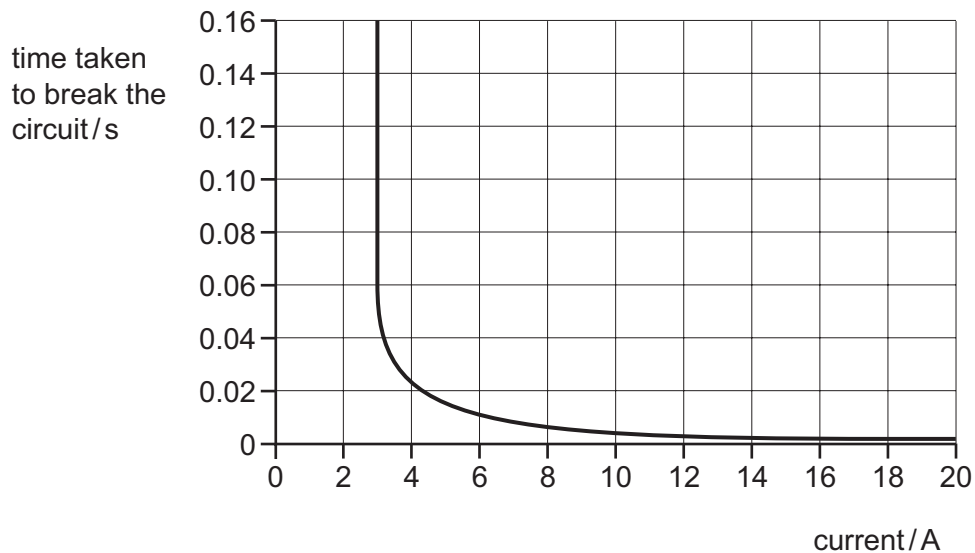


What is the potential difference (p.d.) across the LDR?

- A** 4.0 V **B** 8.0 V **C** 25 V **D** 50 V

34 A circuit-breaker is designed to protect a circuit which usually carries a current of 2 A.

The time taken to break the circuit depends on the current, as shown in the graph.



What happens when the current in the circuit is 2 A and what happens when the current is 18 A?

	when the current is 2 A	when the current is 18 A
A	the circuit breaks in less than 0.01 s	the circuit breaks in less than 0.01 s
B	the circuit breaks in less than 0.01 s	the circuit does not break
C	the circuit does not break	the circuit breaks in less than 0.01 s
D	the circuit does not break	the circuit does not break

35 What is the purpose of a relay?

- A to change a large voltage into a small voltage
- B to change a small voltage into a large voltage
- C to use a large current to switch on a small current
- D to use a small current to switch on a large current

36 A coil of four loops of wire is placed in a magnetic field. When there is a current, the coil experiences a turning effect.

Some extra loops of wire are wound on the coil but the current is unchanged.

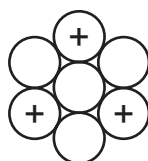
How does this affect the turning effect?

- A It is unchanged.
- B Its direction changes.
- C It decreases.
- D It increases.

37 Which particle has a negative charge?

- A an alpha particle
- B an electron
- C a neutron
- D a proton

38 The diagram represents the nucleus of an atom. The charged particles are shown.



Which row gives the proton number and the nucleon number for this nucleus?

	proton number	nucleon number
A	3	4
B	3	7
C	4	3
D	4	7

39 A sample of radioactive isotope is decaying.

The nuclei of which atoms will decay first?

- A** It is impossible to know because radioactive decay is random.
- B** It is impossible to know unless the age of the material is known.
- C** The atoms near the centre will decay first because they are surrounded by more atoms.
- D** The atoms near the surface will decay first because the radiation can escape more easily.

40 The half-life of a radioactive isotope is 4.0 years.

A sample of this material contains 24 million radioactive nuclei.

How many of these radioactive nuclei remain undecayed after 12 years?

- A** 0.5 million
- B** 2.0 million
- C** 3.0 million
- D** 6.0 million

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PHYSICS

0625/11

Paper 1 Multiple Choice (Core)

May/June 2017

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

MODIFIED LANGUAGE

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

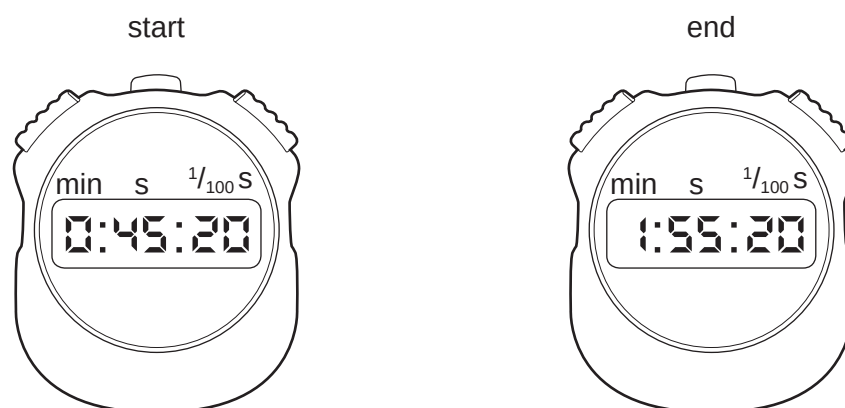
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **19** printed pages and **1** blank page.



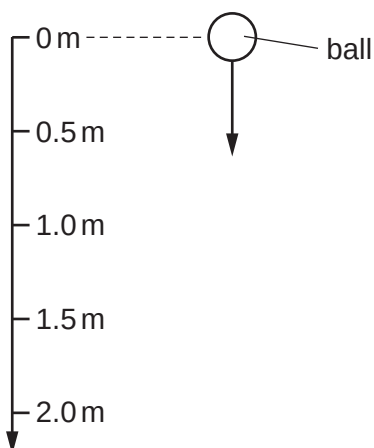
- 1 A stopwatch is used to time a runner in a race. The diagrams show the stopwatch at the start and at the end of the race.



How long did the runner take to run the race?

- A 70.00 seconds
 - B 110.00 seconds
 - C 115.20 seconds
 - D 155.20 seconds
- 2 On Earth, a ball is dropped and falls 2.0 m in a vacuum.

The acceleration of the ball at 1.0 m is 10 m/s^2 .

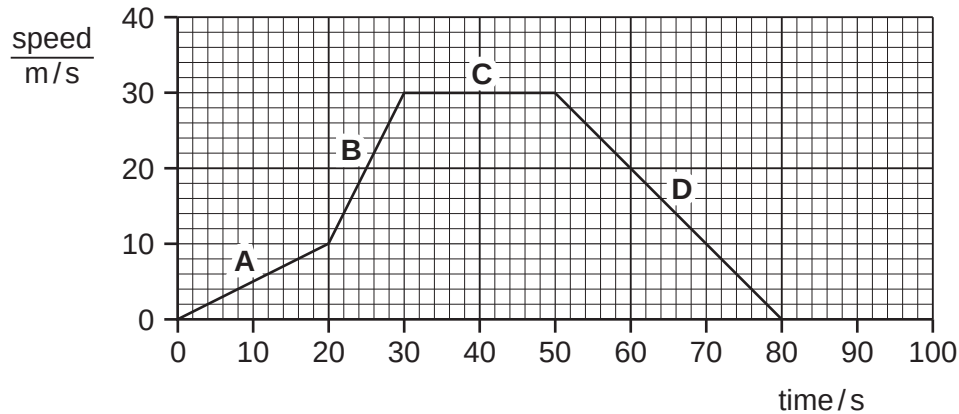


What is the acceleration of the ball at 0.5 m?

- A 5.0 m/s^2
- B 10 m/s^2
- C 15 m/s^2
- D 20 m/s^2

- 3 The speed-time graph represents a motorcycle journey.

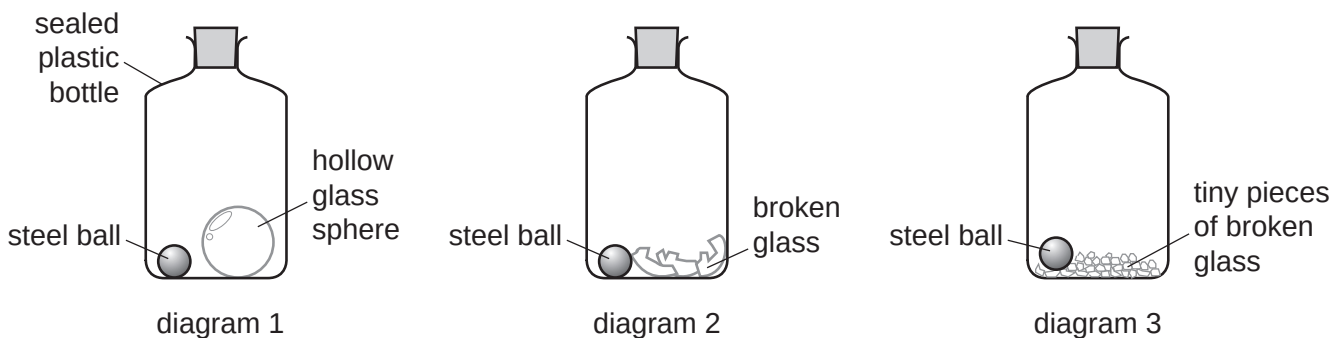
In which part of the graph is the acceleration equal to zero?



- 4 Diagram 1 shows a sealed plastic bottle containing a hollow glass sphere and a steel ball.

Diagram 2 shows the same bottle after it has been shaken.

Diagram 3 shows the same bottle after it has been shaken again until the broken glass is in tiny pieces.



The mass of the bottle and contents in diagram 1 is m_1 .

The mass of the bottle and contents in diagram 2 is m_2 .

The mass of the bottle and contents in diagram 3 is m_3 .

Which statement gives the correct relation between m_1 , m_2 and m_3 ?

- A m_1 is equal to m_2 and m_2 is equal to m_3 .
- B m_1 is greater than m_2 and m_2 is greater than m_3 .
- C m_1 is less than m_2 and m_2 is greater than m_3 .
- D m_1 is less than m_2 and m_2 is less than m_3 .

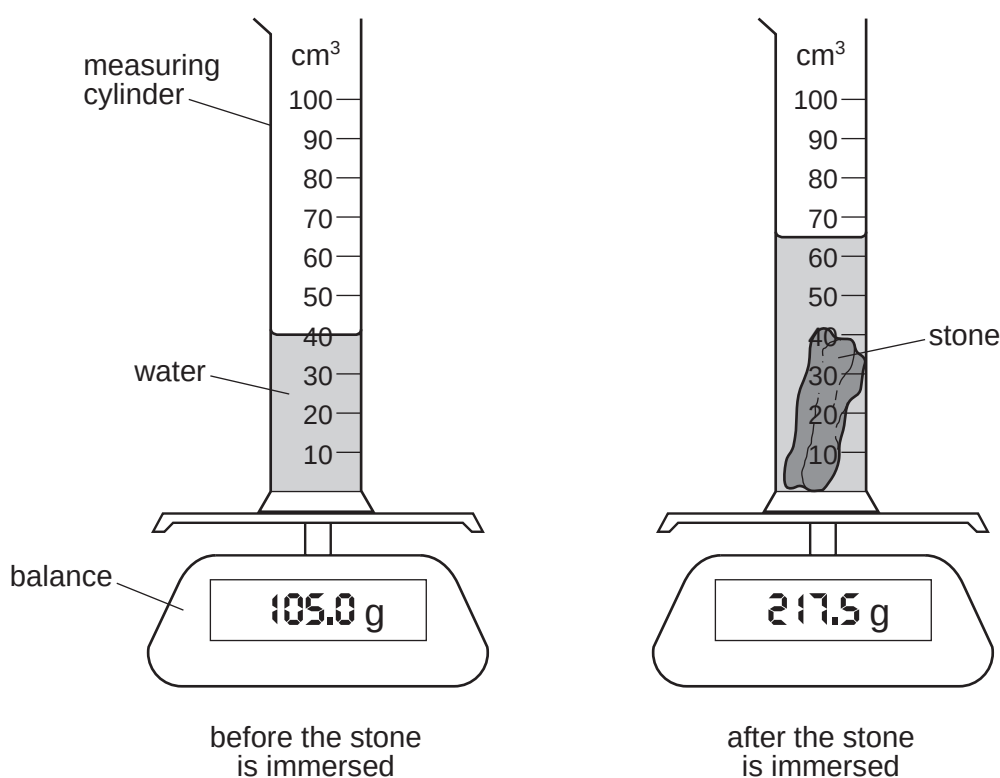
- 5 A student is weighed on laboratory scales.

Which row about weight and mass is correct?

	unit of weight	unit of mass
A	kg	kg
B	kg	N
C	N	kg
D	N	N

- 6 A measuring cylinder containing only water is placed on an electronic balance. A small, irregularly shaped stone is now completely immersed in the water.

The diagrams show the equipment before and after the stone is immersed.



What is the density of the material of the stone?

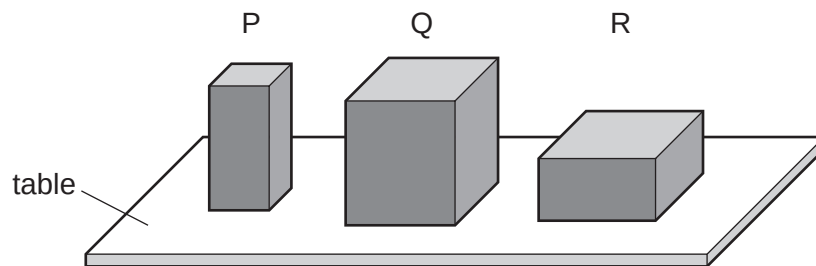
- A** 1.7 g/cm³ **B** 3.3 g/cm³ **C** 4.5 g/cm³ **D** 8.7 g/cm³

- 7 A boat is travelling at a steady speed in a straight line across the surface of a lake.

Which statement about the boat is correct?

- A The resultant force on the boat is in the direction of motion.
- B The resultant force on the boat is in the opposite direction to its motion.
- C The resultant force on the boat is vertically downwards.
- D The resultant force on the boat is zero.

- 8 The diagram shows three uniform, solid wooden blocks with a square cross-sectional area resting on a horizontal table.



Which list puts the blocks in order from the **least** stable to the **most** stable?

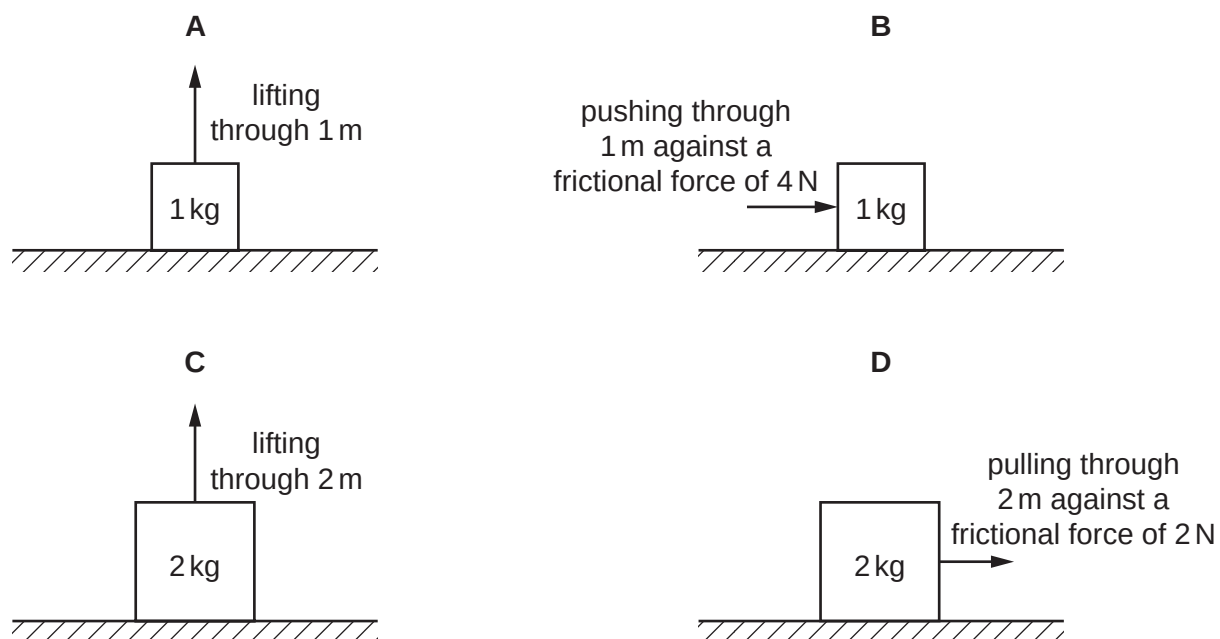
- A $P \rightarrow Q \rightarrow R$
 - B $P \rightarrow R \rightarrow Q$
 - C $R \rightarrow P \rightarrow Q$
 - D $R \rightarrow Q \rightarrow P$
- 9 Energy resources are used to generate electricity.

Which resource is renewable and does **not** release carbon dioxide when being used to produce electricity?

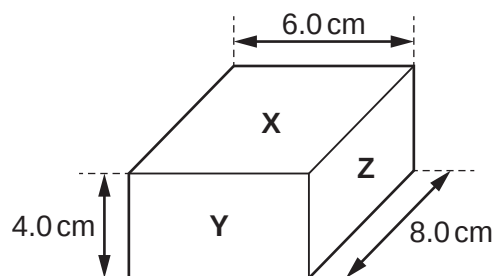
- A biomass
- B nuclear
- C oil
- D wind

10 A student does some simple exercises.

In which exercise is the most work done?



11 The diagram shows a box of dimensions $6.0\text{ cm} \times 8.0\text{ cm} \times 4.0\text{ cm}$.



The box rests on a flat horizontal surface.

On which face must the box rest to exert the least pressure?

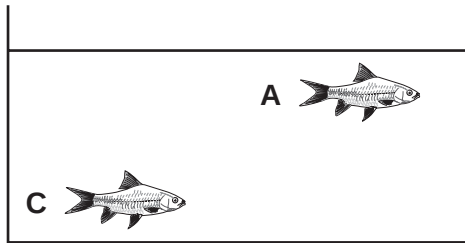
- A face X
- B face Y
- C face Z
- D The pressure is the same for all the faces.

- 12** Two identical fish tanks are filled to the same level with water.

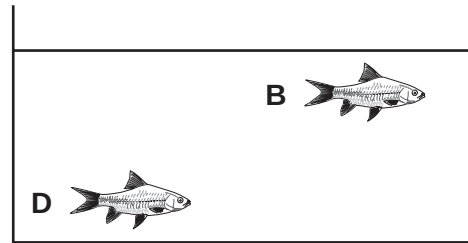
One tank contains fresh water. The other tank contains sea-water.

Sea-water is more dense than fresh water.

Which fish experiences the greatest pressure?



fresh water



sea-water

- 13** A student has a syringe which has its open end sealed. The syringe is completely filled with a certain substance, as shown in diagram 1. The student pushes the piston of the syringe in as far as possible, as shown in diagram 2.

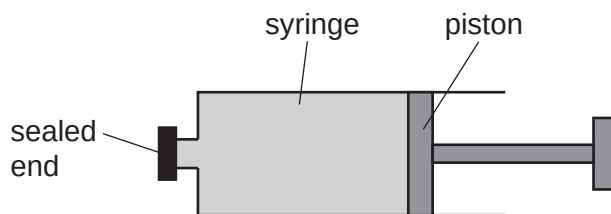


diagram 1



diagram 2

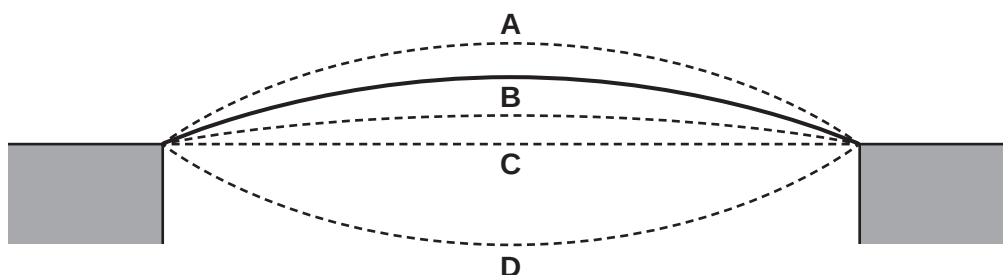
What is in the syringe?

- A** a gas and a liquid
 - B** a solid and a liquid
 - C** a liquid only
 - D** a solid only
- 14** When a microscope is used to look at smoke particles in air, Brownian motion is observed.
- What causes the smoke particles to move at random?
- A** Smoke particles are hit by air molecules.
 - B** Smoke particles are moved by convection currents in the air.
 - C** Smoke particles have different weights and fall at different speeds.
 - D** Smoke particles hit the walls of the container.

- 15 The diagram shows a metal foot bridge located in the Sahara desert where the temperature is much less at night than during the day. The ends of the bridge are firmly fixed to the sides of a narrow valley.

The solid line shows the bridge during the coldest part of the night.

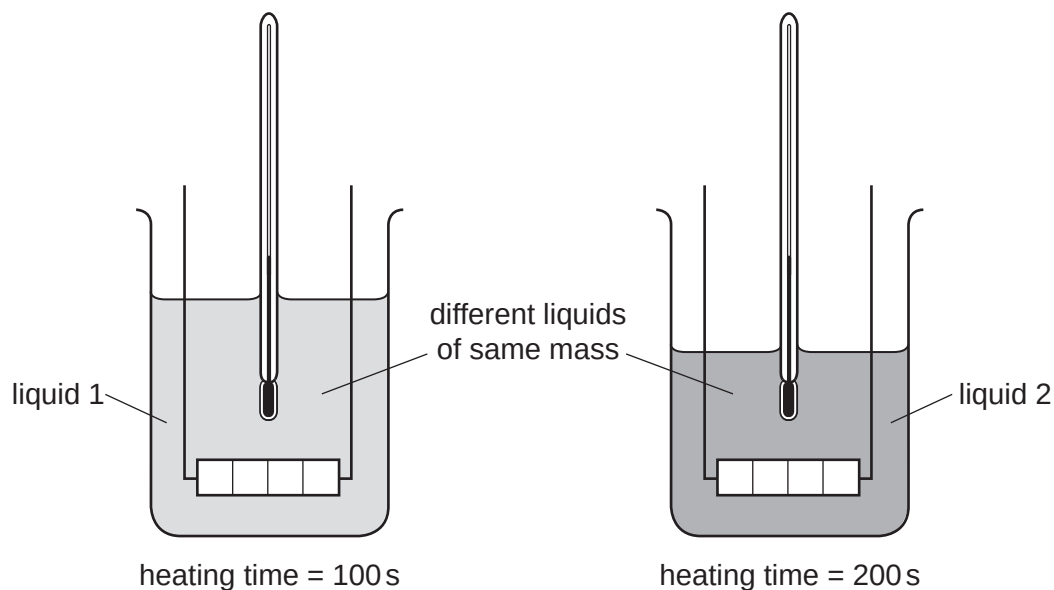
Which dotted line shows the bridge at the hottest part of the day?



- 16 Equal masses of two different liquids are put into identical beakers.

Liquid 1 is heated for 100 s and liquid 2 is heated for 200 s by heaters of the **same power**.

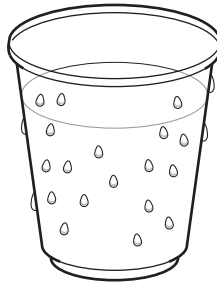
Each liquid has the same rise in temperature.



Which statement is correct?

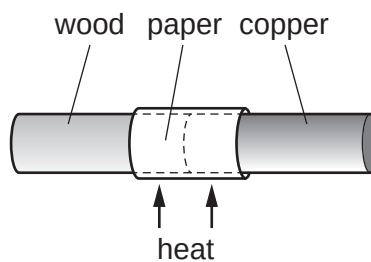
- A Each beaker of liquid has the same thermal capacity.
- B Each beaker of liquid receives the same energy.
- C Liquid 1 receives more energy than liquid 2.
- D The thermal capacity of liquid 1 is less than the thermal capacity of liquid 2.

- 17** It is a warm and humid day. A glass contains an iced drink. Water starts to form on the outside of the glass.



What is the name of the effect by which the water forms?

- A** condensation
B conduction
C convection
D evaporation
- 18** A copper bar and a wooden bar are joined. A piece of paper is wrapped tightly around the join. The bar is heated strongly at the centre for a short time, and the paper goes brown on one side only.



Which side goes brown, and what does this show about wood and copper?

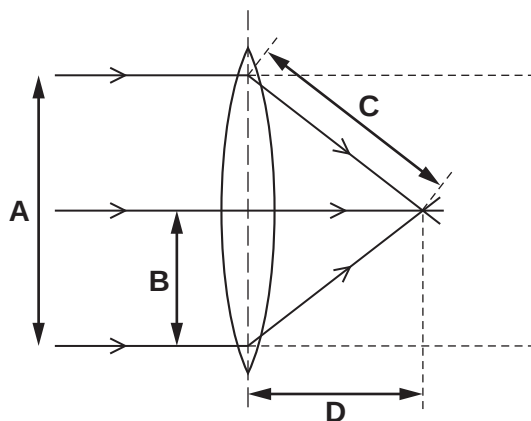
	brown side	wood	copper
A	copper	conductor	insulator
B	copper	insulator	conductor
C	wood	conductor	insulator
D	wood	insulator	conductor

- 19** Four identical metal plates are painted either black or white and have either a dull or a shiny surface.

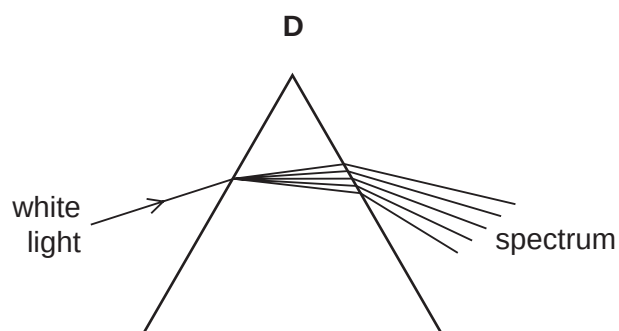
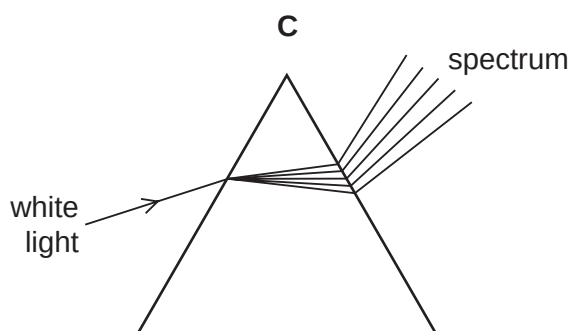
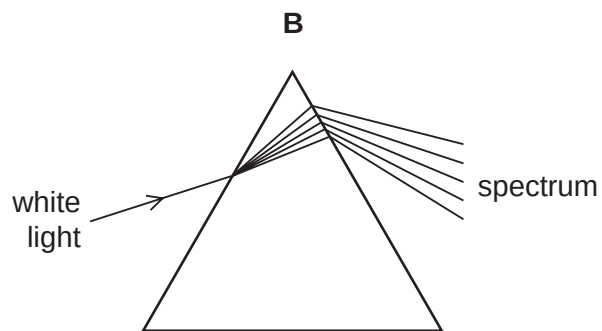
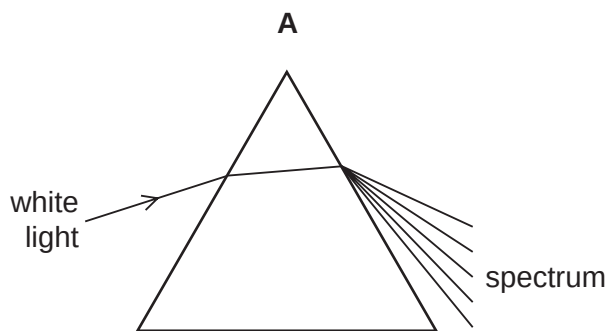
They are heated to the same temperature.

Which plate radiates thermal energy at the greatest rate?

- A** the plate that is dull and black
 - B** the plate that is dull and white
 - C** the plate that is shiny and black
 - D** the plate that is shiny and white
- 20** What causes the change in direction when light travels from air into glass?
- A** The amplitude of the light changes.
 - B** The colour of the light changes.
 - C** The frequency of the light changes.
 - D** The speed of the light changes.
- 21** The diagram shows rays of light passing through a converging lens.
- Which labelled arrow represents the focal length of the lens?

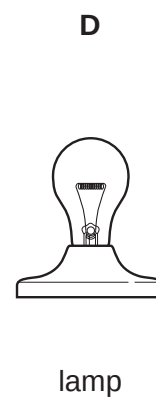
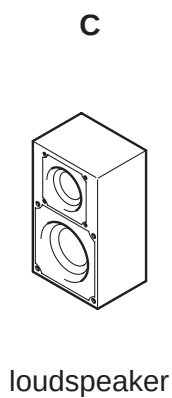
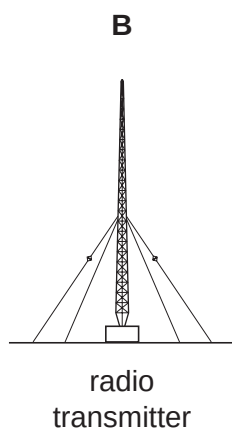


22 Which diagram shows what happens when a ray of white light passes through a prism?



23 The diagrams show four sources of waves.

Which source produces longitudinal waves?



24 The table shows different types of wave in the electromagnetic spectrum.

radio waves	micro-waves	infra-red waves	visible light	ultraviolet waves	X-rays	gamma rays
-------------	-------------	-----------------	---------------	-------------------	--------	------------

Where do all the waves travel at the same speed?

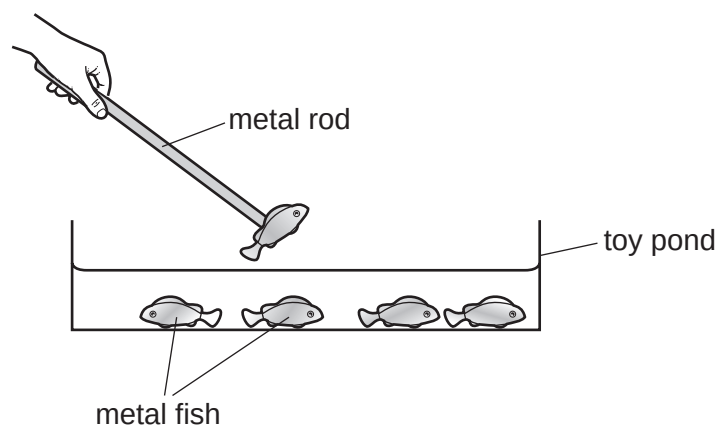
- A** in a vacuum
- B** in diamond
- C** in glass
- D** in water

25 A fire alarm is not loud enough and the pitch is too low. An engineer adjusts the alarm so that it produces a louder note of a higher pitch.

What effect does this have on the amplitude and on the frequency of the sound?

	amplitude	frequency
A	larger	greater
B	larger	smaller
C	smaller	greater
D	smaller	smaller

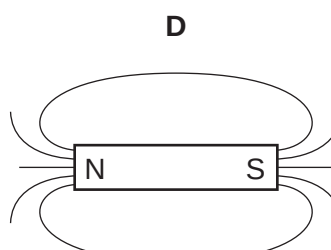
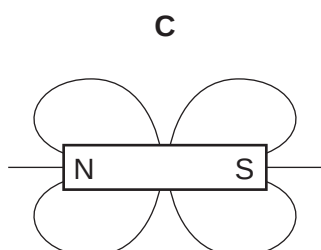
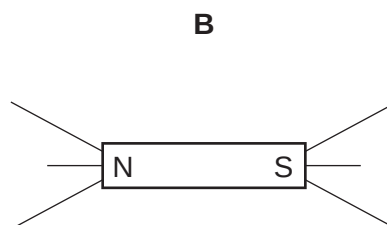
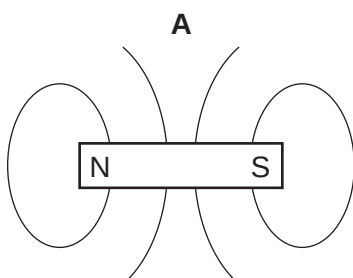
- 26 In a child's toy, a metal rod is used to lift metal fish out of a toy pond. The fish are magnetically attracted to the end of the rod. There is no magnetic force between the fish.



What are possible materials from which the fish and the rod are made?

	fish	rod
A	aluminium	soft iron
B	aluminium	steel
C	soft iron	soft iron
D	soft iron	steel

- 27 Which diagram shows the pattern of magnetic field lines around a bar magnet?



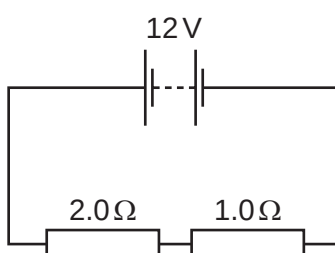
- 28 A student rubs a plastic rod with a cloth.

The rod becomes positively charged.

What has happened to the rod?

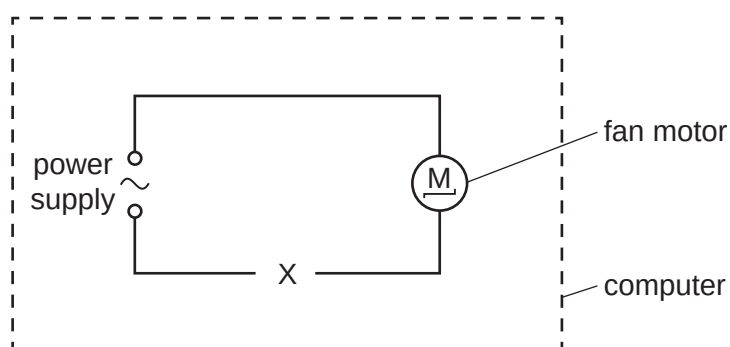
- A It has gained electrons.
- B It has gained protons.
- C It has lost electrons.
- D It has lost protons.

- 29 The circuit shows a 2.0Ω resistor and a 1.0Ω resistor connected to a 12 V battery.



What is the current in the 2.0Ω resistor?

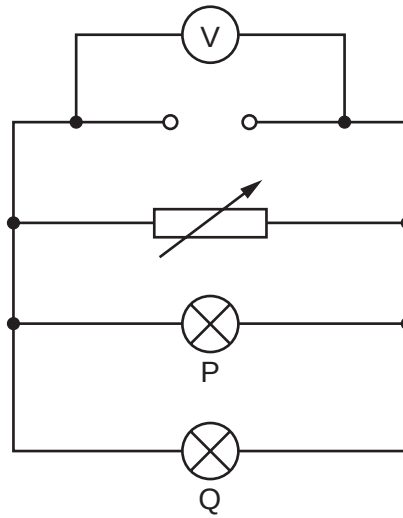
- A 4.0 A
 - B 6.0 A
 - C 24 A
 - D 36 A
- 30 A computer engineer wants the speed of a fan to increase automatically when the temperature inside a computer rises. The engineer knows that a larger current causes the fan to turn more quickly.



Which component should be placed at X to make this happen?

- A a relay
- B a thermistor
- C a transformer
- D a variable resistor

- 31 The diagram shows a power supply with a constant electromotive force (e.m.f.). It is connected to a voltmeter, a variable resistor and two identical lamps, P and Q. Both lamps are lit.

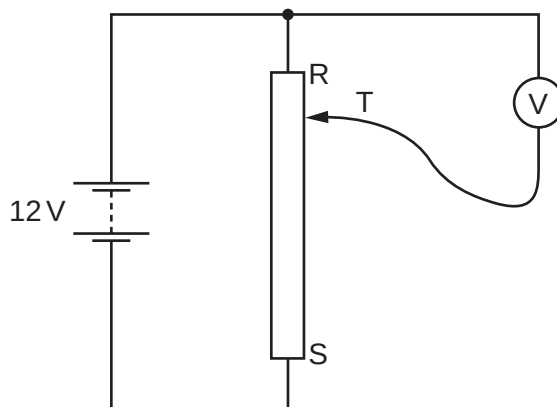


The resistance of the variable resistor is increased. The voltmeter reading remains unchanged.

What happens to the brightness of the lamps?

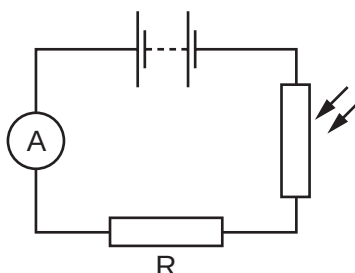
	lamp P	lamp Q
A	brighter	brighter
B	dimmer	dimmer
C	dimmer	unchanged
D	unchanged	unchanged

- 32 A student connects a variable potential divider (potentiometer) circuit.



As the sliding terminal T is moved from R to S, what happens to the reading on the voltmeter?

- A It decreases from 12 V to 0 V.
 - B It increases from 0 V to 12 V.
 - C It remains at 0 V.
 - D It remains at 12 V.
- 33 A light-dependent resistor (LDR) and a resistor R are connected in a series circuit. Light falls on the LDR.



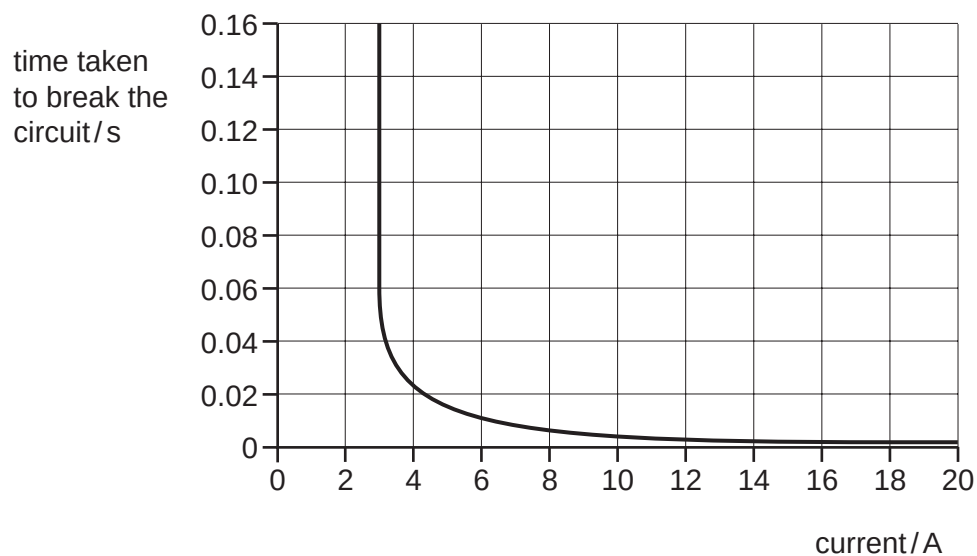
The brightness of the light falling on the LDR decreases.

What happens to the resistance of the LDR and what happens to the reading on the ammeter?

	resistance of LDR	reading on ammeter
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

34 A circuit-breaker is designed to protect a circuit which usually carries a current of 2 A.

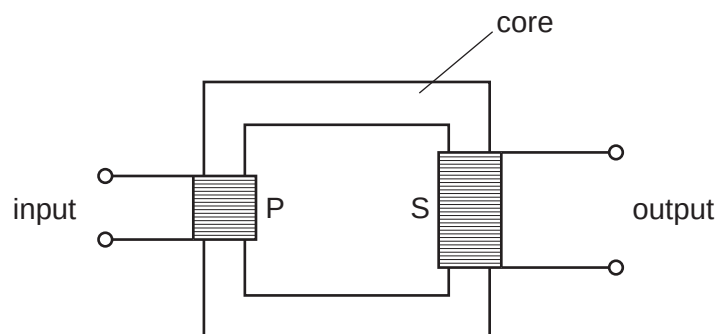
The time taken to break the circuit depends on the current, as shown in the graph.



What happens when the current in the circuit is 2 A and what happens when the current is 18 A?

	when the current is 2 A	when the current is 18 A
A	the circuit breaks in less than 0.01 s	the circuit breaks in less than 0.01 s
B	the circuit breaks in less than 0.01 s	the circuit does not break
C	the circuit does not break	the circuit breaks in less than 0.01 s
D	the circuit does not break	the circuit does not break

- 35** The diagram shows a transformer with primary coil P and secondary coil S. The input potential difference across coil P is V_P . The output potential difference across coil S is V_S . The number of turns in coil P is N_P . The number of turns in coil S is N_S .



Which equation relates the input and output voltages to the numbers of turns in P and in S?

- A** $\frac{V_P}{V_S} = \frac{N_P}{N_S}$
- B** $V_P \times V_S = N_P \times N_S$
- C** $\frac{V_P}{V_S} = \frac{N_S}{N_P}$
- D** $\frac{V_P}{V_S} = \frac{N_P}{(N_P + N_S)}$
- 36** What is the purpose of a relay?
- A** to change a large voltage into a small voltage
- B** to change a small voltage into a large voltage
- C** to use a large current to switch on a small current
- D** to use a small current to switch on a large current
- 37** Which row gives the relative charge of an electron, a neutron and a proton?

	electron	neutron	proton
A	−1	0	−1
B	−1	0	+1
C	+1	−1	0
D	+1	0	+1

38 An element has two isotopes.

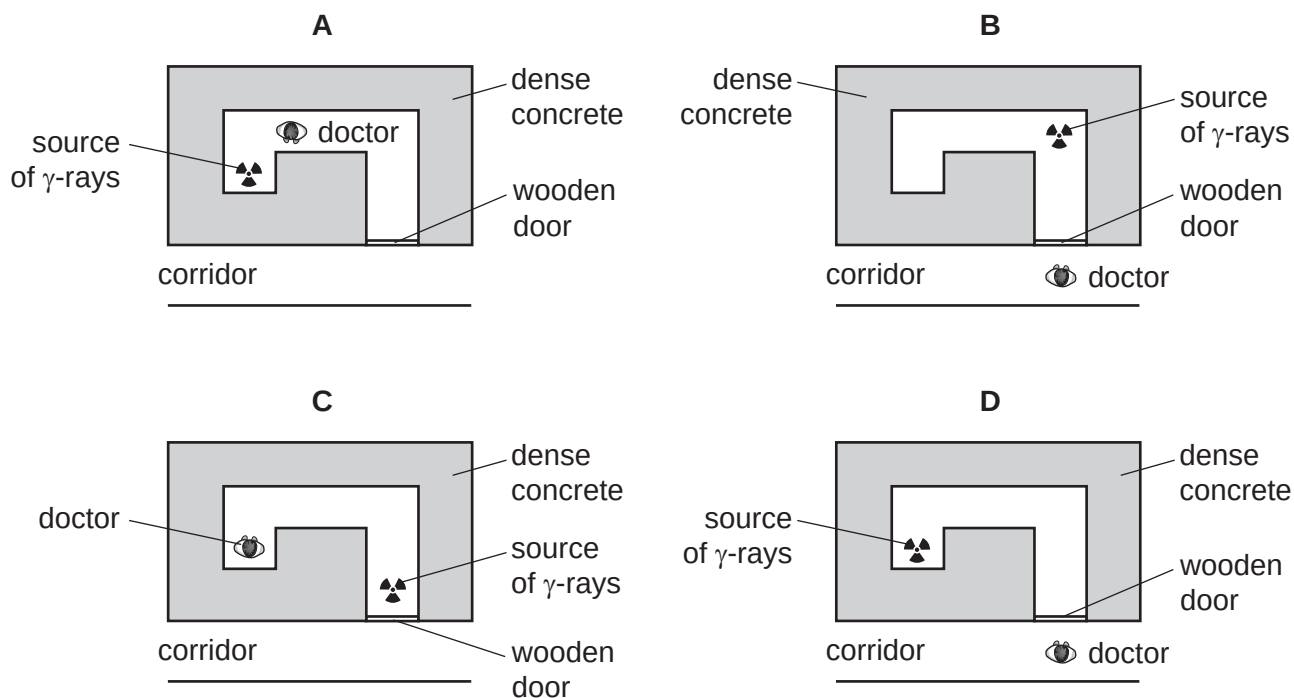
Which row compares the numbers of particles in the nuclei of the atoms of these isotopes?

	number of neutrons	number of protons	number of nucleons
A	different	the same	different
B	different	the same	the same
C	the same	different	different
D	the same	different	the same

39 A hospital doctor is using a source of γ -rays for a medical treatment.

Each diagram shows a view from above of the treatment room.

Which diagram shows the best way to protect the doctor and staff in the corridor from the γ -rays?



40 A sample of radioactive isotope is decaying.

The nuclei of which atoms will decay first?

- A** It is impossible to know because radioactive decay is random.
- B** It is impossible to know unless the age of the material is known.
- C** The atoms near the centre will decay first because they are surrounded by more atoms.
- D** The atoms near the surface will decay first because the radiation can escape more easily.

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0625/13

Paper 1 Multiple Choice (Core)

May/June 2017

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	C	1
2	C	1
3	C	1
4	D	1
5	A	1
6	C	1
7	B	1
8	B	1
9	C	1
10	D	1
11	D	1
12	C	1
13	D	1
14	A	1
15	C	1
16	D	1
17	C	1
18	D	1
19	A	1
20	D	1
21	A	1
22	D	1
23	A	1
24	C	1
25	A	1
26	A	1
27	B	1
28	A	1

Question	Answer	Marks
29	D	1
30	B	1
31	A	1
32	B	1
33	A	1
34	C	1
35	B	1
36	D	1
37	A	1
38	B	1
39	A	1
40	C	1



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PHYSICS

0625/12

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Question	Answer	Marks
1	C	1
2	B	1
3	C	1
4	D	1
5	B	1
6	B	1
7	C	1
8	A	1
9	B	1
10	D	1
11	D	1
12	C	1
13	A	1
14	B	1
15	B	1
16	D	1
17	D	1
18	C	1
19	C	1
20	D	1
21	A	1
22	D	1
23	C	1
24	A	1
25	A	1
26	C	1
27	A	1
28	A	1

Question	Answer	Marks
29	C	1
30	C	1
31	C	1
32	B	1
33	B	1
34	C	1
35	D	1
36	D	1
37	B	1
38	B	1
39	A	1
40	C	1



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PHYSICS

0625/11

Paper 1 Multiple Choice (Core)

May/June 2017

MARK SCHEME

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Question	Answer	Marks
1	A	1
2	B	1
3	C	1
4	A	1
5	C	1
6	C	1
7	D	1
8	A	1
9	D	1
10	C	1
11	A	1
12	D	1
13	A	1
14	A	1
15	A	1
16	D	1
17	A	1
18	D	1
19	A	1
20	D	1
21	D	1
22	D	1
23	C	1
24	A	1
25	A	1
26	D	1
27	D	1
28	C	1

Question	Answer	Marks
29	A	1
30	B	1
31	D	1
32	B	1
33	C	1
34	C	1
35	A	1
36	D	1
37	B	1
38	A	1
39	D	1
40	A	1



Cambridge International Examinations
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PHYSICS

Paper 1 Multiple Choice (Core)

0625/13

May/June 2016

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

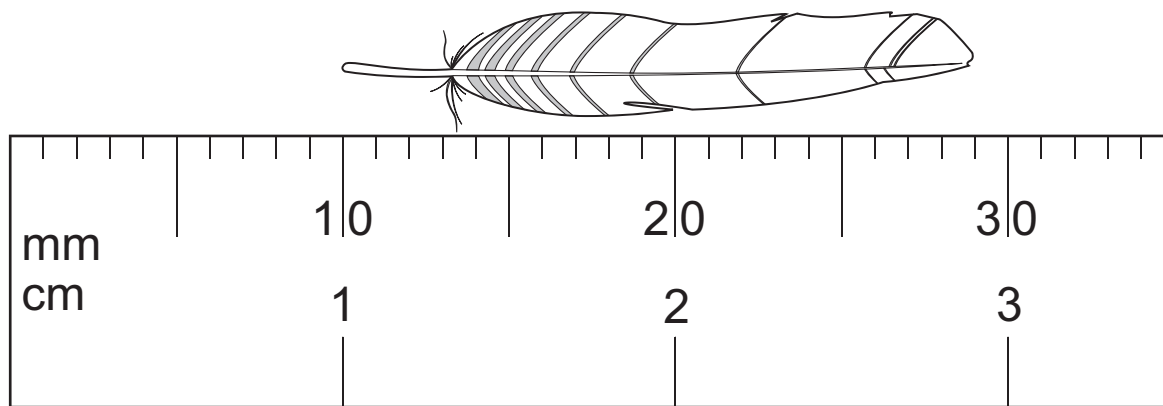
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

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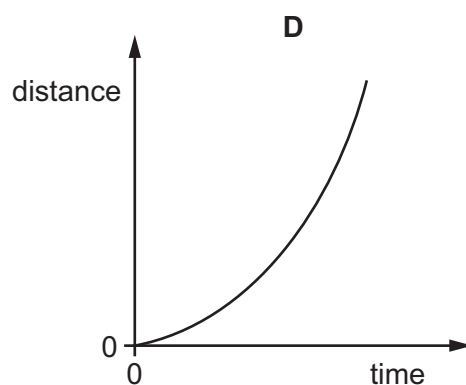
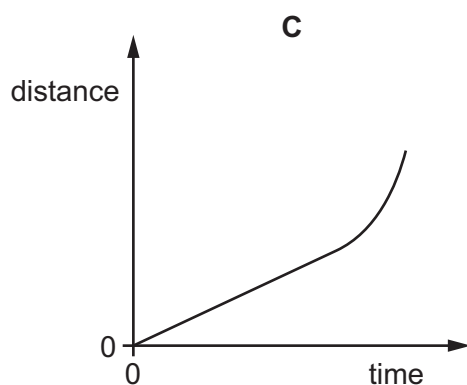
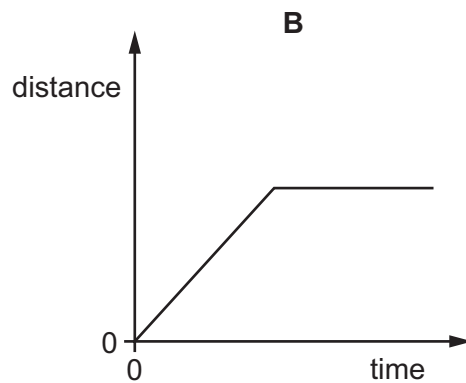
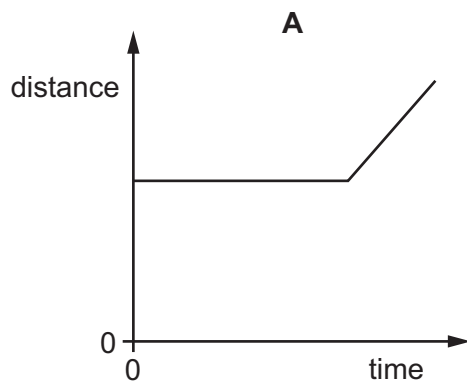


- 1 The diagram shows an enlarged drawing of the end of a metre rule. It is being used to measure the length of a small feather.



What is the length of the feather?

- A** 19 mm **B** 29 mm **C** 19 cm **D** 29 cm
- 2 An object moves at a constant speed for some time, then begins to accelerate.
- Which distance-time graph shows this motion?



- 3** A car travels at an average speed of 60 km/h for 15 minutes.

How far does the car travel in this time?

- A** 4.0 km **B** 15 km **C** 240 km **D** 900 km

- 4** A heavy object is released near the surface of the Earth and falls freely. Air resistance can be ignored.

Which statement about the acceleration of the object due to gravity is correct?

- A** The acceleration depends on the mass of the object.
B The acceleration depends on the volume of the object.
C The acceleration is constant.
D The acceleration is initially zero and increases as the object falls.

- 5** A cup contains hot liquid.

Some of the liquid evaporates.

What happens to the mass and what happens to the weight of the liquid in the cup?

	mass	weight
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

- 6 Diagram 1 shows a measuring cylinder containing water.

Diagram 2 shows the same measuring cylinder and water after 10 identical solid glass spheres have been added.

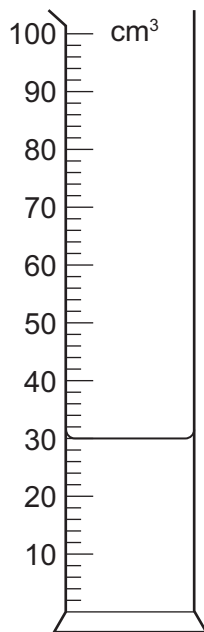


diagram 1

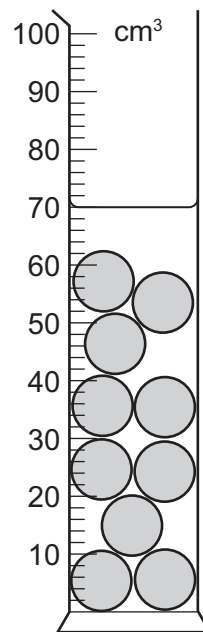


diagram 2

The mass of one of the spheres is 10 g.

What is the density of the glass from which the spheres are made?

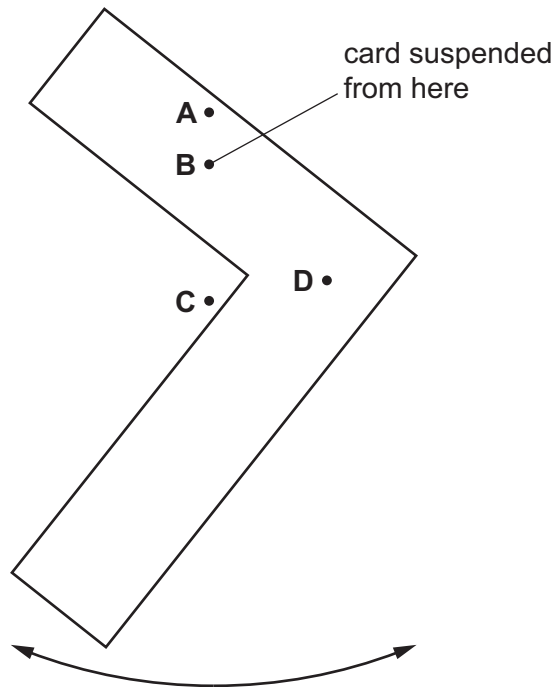
- A** 0.25 g/cm^3 **B** 0.40 g/cm^3 **C** 2.5 g/cm^3 **D** 4.0 g/cm^3

- 7 A feather falls through the air. Its downward speed is constant.

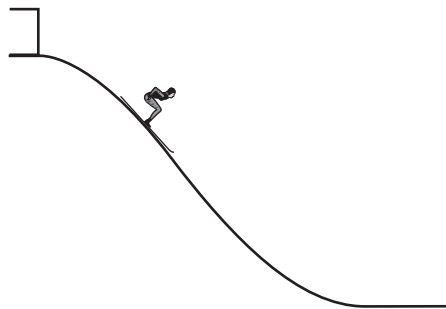
Which statement about the feather is correct?

- A** No energy is lost to the air as the feather falls.
B The air resistance acting on the feather is greater than its weight.
C The feather experiences no resultant vertical force.
D There is a resultant downward force on the feather.

- 8 The diagram shows an L-shaped piece of card suspended freely from a pin at **B**.
When the card is pushed, it swings and then comes to a stop in the position shown.
At which labelled point is the centre of mass of the card?



- 9 A skier walks from the bottom of a ski slope to the top and gains 10 000 J of gravitational potential energy.
She skis down the slope. At the bottom of the slope, her kinetic energy is 2000 J.



How much energy is dissipated in overcoming friction and air resistance as the skier moves down the slope?

- A** 2000 J **B** 8000 J **C** 10 000 J **D** 12 000 J

10 Which energy source is renewable and reliably available at all times?

- A coal
- B geothermal
- C nuclear
- D wind

11 Two workers are stacking cans on to a shelf in a shop. The workers lift the same number of identical cans on to the same shelf from the same level.

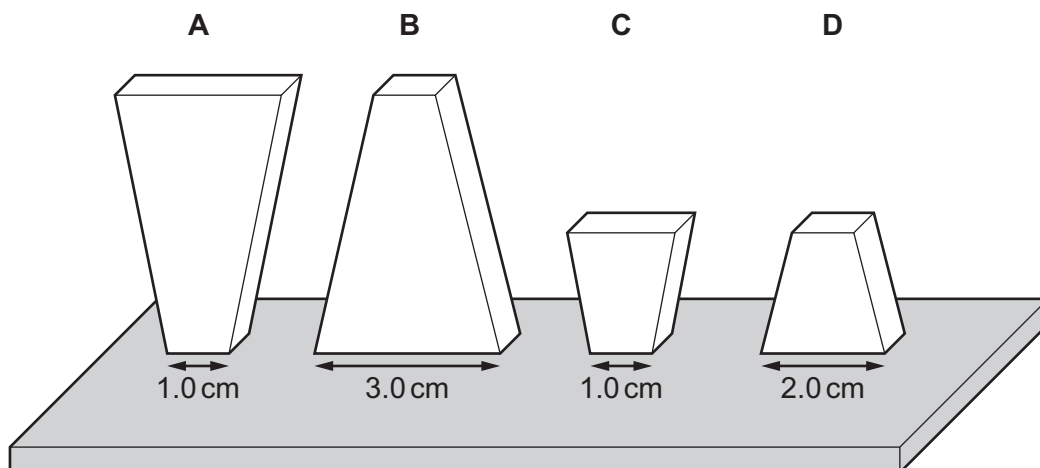
Worker P takes 3.0 minutes to lift the cans. Worker Q takes 4.0 minutes to lift the cans.

Which statement about the workers is correct?

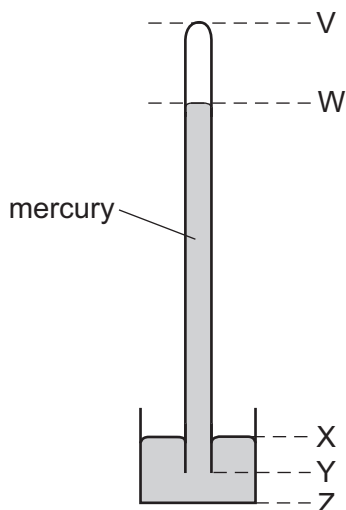
- A Worker P develops less useful power than worker Q.
- B Worker P develops more useful power than worker Q.
- C Worker P does less useful work than worker Q.
- D Worker P does more useful work than worker Q.

12 The diagram shows four solid pieces of the same metal. Each piece has the same thickness and stands on a bench as shown.

Which piece of metal produces the greatest pressure on the bench?



- 13 The diagram shows a simple mercury barometer.



The atmospheric pressure increases.

Which distance increases?

- A** VW **B** WY **C** XY **D** XZ
- 14 What causes the random, zig-zag movement (Brownian motion) of smoke particles suspended in air?
- A** air molecules colliding with smoke particles
B convection currents as the hot smoke rises
C smoke particles colliding with each other
D smoke particles reacting with oxygen molecules in the air
- 15 A sealed bottle of constant volume contains air.

The air in the bottle is heated by the Sun.

What is the effect on the average speed of the air molecules in the bottle, and the average distance between them?

	average speed of air molecules	average distance between air molecules
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

- 16 The melting points of ethanol and mercury are shown.

	melting point/°C
ethanol	−114
mercury	−39

Which of these two liquids is/are suitable to use in a liquid-in-glass thermometer to measure temperatures of -50°C and -120°C ?

- A ethanol only
- B ethanol and mercury
- C mercury only
- D neither ethanol nor mercury

- 17 A jug of water is at room temperature.

Several ice cubes at a temperature of 0°C are dropped into the water and they begin to melt immediately.

What happens to the temperature of the water and what happens to the temperature of the ice cubes while they are melting?

	temperature of the water	temperature of the ice cubes
A	decreases	increases
B	decreases	stays constant
C	stays constant	increases
D	stays constant	stays constant

- 18 Which quantity gives the *thermal capacity* of a beaker?

- A the thermal energy required to change the state of the beaker at constant temperature
- B the thermal energy required to raise the temperature of the beaker by 1°C
- C the total mass of hot liquid that the beaker can hold
- D the total volume of hot liquid that the beaker can hold

- 19 A student suggests some uses for containers made from good thermal conductors and for containers made from poor thermal conductors.

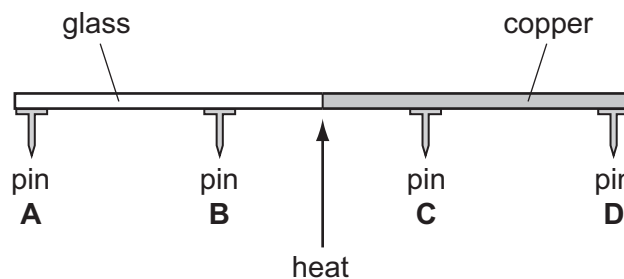
In which row are both suggested uses correct?

	good thermal conductor	poor thermal conductor
A	keeping a cold liquid at a low temperature	transferring thermal energy quickly from a hot liquid
B	keeping a hot liquid at a high temperature	keeping a cold liquid at a low temperature
C	transferring thermal energy quickly from a hot liquid	transferring thermal energy quickly to a cold liquid
D	transferring thermal energy quickly to a cold liquid	keeping a hot liquid at a high temperature

- 20 A rod is made half of glass and half of copper. Four pins **A**, **B**, **C** and **D** are attached to the rod by wax. The rod is heated in the centre as shown.

The pins fall off when the wax melts.

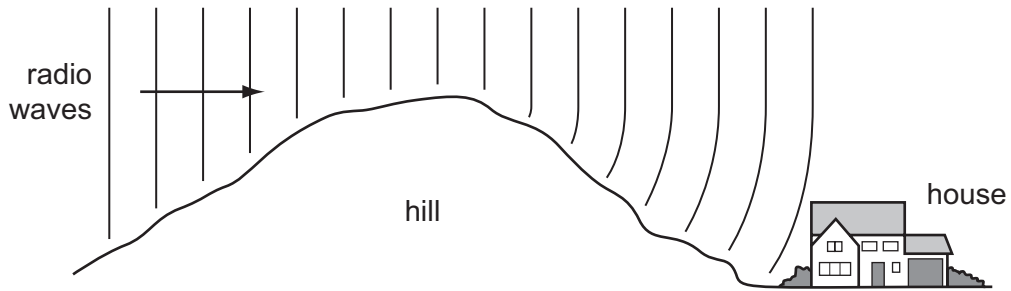
Which pin falls off first?



- 21 Which row shows the natures of light waves, sound waves and X-rays?

	light waves	sound waves	X-rays
A	longitudinal	longitudinal	transverse
B	longitudinal	transverse	longitudinal
C	transverse	longitudinal	transverse
D	transverse	transverse	longitudinal

22 Radio waves are received at a house at the bottom of a hill.

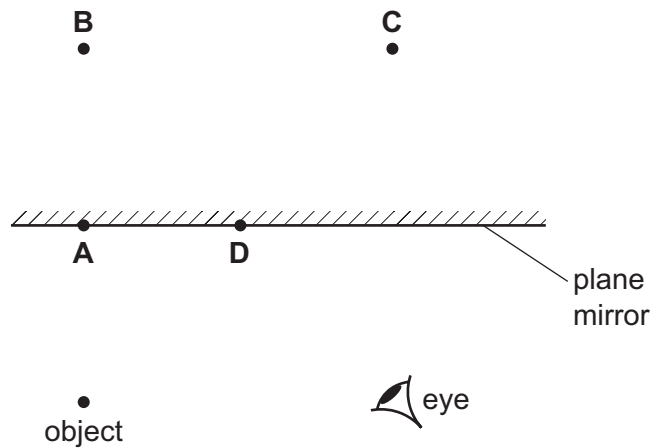


The waves reach the house because the hill has caused them to be

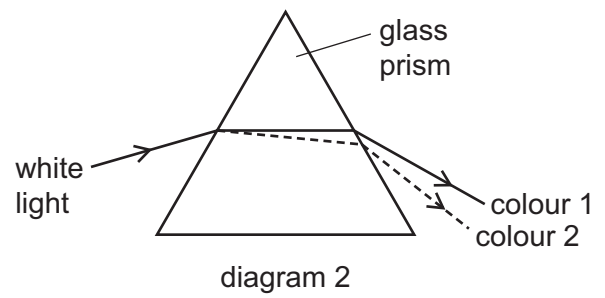
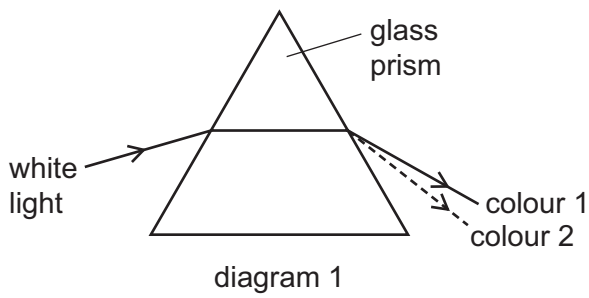
- A** diffracted.
- B** radiated.
- C** reflected.
- D** refracted.

23 The diagram shows an object in front of a plane mirror.

At which labelled position is the image of the object formed?



- 24** White light enters a glass prism. The light leaving the other side of the prism is separated into colours.



Which row correctly describes what happens?

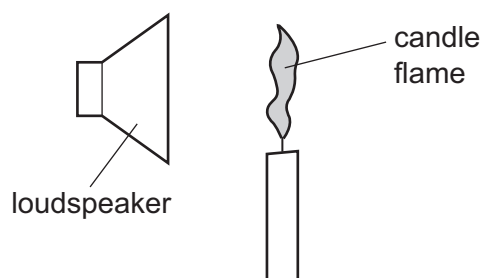
	path taken by the light	colour 1	colour 2
A	diagram 1	red	violet
B	diagram 1	violet	red
C	diagram 2	red	violet
D	diagram 2	violet	red

- 25** Which range of wave frequencies includes only sounds that can be heard by a human with normal hearing?

- A** 3.0 Hz to 300 Hz
- B** 30 Hz to 3000 Hz
- C** 300 Hz to 30 000 Hz
- D** 3000 Hz to 300 000 Hz

- 26 A candle flame is placed in front of a loudspeaker.

The loudspeaker produces a sound wave that causes air particles to vibrate. The vibrating air particles make the candle flame vibrate in the same direction as the air particles.



Which row shows the direction of vibration of the candle flame, and the nature of sound waves?

	direction of vibration	nature of sound waves
A		longitudinal
B		transverse
C		longitudinal
D		transverse

- 27 Which statement about magnetism is correct?

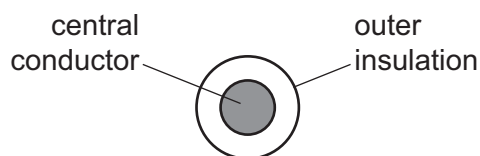
- A** An unmagnetised iron bar becomes magnetised when it is placed near a magnet.
- B** An unmagnetised steel bar can be magnetised by passing a current through it.
- C** The direction of magnetic field lines is from an S-pole to an N-pole.
- D** The N-poles of two magnets attract each other.

- 28 A student wishes to make a permanent magnet. She has an iron rod and a steel rod.

Which rod should she use to make the permanent magnet, and is this rod a hard magnetic material or a soft magnetic material?

	rod	type of magnetic material
A	iron	hard
B	iron	soft
C	steel	hard
D	steel	soft

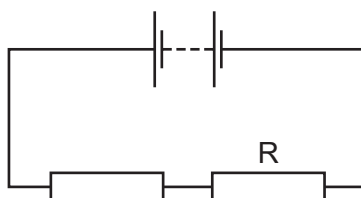
- 29 A flexible electrical cable consists of a central conductor and outer insulation.



Which pair of materials is suitable for the cable?

	central conductor	outer insulation
A	copper	plastic
B	lead	wood
C	plastic	copper
D	wood	lead

- 30 The diagram shows a battery connected to two resistors.



Four students separately measure the electromotive force (e.m.f.) of the battery, the current in the resistors, and the potential difference (p.d.) across resistor R.

Their results are shown in the table below.

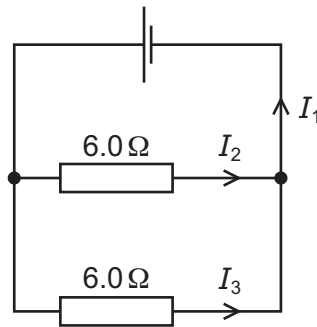
Which row shows values with their correct units?

	e.m.f.	current	p.d.
A	3.0 A	0.30 V	1.5 A
B	3.0 A	0.30 A	1.5 V
C	3.0 V	0.30 V	1.5 A
D	3.0 V	0.30 A	1.5 V

- 31 Why are lamps in a house lighting circuit connected in parallel rather than in series?

- A** If one lamp stops working, the remaining lamps become brighter.
- B** Less current is taken from the power supply.
- C** The lamps can be turned off independently using switches.
- D** When more lamps are added, their brightness decreases.

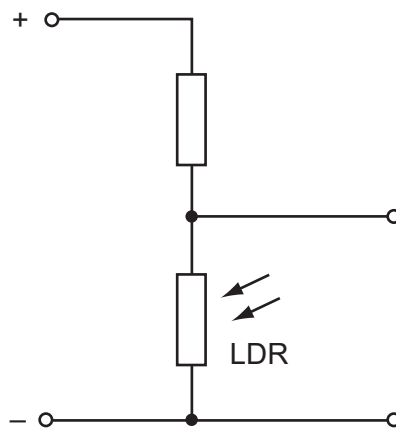
- 32 The diagram shows a cell connected to two 6.0Ω resistors. Three currents, I_1 , I_2 and I_3 , are labelled.



Which row correctly compares I_2 and I_3 with I_1 ?

	I_2	I_3
A	same as I_1	same as I_1
B	same as I_1	smaller than I_1
C	smaller than I_1	same as I_1
D	smaller than I_1	smaller than I_1

- 33 The diagram shows part of a circuit used to switch street lamps on and off automatically.



In the evening it gets dark.

Which row shows the effect on the resistance of the light-dependent resistor (LDR) and on the potential difference (p.d.) across it?

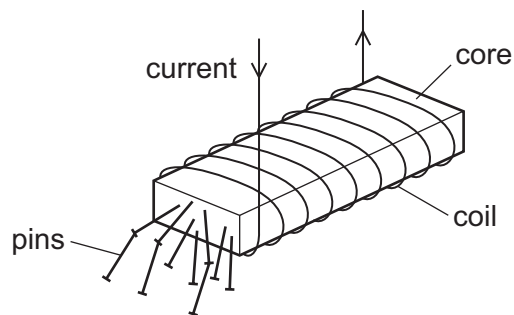
	resistance of LDR	p.d. across LDR
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 34** A domestic circuit includes a 30 A fuse. This protects the wiring if there is too much current in the circuit.

In which wire is the 30 A fuse positioned, and what does it do when it operates?

	position	operation
A	live wire	disconnects the circuit
B	live wire	reduces the current to 30 A
C	neutral wire	disconnects the circuit
D	neutral wire	reduces the current to 30 A

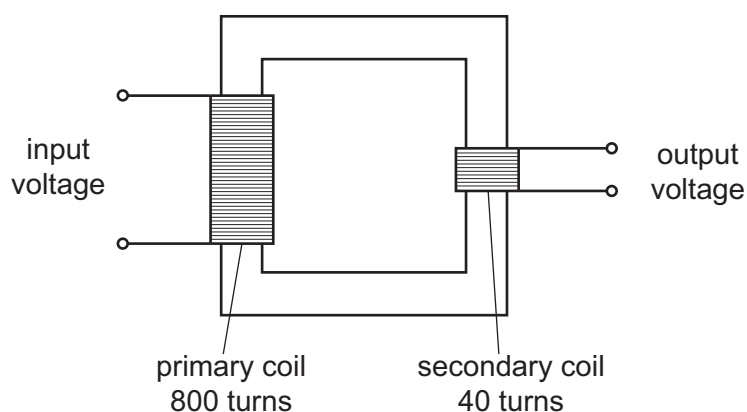
- 35** A strong electromagnet is used to attract pins.



What happens when the current in the coil is halved?

- A** No pins are attracted.
- B** Some pins are attracted, but not as many.
- C** The same number of pins is attracted.
- D** More pins are attracted.

- 36 The diagram shows a transformer.



The input voltage is 240 V.

What is the output voltage?

- A** 6.0 V **B** 12 V **C** 20 V **D** 40 V
- 37 A proton and a neutron are each close to a positive nucleus.



How does the charge on the nucleus affect the proton and the neutron, if at all?

- A** The neutron is attracted; the proton is repelled.
B The neutron is not affected; the proton is repelled.
C The proton is attracted; the neutron is repelled.
D The proton is not affected; the neutron is repelled.
- 38 Below are four statements about isotopes of a certain element.

Which statement about the isotopes **must** be correct?

- A** They are radioactive.
B They are unstable.
C They have the same number of neutrons.
D They have the same number of protons.

39 Radioactive materials should be handled carefully.

Which safety precaution does **not** reduce the risk to people using a radioactive material?

- A keeping the material a long distance from people
- B keeping the material at a low temperature
- C using lead screening between the material and people
- D using the material for only a short time

40 A sample of a radioactive isotope has an initial rate of emission of 128 counts per minute and a half-life of 4 days.

How long will it take for the rate of emission to fall to 32 counts per minute?

- A 2 days B 4 days C 8 days D 12 days

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Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS

Paper 1 Multiple Choice (Core)

0625/12

May/June 2016

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

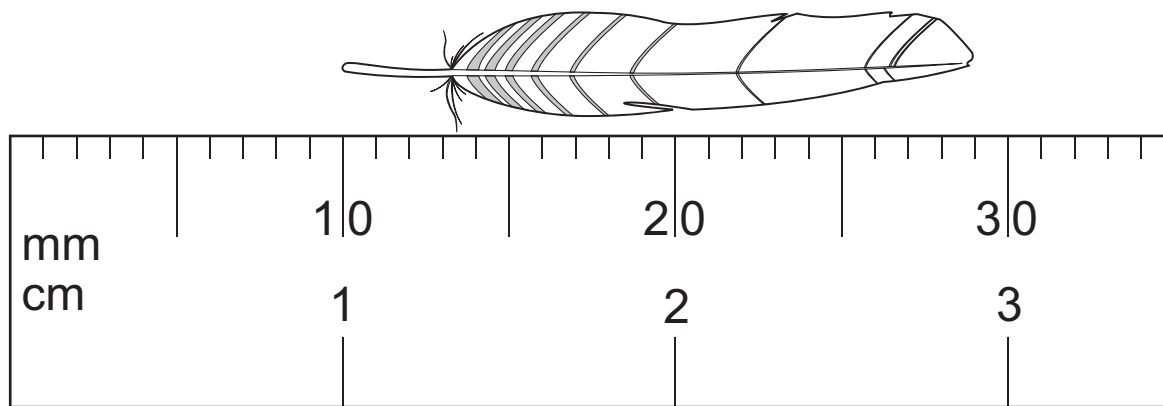
Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **16** printed pages.

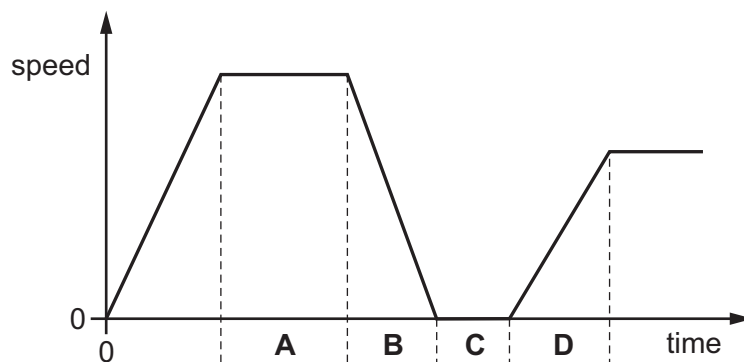


- 1 The diagram shows an enlarged drawing of the end of a metre rule. It is being used to measure the length of a small feather.



What is the length of the feather?

- A** 19 mm **B** 29 mm **C** 19 cm **D** 29 cm
- 2 The graph shows how the speed of a van changes with time for part of its journey.
- In which labelled section is the van decelerating?



- 3 A large stone is dropped from a bridge into a river. Air resistance can be ignored.

Which row describes the acceleration and the speed of the stone as it falls?

	acceleration of the stone	speed of the stone
A	constant	constant
B	constant	increasing
C	increasing	constant
D	increasing	increasing

- 4 A cup contains hot liquid.

Some of the liquid evaporates.

What happens to the mass and what happens to the weight of the liquid in the cup?

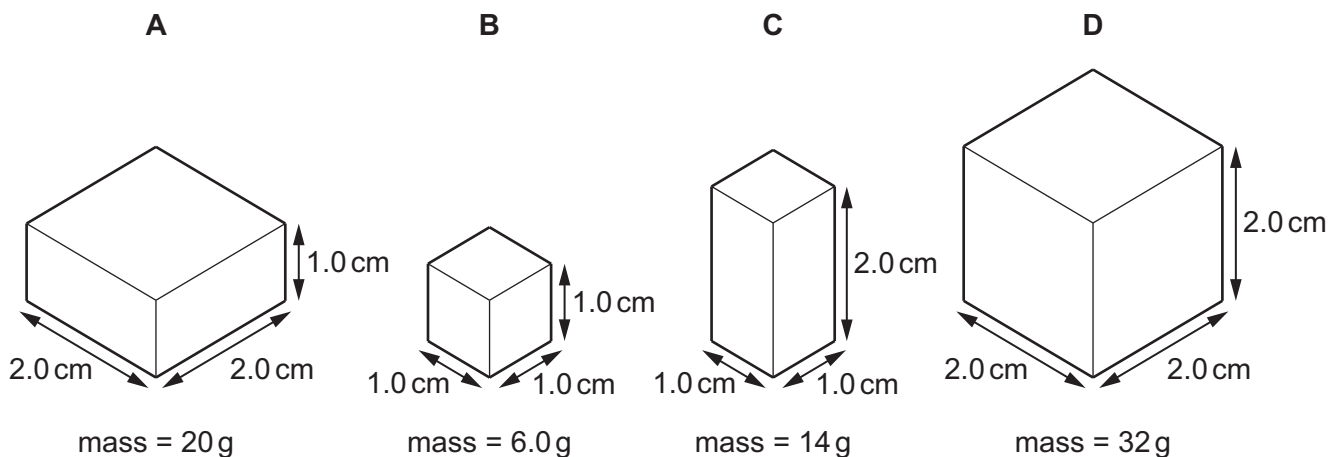
	mass	weight
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

- 5 A boy throws a stone. The stone leaves the boy's hand and moves vertically upwards. Air resistance can be ignored.

How should the force on the stone be described just after the stone leaves the boy's hand?

- A** downwards and constant
B downwards and increasing
C upwards and constant
D upwards and decreasing
- 6 The diagrams show the dimensions and masses of four regular solid objects. The objects are made from different metals.

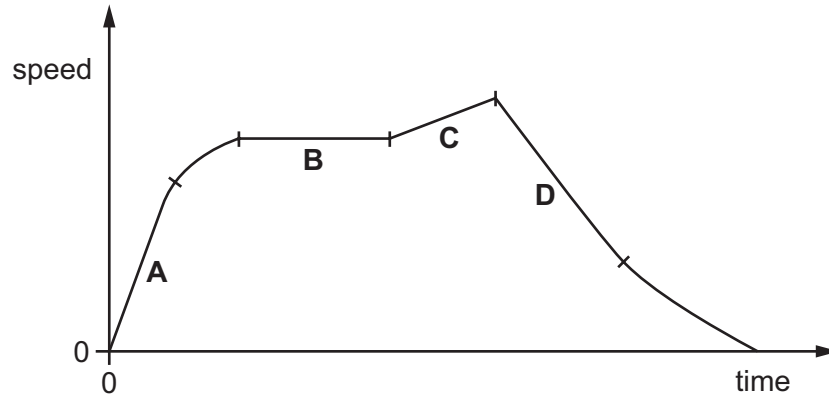
Which metal has the greatest density?



- 7 A car travels along a straight road.

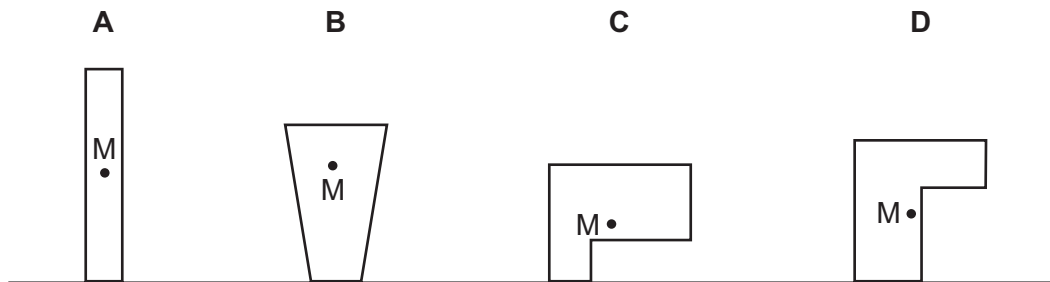
The speed-time graph for this journey is shown.

During which labelled part of the journey is the resultant force on the car zero?



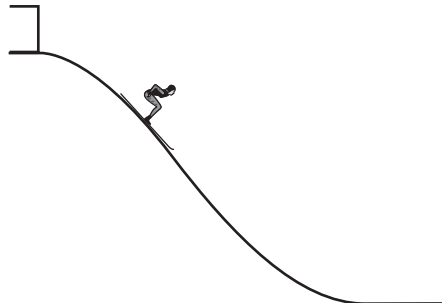
- 8 The diagrams show four objects **A**, **B**, **C** and **D**. The centre of mass M of each object is marked on the diagrams.

Which object is **not** in equilibrium?



- 9 A skier walks from the bottom of a ski slope to the top and gains 10 000 J of gravitational potential energy.

She skis down the slope. At the bottom of the slope, her kinetic energy is 2000 J.



How much energy is dissipated in overcoming friction and air resistance as the skier moves down the slope?

- A** 2000 J **B** 8000 J **C** 10 000 J **D** 12 000 J

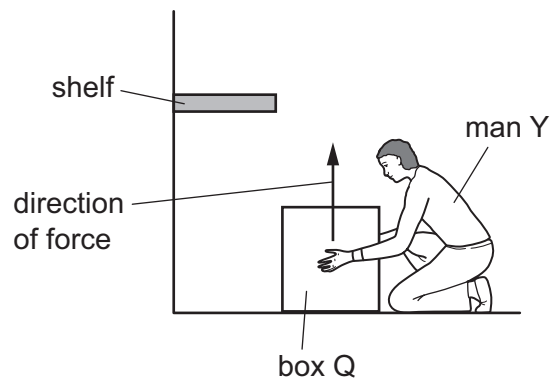
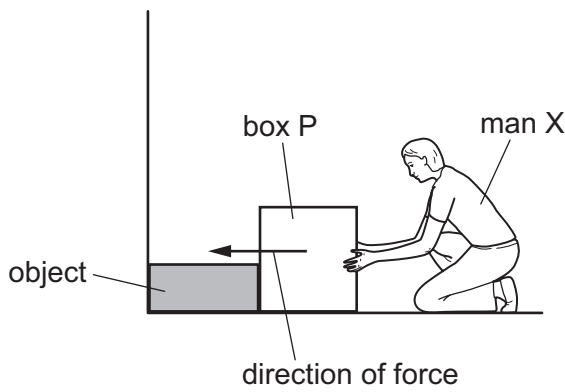
10 Which energy source is one that is used to boil water to make steam in power stations?

- A energy from tides
- B energy from waves
- C hydroelectric energy
- D nuclear energy

11 In a factory, two men X and Y try to move identical heavy boxes P and Q.

Man X tries to push box P along the floor. The box does not move because an object is in the way.

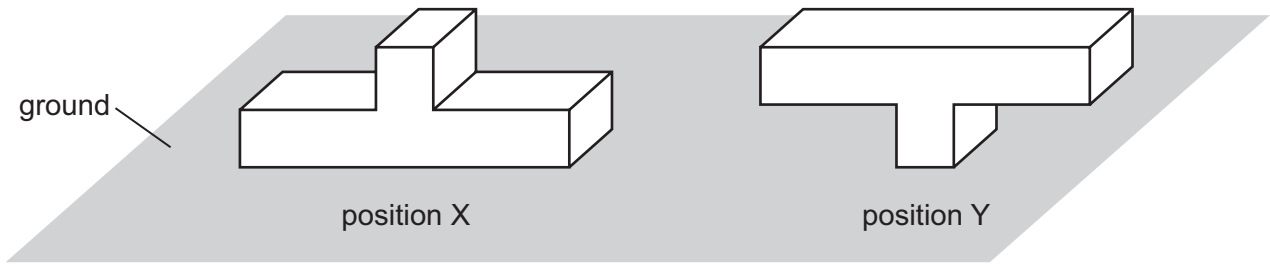
Man Y lifts box Q from the floor onto a shelf.



Which man does the most work on the box, and which box gains the most energy?

	man doing most work	box gaining most energy
A	X	P
B	X	Q
C	Y	P
D	Y	Q

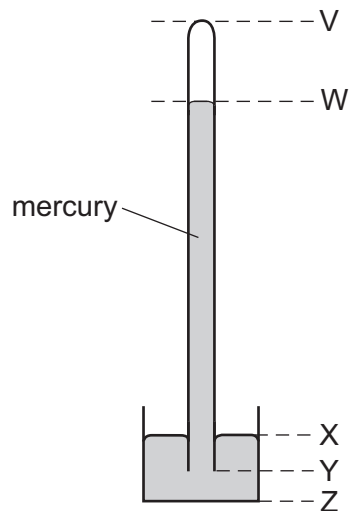
- 12 A T-shaped girder is placed on the ground in position X. It is then turned over to position Y.



What happens to the force on the ground and what happens to the pressure on the ground?

	force	pressure
A	increases	increases
B	increases	remains the same
C	remains the same	increases
D	remains the same	remains the same

- 13 The diagram shows a simple mercury barometer.



The atmospheric pressure increases.

Which distance increases?

- A** VW **B** WY **C** XY **D** XZ